

# S.E.P. Update 78

PUBLIC INQUIRY SUBMISSIONS

FISHERIES & OCEANS CANADA  
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May 1979



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## INTRODUCTION

This is Volume II of the report on the SEP/Update Inquiry 1978 process carried out during the fall and winter of 1978 and 1979. The SEP Update process was sponsored by the Salmonid Enhancement Program (SEP) of Fisheries and Oceans, Pacific Region in response to the public's wish that members of the SEP return to explain how the program incorporated their input, as a result of the initial Inquiry sessions held in the fall of 1976.

This document comprises the written submissions, briefs and letters the public delivered to Update 1978 at the various sessions. It is being published concurrently with the final report, so that it may act as a viable supplement to Volume I, the Report on the SEP Update/Inquiry 1978.

J.H. Boland  
Chairman  
SEP Update/Inquiry, 1978  
May, 1979

VICTORIA



November 1, 1978

SUBMISSION TO: The Salmonid Enhancement Program

FROM: Saanich Peninsula Tribal Council

We, the original Saanich inhabitants, do hereby declare that we will carry on our fishing as formally stated in our treaties.

We, the undersigned make this statement because we are aware of what Public Inquiries can do. Public Inquiries tend to make policies over certain issues, in this case fisheries. If it is the intention of this inquiry to completely do away with Indian Fishing Rights then we shall oppose this inquiry in every possible way.

We, the Native Indian people of this area will not allow the Provincial or Federal government to legislate any more of our rights away from us. We intend to secure and practice those rights as stated in our treaties.

These statements are made because we as Native Indians have not benefitted from the natural resources because they have not been paid for by either government. These natural resources still belong to us.

We wish to make it clear to the Salmonid Enhancement Program that if Native Indian people are not totally involved in this program it will be difficult for fisheries to reach its objectives, reasons being as follows:

- 1.) Most rivers and streams run through Native Indian reserves.

- 2.) There must be Native Indians involved when policies are developed.
- 3.) Native Indian people must be involved in other aspects of planning such as the development of sub-divisions which have a direct effect on the salmon.
- 4.) The logging industry MUST be controlled better than it has been done in the past and present.
- 5.) A closer look must be taken into the farming industry and their use of chemicals. It seems there has been much neglect in how they are used.
- 6.) A tighter control over the Municipal governments as they tend to approve all types of small industry which have an effect on the salmon.

We would like to repeat that the extent of this submission is to inform you that we will carry on our fishing rights as formally as before and we will protect our right.

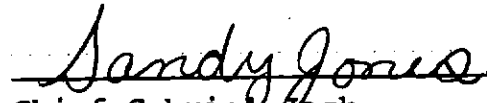
Thank you,



Chief Louis Claxton



Chief Norman Williams



Chief Gabriel Jack



Chief Thomas Sampson

#34 - 1240 Newport Avenue  
Victoria, B.C.  
V8S 5E7  
October 27, 1978

Update '78  
Salmonid Enhancement Program  
Fisheries & Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Gentlemen:

I am not sure I can attend your public meeting November 1st, and hence am forwarding these thoughts for your consideration.

I am a forester and a critic of our present logging oriented forest policy, as compared to a more rational silvicultural and environment-conscious policy. I am very much aware of the economics of logging and the necessity to keep our forest economy going and thriving, but by the same token, I am aware that our frenzy to use bigger and bigger logging equipment and clear-cut ever larger areas is no longer a valid forest management argument. It is a philosophy that has continued by rote from the old days and could not be defended if the public could be made aware of the harm and the alternatives.

Your salmon enhancement program is, in effect, part of an overall good forest management policy, by the fact that preservation of forest streams is part and parcel of multi-use forestry. Clear-cut logging may well be cheaper than selective logging or the cutting of small patches, but considering the harm done by excessive silting of spawning streams, the real economic savings by clear-cutting becomes questionable. To this, I could add other silvicultural arguments against clear-cut logging and also strong arguments to prove the feasibility of selective logging along most creek, river and lake shores to preserve a valid temperature and fish food control along these waterways.

I always feel that Fisheries people do not criticize sharply enough the casualness of the Forest Service and the industry, and fail to provide specific solutions to the conflicting problems. We foresters know that elsewhere in the world, the degradation, that is still taking place in B.C.'s forests, has long been overcome. It is not studies that are required, but the application of well known forestry technology. You weaken your position when you agree to more studies.

But you have also a powerful argument for your enhancement program, in that waterway protection and improvement (restoration) is an ideal Youth Employment opportunity. More so than just forestry improvement work, the rearing of fish makes waterway protection forestry more significant and provides more visible and immediate returns. Adding special fishing rights to a project employing unemployed youth, could make the work significantly more enticing. Don't under-rate the appeal, even to today's youth.

Sincerely,



D.M. Trew



Mr. H. English  
Advisor  
B.C. Wildlife Federation  
November 1, 1979

### Major Critical Factors in Salmonid Enhancement

Effective changes in the harvesting, that provides more adequate spawning escapements for depleted and unenhanced salmonid stocks, is a critical major requirement in salmonid enhancement.

Management and fishermen have to implement more effective ways, to reduce the destructive impact on unenhanced and lesser stocks, of the increased intensity of harvesting enhanced and larger stocks of fish.

Otherwise, at great cost, enhancement will simply displace large amounts of natural production with more vulnerable artificial production without increasing the total harvest.

An increase in the rate of community involvement in small stream rehabilitation is critical in the conservation of their resources.

Small streams will continue to lose their resource productivity - at the hands of other types of development oriented groups in the community; until they are effectively balanced by sufficiently highly motivated stream resource oriented groups in those communities.

Until responsible concerned community groups are given a share in the responsibility for their streams at a level that is meaningful to them, they will not respond effectively and develop their potential to care for the resources in their streams and SEP will be ineffective at the community level, without their assistance.

A BRIEF PRESENTED TO THE  
SALMONID ENHANCEMENT PROGRAM  
PUBLIC INQUIRY

Resulting from the public meeting held at the  
Empress Hotel, Victoria, B.C.  
Wednesday, November 1, 1978

by the

ROYAL CANADIAN NAVY ANGLERS ASSOCIATION  
C/O Ross Alexander  
Secretary  
#3 Rocky Point  
Victoria, B.C.

## I. BACKGROUND

The Royal Canadian Navy Anglers Association is a registered society under the British Columbia Societies Act and was established in 1954. The membership is comprised of members of the Canadian Armed Forces, former members of the Canadian Armed Forces, Civilian Employees of the Department of National Defence, their spouses and dependents.

Our membership during the year 1978/79 is 614 members. The Association is primarily involved in providing an organization for fostering interest in salmonid sportfishing among members of the Canadian Armed Forces stationed in the Greater Victoria Area. The Club has been associated with the Amalgamated Conservation Society for over 20 years, our members have been actively involved in Salmonid Enhancement projects on the Goldstream, Sooke River and DeMAMIEL Creek for many years.

## II. INTRODUCTION

As a result of recommendations made by the United Fishermen and Allied Workers Union to the Public Enquiries on Salmonid Enhancement in 1976 and the commentary made by Mr. KENT of the Fisheries and Marine Service during the 1 November Public Enquiry Meeting in Victoria, we of the R.C.N.A.A. now believe that it is no longer appropriate to adopt the "MOTHERHOOD" attitude towards S.E.P., it is now incumbent upon us to declare what we see as appropriate management of Salmonids to protect and foster our interests.

We do not advocate a reduction of catch by any segment of the user groups, be they Seiner, Gillnetter, Troller, Native Food Fishermen, or Sport/Recreational Fisherman. We hold the belief that a restructuring and relocating of the harvest is necessary for the benefit of all. We believe the Net Fishery is the area that requires the most in the area of restructuring and the following proposals will be aimed at making that point.

It is our overall basic belief that Salmonid Enhancement should mean more Salmonids for every user group and not less for some and more for others.

### III DISCUSSION

a. Mr. KENT declared at the 1 November, 1978 meeting that the Fisheries and Marine Service were considering reductions in Recreational Fishery catch limits and other restrictive regulations regarding that fishery, --- this action is being taken because -- among other things -- presentations were made at other previous Public Enquiries advocating this action.

These presentations were made by representatives of the United Fishermen and Allied Workers Union.

We believe that these proposals were spiteful and were prompted by greed, by an organization who has as its basic aim, to claim and control all of the Pacific Canadian Salmon Fishery. We irrevocably reject those proposals, the spirit of those proposals and the Fisheries and Marine Service's intention to act on them.

b. We believe that the June/July Net Fishery in Area 20 (Juan De Fuca Strait) should be eliminated immediately. The reasons for retaining this fishery are invalid.

This fishery was started in 1961 with the purported purpose of harvesting an early run of Sockeye -- Only once in 18 years has the sockeye run materialized and this fishery has been sustained by the harvest of immature Coho and an incidental catch of Chinooks.



III DISCUSSION (b. Continued)

We have heard Fisheries and Marine Services, Senior Personnel refer to the June/July Net Fishery in Area 20 as the "U.I.C. Fishery" and we believe that statement to be an indication that the fishery was basically started to allow Commercial Net Fishermen to qualify for U.I.C. benefits by assisting them to fulfill the minimum basics of dollar value landed and number of weeks employed -- this is hardly a good fisheries management principle.

We feel that if the Commercial Net Fisherman's lot is so tenuous as to require the collusion of the Fisheries and Marine Service to circumvent U.I.C. Regulations, then these fishermen should seek another line of work.

c. Mr. Les EDGEWORTH in his 1 November 1978 presentation stated that the Salmonid Enhancement program has opted for the concept of obtaining "the biggest bang for the buck". This statement implies that the system that is the cheapest and that provides the most salmon will be pursued. This we project, as meaning that the "Lake Fertilization Program" will receive pre-eminence.

We believe that this concept violates the basic axiom of Salmonid Enhancement and will result in the flooding of the fisheries with Sockeye, the harvesting of which will adversely effect the unenhanced species, due incidental catches in a stepped up net fishery.

We believe that restructuring and relocation of the harvest is necessary so that the net harvest takes place after

III DISCUSSION (c. Continued)

species separation has occurred and the target species can be harvested without undue incidental interception.

The second aspect of our belief is that S.E.P. should, as far as possible, and within the limits of reason, embark upon programs to enhance species in trouble, namely Chinooks of the Fraser River and Gulf of Georgia. The Concept of restricting the Recreational Fishery, to protect Fraser River and Gulf of Georgia Chinooks is unacceptable to this Association, in view of S.E.P.'s lack of a programme to enhance these areas.

The Fisheries and Marine Service must come to the realization that the Recreational Fishery is a bonafide "user group" and that the recreational fisherman is exercising a basic Canadian right to capture a Pacific Salmon. It is no longer sufficient for the FMS to pay "lip Service" <sup>to</sup> recreational fisheries recognition.

We believe that it is the duty of S.E.P. to provide salmon for the Recreational/Sportsfishery in the Gulf of Georgia in sufficient quantities to provide a satisfactory fishing experience for its participants. The level of satisfaction to be determined by the "user group" and not FMS.

We believe that the present recreational catch limits are the minimum acceptable level. They have been arrived at by several successive upper limit cuts, a reversal of the downward trend is now considered to be a priority.

III DISCUSSION (Continued)

d. Consistent with our contention that the net harvest need be restructured, it is our opinion that, when all of these net species projects come on line, that --- the "gauntlet" fishery in both Juan De Fuca and Johnston Straits be eliminated. More immediately, we believe ~~that~~ when the major runs of Sockeye, Chums and Pinks have past their peak of harvest, these gauntlet fisheries should cease.

If there is any degree of wonderment as to why we have adopted the foregoing position, we need only remind the FMS of the interception problems in those gauntlet areas. We fail to see how the protection of unenhanced Gulf of Georgia and Fraser River Chinooks can be appropriately managed when they are subject to such large incidental catch pressure so far from their home rivers.

We also believe that the net fishery targeted on Chinook salmon, particularly in Johnston Strait, Queen Charlotte Sound and the Fraser River should cease or at least be drastically restricted.

The Gill Net Fishery in "Rough Water" areas such as Juan De Fuca Strait should cease because of the "Drop Out" of gill netted fish, this loss amounts to a waste of as much as 30%.

#### IV GENERAL

It seems, from our point of view, that the FMS sees their success manifested in how much in dollar value is realized from the Commercial Fishery. We hold that the Recreational/Sport Fishery and Tourist Fishery is a major contributor to the economy of British Columbia and Canada as well as being an integral part of the quality of life.

A healthy West Coast recreational fishery causes many Canadians to vacation in Canada and reduces our unfavourable national position regarding tourist outflow. Our 1.4 Billion dollar balance of tourism deficit could be a lot higher. The prospect of good sportfishing is a tremendous lure for tourist inflow. Native Indians Operating Marinas and Guide Services are dependent to a great degree on a healthy recreational fishery.

The potential of tourist spending retention and foreign tourist capital inflow in our opinion, has not been appropriately researched by the FMS. We believe that no advocate of this potential exists in FMS, to present an adequate evaluation to the Treasury Board, to offer justification for Salmonid Enhancement monies for Chinook and Coho. As a consequence we believe the "Biggest Bang for the Buck" approach, is a short sighted and short range concept and is unworthy of being considered as S.E.P.'s best effort, saleable to the Treasury Board yes, but far from adequate from the recreational/sportfishing point of view.



# SOOKE CONSERVATION SOCIETY

P. O. Box 688 Sooke, B.C. V0S 1N0  
947

November 24, 1978

Update '78  
Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sirs:

The annual meeting of the Society held on November 15th instructed me to write to you on two subjects pertaining to the Sooke River.

The first is the necessity for the building of a bypass around the falls to give spawning fish access to a further 11½ miles of good spawning and rearing area. This would be about triple the lower spawning area.

The second is the need for enhancement of the chinook run to the Sooke River, which in the last six years has dwindled from a high of 3,000 spawners in 1972 to six this year. In each of the last three years there have been less than 20 chinook spawning.

We have access to graphs showing a long, slow increase from 1952 to 1972, and then this unexplained sharp decline.

However, we do know that latterly there was a noticeable shortage of males on the spawning beds and a similarly noticeable preponderance of males in the local catch.

In any case, the Sooke River certainly needs enhancement of its chinook and coho stocks.

Your program has asked for public participation. We applied for 50,000 chinook eggs for our incubation boxes this year and were recently refused with no proper explanation from the transplant committee. Our suggestions that we be allowed to take some of the few chinooks for our incubation boxes have been similarly frowned on in the past.

This Society has been doing its part for the past six years by summer fry feeding of coho on Demamiel and Charters tributaries, and for the past four years by the operation of coho incubation boxes on Mary Vine Creek using spawners from Demamiel Creek. Some of our members were also instrumental in the construction of a reservoir on Bradford swamp that ensures a viable summer flow for Demamiel Creek. The coho run to this creek is in fairly good shape in contrast to the rest of the system.

We feel that it is about time that Fisheries bestirred themselves and did some work on the Sooke River instead of these everlasting studies with no ensuing action.

A good start would be the provision of 150,000 chinook for our incubation boxes this winter.

Yours truly,

A handwritten signature in cursive script, appearing to read "W. James".

William James, Secretary

Mr. & Mrs. Fred Kockott  
5300 Santa Clara Avenue  
Victoria, B.C.  
V8Y 1W4  
January 2, 1979

To Whom It May Concern:

Regarding salmon fisheries programs on the West Coast of B.C. in general and the Victoria area in particular.

I have lived on the Saanich Peninsula for 40 years, over 20 of which I have been actively interested in salmon fishing. It would be pointless to further mention the already acknowledged decline in salmon catch in this area despite attempts to improve it.

I have attended the SEP meetings held in Victoria on both occasions and listened with interest to the aims and proposals put forth in the first meeting (I submitted a letter at that time). At the recent meeting in 1978, I am sorry to say I get the impression of just another government bureaucracy. Building a little empire with the salmon as a front, providing profitable jobs as the salmon are surely but steadily being depleted.

I'd like to point out all this time I have returned no less than half a dozen marked heads and with one exception, they all were released in American rivers. This leads me to believe my primary catch is American release (or maybe USA just marks more fish).

I submit: (a) Since our forest people claim to be willing to work with the Fisheries in re-enhancement, it is past time that every small creek and stream in any re-established logged watershed be given at least a four year repeated seeding of salmon to re-establish runs in minor streams again. It is plain to see where the new forests are now growing because of reseeding and transplanting but when will a new run of salmon be re-established in these creeks? To last another century till it is logged again. The forest industry should be responsible for the cost of this venture. (b) In hatchery programs, why not use fertilized eggs from a mammoth run such as Nahmint or Rivers Inlet chinook to start similar giant strains in rundown areas requiring enhancement? (c) So long as our herring and other salmon food such as shrimp, etc., is being harvested to near depletion, the salmon we do have will be starved from lack of abundant food to minimal size. (d) How can a situation such as Nahmint 1978 be tolerated by any individual when the bay is closed to all fishing and the adjacent water opened to commercial net boats, when the dwindling return to this great river is already acknowledged. I know the extra large return to the Somass River is the excuse, but to harvest them in the canal instead of at the particular river mouth is the idea of an idiot. I know of eye witnesses, to holds full of mild cure chinook taken in that venture. (e) Those interested in salmon at the Sooke River, as noted by others, trying to protect their decreasing return of spawners from predator seals should have had immediate department help and lacking when

necessary, not after it's too late. Apparently a run established to Millstream suffers similarly as the ever growing herds of seals take their toll. It was pointed out also, the damage waterfowl can do to fish holding pools in rivers. (f) Sports fishermen are discussing a licence fee. A few years ago I was very much in favour of such a fee but in recent years, my feeling has changed. I say, "What do I get for my fee I don't already have"? The fact I pay a licence won't guarantee my kid a salmon in the future anymore than if I didn't pay. What it would do, apparently, is pay someone who appears very vague, judging by some of the recent releases, to make alot foolish suggestions about how many fish a sportsman catches compared to a commercial boat, why he shouldn't do this and has to do that. The entire fishery is geared around the commercial harvest and as such let the guy who buys the can of salmon pay for it. In other words, "tack it on up in front where it belongs", (the same applies to the logging companies to cover costs of stream reseedng). (g) They also speak of limit setbacks. One day we are promised as many salmon as we had 20 years ago, the next we get a limit cut. Idiocy! Let me ask this, "As long as a man makes use of what he catches, why should he be restricted to what he takes up to our present limit?" We have experienced some pretty stupid restrictions in the shellfish department also. If a man uses what he catches for his family, he should be entitled to it. (h) We hear also of gear restrictions. I am particularly interested in the use of the heavy trolling gear referred to as meat line or down rigger. I have fished in this manner for years, I fail to see any reason it should be discontinued. I take my 4 fish, I go home. I have considerable investment in my outfit and I resent being forced to change my fishing style without valid reason. The fish I take are always a good size, seldom do I catch grilse. The fish has a better chance to break loose on solid gear than on a spindly rod and strictly nylon line. If any restrictions are placed, in my opinion, they should disallow treble hooks and jigging type lures which have recently flooded the market. The lure is deadly to small fish and once taken on treble hooks, can not be released without fatal damage. (i) Size limits: I am very definitely opposed to people fishing for grilse. I believe the size limit should be increased to disallow the taking of grilse unless a child is aboard where "one may be allowed" to encourage the youngster as it were. In this case, gear restriction may be the answer. I feel it is important to harvest jack chinook though, I am under the impression that when this runt mates with a giant, some of the offspring will be small (ie. Darwin) and I prefer to see an upgrading of our fish.

Though I have many other opinions, and I am sure they may be incorrect, I will not express them at this time. I would request though, that the fishermen be made more aware of the situation as it is. What is required to change it? Give us an explanation of the reasons for the changes and ask for the fishermen's opinions.

In closing, I would like to express one other opinion, though I don't think in this I am alone. Something should be done to restrict fishing derbys. Something should be done to restrict publications like the Daily Colonist from running their contest and what has been hardest of all on the fish in this area is the column written by Alec Merriman and others like him. Good luck.

Yours truly,

A handwritten signature in cursive script that reads "Fred Kockott". The signature is written in dark ink and has a fluid, somewhat stylized appearance with some loops and flourishes.

Fred Kockott





Province of  
British Columbia

OFFICE OF THE  
DEPUTY MINISTER

Ministry of the Lands, Parks and  
Environment- and Housing  
Parliament Buildings  
Victoria  
British Columbia  
V8V 1X5

January 9, 1979

File: 0327536

Update '78  
Salmonid Enhancement Program  
Fisheries & Oceans Canada  
1090 West Pender Street  
Vancouver, B. C.  
V6E 2P1

Dear Sir:

The Land Management Branch wishes to accept your invitation for written submissions concerning the Salmonid Enhancement Program.

The Land Management Branch has jurisdiction over unreserved provincial Crown land, including foreshore. Due to our administrative authority over industrial foreshore leases, the Branch has played a leading management and planning role in attempting to resolve conflicting land use problems associated with estuarine fish habitat and industrial use. The purpose of this submission is to bring to your attention the potential for increased estuarine land use conflicts as a result of the Salmonid Enhancement Program and suggestions for resolution of those conflicts. The Branch has previously expressed its concerns through input to the Provincial Working Group chaired by the E.L.U.C. Secretariat.

The benefits to the fishery resource of the Province by the Salmonid Enhancement Program are recognized and the Provincial Government is generally fully supportive of the program. However, like most programs, there are both positive and negative aspects that have to be carefully weighed and considered in reaching decisions in respect to implementation of the program.

The river estuaries are important areas of fish habitat, but the extent of habitat required to accommodate the fry that hatch in the rivers and subsequently migrate to the estuaries has never been determined. At present, it is not known whether there is sufficient estuarine habitat to support existing fish stocks, let alone whether there is surplus habitat to accommodate enhanced stocks. The question of how much fish habitat in the estuaries must be protected to ensure adequate habitat for the fry spawned in the river

...2/

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has not been quantified. It would not be good planning to produce large numbers of additional fry in upstream enhancement facilities only to lose them in the estuaries due to a lack of rearing habitat. The success of the Salmonid Enhancement Program is dependent upon all phases of the salmonid life cycle and not just freshwater fry production.

Depending on the nature of the operation, the location of industry is not necessarily in conflict with the objectives of the Salmonid Enhancement Program. However, in recent years the Land Management Branch has been involved in various studies and task forces, dealing with estuarine land use problems. In these studies, it was the position of the federal and provincial fisheries agencies that: (1) industries should be relocated from these estuaries, and/or (2) a major reduction in industrial activity should occur. Part of the argument for this position was that the industries were in conflict with the goals of previous or proposed salmonid enhancement projects. The Quinsam hatchery and the Campbell River estuary and the proposed Puntledge hatchery and the estuary in Comox Harbour are two specific examples.

If industry is to be phased out of key estuaries, after it has been scientifically determined that there is not adequate fish habitat present under existing conditions, the Branch is of the opinion that the cost of moving industry must be borne, at least partially, by the Salmonid Enhancement Program. It is unrealistic in light of current economics and unemployment to simply serve an ultimatum on industries located in these estuaries to vacate without providing an economically viable alternative and without providing financial assistance to relocate.

It must be kept in mind that while Provincial Government supports the Salmonid Enhancement Program, it also supports industry and industrial expansion within the Province and the two must go forward on a compatible basis.

There is not general agreement between provincial and federal agencies concerning the cost of financing industrial relocation. It is the position of some agencies that an industry that historically used an estuary with full concurrence of Government, must somehow now underwrite all costs of relocating.

The draft Federal-Provincial Agreement on Salmonid Enhancement recognizes that salmonid enhancement projects could impinge on the legitimate interests of other natural resource users (e.g., forestry, mining, hydroelectric) and provides for compensation for any unmitigated resource interactions resulting from enhancement projects (Section 9). The Land Management

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Branch suggests that at least part of the costs of relocating industries from estuaries should be provided by the Salmonid Enhancement Programs in the form of compensation.

We request your early reply to our suggestion of resolving estuarine land use conflicts between fish habitat and industry.

Yours truly,

A handwritten signature in dark ink, appearing to read "W. R. Redel". The signature is fluid and cursive, with the first letters of each word being capitalized and prominent.

W. R. Redel  
Assistant Deputy Minister  
Lands Department

QUEEN CHARLOTTE CITY

November 8, 1978

To Whom It May Concern:

Hello. I would like to tell you my feelings about the Salmonid Enhancement meeting which was held in Queen Charlotte last night.

The idea of the meetings are great, the people who represented the Program gave a good talk on what, when, who, etc., of the Program but it was very, very badly advertised. Times were confused, it was even told to many people that it was cancelled. I feel that my tax dollars should be spent on these programs but it really makes me angry when something like this is so badly advertised. Many people that did attend only heard about it hours before it started and couldn't make it or went with no organization and didn't speak. So please, next time there are meetings for the public, lets have better notice.

Thank You.

Loa Olsen  
Box 61  
Queen Charlotte  
City

SEP Update '78

Chairman:

Would be good if money continues to be available to hire three people to build boxes, pools and clear creek mouths between Skidegate and Masset, as these creeks represent a potential of 5,000 fish. By the time they are all done (by this crew of three) four years hence, it will be time to start on the original ones again because of flooding, erosion, winter storms, piling driftwood in front of the creeks. These are all easily accessible as they lie along the highway and therefore, they can be maintained and improved at a minimal cost and provide summer employment for a few people on a permanent basis.

John F. Wood  
Queen Charlotte City, B.C.



SMITHERS





November 15th, 1978

Department of Fisheries & Oceans  
Information Branch  
1090 West Pender Street  
Vancouver, B.C. V6E 2P1

Dear Sirs:      Regarding SALMONID ENHANCEMENT UPDATE '78

PART 1. PROGRESS OR REGRESSION

- (1) The Hatchery on the Babine River has enhanced the Sockeye Salmon at the mouth of the Skeena and provided heavier and more successful Commercial Fishing. However this has automatically effected the rest of the species--namely greater numbers of Coho and Steelhead are caught with the Sockeye.
  - (2) This has caused Catch-and-Release regulations on the Kispiox River August 15th to September 30th and Special Rivers regulations on the Morice and Bulkley Rivers. Most rivers on the Skeena Watershed have already been closed to Chinnoks since 1973--thus making them available for Commercial catches only.
  - (3) This has indirectly resulted in the Spring Closures till July 1st on the Kispiox, Bulkley and Morice Rivers to allow for late Steelhead Spawners and smolts to travel safely to the sea. The days when sportsmen could enjoy a relaxing stay by the Rivers in May and June is gone-----.
- Thus the spring sportsman has resorted to Lake Fishing which is the only fishing left.... PROGRESS OR REGRESSION???

PART 2. WHAT'S NEXT???

- (1) Due to the closure of many Rivers till July 1st, the Salmonid Enhancement must provide (a) planting of Trout in all lakes easily accessible from the road (some go by skidoo in wintertoo)

OLGA WALKER, R.R. 1, Kispiox Road, Hazelton, B.C. V0J 1Y0 Phone: 842-5435



(b) keener public awareness for minimal catches to prevent access fishing (I've heard reports of 70 or 80 fish taken at a time).

(c) access roads and bridges to these lakes. Why must the sportsman always hope that the Forestry might fix a road where he ~~wh~~ wishes to go?

(2) Special Rivers possession limits should all be the same. Even with with Catch-andRelease regulations on the Kispiox---the October and November possession limits should be the same as the Morice and Bulkley Rivers.

(3) It is always noticeable by the number of Steelhead in the Kispiox

(a) when Fish Canneries are on strike (as in 1975)

(b) when Commercial Fishing stops

This proves that there are huge numbers of Steelhead caught by Commercial Fishermen at the Skeena Mouth. Since Commercial Fishing has improved in the Skeena with Sockeye enhancement, and since Commercial Fishing is more intense to harvest the returning Salmon, it is only right that there should be substantial consideration of the Sportsfisherman. After all Tourism (and Leisure Time) is British Columbia's THIRD LARGEST INCOME, as confirmed by Honourable Grace McCarthy! It is imperative that the working public has an opportunity to partake of British Columbia's pleasure fishing as much as the visiting tourists! This day and age industry's load is such that the working public has more and more leisure time to partake of outdoor pleasures. PINNAGE CHARGES may bring income to the enhancement program, but it's only planned release of LIVE STEELHEAD that will bring joy to the Sportsman! *O. Walker.*



WHAT NEXT?

- (1) Quoting Interior News, July 7th, 1976, an interview with Mike Whately, biologist at Smithers, "The commercial fishery takes about 50% of the total harvest of Steelhead in the Skeena River system. Figures showing the commercial fishery catch of Steelhead at the mouth of the Skeena River, reveal an 84% decrease since the early 1950's. The commercial fishery reported a catch of 25,000 steelhead in the early 1950's, compared with 4,000 in 1974."
- (2) The sportsfishermen at Kispiox Steelhead Camp got together a sheaf of letters, showing their concern and suggestions, which I submitted at the Salmonid Enhancement Program, November 24th, 1976. Several of these fishermen have continued to correspond with your Department, especially a Fay Davis, a very knowledgeable and concerned sportsman.
- (3) In 1977 the sportsfishermen formed Kispiox River Steelhead Committee. The whole group follows your Salmonid bulletins and continues to express concern regarding Steelhead Fishing in the Skeena River watershed, in the years to come.  
  
I shall now quote two paragraphs from the Salmonid Bulletin, Volume II No. 6 1977.-----  
We, of the Kispiox Steelhead Committee, are asking,  
"WHAT HANDOUT ARE THE SPORTSFISHERMEN ON THE SKEENA WATERSHED TO EXPECT FROM THE COMMERCIAL FISHERY IN THE NEXT 5 TO 10 YEARS?"

Yours very truly,  
*O. Walker*

## Commercial Steelhead Catch Studied

A study carried out by the British Columbia Marine Resources Branch last year has confirmed that the commercial catch of steelhead trout bound for the Skeena and Fraser River systems is greater than past statistics had indicated. 30

The finding is not unexpected. Figures on the commercial catch have been based on reported sales of steelhead, and it was known that many steelhead are sold as salmon, and others are kept by the fishermen for their own use. More accurate assessment of the commercial take is one step toward ensuring that successful salmon enhancement will not mean inevitable overexploitation of steelhead stocks. The report, Incidental catches of steelhead trout in the commercial fisheries of the Fraser River, Skeena River, and Juan de Fuca Strait, by E. Oguss and T.R. Andrews, describes the basis of concern: "The history of the Pacific salmon fishery has repeatedly demonstrated that maximum exploitation of large salmon runs leads to over-harvesting of non-target stocks that are in the fishing area at the same time as the target commercial stock. Many small runs of salmon and sea trout were fished to extinction during the early years of this century when the fishery was virtually unlimited..." The fear is that heavy fishing stimulated by the presence of enhanced salmon stocks could result in the much less numerous steelhead being wiped out. 1

The report is based on a study that was undertaken by a research team who interviewed commercial fishermen as they delivered their catch. In addition, tallymen and skippers on

collector boats were asked to record the number of steelhead that fishermen reported catching, regardless of whether the fish were kept or not.

Commercial catch of steelhead in the 1976 season in the Skeena area was estimated to be  $5,000 \pm 500$ ; in the Strait of Juan de Fuca (the "San Juan" or Area 20)  $3,000 \pm 500$ , and on the Fraser (Area 29, from the river mouth to Mission)  $1,000 \pm 200$ . Accuracy of previous data ranged from "a relatively large portion of steelhead caught (being)...accurately reported" in the Skeena area, to "only a small fraction of the steelhead catch" being represented in the earlier data for San Juan, "due to the practice of selling steelhead for coho". For the Fraser River only one-third to one-half of the late summer and fall steelhead caught had appeared in the published figures, the researchers judged.

continued p.2

## New Fact Sheet

Second in the SEP series of background information sheets is now available. "Lake Enrichment" describes one of the newest salmon enhancement techniques. An experiment was begun at Great Central Lake on Vancouver Island in 1969 to test the hypothesis that production of sockeye salmon could be increased by adding chemical fertilizer to the waters of sockeye nursery lakes. Some of the freshwater lakes in which sockeye fry rear before migrating to the ocean are lacking in nutrients that support the organisms on which the fry feed. The initial attempt to supplement natural conditions at Great Central Lake proved highly successful, increasing the average return of adult sockeye nearly tenfold. Now the enrichment technique, repeated at Great Central and four other lakes, is expected to bring substantial early returns in the Salmonid Enhancement Program.

Copies of "Lake Enrichment" are free, from the Information Centre, 1090 West Pender, Vancouver.

### Inside



#### Construction and Production

— The Big Qualicum Project



#### Mr. Salmon Enhancement

The purpose of the study was not, of course, to discredit existing statistics, but to improve on them and give Salmonid Enhancement Program planners a sounder base on which to make their predictions of impact of proposed enhancement projects on steelhead stocks. The study also undertook to assess possible means of reducing the impact of commercial fishing on steelhead. Sport fishermen and others have commonly suggested measures that would

- (1) restrict gillnet mesh sizes,
- (2) require the use of weedlines, sunken nets, or shallower nets,
- (3) close small areas where steelhead catches are disproportionately high,
- (4) develop salmon harvesting methods that would discriminate on the basis of the fishes' schooling behavior, and
- (5) require immediate release of all commercially caught steelhead.

The authors stress that "definitive evaluations of these management techniques were not attempted", but they felt the interview data collected gave at least "a basis

for a preliminary assessment of techniques, given certain limitations...".

They found that "steelhead of all sizes were caught in gillnets of all sizes", and that regulations to restrict net mesh sizes were therefore "very unlikely" to reduce the incidental commercial steelhead catch. To determine if regulations requiring weedlines, sunken or shallower nets might let more steelhead pass, they asked the gillnetters who had caught steelhead if the fish were taken in the top, middle, or bottom of the net. Lack of definition of 'top', 'middle' and 'bottom' led to fishermen's answers being "uncertain and facile", and it was concluded the data were not adequate to determine whether or not such regulations might be justified. According to the report, "There may be some potential for a more selective harvest through modified net regulations, but the effect on salmon catch and the degree of increased steelhead escapement must first be properly researched."

see "Steelhead Study" p.4

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## Construction and Production

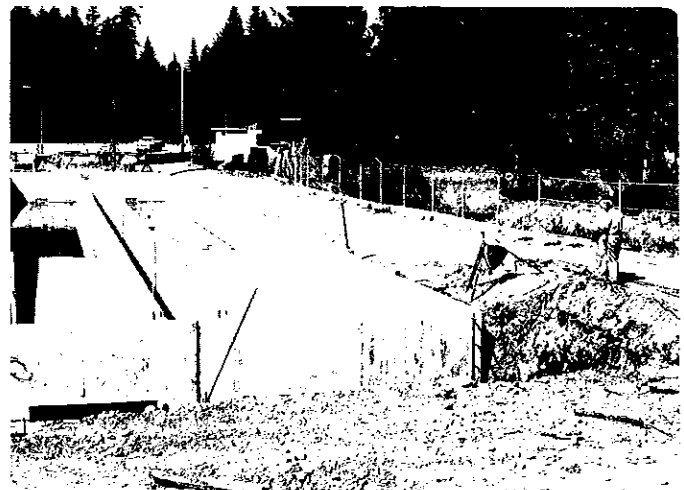
### The Big Qualicum Project

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New holding and rearing facilities and an improved water intake installed at the Big Qualicum project will double the project's capacity to produce chinook and coho salmon and steelhead trout. The new facilities will be ready for salmon spawners returning to the river in early September.

Progress is not without its price, however, and in this instance construction activities at the site of the original rear-channels displaced the current year's entire stock of rearing coho and steelhead fry. Thanks only to the ingenuity of project manager Dick Harvey and his staff, displacement of the little fish did not mean their loss.

The steelhead, numbering 80,000 were not a great problem. They were transferred to floating holding pens in the river, rigged with automatic feeders. Floating pens could not be provided economically to accommodate 1,100,000 coho fry, but the innovative approach that has become a trademark of the Big Qualicum project found another answer. Upstream from the main centre of operations is one of a number of settling ponds on the river designed to catch and hold silt that otherwise might clog spawning gravels. With the addition of a fish barrier, catwalks for feeding the fish and servicing the channels,



*Some salmon and steelhead returning to the Big Qualicum River this fall will be directed into new paired holding tanks, each 220 feet long, where they will be held until their eggs are ready to be stripped and fertilized.*

wire fencing to discourage predators, the settling pond was converted to a workable substitute rearing channel. A small equipment shed, like most of the other material that went into the temporary channel, was moved from the old site. Total cost of the conversion that will keep Big Qualicum in coho production this year: under \$1,000.



# Mr. Salmon Enhancement



The unofficial title of "Mr. Salmon Enhancement" conferred on J.R. (Ron) MacLeod by Fisheries Minister LeBlanc pretty well sums up the SEP Director's involvement. Salmon enhancement is Ron MacLeod's work and hobby all wrapped up into one. In fact, as he puts it, since he joined the Department in 1956 his job in Fisheries has been, "the total focus of my energies and interests".

Asked if any particular incidents stand out in his 21 years with the Department, Ron says: "I've found every bit of it exciting. But I think the most exciting years lie ahead now that Cabinet has approved the Salmonid Enhancement Program. There's an opportunity here to contribute very directly and very vigorously to our fisheries resource. But particularly my interest lies with the program's impact on coastal communities and the people in those communities. That's where most of my sympathies lie - it arises from my background I suppose."

Ron MacLeod's background has its roots at Tofino, on the west coast of Vancouver Island, where he spent his early years working on fishing boats and logging. After a stint as a fishery officer at Alert Bay, he served in a number of subdistricts along the coast and then moved to Prince Rupert in 1963, eventually becoming district supervisor there.

In 1965 he transferred to Vancouver and for a period served as Assistant Chief of Protection. He was appointed Manager of the Northern Operations Branch in 1971, and in 1974 was seconded to head up the regional planning group to work on plans for the salmonid enhancement program.

How did he react when he heard of Cabinet's go-ahead for SEP after the long months of planning? "Well, its very gratifying to have put a program before the government that was of such a quality to satisfy them and win their approval, but I'm more excited by the fact that we've managed to pull together a tremendous team of young people on this program. They're people who are creative, have great initiative and tremendous energy.

"I'm really looking forward to working with them over the next few years with a tool like the enhancement program. With its scope, the program has the power to transform coastal British Columbia. In some quarters this may be a cause for concern - there's always a human tendency to resist change - but it's also a reason for hope and high expectations about the future."

The SEP director has pushed public participation to promote some understanding of the potential of the program, so that communities in British Columbia can control the pace of change themselves. By participating in the decision-making process they can ensure that change doesn't over-run them.

"If people get involved in this way, they can very quickly relate the protection of salmon to the protection of their particular quality of life," he added.

"It's inevitable that there will be conflicts over resource use. There'll be decisions for salmon and decisions against salmon. My hope is to have these decisions made in a climate where everyone understands the implications. If there's a decision to give up a particular salmon stock or add a new one, I want people to understand what is involved, and what the alternatives are."

Ron hastened to add that there was more to public participation than that. "It's the reward of doing useful things that's most important."

And how does he view the major task of doubling the production of B.C. salmon by the 1990's.

"Well, it's going to take a tremendous marshalling of our technical capabilities to produce the quantity of salmon we aspire to. We will need resources that do not yet exist in the private sector. We must get the universities involved. To design, build and operate up to 170 major facilities for salmon enhancement within the next 15 years will be no mean feat. But it's this kind of challenge that's so exciting to our technical people."

"By the end of the fifth year of the program we should have boosted production by 50 million pounds...by the 22nd year we should have reached 95% of our target..."

"The real pay-off will come, not just through the production of an additional 190 million pounds of salmon, but from the tremendous spin-offs that will come through the generation of knowledge in enhancement techniques."

"We will soon reach the stage where we can start to farm our fish and I can visualize at some point in time where we, the Alaskans, the U.S.S.R. and Japan will be pumping out teeming millions of salmon. We will get to the point where we will have to set quotas

on what we send out to the common pasture in the ocean and the remainder will have to be held in fish farms in inshore areas."

"This form of fish farming will bring stability to coastal communities. This is where the real transformation will come. And through the Salmonid Enhancement Program these fish farmers of the future will have developed the technical capabilities to run their own businesses."

There was something of a far-away look in Ron MacLeod's eyes as he made these predictions about the future of the B.C. salmon industry. But if there's anyone who can make these dreams come true it's "Mr. Salmon Enhancement".

(Condensed from an article  
in FISHERIES & MARINE NEWS)

## Steelhead Study

Continued from p. 2

The research team found no area with disproportionately high steelhead catches. Where there were higher steelhead catches, there were also higher salmon catches, and the proportion of steelhead to salmon in the catches was virtually the same as elsewhere. The report suggests future studies should be designed to check distribution and density of steelhead in all commercial fishing areas, as well as the rate of movement of steelhead through the fishing areas. Such data "would be useful in determining the value of controlling gear density or duration of weekly openings as a means of harvesting target salmon stocks without over-exploiting steelhead stocks."

Schooling behaviour of the steelhead was "crudely" assessed by checking how frequently the trout were caught with other salmonids. Over 90% were taken with the salmon species the fishermen sought primarily, and it was concluded that either the steelhead school with the salmon, or, more likely they are distributed randomly through a salmon fishing area. Either way, selective harvesting based on schooling behavior does not appear feasible.

Nor does requiring immediate release of all commercially caught steelhead appear likely to increase steelhead escapements, according to Oguss and Andrews. Washington State has such a requirement, and it has often been advocated for British Columbia. The researchers point out that, "The real value of requiring immediate release depends on survival in the net and the subsequent survival of released fish." They found that gillnet survival seems to reflect how long the nets are left in the water: the longer the set, the smaller the proportion of fish that survive. They comment that "Since there

is nothing to be gained in throwing back a dead fish, it seems likely that Washington-style regulations have little effect on escapement in areas where gillnetters make lengthy sets." Though survival in some seine nets is "relatively high" (in contrast to gillnets), they report, "...the possible contribution to steelhead escapements from seine releases would be insignificant because the seine fishery takes such a small portion of the steelhead catch in areas other than San Juan."

The authors suggest that the value of the incidental commercial catch of steelhead as a large random sampling should not be overlooked. Steelhead, they note, "are very difficult and expensive to census at any stage of their life cycle".

The study brought Oguss and Andrews to the conclusion that "No restriction on commercial salmon fishing gear or fishing methods for the protection of steelhead appear to be justified or warranted." On that premise they "strongly" recommend that "the catch of steelhead by the commercial fishery continue to be legal and unrestricted." They do recommend that "The practice of recording steelhead as coho on sales slips, which is prevalent in some areas, should be corrected." The point might seem trivial, but the practice has been a prime source of distortion in past commercial catch statistics.

Looking to the immediate future, the researchers make a number of recommendations toward action that would help fill in some of the many gaps in information about steelhead. They recommend further data collection on "the time sequence of steelhead catch during (commercial fishing) openings of different lengths"; and on the distribution and density of steelhead in salmon fishing areas. They also recommend consistent test fishing on the Fraser "to measure the catchability and replacement times of steelhead and other salmonids in the fishing area." Tissue and scale sampling of commercially caught steelhead, they suggest, "would provide a large and inexpensive source of data which could be valuable for the management of the freshwater fishery." Finally, they recommend that "High priority should be given to the development of stock identification techniques capable of distinguishing separate stocks of untagged salmonids caught in salt water."

### Salmonid



Fisheries and Environment  
Canada

Pêches et Environnement  
Canada

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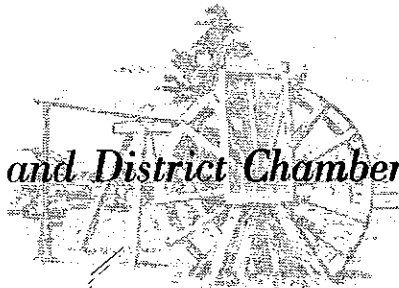
1090 West Pender Street Vancouver 1, B.C.

PRINCE GEORGE





QUESNEL



## *The Quesnel and District Chamber of Commerce*

TÉLÉPHONE (604) 992-8716  
QUESNEL, BRITISH COLUMBIA, CANADA

November 21, 1978.

Salmonid Enhancement Program,  
Fisheries and Oceans Canada,  
QUESNEL, B.C.

Dear Sirs:

During 1976, the Town of Quesnel submitted a proposal for redeveloping the natural salmon spawning area in the Quesnel River within the Town's boundaries.

The Mayor and Council are once again presenting a Brief with this proposal. The Quesnel and District Chamber of Commerce fully support the Town's position and brief. We feel that it would be beneficial to this area and the entire Fraser River system to revitalize these spawning grounds. Would you please give favourable consideration to the Town of Quesnel presentation?

Yours very truly,

Martin McLeod  
Director



November 17th, 1978

BRIEF TO SALMONID ENHANCEMENT INQUIRY II

GREENLEAF RESTAURANT, QUESNEL, B.C.

NOVEMBER 21st, 1978

- Submitted on behalf of the Mayor,  
Council and Citizens of Quesnel -



BRIEF TO SALMONID ENHANCEMENT INQUIRY II  
Greenleaf Restaurant, Quesnel, B.C.  
November 21st, 1978

Members of the Inquiry:

On November 12th, 1976, the Mayor of Quesnel attended and submitted a brief to the members of the Salmonid Enhancement Inquiry at a public meeting held at the Valhalla Motel.

The brief proposed that an area between the B.C. Railway bridge and the Highway 97 bridge on the Quesnel River be developed as a salmon spawning area. The brief also suggested that in the off-year of the salmon run, that major work in the river could be undertaken to prepare the spawning areas for the fish well before they are anticipated in the area. The brief further suggested that improvements to the spawning beds for pink salmon would be easily accessible to the general public and could be used as an educational tool for the district schools.

Since the submission of the brief the Town of Quesnel forwarded a copy to the then Minister of State for Small Business and now Minister of State for the Environment, and MP for Kamloops/Cariboo, the Hon. Len Marchand. Copies of the brief were also forwarded to our local M.L.A. and Minister of Highways and Public Works, the Hon. A.V. Fraser. The Hon. Len Marchand was very interested in the proposal suggested in the brief from the Town of Quesnel and although not a provincial matter, our local M.L.A. was also most interested particularly as problems of riverbank erosion, a Provincial matter, may be lessened by any improvements made in the area between the B.C. Railway bridge and the Highway 97 bridge crossing the Quesnel River.

On at least two occasions during 1977, staff members visited Quesnel and undertook a study of the Quesnel River to ascertain the feasibility of proceeding with a program to improve the natural spawning beds on the Quesnel River. In September 1977 the Town of Quesnel attempted, by telephone, to obtain approval to expend approximately \$400 to make minor



BRIEF TO SALMONID ENHANCEMENT INQUIRY II  
PAGE (2)

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improvements to the natural spawning beds in certain areas of the Quesnel River to encourage spawning of the pink and sockeye salmon. Approval was not granted. Through lack of approval on this very minor project, two years and countless number of dollars in fish harvesting have been lost. We were advised in a letter dated March 7th, 1978 from the Special Projects Branch, Salmonid Enhancement that "jurisdiction over pink and sockeye salmon in the Fraser system is administered by the International Pacific Salmon Fisheries Commission under a convention between Canada and the United States". The letter further advises that a Fisheries agreement with the United States is being negotiated and that pending such an agreement it is Canada's policy not to enhance sockeye or pink salmon stocks, since, under the existing treaty, Canada would have to share the benefits with the United States. The Council of the Town of Quesnel was disturbed to learn of this policy and requested an explanation for the rationale for this policy. No reply was received to our letter of request for an explanation dated March 17th, 1978. Possibly some answer could now be given.

It has been suggested that funding may be available in 1979 for a project on the Quesnel River and it is the request of this brief that consideration and early approval be given to providing funding in 1979 so that an improvement to the natural spawning beds in the Quesnel River in the area suggested will be undertaken prior to the spawning of the pink salmon early in September 1979.

The Council of the Town of Quesnel wishes to thank the Salmonid Enhancement Inquiry members for returning to Quesnel and providing a progress report and providing details of plans for your program in the future.

Respectfully submitted,

Mayor and Council  
Corporation of the Town of Quesnel

EAG/lc



WILLIAMS LAKE





KAMLOOPS

## Further Recommendations for Enhancement of the Salmonids of the Thompson River System

1. The radio tagging program of steelhead on the Thompson has taught us a great deal that in the past was never substantiated, only rumoured. We now realize the full value of the small tributaries, such as Spius and Tank Creeks. We are now aware that a large number of steelhead move into Deadman's, Nicola and Bonaparte Rivers.

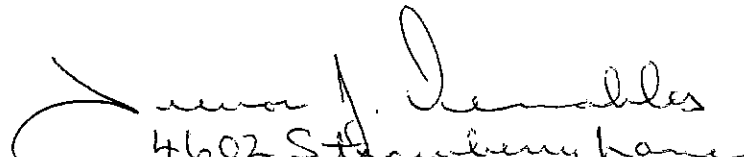
The area of Deadman's that is available to the steelhead and the spring and coho is very extensive and should be developed to realize its full potential particularly for the small run of springs that is left. A comprehensive study of the system should be started immediately to best comprehend the best type of enhancement for the system, be it habitat improvement, spawning channel, water storage for flow control, etc.

2. The Bonaparte is virtually untapped as a producer of salmonids. There is an impassable set of falls shortly upriver from its confluence with the Thompson. There is a good number of steelhead and springs utilizing the small area available. The Fish and Wildlife Branch is looking into the cost of removing the obstacle and the cost of removal would appear to be in the hundreds of thousands of dollars. The funding of this project should be approved under SEP immediately as it will open up a minimum of 30 miles of ideal habitat for salmonid production with a hatchery for rearing purposes already in existence on Hoon Creek. This facility could be used to rear the juveniles until they were of sufficient size to be released into the system. This could cut down greatly on the high mortality suffered from the egg to alevin stage in nature. The benefits of this undertaking would be immediate.
3. I still feel that more money should be allocated to hire enforcement personnel. The main Thompson is well policed by Fish and Wildlife people but it is still imperative that the tributary areas be policed. I know of one person that makes one or two trips a year to the Coldwater to "shoot" spring salmon. Our priorities should be enhancement and enforcement, not strictly enhancement.
4. The fall of 1977 and 1978 have been two of the best years I have seen on the Thompson for the numbers of steelhead taken by sports fishermen. The larger catch has to be due to the larger returns allowed by the restriction on gillnetting in the lower Fraser. I believe they have been cut back to 5 days in the past two Septembers. I would like to see this cut back further to a "blue line" restriction, thereby keeping the gillnets completely out of the main channels of the Fraser. At the most they should, as I submitted in my last brief, be allowed to fish upriver to the Patullo bridge and no further when there are major runs of steelhead and coho moving upriver.

There are other points I would like to make but others must have their say, so in closing I would like to say that I still feel too much emphasis is going into the species that are of the "non sport category" such as the sockeye and dog salmon. I would like to see more programs relating to steelhead and cutthroat rather than accepting that they will benefit as a "spin-off" of the programs for the commercially important species. This is my only criticism of the SEP.

Thank you for the chance to again express my point of view and congratulations on the progress that has been made in a very short two years.

Sincerely,

  
4602 Strawberry Lane  
Kamloops, B.C.  
V2C 5K1

Trevor J. Venables  
4602 Strawberry Lane  
Kamloops, B.C.  
V2C 5K1



KAMLOOPS AND DISTRICT  
*Fish and Game Association*



P.O. BOX 164  
KAMLOOPS, B.C.  
V2C 5K6

Re: "Salmonid Enhancement - 1978 Update" - Kamloops Meeting, November 23, 1978

Mr. Chairman:

Our club does not feel that we can make a submission under the heading of "Where should Salmonid Enhancement Program go from here?" until we know what has happened since the 1976 meeting.

This memo is therefore, just a restatement of the main concerns previously submitted. I think it is likely that our club will be writing to you by the January 12th deadline.

1. What is the present situation regarding:
  - a) The thorough environmental study of the South Thompson River system which has been agreed to by both Ottawa and Victoria?
  - b) Our request for a determination of all the spawning areas in the whole of the Thompson River system?
  - c) Clean-up of the Fraser estuary area?
  - d) Are all government agencies (such as the Department of Highways) now subject to the same restrictions as industry regarding water pollution?
2. We urged enhancement by:
  - a) Restoration or introduction of salmonids to now non-productive waters such as the Upper Adams River.
  - b) Reduction of pollution.
  - c) Artificial spawning channels.
  - d) We also urged that hatcheries should be a very low priority.
3. We recommended more education about environment. Also more public involvement, especially by kids, in worthwhile enhancement projects.

4. In our 1976 submission, we asked that all saltwater fishermen (whether sport or commercial), be licenced. In our follow-up of November 19th, we urged that the full cost of at least the enhancement program be recovered from the fish takers, both commercial and sport.

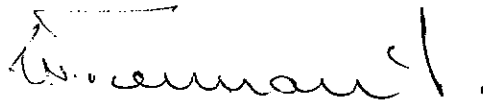
This principle was agreed to and I refer you to a release dated September 21st 1977 entitled, "Proposed Recovery Scheme for Salmonid Enhancement Program." Paragraph 3 (1) on page 5 reads, "All three groups of beneficiaries, commercial and recreational, as well as processors, will be expected to bear the costs of salmonid enhancement in proportion to potential additional benefits received."

However, in January of this year, Mr. Le Blanc is reported to have said, after discussions with B.C. Cabinet Ministers, that saltwater sports fishermen would not be licenced - apparently because of harm that such action might do the B.C. tourist industry.

There is something haywire in this sort of decision making. Freshwater sports fishermen, both locals and visitors, pay licences and special licences for steelhead. Why shouldn't saltwater fishermen also pay licences?

The Americans are already getting about half of the Fraser River salmon, so what is wrong with requiring that American tourists pay for the Canadian salmon that they catch?

5. Further on this question of sport fishing, I hope that you can tell me just how many salmonid are taken each year by saltwater sportsmen. A 1975 survey report says there were 210,000 saltwater fishermen. If this is still a reasonable figure, at \$10.00 average fee per angler, the yield would be over two million dollars. This would be at least something in the way of a direct payment toward the cost of a direct benefit.



Walter Tennant  
Chairman  
Environment Committee

Jay Springs Ranch (Pinantzn)  
R.R. #3  
Kamloops, B.C.  
V2C 5K1  
November 28, 1978

Mr. J. Boland  
Salmonid Enhancement Program  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sir:

Once again, unforeseen events prevented me from attending your recent Salmonid meeting in Kamloops last week.

Since there is one mile of Paul Creek meandering through my ranch property, I wished to bring out the following points:

1. We have developed this ranch since 1965 and have been very conscious of the spawning trout ascending to the small falls below the main buildings. We have noticed a very definite deterioration of the trout, both in size of the spawners and numbers of fish active in the riffles.
2. Along the stream bed, one of the Ips borers has been active for at least two years. This decimation of willows, principally, has meant the riddled trees (mostly smaller diameters so far) are breaking off and damming the stream. After the infestation has run its course, these falling trees will present a real problem.

We feel normally obligated to keep Paul Creek healthy but I know it will be impossible for me to keep ahead of the dying willows along with the natural mortality of the other deciduous trees. Incidentally, our home quarter sections are:

S.E.  $\frac{1}{4}$  Section 25, R20, Twp. 15, W6M  
N.E.  $\frac{1}{4}$  Section 24, R20, Twp. 15, W6M

I believe Paul Creek deserves attention, most probably for its full length, but specifically would it be possible to get some assistance to enhance the stream through my property?

Yours truly,

*G. F. Bodman*  
**G. F. BODMAN**

G.F. Bodman

COURTENAY



Salmon fisheries in our area play a major role in our economy. Tourism and commercial fishing rank as our major industries. In addition to this, the recreational benefits for all of our citizens cannot be measured in dollars and cents. Therefore an ensured continuation of all programs to enhance our fishing stock must receive high priority from all levels of government and the Community.

I regret that no hearings were held in area 14 during 1976 so I therefore commend you on holding a public hearing at this time in an area that has such interest in this endeavour.

Much interest has been expressed in the hatchery being built on the Puntledge River in an attempt to enhance the salmon and steelhead runs on the Puntledge and Tsolum systems. I am in full agreement with this phase of the program. Yet further refinements on these major river systems are needed to bring back the runs to their original numbers. I'm sure others will be speaking on this phase of the program.

However, I wish to direct my brief to you tonight on the matter of the many small streams in our area that were once great salmon and trout streams. As one who was born and raised in this valley and has enjoyed fishing for many years, I wish to point out that many of these waterways have been temporarily, and in some cases permanently ruined during my lifetime. ( I can elaborate on this and show you such streams if you wish. Many reasons can be cited for this that are well known to all of us.

Many of these small streams lie within municipal boundaries and in developed rural areas. One such stream lies almost entirely within the boundaries of the Town of Comox and I refer to Brooklyn Creek.

As the Mayor of the Town of Comox I would like to present to you some of the concerns and recommendations I have regarding the enhancement program.

Councils are constantly and will continue to be faced with the matter of subdivision development bordering on waterways. During the past couple of years steps have been taken by my council, and on my urging to ensure that a green belt strip be created along such watercourses, and that this strip be returned to the crown. I urge you to recommend that this practice be a standard requirement on all salmon and trout producing streams where development is occurring, and further recommend that funds be made available for the purchase of such strips in some pre-existing subdivisions where sensitive rearing areas exist.

I wish to point out to this hearing that very little information and few terms of reference of your program have been made available to Municipalities and Regional Districts that are dealing with development. I recommend that information packages should be prepared and distributed to us. These would include information and guidelines for all capital work projects such as bridging of streams, surface water drainage systems, clearing practices, etc. etc. that relate to development in relationship to preserving these natural spawning waters.

On our particular creek, and I know this applies to many, a man made culvert has blocked the run of salmon and trout beyond this point. Two weeks ago during a visit to the creek with officers from the Fisheries Department,

.../2

I witnessed many salmon and a few cutthroat trout in the lower system unable to get to the more desirable rearing areas above the obstruction. Funds should be made available at an early date to correct such problems. The cost would be very little when measured against the benefits of the future.

Much of this work can be done by interested volunteers. In my community, a couple of such programs have been started and I refer to the work done by our student body under the direction and guidance of Mr. Barry Thornton and also by the commitment given the Rotary Club of Comox of which I am a member. A greater community involvement could become a reality if more information is made available to the news media, to municipalities and to the community at large. I urge you to distribute such information to clubs and organizations with suggestions and guidelines on how they could become involved in this restoration program.

As a member of a pioneer family that has seen six generations enjoy the benefits of fishing in this area, and as a citizen who is very concerned with the future generations yet to come, I wish the Program every success. It must succeed.

I respectfully submit this brief to you and wish to advise you that my Council will be submitting a formal brief prior to the January 12th deadline.

G. I. Piercy  
Mayor, Town of Comox

# Corporation of the Village of Cumberland

POST OFFICE BOX 340

TELEPHONE 336-2111



Our File# .....

2673 Dunsmuir Avenue  
CUMBERLAND, B.C.  
V0R 1S0

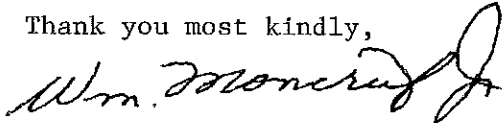
November 27th, 1978

Dear Mr. Chairman:

Please accept my apologies for being unable to attend tonights meeting,  
due to a Regional District Board meeting.

I would sincerely request that you accept this short brief and have it  
read into the records.

Thank you most kindly,

A handwritten signature in cursive script, appearing to read "Wm. Moncrief Jr.", written in dark ink.

Wm. Moncrief Jr.  
Mayor, Village of Cumberland

WM:es  
encl.

The value of the Puntledge, Oyster and Tsolum river systems to the economic well-being of the Comox Valley and the Upper Vancouver Island area is unlimited. Sports and commercial fishermen alike have benefitted directly from the production of these river systems, with indirect benefits resulting in many of our service industries.

The combination of poor logging practices in the upper reaches of the systems and the construction of the Hydro Dam on the Puntledge River have had disastrous effects on the production of fish in these systems.

The spawning channel installed on the Puntledge River has apparently contributed very little to the return of the salmon, steelhead, and sea-run cutthroat to the system. The construction of hatchery facilities will hopefully contribute significantly to the effectiveness of the Puntledge. It is our submission that this is only the first step in restoring fish runs in the Puntledge system. Some means must be found in utilizing the Comox Lake and tributary systems of the upper Puntledge system. It is our understanding that this area has been under consideration for some time and in support of this submission that Comox Lake offers a great potential for future salmon production, the following quotation from the Provincial Fisheries Branch publication "Catalogue of Salmon Spawning Streams and Escapement Population in Statistical Area No. 14 Pacific Region" in 1970 is submitted:

"The theoretical potential for the Puntledge River system exceeds by a wide margin that for any other stream in Area 14. The potential exists in the spawning and rearing opportunities in Comox Lake and the tributary streams. (Upper Puntledge River and lakes and the Cruickshank River system) for coho and sockeye salmon. The problems to overcome to realize this potential area migration of adult fish over Stotan and Nib Falls and migration of juvenile fish from diversion dam to lower stream. Successful transplantation of coho and sockeye salmon is a necessary part of realizing this potential."

We also believe that potential increases in steelhead and sea-run cutthroat trout production also exist in the Comox Lake system

under the above circumstances. This system is capable of providing a real boost to the fresh water sport fishery here in the Comox Valley providing it is recongnized and improved by this Commission.

The Tsolum and Oyster Rivers both suffer from flash flooding, erosion and changes in its water course, which we believe is due to poor logging practises in the upper reaches. Unless drastic steps are taken and significant recognitions given to these river systems by this group, then we believe much will be lost.

No doubt you are aware that even if some rehabilitation work on these rivers takes place now, it will be many years before results are forthcoming.

The Comox Valley area as you know has little in the way of industry to solidify its tax base, and we believe that a strong and productive rehabilitation programme on these areas will do much for the economy.

It is hoped that the Commission takes these comments under consideration and fully support the people of our area whom have demonstrated much concern over the future of these three most important river systems.

Thank you most sincerely;

Wm. Moncrief Jr.,  
Mayor, Village of Cumberland  
Chairman, the Comox Strathcona Regional  
Economic Development Committee

BRIEF

TO COMMITTEE ON SALMONID ENHANCEMENT

SUBMITTED BY

COURTENAY AND DISTRICT FISH AND GAME PROTECTIVE ASSOCIATION

NOVEMBER 27th, 1978

Mr. Chairman, the Courtenay and District Fish and Game Protective Association, a conservation oriented organization of some 180 members, welcomes this opportunity to address your committee, particularly in light of the fact that no opportunity was presented for a local hearing in the initial stages of the programme.

Our organization is no stranger to the salmonid enhancement programme, Mr. Chairman, having been one of the first public organizations in B.C. to take on an enhancement project under S.E.P., this being the clean up of Morrison Creek, an important salmon spawning tributary of the Puntledge.

In the ensuing paragraphs of this brief, Mr. Chairman, we believe we will conclusively show the importance of the salmon fishery, both as a source of commerce and as an important source of recreation for the 30,000 plus persons residing in the area bounded by the Oyster River on the north and by Deep Bay on the south, in which are contained the communities of Courtenay, Comox and Cumberland.

Additionally, we will attempt to inventory the salmon producing rivers and streams in this area, identify and historically examine three of the major watersheds in this group, along with attempting to show the causes of the decline in their fish

populations. Finally, we will recommend corrective action to assist in overcoming these problems.

The main theme throughout our brief, Mr. Chairman, will be the protection and enhancement of our natural stocks of anadromous fish and their habitat.

Looking at the previously designated area, we find much sports fishing activity, and many businesses dependent on a healthy fishery, some of which are as follows:

There are 25 plus salt water resorts directly dependent on this fishery.

There are 15 boat launching ramps all of which, except one operated by the military, are open to the public. Nine of these are free of charge and the remaining 5 are operated commercially. One of the latter group, the ramp operated by the municipality of Comox, at Comox Bay, had some 5,500 launchings in 1976 we are told.

Additionally, there are 2 marinas ( with another in the planning stages ), 2 boat building firms, 4 boat and motor dealers, 9 sporting goods stores, 1 sports fish cannery and numerous motels, hotels and restaurants who rely on this sports fishery for a substantial portion of their business.

Looking at the commercial fishery, we find an average of 75 commercial boats operating in this area during salmon season. Allied with this fishery are 3 fish buying camps and 2 commercial fish markets.



So it can be seen Mr. Chairman, that along with the intangible benefits of many fine hours of recreational enjoyment, for both young and old, this total fishery is a vital source of commerce for this community and surrounding area.

If we examine the salmonid producing streams in this vicinity, Mr. Chairman, we find a total of 38 streams in the area from the Oyster River to Deep Bay, which comprise 25 water courses entering salt water. Three of these prime producing river systems which were major contributors to the fishery in this area, are the Puntledge, Tsolum and Oyster.

In order to fully understand what has happened on these three rivers, Mr. Chairman, we will take an historic look at the numbers of fish produced in each:

- (1) PUNTLEDGE - In the late 1940's and early 1950's there were in excess of 10,000-15,000 chinooks; 5,000-10,000 coho; 10,000 plus pinks; in excess of 30,000 chums; 3,000 plus steelhead; along with a substantial number of sea-run cutthroat trout.

Presently, in this system, Mr. Chairman, we find chinooks number in the high 100's, coho in the low 1,000's, pinks in the low 100's; chums rebuilding and approaching original levels, steelhead in the low 100's and sea-run cutthroats approaching extinction.

- (2) TSOLUM - In the late 1940's and early 1950's we found in excess of 10,000-15,000 coho; 100,000 plus Pinks; 1,000-1,500 chums; and 3,000-5,000 steelhead.

Presently in this watershed, Mr. Chairman, we find stocks number 1,000-3,000 coho, in the low 100's for pinks and chums; and in the low, low 100's for steelhead.

- (3) OYSTER - In the late 1940's and early 1950's there were 10,000-15,000 coho; 100,000 plus pinks; and 5,000-10,000 chums; and 2,000 steelhead.

Presently in this watershed, Mr. Chairman, we find stocks number 1,000-3,000 coho; in the low 100's for pinks and chums; and in the low, low 100's for steelhead.

We will now attempt to isolate the major contributing factors for the dramatic decline in fish populations on these three river systems, Mr. Chairman.

Looking first at the Puntledge System, we find that a major problem started to develop in 1913 with the construction of a hydro dam on this watershed, and was further compounded by the construction of a larger dam in 1954, which made passage impossible for upstream migrating fish. This has resulted in approximately 3/4 of this system not being utilized for spawning.

Construction of these dams also prevented natural movement of gravel, resulting in depletion of spawning areas downstream of the dam.

Additionally, poor upland resource management in such areas as logging and road building, have also been significant factors.

Next, examining the Tsolum System, Mr. Chairman, it is found that there was large scale gravel removal from this water course, in the early 1940's for airport construction. Coupled with poor resource management in the upper reaches, through the removal of timber from large tracts of land and improper logging techniques, this has caused massive run-off and scouring of spawning areas. Subsequent low water levels resulting from this condition, have been further aggravated by an increase in agricultural and domestic water consumption, thereby reducing available habitat and impeding upstream migration.

Turning now to the Oyster watershed, Mr. Chairman, we find this system has been subjected to large scale flash flooding and serious erosion. This again attributable, in large extent, to poor upland resource management.

So it is evident, Mr. Chairman, that there are serious problems on each of these three major river systems, problems which have resulted in a drastic decline in their fish populations and a very substantial economic loss to this community.

While we do not profess to be economists, Mr. Chariman, we have attempted to document in the following table, the potential economic impact on this area, from the commercial fishery alone, had the salmon runs in these three systems been maintained at something approaching historic levels. For the purposes of calculating the figures in this table, we have used the multiplier concept whereby, starting with the fisherman and working through the various businesses and members of the

community, a portion of income will be saved and the major portion passed on. In this case we have assumed a multiplier of 10.

We acknowledge that the figures in this table are only approximate, Mr. Chairman, but, we nevertheless, feel they are significant. It can be readily seen that based on the economic impact of the 1954 and 1976 runs, if the runs had been maintained at the 1954 level, an extra income with an economic impact value of between \$2,473,640 and \$4,919,200 would have accrued to the community for each of the 24 years in this period. Or very conservatively, Mr. Chairman, a total economic impact for this period, of approximately \$88,000,000. We think you will agree, Mr. Chairman, a very substantial figure.

POTENTIAL ECONOMIC IMPACT IN 1978 DOLLARS

| Fish Type          | Year | Nos. of Fish | Avg. Wt. | Total lbs. | Min. Price | Max. Price | Multiplier | Economic Min. | Impact Max. |
|--------------------|------|--------------|----------|------------|------------|------------|------------|---------------|-------------|
| Tsolum Pinks       | 1954 | 70,000       | 3 lbs.   | 210,000    | \$ .75     | \$1.70     | 10         | \$1,575,000   | \$3,570,000 |
|                    | 1964 | 200          | "        | 600        | .75        | 1.70       | 10         | 4,500         | 10,200      |
|                    | 1976 | 10,000       | "        | 30,000     | .75        | 1.70       | 10         | 225,000       | 510,000     |
| Puntledge Chinooks | 1952 | 4,000        | 7 lbs.   | 28,000     | \$1.78     | \$3.40     | 10         | \$ 498,400    | \$ 952,000  |
|                    | 1969 | 500          | "        | 3,500      | 1.78       | 3.40       | 10         | 62,300        | 119,000     |
|                    | 1976 | 600          | "        | 4,200      | 1.78       | 3.40       | 10         | 74,760        | 142,800     |
| Tsolum Coho        | 1954 | 8,000        | 5 lbs.   | 40,000     | \$2.00     | \$3.00     | 10         | \$ 800,000    | \$1,200,000 |
|                    | 1969 | 3,000        | "        | 15,000     | 2.00       | 3.00       | 10         | 300,000       | 450,000     |
|                    | 1976 | 1,000        | "        | 5,000      | 2.00       | 3.00       | 10         | 100,000       | 150,000     |

POTENTIAL ANNUAL ECONOMIC IMPACT FROM COMMERCIAL FISHERY

( If salmon runs had been maintained at 1954 levels )

|                    | Min.               | Max.               |
|--------------------|--------------------|--------------------|
| Tsolum Pinks       | \$1,350,000        | \$3,060,000        |
| Puntledge Chinooks | 423,640            | 809,200            |
| Tsolum Coho        | <u>700,000</u>     | <u>1,050,000</u>   |
|                    | <u>\$2,473,640</u> | <u>\$4,919,200</u> |

Turning now, to what direction the Salmonid enhancement Programme could take to solve the problems of each of these three river systems, Mr. Chairman, we would make the following recommendations:

Puntledge River -

In order to ensure the preservation of natural fish stocks, Mr. Chairman, a percentage of returning chinooks, coho, steelhead and cutthroat trout, should be allowed safe passage to traditional spawning areas in the upper reaches. I.e; the waters above Comox Lake and their feeder streams.

We recommend that at least 20% of the juvenile fish from the present hatchery facility should also be released in the upper watershed. This action would help guarantee natural stocks in the upper river system.

In order to expedite the rebuilding of Puntledge salmon runs, Mr. Chairman, it is strongly recommended that these two actions be considered for 1979.

It is further recommended that negotiations between B.C. Hydro and fisheries personnel proceed immediately, with a view to implementing measures at the diversion dam to prevent downstream migrating fingerlings from entering the turbines.

Additionally, Mr. Chairman, in view of Hydro's recent announcement of a proposed new transmission line to supplement Vancouver Island's power needs, the possibility of shutting down the power dam during peak

migration periods should be investigated.

In order to ensure safe passage of summer fish over Stotan Falls, it is recommended that work be carried out to improve the fishway over this obstacle. These falls, Mr. Chairman, have been a source of injury, and in some cases, death, for fish attempting to reach upstream spawning areas.

Dredging in the Courtenay River, in the early 1940's, Mr. Chairman, has resulted in the loss of valuable spawning gravel. Due to the dams, recruitment of new gravel by natural process, is no longer possible. It is therefore recommended that suitable gravel be distributed at strategic points in the River, in such a manner that the natural processes will again renew the spawning beds downstream of the dams.

We feel that this work is of vital importance to the presently rebuilding chum runs, Mr. Chairman, as these fish are currently utilizing all available spawning habitat below the dams and unless more spawning areas are created shortly, these salmon stand to be wiped out from overcrowding.

Tsolum River -

In order to meet the increased demands for both agricultural and domestic water consumption on this system, Mr. Chairman, and still provide adequate flows for fish, it is absolutely essential that some sort of water storage facilities be created on Wolf Lake and the headwaters of some of the tributary streams, if any sort of a fishery is to be maintained in this river. Since the agricul-

ural industry has a vested interest in this watershed, it is recommended that the Salmonid Enhancement Programme and this industry negotiate a cost sharing arrangement for the construction of any water storage facilities on this system.

In conjunction, with this, Mr. Chairman, we further recommend, that all future demands by the agricultural industry be closely monitored by the Provincial Water Rights Branch to ensure a more equitable distribution of stored water.

As a means of expediting the return of pink and coho salmon, steelhead, and cutthroat trout to this system, we recommend the immediate construction of up-welling incubation boxes. If necessary, Mr. Chairman, this club is willing to undertake the construction of these boxes, as an interim measure, until such time as a more permanent facility can be built.

In order to assist upstream migrants in reaching historical spawning areas, we would recommend that a programme be initiated to improve culverting on this watershed.

As a final recommendation on this river Mr. Chairman, we suggest that young coho, steelhead, and cutthroat trout, be released in the upper tributary streams, for rearing purposes.

OYSTER -

As a means of rebuilding the runs of coho, steelhead and cutthroat trout on this watershed, Mr. Chairman, we



would recommend the construction of incubation boxes, in particular on the Little Oyster, and Woodhus and Bear Creeks.

Additionally, we recommend that the feasibility of reinstating the fish ladder on Woodhus Creek be examined as a possibility of providing additional access to spawning areas on this system.

Any recommendations on salmonid enhancement, Mr. Chairman, would be incomplete without examining the factors which affect these fish while at sea. We will now take a brief look at this area.

Food sources of salmonids, Mr. Chairman, is one of the areas of our concerns. We would recommend that the S.E.P. undertake a study of the relative importance of shrimp, herring, needlefish, squid and other marine life, in the diet of salmonids. We feel that any such study would indicate the importance of herring in this respect, and the need for conservative harvesting of our herring stocks. We feel this is particularly important, Mr. Chairman, in light of the large amounts of herring being taken for the roe fishery.

Kelp harvesting, is another of our concerns, Mr. Chairman. The importance of kelp as a source of protection, in the rearing of herring, is a known fact. We feel this concern is justified locally, Mr. Chairman, in that where we used to have large kelp beds in the area off of Willemar Bulff and Point Holmes, presently, some 2 years after harvesting was carried out there, we find very little sign of this important marine plant. For this reason, Mr. Chairman, it is essential that the Salmonid Enhancement Programme work very closely with the provincial authorities in the protection of kelp beds from pollution, and in determining what

areas, if any, should be harvested.

Lastly, Mr. Chairman, but an area of vital concern is our estuaries. The values of estuaries are now becoming well documented. In addition to serving as critical nursery grounds for salmonids, they play an important role in the existence of many other forms of wildlife. One of the key areas in these roles, Mr. Chairman, is that played by the tidal interface, the exposed area between high and low water, an area very rich in marine life.

A prime example of this type of estuary locally, the Courtenay River Estuary, has been singled out as one of the most productive and important, on the east coast of Vancouver Island, and perhaps, Mr. Chairman, even in all of Georgia Strait.

We strongly urge, Mr. Chairman, that the S.E.P. work in conjunction with municipal and provincial authorities to maintain and enhance the integrity of this valuable estuary, through protection from such forms of pollution as sewage, marinas, landfill and other forms of water pollution.

In this regard, Mr. Chairman, we were most encouraged by Comox Council's recent decision to protect Comox Bay from further development, and strongly recommend that the Courtenay River Estuary be treated in the same manner.

In summary, Mr. Chairman, we feel that the Salmonid Enhancement Programme can play a major role in increasing the numbers of our salmonids. However, we would caution against attempting to accomplish this primarily through the construction of highly visible concrete fish factories, such as hatcheries and spawning channels, the direction the early stages of the programme appear

to have taken. While we do feel that hatcheries can play a limited role in attaining this objective, we would for the most part recommend a strong programme of natural rehabilitation and enhancement of our streams, supplemented by some hatchery facilities in the cases where fish populations have fallen beyond the level of self rehabilitation. This programme, Mr. Chairman, while not as readily visible in the public eye, is one in which we feel great potential is offered for public involvement and one which we believe will go further towards solving the long term problems facing our salmonids, and providing the maximum of economic benefits for our communities.

While we do not profess to fully understand the ramifications of maintaining the natural generic strain of the native fish in each of our streams, Mr. Chairman, we do feel this is a desirable feature, and recommend that both for the taking of eggs for artificial rearing purposes and for the seeding of young fish in streams, that native fish be utilized as much as possible, throughout the Salmonid Enhancement Programme.

We believe the time is opportune, Mr. Chairman, for a good portion of S.E.P. funds to be utilized in re-establishing the fishery in this area, and that any investment here will by far, be outweighed by the economic and social benefits returned to this community and the country as a whole.

To expand on this, Mr. Chairman, much of the forest cover has now returned to the upper reaches of the Puntledge and Tsolum watersheds and there should be a limited impact on any re-established fishery in these systems.

Secondly, Mr. Chairman, any monies spent on enhancement of local salmonid runs should benefit the fishery as a whole, and carry a wide economic impact, as Fisheries Department tag returns have indicated that fish produced in local rivers are caught over an area bounded by Johnstone Strait on the north, the Gulf Islands on the south and the Sunshine Coast on the east.

Lastly, Mr. Chairman, it is widely known that this area is currently suffering from substantial economic recession, and an early start on some of these projects would do much to assist in off setting these economic difficulties.

As a final note, Mr. Chairman, we would point out that unless the protection and enhancement of our natural fish stocks, streams, spawning beds and estuaries become an integral part of the Salmonid Enhancement Programme, all the enhancement in the universe will be but an exercise in futility.

Keith G. Seguin

Chairman Conservation Committee

SALMONID ENHANCEMENT PROGRAM

A BRIEF CONCERNING SALMONID ENHANCEMENT IN THE COMOX VALLEY

Submitted by:

Art and Joy Johnston  
Killarney Resort  
Saratoga Beach  
R.R. #1  
Campbell River, B.C.  
V9W 3S4

It would be very beneficial for the salmonid streams in the Comox Valley to be rehabilitated and returned to near optimum production. The benefits that would accrue are:

1. An increase of fish available for local and tourist sport fishermen. While hatcheries release large numbers of fish, the return movements of these fish are fairly predictable and the runs can be intercepted by the commercial fleets. The return of natural fish to local streams is obviously more diffuse and less predictable making more fish available along the length of the coastline.
2. An increase in tourist revenues. Better local fishing would lengthen the tourist season particularly in the months of September and October which at present see a drop off in visitor numbers. A longer season would lengthen the employment time of our staff and an increased cash flow would allow more renovation and/or expansion to our facilities which would also benefit the local economy.
3. An increase in local employment opportunities. Rehabilitation of salmonid streams is labour intensive and with competent leadership can be done by a relatively unskilled work force. Such employment is badly needed in this area and given the reasonably mild climate the employment generated could be almost year round which would greatly benefit local communities.

Should a program of enhancement be undertaken, we must also remember that estuaries play an important part in the salmonid life cycle. Urban or commercial encroachment on estuary habitat must be regulated to maintain a suitable environment for developing fish. Some may feel that this factor imposes undue restriction on local growth. Let us remember that growth may be qualitative rather than quantitative and opt for quality when we consider the environment in which we live and hold in trust for those who will come after.

COMMUNITY PLANNING ACTION COMMITTEE

Submission to

THE SALMONID ENHANCEMENT COMMITTEE

Courtenay, B.C.  
November 27th, 1978

Submission by Ruth Masters, representing COMMUNITY PLANNING ACTION COMMITTEE to the SALMONID ENHANCEMENT HEARING at Courtenay, B.C., Monday, November 27th, 1978:

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## INTRODUCTION

Before making our pitch for Salmonid Enhancement money, I would like to review briefly what I believe have been the factors which have brought the fish runs in this District to a mere shadow of their former fantastic abundance, resulting in our present pleas for restoration funds in the attempt to recreate artificially what we have carelessly destroyed.

## RIVERS AND STREAMS IN COMOX VALLEY

Having lived in the Comox Valley most of my life, I can say that I am appalled at the attrition rate of our once beautiful salmon and trout streams. A resume of their present status reads like an obituary column:

The Puntledge River system, including Comox Lake, Willemar and Forbush Lakes and the Cruickshank River once supported a run of between eight and ten thousand spring or tyee salmon, as well as huge runs of coho, pinks, chums, steelhead and cutthroat trout. Now a mere two hundred or so springs trickle into the river to spawn in the artificial spawning channels constructed in a last ditch attempt to avoid complete annihilation.

I can remember what I saw as a child in the 1930's, the Puntledge literally boiled with spawning fish, commencing with summer springs in July, pinks in August, then coho, sea run cutthroat, chums and tyees until November when the steelhead arrived. Standing on the old Courtenay Bridge at the end of August, people felt they could walk across on the backs of the thousands of humpbacks which packed the river. Morrison Creek was alive with them and then coho. All that is now history. The few salmon which do return to the Puntledge and Tsolum Rivers to complete their life cycle are eagerly counted by Fisheries Officers.

Even little Royston Creek once supported a large run of pink salmon and coho. Housing development wiped this one out.

Little River, a lovely bubbling spring-fed stream used to be choked with salmon during the autumn. Formerly a year round stream, it virtually dries up in the summer months, thanks to the diversion of its flow to the Air Base, and farm irrigation sprinklers. Only a few token salmon survive the odds and complete their cycle.

Ten years ago, Trent River was well on its way to recovery from the massive clear cut logging of Van West, but has now been virtually wiped out as a fish bearing stream by Cumberland's sewage, which flows out of the pond on the former Montgomery farm, down Shite Creek into the Trent River. Don't take my word for it. Walk down the Trent, and you will find rocks coated with a fine slime, so that you have difficulty crossing without falling in. Fishermen tell me they belly flop continually in their hip waders. The formerly splendid coho run is gone, and even expert fishermen are hard put to get a steelhead, if they can stand up to fish. All this happened in the last seven years.



Fish in Millards Creek were wiped out several times when a Minto farmers barn manure tank flowed into the stream. A developer has plugged a good part of the Millards Creek estuary on Sandpiper Drive with rough landfill, and many more truck loads are piled awaiting spreading, enough to wipe out the beautiful little estuarine creek mouth. Mercifully, the local Fisheries Officer had the spreading job halted, but will all that rough fill material ever be trucked out of there?

The list is long. Rivers like the Oyster, Tsable and Black Creek have met the same fate, and their runs of anadromous fish are only a fraction of what they were, even in my lifetime.

Fishing at Wolf Lake at the head of the Tsolum appears to be slowly recovering after the big set-back it got when Mount Washington Copper poured its tailings into the watershed at the north of Mount Washington.

#### PROTECTION OF COURTENAY RIVER - COMOX BAY ESTUARY

We believe that prohibiting further destruction and proceeding to restoration of the Courtenay River estuary has to be foremost in any recommendations dealing with our local fish resource.

In spite of what is known about estuaries, we continue to allow damaging intrusions into nearly all our estuaries. Courtenay River and Comox Bay are known to have outstanding environments for salmonid rearing capacity. Here we have an area of high biological production due to a combination of various favourable factors, one of which is its shallow foreshore, but this feature is gradually being lost by filling, silting and dredging. Given the treatment it has received, it is a near miracle it is able to function even partially as an estuary.

No one has yet found a way to build an estuary or any equivalent, and the cost of restoring one which has been extensively filled and damaged is astronomical. This estuary is vital for small pinks, springs and chums during their transition period from river to ocean environments.

Yet, in spite of all the warnings from biologists and environmentalists, in spite of Regional Board plans and public desire for greenbelts and natural foreshores, our Courtenay River estuary area is continually being down-graded.

For years, Fields Sawmill filled the tidal marshes with bark, hog fuel, oil cans, wire and planer shavings. This illegal filling covering several acres went on for nine years until stopped by Provincial Fisheries Biologist, George Reid. This company's pile of planer shavings at the river's edge still spills into the river at high tide, and their mill debris pours daily into the river, alongside of their pilings which were placed to protect their landfill.

Portuguese Joe continues to fill the dyke foreshore with a wonderful variety of rocks, dirt, cement and junk, and he too has driven in pilings to protect his encroachment.

Comox Marine Industries adjacent to the Comox Marina (that being also on a massive landfill job), last spring dumped many more tons of earth fill into Comox Bay tidal flats for car parking, of all things, and this stuff washes over the life rearing zones.

Midway Electric on the dyke dumped tons of asphalt rubble and pushed it onto the foreshore for parking.

Courtenay City spread many truck loads of asphalt road rubble on the west side of the estuary into the salt marshes for airport runway expansion.

Canadian Forces destroyed acres of waterfowl and small fish feeding ground at the Spit by bulldozing valuable sand dunes into the Bay, and no one did anything.

We can cite plenty more, or we could invite you on an inspection tour of our estuary - which has always been treated as a mudhole, which only needs filling for industry or housing, dredged for marinas and parking lots, used for booming grounds and as a cesspool. Presently mill bark and debris on the river bottom help choke off the small marine life needed for supporting the life systems in the river and estuary areas.

City people pour a variety of detergents, solvents, caustic and toxic materials down their sinks, and this with the City's weed killer (Ureabor) along with ever increasing quantities of human waste as the City grows, flow into the City's sewage lagoon on the west side of the estuary. Hunters add a good spray of lead shot each year.

From the foregoing, it will be apparent that there is no one single culprit to point at to explain the present unhappy state of our fish resource. Certainly irresponsible clear cut logging at the headwaters (remember when Brown's River was really brown?), farming and building right to stream banks, use of pesticides, herbicides, and fungicides, sewage outfalls, filling and log booming, overfishing, Department of Transport dredging Courtenay River (which practice draws needed spawning gravel from the entire system), Hydro development, oil pollution from boats and garage storm drains and road run-off, coupled with landfill for buildings, marinas, draining of swamps and wiping out of tributary creeks by developers ditches all have to be contributing factors.

The estuary study, "Comox Harbour - a Plan for Conservation", by Biologist, Ted Burns, which enlarges on our poor estuary conduct and makes recommendations for its restoration was kept neatly under wraps from the public who paid for it incidentally, for a year or so, until someone leaked it to the press. In the event that the panel does not have a copy, I have prepared one, and it is submitted herewith.

#### LAW ENFORCEMENT AND EDUCATION OF COUNCILS AND BOARDS

One of the principal problems, as I see it in our attempts to restore the fish runs is in getting action to stop environmental offences which undermine the

health of streams and estuaries, putting us into the position of having to cry for federal help.

Every time I turn around something else which is precious, unique and essential for fish is being nibbled at, often with approval of the Lands Branch, Pollution Control, as well as by Regional Board and Councils, who permit industry and developers to do what they want with our water resource and systems, with little environmental understanding, or conscience, or respect for the law, many of the offences taking place on public land - Crown foreshore which belongs to the people of British Columbia.

May I call your attention to Section 54 of the British Columbia Land Act, which provides that no one without authority may place material on Crown land and if he does he may be forced to remove it at his own expense. This section would seem to give clear direction for the clearing out by the offenders of most of the estuary and foreshore filling we witness regularly. Pollution Control regulations and the Fisheries Act if enforced would put an end to much of the dumping problem.

We environmental worriers who take an interest in the rich productive and beautiful estuary area are constantly being brushed aside as our cries of anguish over further encroachments are raised to Councils, Pollution Control, Lands Branch and Regional Board, as well as Federal Fisheries.

This submission is intended to draw your attention to causes contributing to the decline of our former vast fish runs, and to suggest that with proper stewardship some of the damage could be reversed, while there are still sufficient stocks of wild fish to restore the runs naturally.

#### CONCLUSION

Summarizing the foregoing, we request that Federal money be directed towards reforming our rivers and estuary management policies, and that laws in respect of the same be enforced.

Federal bucks would be especially helpful in assisting with proper sewage studies, leading to facilities which would see solids and nutrients returned to the land, rather than being flushed into any body of water, especially an enclosed fish producing river estuary like ours.

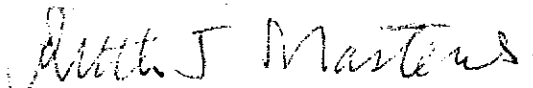
The message from here is that after taking care of fundamental problems, and helping us correct past mistakes, money is available for spawning channels, resting ponds and the like, that the work be done in as natural a fashion as possible, since hatchery fish have not proven a match for wild fish.

May I thank you for this opportunity of presenting information, which I believe could help towards a changed approach in conservation of our fish resource.  
DATED at Courtenay, B.C., this 27th day of November, 1978.

#### Appendices

Comox Harbour - A Plan for Conservation by Ted Burns, and the Case for the Low-Gradient Estuaries by Consulting Engineer, Wolf Bauer

COMMUNITY PLANNING ACTION COMMITTEE,



per: Ruth J. Masters

A Brief to:

Update '78  
Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

My first knowledge of the Comox Valley came many years prior to 1965, the year we chose to live here.

It was a fisherman's recollections of fantastic sea-run cutthroat trout fishing on the Tsolum River that first made me aware of the valley. The fisherman was my father-in-law, Mr. G.T. Bullen. He and his friend, Gilbert Thompson spent several marvelous holidays fishing the Tsolum about 45 years ago. Mr. Thompson built a sailboat later which he called "The Tsolum". The boat still sails - the fishermen are gone now and so are most of the fish. However, through Salmonid Enhancement Program, the local rivers may again become a fisherman's dream.

Salmonid Enhancement Program would mean a great deal in many ways to the Comox Valley. We are graced within the valley with several rivers whose past history of fish production would indicate a great potential for reclaiming as fish producing streams. The Tsolum River in its prime had 200,000 pinks, cohos and 3 to 4,000 steelhead. The Puntledge had runs of significant numbers. The Trent River, Brooklyn Creek and both natural salmonid rearing waterways.

With our rivers producing more fish, there would be benefits from:

- a) increased tourism - sports fishing;
- b) increased commercial fishing as a basic industry;
- c) business benefits from spin-off of the tourist - use of resorts, restaurants, etc.;
- c) tourists attracted to see the natural phenomenon of the estuary, the salmon cycle, etc. all related forms of wild life.

Perhaps our bird population and species may also return to former numbers. We have fewer the number of birds here that we had 50 years ago.

Would strongly recommend:

- a) Maximum money available and effort be put into reclaiming the natural water systems and improve the quality of the habitat. Encourage the return of natural species known to have thrived in our area.

- b) Hire additional staff - both in the field of trained habitat technologists and conservation officers.
- c) Prosecute citizens violating fishing regulations or violating rules regarding protection of salmonid habitat.
- d) Action be taken to return the Puntledge River to its natural and past habitat potential. Get B.C. Hydro out of there. Use artificial fish hatcheries where deemed desirable.
- e) Give maximum support to school programs - when the young learn conservation, we can begin to hope for the future. The present program at Brooklyn Creek gives youngsters a real concern for the value of Brooklyn Creek and the fish it nurtures.

The Comox Municipal Council recognizes the value of this creek and has passed legislation protecting it. Comox Rotary Club are contributing to the installation of fish ladders to help the salmon through the culvert under Balmoral Street.

Comox Council also has recently recognized and encouraged protection of the Comox estuary and the role it plays in enhancing the life of salmonids. This involvement of students and citizens in protecting and rehabilitating a salmon producing creek running right through an urban area - this opportunity to see the miracle of the salmon cycle firsthand in your own back yard - this is the quality of life that attracts people to live in the Comox Valley. Yes - we want the Salmonid Enhancement Program here.

Let's rehabilitate our Brooklyn Creek, Trent River, Puntledge and Tsolum Rivers. May George Bullen's great grand children (he already has two) find in the near future, the fishermen's paradise he once found here.

Mrs. Alice Bullen  
1865 Queens Avenue  
Comox, B.C.  
V9N 4H2

# COAST STEELHEADERS

## Island chapter



"Update '78"

Salmonid Enhancement Program,  
Fisheries and Oceans Canada,  
1090 West Pender Street,  
Vancouver, B.C.  
VGE 2P1

This Brief represents the feelings of the members of the Coast Steelheaders Club.

1. We congratulate you for your efforts to date.
2. The Puntledge and Tsolum River system must receive immediate priority within the federal enhancement program, with special emphasis given to the problems of allowing natural spawning of anadromous fish in the tributaries and streams feeding into Comox Lake. The problems of downstream migrants being diverted around the B.C. Hydro turbines should receive top priority. IF the above problems can be solved, why is a multi million dollar hatchery necessary?
3. Enforcement agencies are still understaffed in some management areas. Why?
4. The Johnson Strait Commercial fishery is the best example of steelhead interception, and we suspect that the estimated 3000  $\pm$  500 commercially caught steelhead in 1977 was much higher.
5. The old problems are still with us:
  - (a) Protection of esturine areas
  - (b) Poor logging practices
  - (c) Pollution
6. We see little evidence of funds expended in the area of SEP Educational programs in B.C. schools. Was there not a budgetted amount of some .25 million dollars for this project?
7. We are concerned with the outcome of the Indian net fishery on the Squamish River, and wonder if, in the final outcome of this problem, the Boldt decision in the States will look like a good deal!
8. Why has the federal/provincial SEP agreement not been signed as of this date?

Respectfully submitted

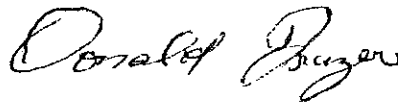
Coast Steelheaders (Vancouver Island Chapter)

Hearing on Salmon Enhancement  
Courtenay, B.C.  
November 27, 1978

Gentlemen:

I had planned to speak personally on behalf of this program, being considered for the local area. Due to an unforeseen commitment, I find this action impossible, so shall try to communicate to you, in this note, my feelings.

I am aware, as are many others, of the importance of this program, from the point of view of business, sportsmen, fishing, tourism, etc., to Comox Valley. More important still, is the fact that we must demonstrate, to the younger generation, the fact that as adults, we are concerned with trying to maintain recognition of other life forms. Our society will react to one another, on a plane, parallel to the way that we react to other life forms. I am sure that younger people learn by example, and no amount of money or talk can take its place. This program, in this area, is badly needed as one good example of our concern.

A handwritten signature in cursive script, reading "Donald Frazer". The ink is dark and the handwriting is fluid, with a large initial 'D' and a stylized 'F'.

Donald Frazer

Mr. W.M. (Bill) Lechasseur  
R.R. #1  
Black Creek, B.C.  
V0R 1C0  
November 27, 1978

Update '78  
Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Gentlemen:

I am very pleased to have this opportunity to speak on the Salmonid Enhancement Program. Particularly since I've been associated with the Sports Fish Advisory Committee for several years, representing the B.C. Resorts, Trailer Parks and Motels Association.

It was of extreme pleasure to me that the Black Creek Recreation Association applied to the Salmonid Enhancement Program last spring (1978) for a grant to take a count of the young salmon migrating to sea.

This project, in conjunction with Federal Fisheries and the Provincial government, doing a count of the returning salmon and trout species in the fall is, I think, a perfect example of public involvement.

Credit should be given to our local Fisheries Officers because of their dedication, this program has been accepted by the community of Black Creek, with much enthusiasm.

The consent of the Department of Fisheries to build a counting weir in Black Creek did not happen overnight, it took years of letter writing, talking to the Department of Fisheries people, this writer, in conjunction with some of these Fisheries Officers mentioned above, were successful in obtaining the very beginning of this project.

Because Black Creek has difficulty in providing sufficient water for the upstream escapement, as well as the downstream smolt migration to the sea is where the problem is.

We do have a satisfactory amount of water in the swamps and Northy's Lake to create the finest brood ponds available, this is obvious, by the condition of these young fish when they finally arrive at the mouth of the water course.

The indiscriminate issuing of water licences by the Comptroller of Water Rights in Victoria, also the irresponsible use of this water by certain licence holders, makes a two-fold problem for the people and creatures who depend on this water downstream.



It is simple enough to realize that a stream such as this, which is able to produce 52,000 young salmon, from a spawning of 3,500 adult salmon in 1976, and then watching this water course go dry during the summer months. It is imperative that something be done to conserve the excess winter water, that it may produce a constant minimum flow.

In order to complete and evaluate this stream, it is most important that we continue the count of fish, both spring and fall through the four year cycle.

The surface of this project has merely been scratched, but the completion of the work that should be done, causes a vision which would be most rewarding and exciting.

Using Black Creek, as an example of public involvement in what could be done, a duplication of this could be carried out in many areas, where our recreational fish are having difficulty in re-propagating themselves such as the Oyster River.

Siltation problems in this river rendered it very difficult for a salmon hatch to develop, however there are several tributary streams which flow into the Oyster, could be diverted into artificial spawning beds.

In view of this, a tremendous opportunity presents itself to the rather large and active community of Saratoga Beach, to become involved in the re-establishing the population of trout and salmon species for the recreational fishery.

*W. M. Lechasseur,  
R.R. #1 Black Creek, B.C.*

*VOR-1C0.*

*Ph. 337-5171.*

W.M. Lechasseur  
R.R. #1  
Black Creek, B.C.  
VOR 1C0  
Phone: 337-5171

# COAST STEELHEADERS

## Island chapter



Update '78  
Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

The Comox Valley Needs S.E.P.

The Puntledge and Tsolum Rivers must receive immediate attention from the S.E.P. program. The best way to enhance both streams would be as follows:

1. A hatchery facility on the Tsolum River using Wolf Lake as a source for stream control.
2. The upper Puntledge and Crookshank Rivers to have spawning channels. Thereby utilizing the entire Puntledge system, Comox Lake, etc., as a tourists fishing mecca.

The impact of this type of development would be what is needed in the Comox Valley to insure the future prosperity for the community.

Past President  
Courtenay Datsun Ltd.

M.T.R. Harris

"LET COMMON SENSE PREVAIL"

A brief presented to members of the Federal Salmonid Enhancement Committee, at the Mexicana Motor Inn, in Courtenay, B.C. on November 27, 1978.

Written by: Mr. T.R. Scress, R.R. #3, Courtenay, B.C.

For the Federal authorities to withdraw even a single dollar from the Puntledge River salmon enhancement program does not make sense. All one has to do is conduct a random survey of residents of the Comox Valley to find out that they are deeply concerned about this important commercial and recreational resource.

The reasons for this are simple to understand. People here realize that a once bountiful harvest of fish has dwindled to a mere trickle of the salmon that once enhanced our lives. The spring, coho and sockeye runs can be, and must be restored. We all realize that the quality of life in our region cannot, and should not, be compromised.

When viewed from an economic standpoint, the continuation of the program makes even more sense. It is a matter of record that the economic base of this valley is a narrow one; consisting of C.F.B. Comox, logging and tourism. The two former activities remain relatively healthy, but the latter, tourism has not experienced nearly the growth that should be expected. Tourists, who once stayed to catch an abundant harvest from the sea, now simply pass through town on their way to more promising pastures further north. This is particularly lamentable when one considers that in the past these fishing waters were second to none in the quality and quantity of landed salmon.

Logic would dictate that not only should the stocks of local salmon be restarted, but every effort should be made to expand the capacity of local streams and rivers. In conclusion, gentlemen, lets let common sense prevail!

121 - 17th Street  
Courtenay, B.C.  
V9N 1Y3  
November 23, 1978

Update '78  
Salmonid Enhancement Program  
Fisheries & Oceans Canada  
1090 West Pender Street  
Vancouver, B.C. V6E 2P1

Dear Sirs:

In answer to your request for public input to your committee re: the Salmonid Enhancement Program. I would like to lend my wholehearted support to the project, particularly regarding the Tsolum, Puntledge and Courtenay River systems.

However, I would like to make the following comments or suggestions that may be worthwhile before the SEP becomes a reality.

1. That no further industrial expansion be permitted in, on or around the Courtenay River estuary and Comox Bay.
2. That present industry already established in the above areas be encouraged, or forced to relocate - possibly on a cost sharing basis with the government.
3. That no raw sewage be allowed to be discharged into the Courtenay River, Comox Bay and adjacent areas of the Georgia Strait.
4. That the tremendous amount of wood bark, primarily cedar bark, that is being deposited by Field Sawmills into the Courtenay River and thus throughout the entire estuary and which is forming a very effective but very damaging blanket on large portions of the estuary bottom be stopped immediately.
5. That the present hair seal population be brought under control, as there has been a dramatic increase in their numbers, with a particularly alarming increase in the whole of the Courtenay River and estuary. The seals are definitely contributing to the decline of our salmon stocks once the fish enter the restricted confines of the river and even above the junction of the Puntledge and Tsolum Rivers.

I feel the foregoing suggestions, with others that may be forthcoming from interested parties, would definitely contribute to the success of the Salmonid Enhancement Program.

Thanking you for your time and attention.

Yours truly,

  
( V.H. EVERETT )



# THE STEELHEAD SOCIETY OF BRITISH COLUMBIA

c/o L.W. Peterson  
1625 McLauchlin Drive  
Courtenay, B.C.  
V9N 3V9

Vancouver Island Chapter  
Steelhead Society of B.C.  
Brief to Salmonid Enhancement Program  
Courtenay, B.C.  
November 27, 1978

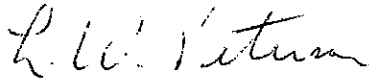
The Vancouver Island Chapter of the SSBC is pleased that the Salmonid Enhancement Program seems to be proceeding as planned. However, we have several concerns which we feel must be attended to if enhancement monies are to be utilized efficiently and fairly.

1. Lack of support field staff supplied by SEP and paid by SEP. As it is now, local Fisheries Officers have all their own work plus whatever SEP work which is thrust upon them.
2. The report on the cost recovery licencing program, headed by Dr. Sol Sinclair, is due by the end of November and he has had no formal contact with sports fishermen. We feel no such report can be fair and reasonable until such time as there has been public input.
3. Lack of public information. Even this public hearing was poorly advertised by other than local agencies.
4. Rumour has it that one Federal Minister gave away fishing rights without consulting the Federal Fisheries Minister. This rumour needs clarification and raises the whole question of the Indian fishery. Salmonid Enhancement officials should get together with the Indian peoples of B.C. and work toward a reasonable and equitable resolution of the problems of Indian fishing rights and right of access to Indian lands.
5. According to our information, the Federal/Provincial agreement is not yet signed. If it is not, we would like to know why not and when it will be as Provincial involvement is vital to steelhead and cutthroat management.

6. An Educator's Package has been produced at a cost of roughly a quarter of a million dollars. How will it be distributed to schools and when?
7. Guaranteed minimum flows are required to ensure that enhanced stocks can survive during periods of extreme low water. This is especially true for coho, steelhead and cutthroat, whose young spend a year or more rearing within the freshwater system.
8. Genetic pollution can become a problem if stocks are transplanted from system to system or even stream to stream and safeguards should be set up to prevent this problem.
9. The commercial catch of steelhead is a continuing problem, especially in such areas as Johnstone Strait and Alberni Inlet. We feel that steelhead should be classed as a trophy fish and that all commercially caught steelhead should be released, whether dead or alive, as is done in Washington, Oregon and California.
10. Continuing emphasis on commercial fisheries and commercial species such as pinks, chums and sockeye. Enhancement of those species is good only where it is not to the detriment of coho, chinook, steelhead and cutthroat.
11. Any kelp harvesting should be carefully investigated before being carried out. Studies should be carried out concerning the size and importance of kelp beds with regard to juvenile fish, bait fish and maturing fish. These studies should include whether the kelp beds are growing, diminishing or maintaining themselves and should include the question of the effect of sea urchins and other marine creatures on the size of kelp beds.
12. The monitoring of the herring fishery must continue on a careful and conservative basis. It is rumoured that 200,000 tons of a possible 290,000 tons of Georgia Strait herring are to be harvested this winter. We hope this is a realistic and not a threatening quota.
13. Lack of concern for the rivers of northern Vancouver Island. Projections list only one north island river as a candidate for protection and enhancement.

14. Finally, protection of estuaries is of the utmost importance or enhanced stocks will have nowhere to rear and acclimatize for ocean conditions and mortality would be extreme. Such classic marine nurseries as the Courtenay estuary, the Cowichan estuary, the Nanaimo estuary and the Campbell River estuary must be protected at all costs.

Respectfully submitted,

A handwritten signature in cursive script, appearing to read "L.W. Peterson".

L.W. Peterson  
Chairman, Vancouver Island Chapter  
Steelhead Society of British Columbia

## *Comox Junior Secondary School*

1809 Robb Avenue  
COMOX, B.C.  
V9N 4S4  
TELEPHONE 339-2232

December 6, 1978

Mr. James Boland, Commissioner  
Salmonid Enhancement Program  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Mr. Boland:

For the past four years Brooklyn Creek has been part of the educational program of Brooklyn Elementary and Comox Junior Secondary Schools. It is the goal of these two schools to use Brooklyn Creek as an educational tool while trying to improve the natural conditions of the creek. It is hoped that any improvements to the spawning conditions in the creek will enhance Brooklyn Creek in a number of ways.

Comox Junior Science classes have constructed dams to create pools and together the two schools have planted vegetation along unprotected areas of the creek. This has all been possible because of the co-operation from the residents who owned property, Crown Zellerbach who supplied the logs, and the Department of Highways who have supplied the shot rock.

This year Comox Junior has an outdoor educational program for the first time. Brooklyn Creek will be a major focal point for the course. The class is presently working on an inventory of the flora and fauna adjacent to the creek. A detailed map is being prepared to locate the present conditions in the creek. For example:

- location of ponds and riffles
- location of barriers
- location of areas suited for salmon spawning
- a description of the various sections of the creek

One of our aims of adopting Brooklyn Creek is to make the students and public aware of the potential and beauty of such a creek. It becomes far too easy to eliminate such areas as an inconvenience to urban development.

From an educational point of view, the creek offers us a place to meet and have meaningful discussion on:

- pollution
- the value of fish and wildlife
- the value of parks and recreational areas



- the value of urban development
- the value of the Salmonid Enhancement Program

Hopefully, in the years to come, Brooklyn Creek will have a walking trail from the estuary to Noel Avenue (approximately one and one-half miles). The public and students will then be able to enjoy this area as a park land in their backyard. Years from now, all of us will appreciate the fact that people cared for such areas. Parks like Brooklyn Creek can be incredibly beautiful and productive.

Yours sincerely,

A handwritten signature in black ink, appearing to read 'J. Milne', with a long horizontal flourish extending to the right.

James Milne  
Outdoor Education Teacher

JM/b

# Rotary Club of Comox



December 21, 1978

'Update 78'  
Salmonoid Enhancement Programme,  
Fisheries and Oceans Canada,  
1090 West Pender Street,  
Vancouver, B.C.  
VGE 2P1

Attention: Mr. J. Boleau - Public Hearing Chairman

Dear Mr. Boleau:

The Rotary Club of Comox is pleased to endorse the attached Brief prepared and presented to 'Update 78' in Courtenay by the Courtenay & District Fish & Game Protective Association. Our Club is in full agreement with the principles and intent of this excellent document.

Our Club congratulates all those connected with the Salmonoid Enhancement Programme for their efforts to date. We see hope for the future of salmon and steelhead runs and, indeed, for a greatly improved environment of the future.

We all understand more about anadromous trout and salmon, and realize how precious and fragile is the ecosystem that supports them. We can repair past damage and increase their natural abundance. The future can and will be better. We are now moving towards technical excellence and a period of constructive rebuilding that will enhance our fish stocks instead of destroying them.

Good luck for the future!

Yours truly,

A handwritten signature in dark ink, appearing to read "A.H. Linder".

A.H. Linder,  
President,  
Rotary Club of Comox

/agr

BRIEF

TO COMMITTEE ON SALMONID ENHANCEMENT

SUBMITTED BY

COURTENAY AND DISTRICT FISH AND GAME PROTECTIVE ASSOCIATION

NOVEMBER 27th, 1978

Mr. Chairman, the Courtenay and District Fish and Game Protective Association, a conservation oriented organization of some 180 members, welcomes this opportunity to address your committee, particularly in light of the fact that no opportunity was presented for a local hearing in the initial stages of the programme.

Our organization is no stranger to the salmonid enhancement programme, Mr. Chairman, having been one of the first public organizations in B.C. to take on an enhancement project under S.E.P., this being the clean up of Morrison Creek, an important salmon spawning tributary of the Puntledge.

In the ensuing paragraphs of this brief, Mr. Chairman, we believe we will conclusively show the importance of the salmon fishery, both as a source of commerce and as an important source of recreation for the 30,000 plus persons residing in the area bounded by the Oyster River on the north and by Deep Bay on the south, in which are contained the communities of Courtenay, Comox and Cumberland.

Additionally, we will attempt to inventory the salmon producing rivers and streams in this area, identify and historically examine three of the major watersheds in this group, along with attempting to show the causes of the decline in their fish

populations. Finally, we will recommend corrective action to assist in overcoming these problems.

The main theme throughout our brief, Mr. Chairman, will be the protection and enhancement of our natural stocks of anadromous fish and their habitat.

Looking at the previously designated area, we find much sports fishing activity, and many businesses dependent on a healthy fishery, some of which are as follows:

There are 25 plus salt water resorts directly dependent on this fishery.

There are 15 boat launching ramps all of which, except one operated by the military, are open to the public. Nine of these are free of charge and the remaining 5 are operated commercially. One of the latter group, the ramp operated by the municipality of Comox, at Comox Bay, had some 5,500 launchings in 1976 we are told.

Additionally, there are 2 marinas ( with another in the planning stages ), 2 boat building firms, 4 boat and motor dealers, 9 sporting goods stores, 1 sports fish cannery and numerous motels, hotels and restaurants who rely on this sports fishery for a substantial portion of their business.

Looking at the commercial fishery, we find an average of 75 commercial boats operating in this area during salmon season. Allied with this fishery are 3 fish buying camps and 2 commercial fish markets.

So it can be seen Mr. Chairman, that along with the intangible benefits of many fine hours of recreational enjoyment, for both young and old, this total fishery is a vital source of commerce for this community and surrounding area.

If we examine the salmonid producing streams in this vicinity, Mr. Chairman, we find a total of 38 streams in the area from the Oyster River to Deep Bay, which comprise 25 water courses entering salt water. Three of these prime producing river systems which were major contributors to the fishery in this area, are the Puntledge, Tsolum and Oyster.

In order to fully understand what has happened on these three rivers, Mr. Chairman, we will take an historic look at the numbers of fish produced in each:

- (1) PUNTLEDGE - In the late 1940's and early 1950's there were in excess of 10,000-15,000 chinooks; 5,000-10,000 coho; 10,000 plus pinks; in excess of 30,000 chums; 3,000 plus steelhead; along with a substantial number of sea-run cutthroat trout.

Presently, in this system, Mr. Chairman, we find chinooks number in the high 100's, coho in the low 1,000's, pinks in the low 100's; chums rebuilding and approaching original levels, steelhead in the low 100's and sea-run cutthroats approaching extinction.

- (2) TSOLUM - In the late 1940's and early 1950's we found in excess of 10,000-15,000 coho; 100,000 plus Pinks; 1,000-1,500 chums; and 3,000-5,000 steelhead.

Presently in this watershed, Mr. Chairman, we find stocks number 1,000-3,000 coho, in the low 100's for pinks and chums; and in the low, low 100's for steelhead.

- (3) OYSTER - In the late 1940's and early 1950's there were 10,000-15,000 coho; 100,000 plus pinks; and 5,000-10,000 chums; and 2,000 steelhead.

Presently in this watershed, Mr. Chairman, we find stocks number 1,000-3,000 coho; in the low 100's for pinks and chums; and in the low, low 100's for steelhead.

We will now attempt to isolate the major contributing factors for the dramatic decline in fish populations on these three river systems, Mr. Chairman.

Looking first at the Puntledge System, we find that a major problem started to develop in 1913 with the construction of a hydro dam on this watershed, and was further compounded by the construction of a larger dam in 1954, which made passage impossible for upstream migrating fish. This has resulted in approximately 3/4 of this system not being utilized for spawning.

Construction of these dams also prevented natural movement of gravel, resulting in depletion of spawning areas downstream of the dam.

Additionally, poor upland resource management in such areas as logging and road building, have also been significant factors.

Next, examining the Tsolum System, Mr. Chairman, it is found that there was large scale gravel removal from this water course, in the early 1940's for airport construction. Coupled with poor resource management in the upper reaches, through the removal of timber from large tracts of land and improper logging techniques, this has caused massive run-off and scouring of spawning areas. Subsequent low water levels resulting from this condition, have been further aggravated by an increase in agricultural and domestic water consumption, thereby reducing available habitat and impeding upstream migration.

Turning now to the Oyster watershed, Mr. Chairman, we find this system has been subjected to large scale flash flooding and serious erosion. This again attributable, in large extent, to poor upland resource management.

So it is evident, Mr. Chairman, that there are serious problems on each of these three major river systems, problems which have resulted in a drastic decline in their fish populations and a very substantial economic loss to this community.

While we do not profess to be economists, Mr. Chairman, we have attempted to document in the following table, the potential economic impact on this area, from the commercial fishery alone, had the salmon runs in these three systems been maintained at something approaching historic levels. For the purposes of calculating the figures in this table, we have used the multiplier concept whereby, starting with the fisherman and working through the various businesses and members of the



community, a portion of income will be saved and the major portion passed on. In this case we have assumed a multiplier of 10.

We acknowledge that the figures in this table are only approximate, Mr. Chairman, but, we nevertheless, feel they are significant. It can be readily seen that based on the economic impact of the 1954 and 1976 runs, if the runs had been maintained at the 1954 level, an extra income with an economic impact value of between \$2,473,640 and \$4,919,200 would have accrued to the community for each of the 24 years in this period. Or very conservatively, Mr. Chairman, a total economic impact for this period, of approximately \$88,000,000. We think you will agree, Mr. Chairman, a very substantial figure.

POTENTIAL ECONOMIC IMPACT IN 1978 DOLLARS

| Fish Type          | Year | Nos. of Fish | Avg. Wt. | Total lbs. | Min. Price | Max. Price | Multiplier | Economic Min. | Impac Max.  |
|--------------------|------|--------------|----------|------------|------------|------------|------------|---------------|-------------|
| Tsolum Pinks       | 1954 | 70,000       | 3 lbs.   | 210,000    | \$ .75     | \$1.70     | 10         | \$1,575,000   | \$3,570,000 |
|                    | 1964 | 200          | "        | 600        | .75        | 1.70       | 10         | 4,500         | 10,200      |
|                    | 1976 | 10,000       | "        | 30,000     | .75        | 1.70       | 10         | 225,000       | 510,000     |
| Puntledge Chinooks | 1952 | 4,000        | 7 lbs.   | 28,000     | \$1.78     | \$3.40     | 10         | \$ 498,400    | \$ 952,000  |
|                    | 1969 | 500          | "        | 3,500      | 1.78       | 3.40       | 10         | 62,300        | 119,000     |
|                    | 1976 | 600          | "        | 4,200      | 1.78       | 3.40       | 10         | 74,760        | 142,800     |
| Tsolum Coho        | 1954 | 8,000        | 5 lbs.   | 40,000     | \$2.00     | \$3.00     | 10         | \$ 800,000    | \$1,200,000 |
|                    | 1969 | 3,000        | "        | 15,000     | 2.00       | 3.00       | 10         | 300,000       | 450,000     |
|                    | 1976 | 1,000        | "        | 5,000      | 2.00       | 3.00       | 10         | 100,000       | 150,000     |

POTENTIAL ANNUAL ECONOMIC IMPACT FROM COMMERCIAL FISHERY

( If salmon runs had been maintained at 1954  
levels )

|                    | Min.               | Max.               |
|--------------------|--------------------|--------------------|
| Tsolum Pinks       | \$1,350,000        | \$3,060,000        |
| Puntledge Chinooks | 423,640            | 809,200            |
| Tsolum Coho        | <u>700,000</u>     | <u>1,050,000</u>   |
|                    | <u>\$2,473,640</u> | <u>\$4,919,200</u> |

Turning now, to what direction the Salmonid enhancement Programme could take to solve the problems of each of these three river systems, Mr. Chairman, we would make the following recommendations:

Puntledge River -

In order to ensure the preservation of natural fish stocks, Mr. Chairman, a percentage of returning chinooks, coho, steelhead and cutthroat trout, should be allowed safe passage to traditional spawning areas in the upper reaches. I.e; the waters above Comox Lake and their feeder streams.

We recommend that at least 20% of the juvenile fish from the present hatchery facility should also be released in the upper watershed. This action would help guarantee natural stocks in the upper river system.

In order to expedite the rebuilding of Puntledge salmon runs, Mr. Chairman, it is strongly recommended that these two actions be considered for 1979.

It is further recommended that negotiations between B.C. Hydro and fisheries personnel proceed immediately, with a view to implementing measures at the diversion dam to prevent downstream migrating fingerlings from entering the turbines.

Additionally, Mr. Chairman, in view of Hydro's recent announcement of a proposed new transmission line to supplement Vancouver Island's power needs, the possibility of shutting down the power dam during peak.

migration periods should be investigated.

In order to ensure safe passage of summer fish over Stotan Falls, it is recommended that work be carried out to improve the fishway over this obstacle. These falls, Mr. Chairman, have been a source of injury, and in some cases, death, for fish attempting to reach upstream spawning areas.

Dredging in the Courtenay River, in the early 1940's, Mr. Chairman, has resulted in the loss of valuable spawning gravel. Due to the dams, recruitment of new gravel by natural process, is no longer possible. It is therefore recommended that suitable gravel be distributed at strategic points in the River, in such a manner that the natural processes will again renew the spawning beds downstream of the dams.

We feel that this work is of vital importance to the presently rebuilding chum runs, Mr. Chairman, as these fish are currently utilizing all available spawning habitat below the dams and unless more spawning areas are created shortly, these salmon stand to be wiped out from overcrowding.

#### Tsolum River -

In order to meet the increased demands for both agricultural and domestic water consumption on this system, Mr. Chairman, and still provide adequate flows for fish, it is absolutely essential that some sort of water storage facilities be created on Wolf Lake and the headwaters of some of the tributary streams, if any sort of a fishery is to be maintained in this river. Since the agricul-

ural industry has a vested interest in this watershed, it is recommended that the Salmonid Enhancement Programme and this industry negotiate a cost sharing arrangement for the construction of any water storage facilities on this system.

In conjunction, with this, Mr. Chairman, we further recommend, that all future demands by the agricultural industry be closely monitored by the Provincial Water Rights Branch to ensure a more equitable distribution of stored water.

As a means of expediting the return of pink and coho salmon, steelhead, and cutthroat trout to this system, we recommend the immediate construction of up-welling incubation boxes. If necessary, Mr. Chairman, this club is willing to undertake the construction of these boxes, as an interim measure, until such time as a more permanent facility can be built.

In order to assist upstream migrants in reaching historical spawning areas, we would recommend that a programme be initiated to improve culverting on this watershed.

As a final recommendation on this river Mr. Chairman, we suggest that young coho, steelhead, and cutthroat trout, be released in the upper tributary streams, for rearing purposes.

OYSTER -

As a means of rebuilding the runs of coho, steelhead and cutthroat trout on this watershed, Mr. Chairman, we

would recommend the construction of incubation boxes, in particular on the Little Oyster, and Woodhus and Bear Creeks.

Additionally, we recommend that the feasibility of reinstating the fish ladder on Woodhus Creek be examined as a possibility of providing additional access to spawning areas on this system.

Any recommendations on salmonid enhancement, Mr. Chairman, would be incomplete without examining the factors which affect these fish while at sea. We will now take a brief look at this area.

Food sources of salmonids, Mr. Chairman, is one of the areas of our concerns. We would recommend that the S.E.P. undertake a study of the relative importance of shrimp, herring, needlefish, squid and other marine life, in the diet of salmonids. We feel that any such study would indicate the importance of herring in this respect, and the need for conservative harvesting of our herring stocks. We feel this is particularly important, Mr. Chairman, in light of the large amounts of herring being taken for the roe fishery.

Kelp harvesting, is another of our concerns, Mr. Chairman. The importance of kelp as a source of protection, in the rearing of herring, is a known fact. We feel this concern is justified locally, Mr. Chairman, in that where we used to have large kelp beds in the area off of Willemar Bulff and Point Holmes, presently, some 2 years after harvesting was carried out there, we find very little sign of this important marine plant. For this reason, Mr. Chairman, it is essential that the Salmonid Enhancement Programme work very closely with the provincial authorities in the protection of kelp beds from pollution, and in determining what

areas, if any, should be harvested.

Lastly, Mr. Chairman, but an area of vital concern is our estuaries. The values of estuaries are now becoming well documented. In addition to serving as critical nursery grounds for salmonids, they play an important role in the existence of many other forms of wildlife. One of the key areas in these roles, Mr. Chairman, is that played by the tidal interface, the exposed area between high and low water, an area very rich in marine life.

A prime example of this type of estuary locally, the Courtenay River Estuary, has been singled out as one of the most productive and important, on the east coast of Vancouver Island, and perhaps, Mr. Chairman, even in all of Georgia Strait.

We strongly urge, Mr. Chairman, that the S.E.P. work in conjunction with municipal and provincial authorities to maintain and enhance the integrity of this valuable estuary, through protection from such forms of pollution as sewage, marinas, landfill and other forms of water pollution.

In this regard, Mr. Chairman, we were most encouraged by Comox Council's recent decision to protect Comox Bay from further development, and strongly recommend that the Courtenay River Estuary be treated in the same manner.

In summary, Mr. Chairman, we feel that the Salmonid Enhancement Programme can play a major role in increasing the numbers of our salmonids. However, we would caution against attempting to accomplish this primarily through the construction of highly visible concrete fish factories, such as hatcheries and spawning channels, the direction the early stages of the programme appear

to have taken. While we do feel that hatcheries can play a limited role in attaining this objective, we would for the most part recommend a strong programme of natural rehabilitation and enhancement of our streams, supplemented by some hatchery facilities in the cases where fish populations have fallen beyond the level of self rehabilitation. This programme, Mr. Chairman, while not as readily visible in the public eye, is one in which we feel great potential is offered for public involvement and one which we believe will go further towards solving the long term problems facing our salmonids, and providing the maximum of economic benefits for our communities.

While we do not profess to fully understand the ramifications of maintaining the natural generic strain of the native fish in each of our streams, Mr. Chairman, we do feel this is a desirable feature, and recommend that both for the taking of eggs for artificial rearing purposes and for the seeding of young fish in streams, that native fish be utilized as much as possible, throughout the Salmonid Enhancement Programme.

We believe the time is opportune, Mr. Chairman, for a good portion of S.E.P. funds to be utilized in re-establishing the fishery in this area, and that any investment here will by far, be outweighed by the economic and social benefits returned to this community and the country as a whole.

To expand on this, Mr. Chairman, much of the forest cover has now returned to the upper reaches of the Puntledge and Tsolum watersheds and there should be a limited impact on any re-established fishery in these systems.



Secondly, Mr. Chairman, any monies spent on enhancement of local salmonid runs should benefit the fishery as a whole, and carry a wide economic impact, as Fisheries Department tag returns have indicated that fish produced in local rivers are caught over an area bounded by Johnstone Strait on the north, the Gulf Islands on the south and the Sunshine Coast on the east.

Lastly, Mr. Chairman, it is widely known that this area is currently suffering from substantial economic recession, and an early start on some of these projects would do much to assist in off setting these economic difficulties.

As a final note, Mr. Chairman, we would point out that unless the protection and enhancement of our natural fish stocks, streams, spawning beds and estuaries become an integral part of the Salmonid Enhancement Programme, all the enhancement in the universe will be but an exercise in futility.

Keith G. Seguin

Chairman Conservation Committee

A.E. Molyneux  
1931 Comox Avenue  
Comox, B.C.  
V9N 4A4  
January 4, 1979

Salmonid Enhancement Program  
Public Hearing  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sir:

I wish to make a written frief to you concerning a few points arising from my fishing experiences over the last six years here in the Comox area.

Firstly, the kelp beds; I notice these are havens for needlefish and young herring and I've seen a whole kelp bed removed or harvested. This year, the middle kelp bed (#4) had gone from Comox Bluff area - no protection for needlefish etc., hence no salmon. Why can't the people who allow kelp harvesting, allow only 25% of a kelp bed to be removed from one bed? P.T.O. This would allow faster recovery and not a total disruption of the habitat.

Secondly, river estuary; this is a delicate area which provides food for fry and an area of adjustment for young fish to salt water. Areas on each side of the river mouth must be set aside that correspond to the volume of water in the river and the river's spawning channel potential. No development in terms of dredging, filling, dyking shall be allowed. Plus the river channel itself must be protected from rapid run-off due to clearing from logging, farming and subdivision. What we need are sediment laws - that provide stiff fines for individuals and firms that allow excessive soil (etc.), sediment into the river channels. Some of the States already do this. This makes a farmer grow cover crops to prevent soil erosion from bare fields and encourages better forestry management.

Thirdly, spawning and hatcheries; all rivers that have spawning channels and hatcheries should have a policy to allow the excess salmon to escape to rivers above their channels in numbers that will enhance the natural or wild stock - even if a large percentage of the fingerlings might be killed by hydro dams, etc. When the optimum number have been allowed to escape, the remaining fish held back by fences should be harvested and sold in numbers of 10 to the general public at the going commercial fishermen rates. This commercial fisherman has been given a chance to harvest the excess just off the river mouths. In fact, all sport fishermen should be allowed to fish for all unwanted salmon species held below a fence regardless of species.

Fourthly, sport fishing limits; I would like to see this done after sufficient evidence has been gathered that is factual and not projected by the catching of marked fish. I propose that all fishermen buy a fish card good for 20 salmon for \$2.00 and be allowed to catch 4 a day. When the card is used up, he/she must turn it in to get the next card. Each card must carry name, address, dates fish caught and be punched up to date. All cards must be returned by January 1, 19\_\_ to the Federal Fisheries. This will give accurate data, control individuals from going over limits and provide funds to take care of this extra clerical matter. This will tell our officials the exact location of catches and what areas should be closed, ie. 6 out of 7 days a week to protect key runs.

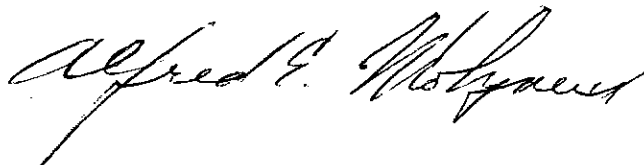
Fifthly, commercial harvesting; should occur later in the season when the salmon are more mature - catches will be more fish with more weight. Also, herring catching used, be reduced to allow for an increase in salmon food especially in the Strait of Georgia. Could we not develop runs that would permanently stay in our inside waters if we had lots of herring? Should our whales be systematically harvested to prevent their excessive multiplication and competition?

Sixthly, hatcheries and food; I'm for this, hatcheries however the size of population, depends on food in the salt water, therefore, as we increase fingerling population, so should steps be taken to ensure an increase in salt water food.

Seventhly, size of fish harvest by sports fishermen; I feel the early harvest of bluebacks is a crime in May and June and the season should be opened at the beginning of July when the fish are 4 to 6 lbs. in size. It is a shame to see boat after boat coming in with 2 lb. bluebacks and grilse. Spring salmon fishing should be allowed from December 1 on to spawning time, then closed.

I hope you will consider my suggestions in your public hearing.

Sincerely yours,

A handwritten signature in cursive script, reading "Alfred E. Molyneux". The signature is written in dark ink and is positioned below the typed name.

Alfred E. Molyneux

Kye Bay.

Bed 1  
Kelp system.

Point  
Holmes.

Kelp  
Bed 3

Bed 2  
Kelp

Lanark.

all of this  
was removed!

Kelp Bed 4

Core boring.

Comox, B.C.  
9 January, 1978.

BRIEF FROM THE COMOX VALLEY FARMERS INSTITUTE  
TO THE SALMONOID ENHANCEMENT COMMITTEE  
c/o Federal Fisheries Department, 1090 West  
Pender, Vancouver.

The farmers represented by this Institute wish to express support for the concept of improving rivers and streams for increased fish production. At the same time we are concerned that such projects may be undertaken with detrimental effects on agriculture, or that an opportunity to develop a scheme beneficial to both fishing and farming might be missed.

A very small percentage of our Province is arable. Most of the best farmland is in river valleys. We wish to remind the Committee members that this land is in the A.L.R. and farmers have a legitimate right to farm this land to obtain maximum agricultural production. As one of only three or four farming areas on Vancouver Island, the Comox Valley plays an important role in feeding the population in the north half of the island. We hope that any enhancement program in this region would choose, as much as possible, those many watersheds where a conflict with agriculture does not exist. In addition we feel an enhancement program

- (a) should not reduce the existing and future requirement of water allotted for irrigation;
- (b) should not decrease the present or future development of scarce arable land;
- (c) should not interfere with good farm management practices necessary to maximize agricultural output. These practices include drainage, irrigation, land clearing, etc.;
- (d) should not expect the farmer, whose land happens to be traversed by a fish bearing stream, to shoulder the economic burden of an enhancement program designed to benefit society at large.

Strict enforcement of existing and proposed water and fisheries legislation may result in harrassment of farmers and loss of actual or potential farm income. Therefore, we would like to see all proposed fisheries projects in this area submitted to a local committee representing all affected parties.

If the Tsolum River is to be improved for fish rearing, we hope the project will include provisions to reduce flooding and increase the amount of water available for irrigation. A large reservoir at Wolf Lake would provide thousands of acre feet of water for irrigation. It would also enhance the production of fish by reducing summer water temperature, by increasing summer flow for juvenile rearing, and by increasing winter flow during periods of freezing weather. Reduced flooding would cut down loss of arable land and reduce the deposition of the silt which kills salmon eggs.

We sincerely hope that the recommendations of the Salmonoid Enhancement Committee lead to projects, which benefit both fishing and farming. Since we are both interested in the production of food, we trust that the increased production of fish will not be done at the expense of agricultural production.

President: 'Mr. George Prothero,  
R.R. # 1,  
Comox, B.C.  
V9N 5N1

Secretary- Mr. L.M. Beks,  
Treasurer: R.E. # 3,  
Courtenay, B.C.  
V9N 5M8

Mr. Alfred Recalma  
R.R. #3  
Qualicum Beach, B.C.  
V0R 2T0  
January 10, 1979

Update "78"  
Salmonid Enhancement Program  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sirs:

The Big Qualicum River has an advantage over other rivers for SEP because it has flood control. Therefore, there should be as many spawning channels as, "The traffic can bear." We need natural spawned salmon. It is the flash floods that do the damage.

The Provincial government has a large responsibility for rivers, streams, creeks and estuaries damage. They allowed the forest companies to rape the land. Denude our mountain sides which destroyed our watersheds. Reforestation came too little and too late. Salmon is a resource that can be replaced every few years and could be in years to come, our number one industry. When the trees are all gone.

The salmon runs were once fantastic, beyond our present day comprehension. The Big Qualicum Hatchery should supply the Southern Island Indian Bands with food fish. This way it is protecting the spawning salmon from being disturbed in all these areas.

Roe should be sold from the salmon. It would create work and help pay the expenses for food fish delivery.

Spawning salmon should not be disturbed, therefore, all rivers and streams should be closed during the spawning season.

Five dollar licence fee is much too small. Fee should be according to size of boat.

The limit of four salmon to sports fishermen should be cut and a boats limit regardless of number in boat, should be six salmon per day. One day closure a week should be enforced, lets say Wednesdays.

If they are true sports fishermen, they should realize that the reduction in salmon take is a must.

What do they do with all this salmon? Taking small salmon in early spring should be discontinued. Community input should be encouraged, such as clearing streams and reporting violations of Fisheries regulations.

Incubation boxes should be installed in all creeks near a school where the students could care and tend to the projects, thus they would learn and respect nature.

In conclusion, I would like to remind you of the promises of the Federal government that we, Indians, would be allowed to fish as long as the sun shone and the rivers flowed. No where in these treaties and promises that we were to be licenced and that we were not allowed to sell the fish. The early settlers and ships bought and traded salmon and were happy to do so.

Sincerely,

*Alfred Recalma*

Alfred Recalma



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of the Salmon fishing area!*



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R.R. 2, COURTENAY, British Columbia, V9N 5M9

Phone (604) 334-4154

January 10, 1979

Update '78  
Salmonid Enhancement Program  
Fisheries & Oceans Canada  
1090 West Pender Street  
Vancouver, B. C. V6E 2P1

Dear Sir;

We wish to express our appreciation of your Salmonid Enhancement Program. To a large extent the increased number of salmon available to local sports fisherman during the past few years can be credited to production in hatcheries on Big Qualicum, Capilano and Quinsam Rivers.

We feel, however, it is equally vital to salmon stocks that measures be taken to promote natural salmon production. Thus, we heartily endorse any measures which would encourage the return of the Puntledge River system to a higher level of productivity.

We also appreciate your stream programs and request that steps be taken to maintain Kitty Coleman Creek by keeping it clear for return of spawning coho and cutthroat trout and for the migration of fry downstream. For the past thirty year we have regularly checked Kitty Coleman Creek and have noticed a gradual decline in the numbers of coho returning to spawn. We believe that productions in this creek could be increased by regular maintenance which would not be costly in time or money.

When considering licenses for sports fishermen we ask you to bear in mind that in our thirty years experience in boat rental operations, the number of salmon caught by fishermen renting our boats averages one per three or four rentals. For this reason we feel the license should not be prohibitive and also that the license requirements remain for non-Canadian-owned sports fishing boats which contribute much less to our economy.

We view with great alarm the herring roe fishery and strongly request close monitoring of herring stocks. In this area our past experience has been most distressing because our local salmon fishing declined so greatly as a result of herring seining on our door step. It is obvious that both natural and hatchery produced salmon must have a good supply of food, the bulk of which is herring.

The provision of a pollution free environment is basic to any salmonid enhancement program. With this goal in mind we believe the time has come to discontinue the discharge of sewage both treated and untreated in to the Straits of Georgia.

It is our understanding that the water in the straits is relatively constant with little exchange with water outside its area thus becoming progressively sensitive to the addition of foreign matter.

Thank you for this opportunity to express our opinions.

Respectfully

A handwritten signature in cursive script, reading "George H. Bates". The signature is written in dark ink and is positioned below the word "Respectfully".

GEORGE H. BATES

Mrs. A.M. Smith  
R.R. #1  
Anderton Road  
Comox, B.C.  
V9N 5N1  
January 10, 1979

Brief on Salmonid Enhancement Program

Dear Sirs:

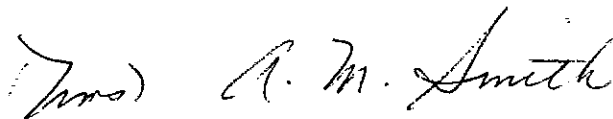
It seems to make sense to sort out the fish at the mouths of rivers, when they are returning to spawn. Then pick out the best salmon to use for artificial hatching, selling the surplus meat collected, at that time.

After artificially hatched, the young could be put back into the headwaters of the rivers they were collected from. If these rivers hadn't any rearing ponds, it wouldn't take very much effort, in many cases, to make small lakes that these young fry could be kept till they were ready for migration.

There is one creek, in particular, Washer Creek, at Union Bay area, that has a swampy lake that could be improved in order to handle many fish in the manner mentioned.

This is a rather simplified plan but I'm sure it could be worked.

Yours truly,

A handwritten signature in cursive script that reads "Mrs. A.M. Smith". The signature is written in dark ink and is positioned below the typed name.

Mrs. A.M. Smith

D.R. Goodsell  
R.R. #1  
Comox, B.C.  
January 20, 1979

Salmonid Enhancement Program  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sir:

It seems to me that the S.E.P. is at present a hell of a lot more political and non-productive as is usual with a government program, ie. I visited the Adam's River this year and saw thousands of wasted money in the form of parking lots, attendants, R.C.M.P., B.C. Forest Service persons and Fisheries persons, all on high salaries and standing around doing practically nothing.

I saw wasted monies in all the Rod Haig Brown conservationist B.S. Mr. Brown and I fished the Campbell River together at times and he continually, almost constantly fished until the fish became so scarce that they were practically non-existent and for the lack of fish, he became a loud mouthed conservationist. Both the last newsletters had a considerable amount of space and time wasted on this subject and I hope this is the last of it.

Pond rearing of S.E.P. anadromous species should be either discontinued or the release times set so as not to coincide with hatching time or release times of fry in the affected streams, ie. super chinook smolts were released into the Puntledge River system at the same time as the natural hatch was in full bloom and it was observed by myself and others, including Mr. Trent (a Fisheries Officer) these super smolts eating fry by the thousands. All species suffered from predation by these super smolts and undoubtedly all streams where super smolts, pond raised, are released suffer the same fate. Another factor in the pond raising is the disease noticed in the Quinsam Hatchery. It seems to me that these hatcheries are not being kept very clean, holding ponds are very dirty and the main object of the staff working is to do very little about keeping the nursery areas in a scrupulously clean condition as they should be. More staff is not needed, more work by the existing staff should be the order of the day.

Another item noticed seems to be the lack of supervision over the expenditures by the engineering division. It would almost seem that the engineering division is running the show according to the changes in the hatcheries noted every year involving millions of dollars in some cases only to be changed in very short order even when the existing setup is working satisfactorily. Yet with all the engineers ability (?) such fiascos such as the Capilano kill could have been avoided had some thought been given to the possibility of chlorine poisoning been eliminated from the hatchery system.

A small pressure sensitive valve could have been installed on the Rosewall system to either sound an alarm or activate a pump to solve the water supply problem. Other errors I saw or heard about are washouts on streams where spawning channels have been established, ie. Weaver Creek where an argument seemed to be who would remove the log jam. On the Fulton River a mound of gravel used to stop a flooding river and I am given to believe this happened more than once. Also, on the Fulton River only sockeye were allowed in the spawning channel, other species of which we are now short were braided out of the channels.

The retention of beaver dams in most streams is more desirable than the removal of same except in cases where salmon are completely stopped by them in their migration upstream and only at migration time should the salmon be assisted. The dams are of far greater value as rearing ponds for all species that remain in the streams over dry and freezing spells of weather. This is based on lifetime observation of local streams, especially the Kitty Coleman and the Black Creeks. Both of these streams have had continuous runs of coho, steelhead and trout. The steelhead and trout were eliminated by the stupidity of the biologists in allowing unrestricted fishing in all streams. I think a mistake was made in Black Creek this year by biological teams stopping the coho in their migration up the creek. I observed fish migrating, accompanied by the usual birds of prey and several hundred coho were seen, two days later no birds were observed so approximately a mile of stream was checked and no salmon were observed and a subsequent check also revealed no salmon in evidence. It seems the gates installed did alot more harm than good as spawners seemed to be eliminated from this valuable stretch of spawning area. It seems to me, the biologists who have never spawned, are trying to tell the fish how to do it, with disastrous results to the fish. I do not think that biologists should be allowed to make any decisions until they have practical knowledge, not book knowledge, of what they are trying to accomplish. A good way to get this knowledge is to hire the biological persons and combine the practical with the technical knowledge.

Gates were put in on the Tsolum River with disastrous effects on the resident runs, ie. gates were put on above the junction of the Puntledge and the Tsolum two years running which really cut into the pink salmon runs. Further gates were put on the Tsolum above the junction of the Wolf Creek and in the past two years, I have not observed any salmon in the Tsolum system and the Tsolum has practically unlimited spawning potential whereas the Wolf Creek has a very limited spawning potential. The personnel encountered on the Wolf Creek station left me quite concerned for the welfare of this valuable resource. I encountered him a few times and was quite surprised to have him hail me from his position on the stream, during the late fall to have a look at the run of steelhead in the creek, this seemed unusual at this time of year and as this man had worked on the stream in previous years, I thought he must know something, so went to look and found several coho and some trout in evidence. I told him they were not steelhead but did not seem to convince him of the fact. Later, while in the local Fisheries office, I asked what kind of duds they hired and I found out almost immediately from the patrol officer that answered when I was asked to explain my question. He said, "Oh! You mean those sockeye that are in there." The other day I phoned a local government official to ask to pass my observation of super smolt predation on existing wild fry especially the chum fry and I received the usual bright answer and this one was, "How did you know they were chum fry?"

Another thing that will have to be curtailed is the Indian food fishing in spawning streams with modern nets. The Indians may have aboriginal rights if they want to go back to the aboriginal ways of catching fish. Give them back their spears and bows and canoes and not unlimited potential such as high speed boats and gillnets or seiners. There is a hell of alot of cheating in the food fish, not only by Indians and alot of this is going to market and not for its intended purpose. I think netting in any stream should be abolished above certain distances upstream and especially the Fraser River, where you see nets in every backeddy which can be reached by high powered boats. I also think that a Fisheries Officer should be accompanied on his patrol, especially after firearm incidents, by army or police personnel to stamp out this law breaking and the use of machine guns should be made to stamp this out.

Every effort should be made to restore the runs of fish in the Tsolum River. If possible the silt deposited between the dam and intake on the Puntledge should be cleaned up. All silt from the gravel washing for the dam was allowed to enter the stream thereby covering the available spawning gravel and I am certain a considerable amount of trout and salmon spawned in this area. A suit-deterrent should be found to reduce the number of mortalities or the powerhouse closed down at the time of the downstream migration.

Hatchery processes should be started on the West Coast and the North Coast, these would not bring many votes to the politicians but they would produce alot of fish and larger fish in most cases than Gulf hatcheries. The Netarts boxes seem to do a great job (upwelling systems) lets work on them and not these million dollar spectaculars, save alot of money and produce alot of fish.

Thank you for your time and I hope this long winded missive is of some value in helping to restore the salmon in the Pacific Coast. These are my opinions and based on observations of a lifetime in this area and as a commercial fisherman.

Another observation is the fact that most Fisheries jobs are being taken by ex-military personnel and prairie farmers. It seems to me that we should be training younger persons that need these jobs and training, rather than the largely useless persons we seem to be acquiring through the ex-military agreement.

*D.R. Goodsell*

D.R. Goodsell

As a resident of Comox Valley for 57 years, I have seen the time when the rivers were full of salmon, bank to bank. It distresses me to see such pitiful small numbers of salmon now using our streams. I'm also a farmer who is interested in the economy of the valley. As well, I have travelled some and can visualize the potential for the tourist trade to our area, if fishing were up to its maximum levels. The following recommendations would help bring this about.

The Comox Bay and the estuary of the Courtenay River need to be cleaned up so that the young fry can survive. The sewage and pollution of Courtenay and area should be confined and treated to enable the Comox Bay to come back to its original potential, as a herring spawning ground and fry rearing area. The seal population of this area should also be controlled.

There should be a control-flow dam on the Tsolum River, as well as a fish ladder. Then Wolf Lake could be used for a rearing lake of coho fry. The lake should also be large enough so there could be irrigation water available as needed. This would then ensure hundreds of acres of irrigated farmland, as well as millions of salmon each year in unending supply. Here then, is a plan that provides us with a much needed work employment program and can also return the cost over a short period of time.

A fish ladder on the Puntledge River at Bevan, should be planned so the salmon can get into Comox Lake; thus enabling the fish to spawn in the rivers flowing into the lake. This may necessitate improving the condition of some of these rivers to make the spawning of salmon more efficient by less mortality. It would surely seem possible to screen a passageway for the fry, to keep them away from the intake pipe to the powerhouse. Perhaps locate a screening or air bubble diversion upstream about  $\frac{1}{2}$  to 1 mile distance.

A lake of the size of Comox Lake could easily be used for a rearing site for springs and coho, entailing very few problems re: food and growing areas for fry.

There are more creeks and rivers, from south at Deep Bay to north at Oyster River, that flow into Georgia Strait, than in any other area on Vancouver Island. With a little foresight and ingenuity, they could all be improved to hatch and rear many times the fish that are now reared there.

Consider the small Little River, the air base at present is taking alot of water from a spring which should feed Little River. If they would take their water supply from the Little River supply system, the spring water could be diverted down a ditch along Anderton Road to the river, where the river crosses Anderton Road. This would give plenty of water for rearing salmon.

There is a small waterfall on the Amor de Cosmos River that could be laddered for the salmon to get into McCreight Lake. This would make miles of river above the lake available for spawning coho salmon and the lake would be the rearing pond for the young.

These following rivers, namely Trent, Puntledge, Tsolum, Tsable, Cook, Cougar, Oyster, Black, Washer, Millard, Little and Rosewall are waterways that could be stocked with young coho and spring each year, if it was found too difficult to handle for spawning salmon.

Now if all these rivers and creeks were producing properly, the Strait of Georgia would have to be managed more efficiently, with the likelihood that the herring fishing in the Strait would have to stop so as to let the herring be just for fish food.

With the world problems of food and protein and the pressing problems of unemployment, this program could, if set up properly, create hundreds of jobs and make available tons and tons of high protein food.

Also, it would be a renewable resource that would be here for all time.

Submitted by:

Mr EDGAR A. SMITH  
RR# 1 Comox, B.C.

339-2669.

Mr. Edgar A. Smith  
R.R. #1  
Comox, B.C.  
339-2669



Dr. J.B. Lunam  
R.R. #1  
Fanny Bay, B.C.  
VOR 1W0  
Dec. 8, 1978

Salmonid Enhancement Inquiry  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sirs:

As I live in Fanny Bay, about 16 miles south of Courtenay, and having attended the large public meeting sponsored by S.E.P., was persuaded by some friends to put in a mini-submission re: the estuaries and feeder streams, starting with the Tsable River and going south by the Cougar, Coal, Waterloo and Rosewall Creeks. This entails a distance of about 6 miles.

The estuaries are in excellent condition, partly because a great deal of this area is oyster lease and also because no other industrial development has taken place. I hope this will be preserved for the non-polluting oyster industry, which is economically very viable and should continue to be so in perpetuity. It is too bad that the importance and high nutritive value of estuaries is so poorly understood by a great deal of the public and many politicians. The main aim, then, should be to allow no encroachment on the above estuaries.

The streams themselves have been logged heavily and in the lower reaches it is the deciduous trees, especially the alders that give them some shade in spring, summer and fall. The health of the stream is going to be dependant on logging policy and clear cutting, etc. to the banks should be a thing of the past, but it still goes on.

These five streams are invaluable for the propagation of salmon, steelhead, and trout, if their health is maintained. I think you will find that an aroused public opinion will back you up in any stand you take to prevent any deterioration of the above systems.

Sincerely,

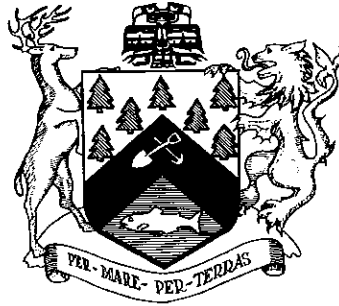
A handwritten signature in cursive script, reading "J.B. Lunam". The letters are fluid and connected, with a prominent loop at the end of the last name.

J.B. Lunam

PORT HARDY

# DISTRICT OF PORT HARDY

P.O. BOX 68,  
PORT HARDY, B.C.  
V0N 2P0



TELEPHONE 949-6665

## ENHANCEMENT POTENTIAL OF SOME STREAMS IN THE DISTRICT OF PORT HARDY

The Municipal Council of Port Hardy is presenting this brief on behalf of the citizens of Port Hardy. The citizens are concerned with the preservation, conservation and enhancement of the salmonid species in Hardy Bay and Beaver Harbour Bay. These two bays are comprised of one major river, the Quatse, and two smaller river/creeks, the Glen Lyon and the Stories Beach Creek:

### HARDY BAY

#### A. Quatse River

The Quatse River seems to have been loosing production steadily over the last ten years due to a variety of factors. Long term residents recall when "You couldn't sleep on the shore of Quatse Lake due to the noise of the fish jumping and now there is no noise at all."

and

"I remember when you could walk across the river on the backs of the spawning fish."

While purely subjective these comments do not reflect what we see today. The question that remains is what has happened to the river to produce this change? It appears that logging patterns with its resultant release of heavy volumns of water in the fall and spring and general industrial and commercial development have resulted in scouring floods. These floods may be the major cause of diminishing returns on pinks, coho and chum salmon.

The Federal Fisheries officers for the area suggest that the Quatse has an optimum number of pinks of 150,000 in any one year. In viewing stream counts since 1950 it can seem that the highest return is in the 78,000 range on a dominant cycle and a lowest return is in the 15,000 in a recessive cycle. With this in mind, the enhancement of the pink run is certainly one which deserves high consideration. Similarly the coho and chum runs are extremely diminished with the steelhead run apparently holding its own.

It would appear on first glance that effective water control to prevent scouring floods particularly in the fall would enhance the pink return and the chum and coho would have sufficient water to reach Quatse Lake to re-establish their return.

#### B. Glen Lyon Creek

The Glen Lyon Creek is a small formerly productive creek which due to a dam built upstream is almost non-productive. Fisheries officers estimate that there is still a small return to the lower end of the creek but this could be successfully enhanced to insure a pink run and a coho run numbering in the 10,000 range. It appears that negotiations are drawing to a successful conclusion regarding the removal of the dam and it appears enhancement techniques could re-establish the return to this creek almost immediately. With the addition of incubation boxes in the upper creek this would encourage the immediate revitalization of the coho and pink run.

### BEAVER HARBOUR BAY

#### A. Stories Beach Creek

This creek which wanders along Stories Beach and is fed by the Seven Hills uplands was a productive coho stream until 1972/73. It appears at that time that gravel removal upstream caused disastrous results with a complete wipeout of the existing salmon run. In the past three years, only 1978 has shown any return and the numbers are less than one hundred. With the Seven Hills Golf Society in the process of developing a nine hole golf course in the area, it is highly likely that the water hazards and streamways which will form part of the golf course could be successfully used to enhance the coho run.

By using incubation boxes and lake enrichment techniques it may be possible to re-establish the coho run and establish a pink run as well.

With a concentration of these three river/creeks, the District of Port Hardy Council would request that the Salmonid Enhancement Committee consider these projects for funding.

While the immediate benefits of enhancement are limited, the long term benefits for the commercial fishing fleet community, the cannery workers of our community and the sport fishing residents of our community are exceptional. The residents of Port Hardy would be reaping directly the benefits of the enhancement program. In addition with the completion of the paved highway from Campbell River to Port Hardy and the initiation of Port Hardy as the southern terminus of the ferry run to Prince Rupert for the summer season, the potential for tourist sport fishing and its resultant spill over to all commercial establishments of the community would once again aid in providing long term benefits to the community of Port Hardy.

With these points in mind we would request your technical assistance and financial support for these projects in the Port Hardy area.

A handwritten signature in dark ink, appearing to read "Brian Welchman". The signature is fluid and cursive, with a large initial "B" and a stylized "W".

Brian Welchman  
Mayor

1. Tex Lyons
2. Larry Burroughs



# GOLD RIVER ROD & GUN CLUB.

P.O. BOX 548

GOLD RIVER, B.C.

Federal Fisheries  
Salmonid Enhancement Program

Hello Again:

We at the Gold River Rod & Gun Club understand the Federal Fisheries have intentions of fertilizing Muchalat Lake to increase the sockeye run. We agree with the Provincial Fish & Wildlife Branch that this should not be undertaken. The Gold River has a substantial run of winter run steelhead and a rapidly declining number of summer run steelhead. If the sockeye run was substantially increased to warrant the existence of seine boats in Muchalat Inlet then these summer runs would disappear. Instead this run should be left alone and made into a sports fishery. These fish are capable of being taken by legal methods and would be of great recreational value to the Gold River area.

We have a S.E.G. for the Heber Creek which runs near the clubhouse. We have tried to get a stream technician to have a look at the creek but have been unable to obtain one. We have done some work on the creek as advised by Neil Armstrong of Tahsis who has been very helpful to us.

Last year we transported about 12,000 fry from the creek to the main Gold River. They had to be transported because of the creek drying up in summer. About 3/4 of these fry were coho. There is a short stretch of the creek which does not dry up and we would like to put the fry there instead of the main river. We would feed these fry with fish at the creek but have been unable to obtain one. We got some holes dug to hold the young fry but they filled in although it did put the creek down one channel instead of splitting.

Last year we transported about 12,000 fry from the creek to the main Gold River, about 2/3 of these fry were coho fry. These fry had to be transported because of the creek drying up in summer. There is a short stretch of the creek that flowed all summer long. We would like to transport the fry to this part of the creek instead of the main river. We were told we would be able to obtain fish meal from the Conuma River Hatchery to feed them. We would like to know how many fry we would have to have concentrated in one area to have people come in and get these fish tagged?

Mr. P.A. Franklin  
P.O. Box 843  
Port Hardy, B.C.  
VON 2P0  
December 21, 1978

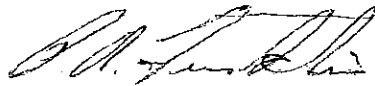
To Whom It May Concern:

My name is Paul Franklin and I am the Chairman of the North Island Chapter of the Steelhead Society of B.C. I presented a brief at the November SEP meeting held in Port Hardy. Enclosed is a copy of that brief. Also, we felt one further point should be made with respect to our strong stand on river estuary protection.

We feel that commercial clam dredging should be stopped in all river estuaries. This method of churning up the oceans floor, disrupts the natural balance of marine life in this, an already sensitive salmonid rearing area.

Could you please incorporate the previous paragraph into our brief and enter them into the minutes of the meeting.

Yours truly,

A handwritten signature in dark ink, appearing to read 'P.A. Franklin', with a stylized, cursive script.

P.A. Franklin

NORTH ISLAND CHAPTER

STEELHEAD SOCIETY OF BRITISH COLUMBIA

SALMONOID ENHANCEMENT PROGRAM



## SALMONOID ENHANCEMENT PROGRAM

I am speaking to you on behalf of the North Island Chapter of the Steelhead Society of British Columbia.

The North Island Chapter of the Steelhead Society and its parent body are primarily concerned with the protection and the preservation of all river systems within the province and their contained ~~nsport~~ sport fisheries. In order to maintain a healthy fisheries we feel certain measures have to be taken.

We need stricter regulations and stronger enforcement. There are many routes to follow.

We could have fishing closures on systems which are in a severely depleted condition.

However, we feel that partial closures might well be a better solution. By this we mean that the rivers would be closed only during crucial spawning times or times in which lower than necessary Salmonoid returns might place any rivers run of fish in jeopardy.

Another solution would be to designate any river system that receives undue fishing pressure to which it cannot support, as a Catch and Release Fishery. This would be best in conjunction with the use of barbless hooks which would ensure a higher survival rate of the released fish. There is also one advantage to the designation of Catch and Release rather than fishing closures on our streams. This is the fact that during river closures the sport fishing recreational value is totally lost.

One other method of protecting anadromous gamefish could be through the designation of flyfishing only. This is a worthwhile idea, however, great care should be taken to ensure that it only be instituted on river systems which are suitable for flyfishing. If too many unsuitable systems are designated as such, then we will be losing part of the

valuable recreational value of sport fishing on these rivers. Mainly due to the fact that although there is a large flyfishing following they are still in the minority when compared to drift and spin cast fisherman. Such a designation has been put on Marble River which we feel is an excellent choice. On this river system one further step could be taken. We feel the minimum size limit should be raised to ten inches on all species. This would allow a better smolt escapement from the river, as many of the resident trout caught are in actuality salmonoid smolts.

Tackle restrictions could be imposed on all systems, such as abolishing the use of all treble prong hooks for fresh water fishing. This will make it more difficult for the art of snagging to be practised.

None of these measures would be effective unless the rivers are patrolled strenuously and all law violators dealt with severely, as an example to potential violators and the public at large.

We feel that there will be a threat in the near future of indiscriminate harvesting of kelp. This must not be allowed until its impact on the balance of nature in our oceans can be completely ascertained. The herring are being harvested and now the kelp in which they seek food, protection from predators, and spawn in, is being removed. The Salmonoid Enhancement Program has plans to drastically increase the numbers of feeding salmon in our coastal waters. However, we must ensure healthy herring and aquatic life populations in order to feed the swelling numbers of salmon we will soon have.

We must protect river estuaries which are the heart of any river system. A valuable salmonoid rearing area of this nature in any river system should not be thrown away. Industrialization or development of any kind which might upset the fragile balance of nature should be studiously avoided. An example of this is the Tsitika River basin which is basically the only unlogged river on Eastern Vancouver Island. To coincide with the possible detrimental effects of the recent plans to log its upper reaches. A plan has been put forth to use its estuary for

booming and dry land sorting. We of the Steelhead Society feel strongly that one of the other alternatives should be implemented such as using already existing facilities elsewhere.

We believe that no minority group should control the fate of any river system. We feel that both the Federal and Provincial fisheries departments should work in conjunction and have complete control over all fisheries management. This would help in avoiding any conflict of interest which might prove detrimental to fisheries in any manner. We feel a potential threat of this nature has recently arisen on the Squamish, Capilano, and Seymour River Systems. We feel the public should be educated about one of our greatest natural heritages, the wealth of fish life both in our oceans as well as fresh water systems. To be effective this must be started in our schooling systems. Perhaps through this our children will grow up with a better understanding and appreciation for Salmonoid species and their right to be protected and enhanced. A good example of this is the environmental program offered to the grade seven students of Fort Rupert Elementary School. Not just in theory but with field involvement to make it more realistic.

We feel that river enhancement such as work being done on the Keough River is perhaps more desirable and economical on our small North Island Rivers than an overkill of hatcheries.

Hatcheries which are turning out large numbers of Salmon draw increasingly heavy commercial and sport fishing pressure. This weighs heavily on the native populations of Steelhead, Cutthroat, Rainbow Trout as well as the totally forgotten Dolly Varden. Meaning that unless we are prepared to enhance all Salmonoid species, then extensive use of hatcheries should be avoided.

The North Island Chapter of the Steelhead Society has been active. Our members have from time to time assisted the fisheries from Camp Keough on their stream enhancement program.

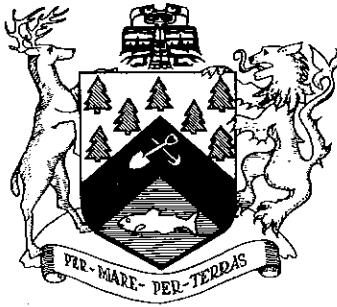
Our Chapter has undertaken a Salmonoid Enhancement Project. We are trying to return healthy runs of Coho to a small tributary of the Goodspeed River. The facility has been completed, consisting of an upwelling incubation box, in a protective shack. Water is gravity fed through a buried plastic pipe from a small nearby lake approximately 350 feet away. On the lake we built a float with a ten by ten foot hole in the middle to accomodate our holding net. Coho were then caught and kept in our holding net to ripen. Then working with the fisheries department, we stripped the fish and fertilized the eggs.

They are now in our incubation box and waiting to hatch. It is only a step but it is in the right direction and it is to be treated as an ongoing project. Each year we hope the Coho population will grow stronger.

We are thankful for the grants we received and the time, material and expert advise by interested individuals and companies as well.

In closing, we would like to issue a challenge to all other Steel-head Society Chapters, private citizens groups, clubs, logging and mining comapnies, and last but not least our local cannery and commercial fisherman who are passing away the winter months waiting for the new season to start, to come up with their own Salmonoid Enhancement project. Let's all get conservation minded and do our part.

# DISTRICT OF PORT HARDY



P.O. BOX 68,  
PORT HARDY, B.C.  
V0N 2P0

TELEPHONE 949-6665

1978-12-14

Salmonid Enhancement Program  
1090 West Pender Street  
Vancouver, B.C.

Dear Sirs:

We in Port Hardy are anxiously awaiting the outcome of your deliberations concerning the Salmonid Enhancement. As you are aware, we submitted a brief showing the needs in this area and I cannot stress enough the importance of this to our future.

We are attaching a copy of a report compiled by Mr. Shannon Dodman of Small Craft Harbours concerning harbour requirements for this area.

Yours truly,

Brian Welchman  
Mayor

BW/le  
Encl.

FISHERIES AND OCEANS  
CANADA

FISHERIES MANAGEMENT

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A STUDY OF THE HARBOURS IN AREA 12,  
RELATING TO THE NEEDS OF THE COMMERCIAL FISHING FLEET

SHANNON JAMES DODMAN, P.Eng.

SMALL CRAFT HARBOURS BRANCH  
PACIFIC REGION

NOVEMBER, 1978

## C O N T E N T S

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A. The Purpose of this Study

The purpose of this study was to determine the relative importance of Port Hardy, Alert Bay, Sointula and Port McNeill to the commercial fishing fleet in Area 12.

B. General

Area 12 encompasses the waters of Queen Charlotte Strait and the western portion of Johnstone Strait. Port Hardy, Alert Bay, Sointula and Port McNeill are the major communities servicing the needs of the commercial fishing fleet in this area. Bull Harbour, Mitchell Bay and Cracroft are minor government ports. The major communities are centrally located in the area. Port Hardy and Port McNeill are situated on Vancouver Island and Alert Bay and Sointula are on separate islands. A provincial ferry operates daily, providing transport between Alert Bay, Port McNeill and Sointula. There is a paved highway between Port Hardy and Port McNeill. To link the upper island area to the lower Vancouver Island communities, the Provincial Government is building a road from Port McNeill to Sayward. The road is expected to be open to the public prior to January 1979 and paved by July 1979. Jet air services operate between Port Hardy and Vancouver daily year-round.

The composition of the commercial fishing fleet using each port characteristically differs from one port to another. The home fleets of Alert Bay and Sointula consist of approximately 90 vessels each. The home fleet at Port Hardy is only half this number and at Port McNeill it consists of only a few vessels. The total home fleet in the area numbers 250 vessels. Three-quarters of the area's home fleet are



gillnetters, trollers and combination vessels and one-quarter are seine vessels. Whereas Alert Bay is favoured by the seiners, Sointula and Port Hardy are favoured by the trollers, gillnetters and combination vessels. In the off season, there is adequate float space in each harbour for the respective home fleets. Port Hardy is used intensively during the spring and early summer when the transient fleet of trollers, draggers and halibut vessels establish their base of operations here. Vessel populations in Sointula and Alert Bay decrease during this period due to the movement of part of the home fleet into other areas. At this time, there are approximately 250 vessels operating in the area. In August and September, conditions change again in all four ports when the transient seine and gillnet fleet move into the area and the itinerant home fleet returns. Now more than 700 vessels may be operating in the area during peak fishing periods. The maximum acceptable levels of occupancy are exceeded at Port Hardy and Alert Bay, but not at Sointula. Port McNeill remains uncrowded even with the transient fleet present. In addition to the commercial fishing fleet, a significant number of recreational vessels are berthed in each of the four harbours. These vessels take up float space far in excess of their numbers of aggregate lengths relative to the commercial fishing fleet.

C. Sources of Information

The major information bases are discussed in this chapter. Most of the information was gained through a survey of commercial fishermen using the area. An interviewer asked a selected number of fishermen in each harbour questions aimed at determining their preferred harbour location. The fishermen were also asked to identify their home port and vessel type, and to rate berthing conditions at the found-in harbour.

Wharfingers at Port Hardy, Alert Bay and Port McNeill counted the vessels in their respective harbour and listed them according to type and length on the most representative day of each of the 14 fishing periods from the end of June to the middle of September. This count was used to determine the level of utilization at each harbour during the study period and for establishing the type of vessels using the harbours. In the absence of factual records of utilization at Soimtula, the author estimated the number and type of vessels there from photographs taken in early September and other sources of information.

The interviewer recorded his impressions of the harbour conditions in Area 12 during each visit. He also recorded impressions offered by the fishermen regarding harbour conditions.

Fisheries Service, through an analysis of sales slip returns, has compiled and constructed records describing the nature and achievements \* of the CFV fleet in Area 12 for 1975, 1976 and 1977. These are hard facts estimating fleet levels conservatively. Vessels in transit only, for instance, or transient but inactive vessels are not counted. These references formed

\* Home Port Study Report

gillnetters, trollers and combination vessels and one-quarter are seine vessels. Whereas Alert Bay is favoured by the seiners, Sointula and Port Hardy are favoured by the trollers, gillnetters and combination vessels. In the off season, there is adequate float space in each harbour for the respective home fleets. Port Hardy is used intensively during the spring and early summer when the transient fleet of trollers, draggers and halibut vessels establish their base of operations here. Vessel populations in Sointula and Alert Bay decrease during this period due to the movement of part of the home fleet into other areas. At this time, there are approximately 250 vessels operating in the area. In August and September, conditions change again in all four ports when the transient seine and gillnet fleet move into the area and the itinerant home fleet returns. Now more than 700 vessels may be operating in the area during peak fishing periods. The maximum acceptable levels of occupancy are exceeded at Port Hardy and Alert Bay, but not at Sointula. Port McNeill remains uncrowded even with the transient fleet present. In addition to the commercial fishing fleet, a significant number of recreational vessels are berthed in each of the four harbours. These vessels take up float space far in excess of their numbers of aggregate lengths relative to the commercial fishing fleet.

D. Fisherman's Hierarchy of Needs

The primary need of any fisherman, home or transient, wishing to moor his vessel is a sheltered harbour with berthing facilities. The favoured type of berthing structure is a float. Land access is also preferred. The fisherman's other port needs can be classified as essential or desirable.

The essential services in a port, listed in descending order of priority are stores, marine fuel outlets and buyer-processors. Basically, no fisherman would use any port, except as a port of refuge, unless it had a marine-oriented store or at least a grocery store so that he could buy supplies to continue fishing. Practically all fishermen prefer having marine fuel outlets at all their major fishing ports. The largest part of the fleet, gillnetters and day-trollers, must be refuelled at a facility near their fishing grounds. Seiners and larger trollers are not as affected by the lack of a marine fuel outlets nearby because their large fuel capacity allows them to roam extensively. Buyer-processors are important to the independent trollers and gillnetters for delivering and marketing their catch and supplying them with ice. Most fishermen operating seiners and gillnetters deal exclusively with one company and deliver their catch to the company packer on the grounds. Some independent and company affiliated seiners deliver their catches directly to Vancouver or Prince Rupert on weekends but this is not the usual practice. A major port is not functional without a buyer-processor.

Hotels, transportation services and boat repair yards are considered desirable but not essential by transient fishermen. A hotel

serving beverages gives them somewhere to go other than their boats when in port. Often the transient finds it necessary to go to a hotel to have a shower. Many transient fishermen fly home during lay-overs, but on occasion their families fly to the area instead. There must be quick access to safe, reliable transportation at reasonable rates to make either exercise attractive. Fishermen are not prone to wander if their own ports are adequately serviced. Fishermen are generally unconcerned about the presence or absence of boat repair facilities at a port. Most fishermen will travel to the port where these repairs are available, if necessary.

To a lesser extent fishermen are attracted to ports where shops, restaurants, theatres, laundromats, banking facilities, medical and dental facilities, and recreational facilities are available.

E. Existing Harbour Conditions

The fishermen rated berthing conditions in Port Hardy, Alert Bay, Sointula and Port McNeill at less than adequate during the fishing periods. For safety and convenience in a harbour, they consider three vessels tied abreast at floats to be the maximum allowable level of utilization.

Berthing conditions in a harbour can be measured in terms of utilization of its available space. Four levels of utilization are considered: (1) under-utilization (single berthing or less), (2) standard utilization (double berthing), (3) maximum acceptable utilization (triple berthing), and (4) full utilization (harbour area full). These are purely abstract terms. There are 3300 feet of available float space in Port Hardy, 2500 feet in Alert Bay, 2300 feet in Sointula, and 1200 feet in Port McNeill. The level of utilization may be determined, as a general rule, by dividing the aggregate lengths of the vessels at a harbour by the appropriate figure in the preceding sentence. Again, each harbour has practical limits, these being a function of the useable space between structures in a basin and the types of vessels using the harbour. This method becomes inapplicable when a significant number of the vessels are seine vessels. Normally we consider a 10 X 35 foot vessel as the standard vessel. Where vessels differ from this shape, a modifying factor should be applied to the aggregate lengths to give a fairer comparison.

Data provided by the wharfingers at Port Hardy, Alert Bay and Port McNeill was used to determine the level of utilization of these harbours. Home port statistics and individual enquiries were used to determine the level of utilization of the harbour at Sointula. Levels of

utilization in the Port Hardy harbour frequently exceed the standard (double berthing) and sometimes exceed the maximum acceptable. Utilization levels in Port McNeill are usually slightly in excess of single berthing and on other occasions they are less. The Sointula harbour is considered to be filled to a level approaching the standard -- by its home fleet. Utilization levels in Alert Bay fall short of the standard more often than they exceed it during the study period. However, in the fall the home fleet fills the harbour to a level exceeding the standard. This condition gives a more accurate measure of the utilization of the harbour. More intense levels of utilization may actually occur in the harbours because the previous evaluations were carried out using ideal conditions.

Departmental records \* indicate that during peak fishing conditions in 1977 there were  $750 \pm$  vessels operating in Area 12. One may assume that a fleet, similar in size and composition, is operating in the area in 1978. Wharfinger's records and home port statistics show that 425 vessels were berthed at government facilities in the major harbours. It is estimated that  $175 \pm$  vessels found accommodation at other facilities in the major harbours. These facilities, which are operated by fishing companies, oil companies, private individuals and indian bands, are chiefly located in Port Hardy and Alert Bay. The remaining  $150 \pm$  vessels in the area are presumed to be berthed in the minor harbours, anchored in the bays or in transit. Calculations show that at the maximum acceptable level there is accommodation for 470 vessels at the government facilities in the major harbours. Allocating space for unloading zones, movement of vessels and errors in calculations, one can assume that the available accommodation equals the demand for berthage.

\* Utilization Study Report

In reality, there are other vessels present in the harbours taking up berthage space. Taking these vessels into account, there is a shortage of accommodation.

Each of the major harbours accommodates a sizeable home recreational fleet. The operators of these vessels tend to waste berthage space by not rafting their boats in the accepted manner. Alert Bay and Port McNeill would have adequate space for commercial fishing vessels at peak fishing periods if no recreational vessels were present. There is adequate space at Sointula for both fleets. Irrespective, Port Hardy would still be crowded.

At least one marine orientated store is located in each community serving the needs of the commercial and recreational fishermen. Marine fuel outlets are present in Port Hardy, Alert Bay and Sointula. Port McNeill is limited to a dry-land facility, and consequently many fishermen avoid this port.

Buyer-processor facilities are located in Port Hardy, Alert Bay and Sointula, but not in Port McNeill. The Seafoods Products plant in Port Hardy has a cannery, a large fresh fish handling line, a cold storage and an ice-producing plant. This facility accommodates a large transient fleet and a smaller local fleet of independent trollers, gillnetters, seiners and halibut vessels. Regular customers are allowed to use the company's net floats and net storage facilities. The Norpac Fisheries barge in Sointula has fresh fish handling line, a cold storage and an ice-producing plant. This facility accommodates the large local fleet and small transient fleet of trollers, gillnetters and seiners. Norpac net floats and private net floats are used by everyone with preference



given to locals and regular Norpac customers. B.C. Packers in Alert Bay operates a large net shed and an ice-producing plant for its large seine fleet and a smaller gillnet fleet. Ice is also available to non-company vessels. No off-loading facilities are needed for the B.C. Packer vessels because their catch is transported to its Vancouver plant by packers.

There are hotels with beer parlours that cater to the needs of the commercial fishermen in Port Hardy, Alert Bay and Port McNeill, but not in Sointula. Most transient fishermen avoid Sointula. Many transient fishermen, wishing to fly home for the weekend tie up in Port Hardy within 10 miles from the airport. Port McNeill is only 20 miles by road from the airport but the limited infrastructure discourages transients from using this port. Transients who tie up in Alert Bay or Sointula must take the ferry to Port McNeill to get to the airport. Therefore, it is not usual for fishermen to travel to Vancouver from Alert Bay or Sointula on lay-overs. There are adequate machine shops for boat repairs in all four harbours. Port Hardy and Alert Bay are the only locations of commercial marine ways in the area. Many local people have their own marine ways in Sointula, but these ways are not known to be open to the public.

Of the four ports, Port Hardy has the largest selection of town services such as shops, restaurants, theatres, laundromats, banking facilities, medical and dental services, and recreational facilities. The remaining ports in ranking order of town services are: Alert Bay, Port McNeill and Sointula.

F. Fisherman's Choice of Harbour

The fishermen resident in Area 12 preferred their own port under all circumstances. Transient fishermen early in the season preferred Port Hardy much above all other ports. Alert Bay was rated second, Sointula third, and Port McNeill fourth. At the height of the fishing season, however (second visit) transients favoured Port Hardy slightly less than before and Alert Bay and Sointula somewhat more. This shift may be attributed to a modal change undergone by some vessels enabling them to operate on grounds near the other ports. Irrespective of the onset of fishing season, fishermen preferred, in order of precedence: Port Hardy (36%); Alert Bay (29%); Sointula (29%) and Port McNeill (6%). For the off season, however, these same fisherman preferred in descending order: Alert Bay (34%); Sointula (32%); Port Hardy (29%) and Port McNeill (5%). The shift in preference may be attributed partly to the accidental polling of a disproportionate number of fishermen resident at Alert Bay and Sointula. By weighting the totals to reflect the actual ratios of population resident at these communities, the results are shifted to favour Port Hardy, once again.

Most transient fishermen preferred to use the same harbour in both seasons. Alert Bay is the most popular refuelling place as it lies on a direct route through the area. Hence, some of the transients, proceeding to or from Port Hardy for instance, refuelled at Alert Bay.

Most home port fishermen think that their harbours should be expanded to provide for the future activities in Area 12. Among the transients, Port Hardy was first choice, Alert Bay second, Sointula third

and Port McNeill fourth. An analysis of all returns revealed that fishermen preferred to see Port Hardy expanded (42%) most and Alert Bay (25%), Sointula (22%) and Port McNeill (11%) to a lesser extent.

G. Recommendations \*

The major harbour expansion should take place at Port Hardy to benefit commercial fishing in Area 12 the most. Relevant data indicates that among the fishermen Port Hardy is the most popular of the four ports and is also the most crowded at peak fishing periods. In assessing the future harbour needs of a port, one must take into account a community's potential growth. At the moment, it appears that Port Hardy will continue to occupy the dominant role as regional centre of Northern Vancouver Island for fishing, transportation and other industries.

No immediate expansion of facilities need be considered for Port McNeill or Sointula. These ports do not experience severe crowding at peak periods. Some expansion may be considered at Alert Bay but not of the same extent as at Port Hardy. The economic growth potential does not seem to exist here. However, Alert Bay does have a large home fleet which fills the harbour to capacity at all times. The breakwater could be extended with the addition of one new string of floats to accommodate some overflows.

The District of Port Hardy under the Canada Works Program is planning to construct a new float for berthing larger vessels. This 300 foot float will take care of the immediate needs of the transient fleet. The long-term requirements of the fleet using Port Hardy could be met by dramatically enlarging the existing harbour facilities. This could involve the construction of a breakwater on the reef just north of Seafood Products and the Esso Oil Dock. More facilities could be installed in the shelter of this breakwater, thus adequately satisfying both the current and foreseeable needs of the fishing fleet in Area 12.

\* These recommendations deal with the needs of the commercial fishing fleet, purely.



CAMPBELL RIVER

To: Salmonid Enhancement Committee

From: Peter Johnson, 302 Birch Street, Campbell River.

Preamble: It is essential that public support and co-operation be obtained if the Salmon Enhancement Program is to be successful and consequently it is necessary for the public to know exactly what is the full range of facts and figures involved in both sports and commercial fishing.

At the present time, no such information regarding sports fishing is available and consequently suggest that the following be done so that accurate data can be obtained:

1. Institute a saltwater licence for anyone over the age of 16 who fishes for salmon along with a "20-fish punch card" system so that an accurate count may be obtained. Perhaps the licences can be of two kinds, namely:
  - a) Canadian resident;
  - b) non-Canadian resident
    - short term (7 days or less)
    - long term (more than 7 days)

The only exception to the above should be customers on a registered guided boat operated by a registered guide who has paid an annual fee, to be a registered guide and who may receive additional punch cards each. Only registered guides should be allowed to purchase additional punch cards. The purpose, of course, is to obtain an accurate record and by giving such cards to guides there could probably be a much better return of data. Also, perhaps a \$500 lottery could be held in order to encourage people to return the punch cards.

2. Eliminate boat licences because they are unfair, for example, a large Canadian or non-Canadian boat may have 50 or more people fish off it during the summer.
3. Summer limits should stay the same, ie. the first four legal size salmon caught per day but the winter limit (October 15 to June 15) should be the first two legal size salmon per day. This would ensure further conservation of both the winter springs and the bluebacks.
4. Possession limits should remain as at present, ie. two days catch but instead of the 25 pound limit on canning, have a punch card limit.
5. Establishment of a SEP Public Advisory Board in Campbell River to give public opinion to the Department. This committee should consist of:

- a) one Estuary Society representative
- b) one Guiding Association representative
- c) one Chamber of Commerce representative
- d) one Municipal Councillor
- e) one Fish and Wildlife Club representative
- f) one local Department Official representative
- g) two members of public at large elected at a public meeting

Such a committee could be a forerunner of similar committees in other coastal centres and perhaps eventually these committees could establish a "West Coast Sports Fishing Commission" which would provide good public input into policy.

- 6. All funds received by the licencing system should be given to a SEP Public Advisory Board in the community in which it was collected so that such money may be used solely for conservation of salmon on a public basis. Such funds should not reflect a decrease of government funds to the SEP.
- 7. The SEP Board would hold local meetings for local suggestions for spending the funds collected. The Board may also decide to hire a local person (secretary-treasurer?) to co-operate with the Fisheries Department on a basis of joint responsibility.
- 8. In order to discourage violators there should be a stiff fine of \$1,000 - this would save much policing.
- 9. Licences should be available at any business establishment which caters to tourists.

Conclusion: The above should be instituted for a two year period and subject to a thorough review and evaluation by the local public and local Fisheries personnel.



A BRIEF SUBMITTED BY THE CAMPBELL RIVER FISHING GUIDES ASSOCIATION

Gentlemen:

Since the introduction of the Salmonid Enhancement Program in which we supported as a desirable objective, and submitted a brief to you, stating our beliefs and suggestions as requested by the Fisheries and Marine Service.

We note that during the original inquiry, the input was summarized, the priorities were established, areas such as more enforcement, natural enhancement, small stream enhancement, public involvement in projects, better protection of estuaries, received alot of input and priority, yet very little has been done in these areas, other which received lesser attention seems to have engaged a larger amount of attention, mainly aimed it seems at controlling, taxing and limiting the sports fishing industry. The amount of regulations that is being imposed or pending brings to mind the following questions.

- A. Why is there a licence for sports saltwater fishermen, who take a minority of the annual catch, and yet no action regarding a levy or finnage fee against the commercial fish companys?
- B. Why is it necessary to cut down the sportsmans salmon limit when the Federal government Fisheries Department is getting into the increased harvesting of salmon every year as is obvious from the increase of newspapers and radio advertisements offering for sale, over-production returns to the hatcheries all over the province of B.C. and only offered for sale in such a way that only large fish canning companies can take advantage of it and not the general public.
- C. That the Fisheries Department allowing almost unchecked the growth of the commercial fleet. In 1976, the total commercial fishing fleet was 6,228 boats of which 582 were seiners. In 1978, the total commercial fleet is now 7,010 boats of which 610 are seiner boats, this is an increase of 28 seine boats, although this is not a dramatic increase in the seine portion of the fleet, replacements and increased use of electronic gear has allowed a 25% increase in efficiency. These figures are from the Federal Fisheries Department statistics and that vast increases of this size leaves us with this thought. Mr. Chairman, what is the question???? It seems that the political expediancies of the commercial fish interests reminds us of the 19th century buffalo hunt.

Another area that it is obvious no attention has been paid to, is the new reduced salmon limit of 1 spring and 2 coho per fisherman as a daily limit. This could be the largest waste program instituted yet, as an example, one could see a fisherman catching a 10 lb. spring salmon upon continuing to fish for his 2 coho catch for more springs, first we all know the spring salmon being the tenacious fighter he is, there would be little doubt that when released these fish would be either dead or dying from excess acid build-up, or not unrealistic is the fact that how many fishermen upon catching a small 5 lb. spring who later boats a 25 lb. spring. Which will get returned to the water? The 25 lb. one will surely die or the already 5 lb. spring which is already dead? This time we say Mr. Chairman, "What is the answer"? It does not take long to decide that the imposition of such a limit is foolhardy to say the least.

Now another area we wish to cover is the herring fishery. If the salmonid enhancement is to work with greatly increased salmonid stocks, they have to feed on something and one major item in the food chain is the herring. Should the annual rape of herring stocks be allowed to go on unabated, there will not be much use of a salmonid enhancement program, as there will be no herring left for the salmon to eat.

When Pacific Area Field Director Mr. Dick Crouter, delayed the netting of herring roe, the commercial industry yelled loud enough and Mr. Crouter was removed to Ottawa and herring fishing resumed. This a great mistake as Mr. Crouter understood the West Coast salmon industry and was not afraid to express his opinion. The lack of herring in local waters seems to be obvious proof to us that once again the wrong decisions are being made, and one wonders how many of the shots are being called by the commercial industry???

In conclusion, we respectfully make these following recommendations for consideration:

1. That all future expansion of the commercial fleet be halted until there are salmonid stocks to support it.
2. That all herring fisheries for roe or reduction purposes be halted.
3. That commercial fish companies be required to pay their share of Finnage or Royalties.
4. That the direction of the Salmonid Enhancement Program be given to people of proven understanding, in both the field of sports and commercial fishing.
5. That more emphasis be placed in areas of high priorities, ie. natural enhancement, incubation boxes and small stream enhancement.

I would like to quote you the following words from the late Roderick Haig Brown. "Restrictive regulations alone will not make a great fishery, on the scale of todays demand. Only an ambitious and intensive program of stream improvement can achieve this. There should come a day not to long from now when every stream in the province is controlled and cared for and guarded as the enormously valuable asset it is, and when that day comes, the fisherman will again find his fish."

I would ask you gentlemen to please heed those words.

We would like to thank you at this time for letting us express our views, while being somewhat critical of the work done to date, our deep concern is the future of our fishing resources.

Thank you.

Respectfully submitted.

Robert Huxley  
Vice President  
Campbell River Fishing Guides Association

BRIEF FROM THE CAMPBELL RIVER ESTUARY ASSOCIATION TO THE SALMONID ENHANCEMENT  
PROGRAMME

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The Association is acutely concerned with the Campbell River Estuary remaining a suitable habitat for salmonids.

It is a small estuary and is a rearing ground for all five species of Pacific salmon, steelhead and sea run cutthroat.

Already approximately 80% of its estuarine zone is being degraded by log handling. (Burns Report 1976). Its entire left bank is almost completely industrialised. The whole of Nunn's Creek beyond the Indian reservation along with its wetlands has been built up and is now a hazard as it drains industrial wastes into the estuary. Nunn's Creek tributary, Camp Creek drains a large sidehill in the municipality and carries run-off from lawns and storm sewers containing fertilizers and herbicides.

The only portion of the estuary which remains as a suitable habitat for fish is that at the mouth of Nunn's Creek. The Estuary Association doubts its ability to support existing stocks in the river. It is extremely doubtful if it can support the extra pressure of several hundreds of thousands of juvenile salmonids being released by the hatchery each year.

On top of all this there is now an active proposal to transport, stockpile and ship coal from this portion.

This Association feels very strongly that in no way must this proposal be allowed no matter what safeguards are offered.

Fifty years ago the Indian people claim that as many as 5000 coho used Nunn's Creek for spawning. (Pers. comm. with Geo. Quocksister) Fifteen years ago 1000 to 1500 used it. (Pers. comm. with J. Fielden) Today it is practically nil. The same can be said for Kingfisher Creek in Campbellton.

It is a well known biological fact that every square foot of swampland that is reclaimed is lost forever. Huge areas of wetlands in the Campbell River flats and in Campbellton have been built on. The same applies to the left bank of the estuary. It is only a matter of time when the last straw will be placed on the camels back. Once the nursery goes then all the enhancement in the world won't bring the salmon back.

The Estuary Association feels at this time that the Federal and Provincial Governments combine to enact a moratorium on any development within the estuary. The Association plans to send further briefs to both the District Municipality and the Regional District in an effort to clean up the streams draining into the estuary.

This Association in the months and years ahead promises to exert every effort within its power to restore this estuary to its rightful owners, the salmonids.

Presented by the Campbell River Estuary Association,



R. Murphy, Chairman.

Van Egan  
2340 Campbell River Road  
Campbell River, B.C. V9W 4N7

A Brief to The SALMONID ENHANCEMENT PROGRAM

at the Public Hearing in the Discovery Inn, Campbell River, 29 Nov. 1978

The first public meeting in Campbell River for the Salmonid Enhancement Program in November 1976 was characterized by a broad public interest in the enhancement of our wild salmonid stocks. I believe it can be truthfully said that a majority of the presentations were concerned with the protection and restoration of our rivers and estuaries for the natural production of salmonid species. The SEP Report on the Public Inquiries dated March 1977 bears this out under the title: Moods of the Inquiries. Apparently this mood prevailed at other meetings in the Province as well.

I am therefore disappointed in reading the SEP Annual Report for 1977 to see that the major expenditures are going toward artificial means of salmonid enhancement, in particular, hatcheries. A good case can be made for the use of hatcheries to restore badly depleted runs, but their use for the purpose of maintaining runs in rivers goes against the prevailing moods of those first public inquiries. The demonstrated risks of genetic damage and wide-spread disease in hatchery-produced fish should be a warning. The best use of SEP funds would be to restore spawning rivers and estuaries and gain prior rights over other users, in particular the logging and lumbering industries. The salmonid resource is too valuable to settle for anything less.

Toward restoring the natural productivity of the Campbell-Quinsam River system, I believe the Fisheries Department should make every effort to re-open and re-negotiate the earlier agreement with B.C. Hydro regarding mean-minimum flows in the Quinsam and Heber Rivers. I'm sure everyone who knew anything of the Quinsam stocks of steelhead and salmon prior to the diversion recognize the severe declines that occurred after that time. Recently, provincial fisheries people have expressed an intention to stock with steelhead the Quinsam River above the falls located a short distance below

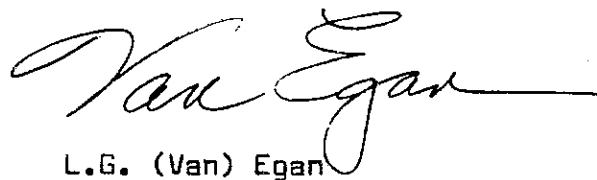
Lower Quinsam Lake. Prior to the diversion of Quinsam water in 1957 an extraordinary run of late winter steelhead occurred there. It seems likely that the restoration of that run, or the re-introduction of a new run, would best be served by having an adequate flow in the river and over the falls during the period of the run.

The present agreement with B.C. Hydro regarding the Heber River, I understand, allows the minimum flow to be exceeded downward in order to maintain a minimum flow through the diversion pipe, which operates on the *siphon* principle. Surely the water brought into Upper Campbell Lake by that little pipe is most insignificant to the generating capacity of the Campbell River hydro system. On the other hand, that same volume of water contained within the Heber's natural flow could be important in the spawning and raising of its salmonids. The Heber diversion system should be turned over entirely to Fisheries for the use of flow control, so far as it can be used in that capacity. Also, little Crest Creek that was diverted into Drum Lakes should be re-diverted back to its original channel to again add to the Heber's flow.

I would like, here, to support the proposals for licensing of salt-water anglers. To me the fundamental principle in this matter is that all resource users should pay something directly toward the management of the resource they use. In North America we seem to still labour under the delusion that resources are here for the taking. Rather, we should feel a debt toward those resources for the high standard of living we enjoy. To pay a fee toward the management and perpetuation of that resource is our responsibility to future generations.

I would like to add too, that I find the recent regulation to restrict the number of canned salmon in the possession of anglers distasteful. I am sympathetic to its intent: To stop the over-kill of salmon by greedy anglers who go far beyond their needs for food and recreation. But I believe a better solution would be a punchcard, preferably as part of a salt-water angler's license, that would set an annual kill limit, and then allow the angler to process his or her fish any way they prefer.

Thank you for this opportunity to express my views. I shall continue to follow your progress with great interest.

A handwritten signature in cursive script, reading "Van Egan", followed by a horizontal line extending to the right.

L.G. (Van) Egan

## CAMPBELL RIVER & DISTRICT FISH & WILDLIFE ASSOCIATION BRIEF

Presented to Salmonid Enhancement Programme, Public Inquiry II - November 29, 1978.

We wrote and presented a brief to S.E.P. on November 4, 1976. Now, November 29, 1978, we present another. The only way we were able to assess the progress or lack of it for S.E.P. was to go to our brief of 1976 and see what had happened locally.

S.E.P.'s #1 question was:

WHAT ARE LIKELY TO BE THE MOST EFFECTIVE WAYS TO GAIN BETTER PROTECTION OF THE SALMON AND STEELHEAD RESOURCES? We had 5 suggestions:

1. Our present day concerns are still the management of watersheds, yet at this very moment there are plans to strip mine COAL in the Campbell, Quinsam and Oyster drainage systems and also to off load coal in the Campbell River estuary. The Quinsam River has a very good and very expensive hatchery which admirably enhances both the Quinsam and Campbell Rivers already high productivity.
2. We would demand that this precious renewable source, which adds more dollars to Campbell River economy than the coal mine with its non-renewable base ever can, be protected to the fullest extent possible.
3. We do not want the environment protection that was afforded the people of North Vancouver Island's Rupert Inlet by Utah Mines and the various Incompetent Government Environmental Agencies.
4. The Federal Government has enacted Bill C-32 which is good. We still don't have adequate legal help or moneys for prosecution of industrial offenders. We have not heard of any really heavy fines arising from convictions.
5. The Campbell River estuary has not improved over the last 2 years and is once more threatened by the off loading of coal, by Welwood Luscar Limited.

## SHOULD FISHERIES AGENCIES GO ABOUT PRODUCING MORE FISH? AND IF SO - HOW AND WHY?

This has been answered by the continuation of S.E.P.

How ought the public to be involved in the S.E.P. programme?

We feel that lack of public involvement is due to communications problems. The only link we have with S.E.P. is the newsletter Salmonid. In our first brief we suggested that S.E.P. show people at the community level how they could get involved. This has not been done. We suggest that there be a joint programme with Fisheries and the public, this has not been done.

There are lots of people that would gladly participate in S.E.P. if they had leadership and were given direction on where to enhance, how and what to do.

We also recommend that any monies that S.E.P. spends should be spent directly and only for production of salmonids, not for another of the endless studies that are non-productive.

One observation we have is that if S.E.P. advertised and promoted in the same manner for participation as this inquiry has done, public participation would be overwhelming. What are some suggestions as to how Fisheries Resource Management might improve its overall performance?

In our first brief, we suggested that they change their attitude toward the sport fisherman. They have not. The S.E.P. programme and its continuation is being used as a lever for a sport fishing licence. We are not asked to participate. There is a new possession limit for non-residents; the sport fishing regulations were printed before the new regulation was even gazetted.

The Sport Fish Advisory Board was not even consulted on this new possession limit. We now hear rumours of new possession limits; chinook and 3 coho. A closure for our winter fishing as well as a sport fishing licence.

We hear that the sport fishermen are catching 40% of the chinook from Campbell River to Victoria.

Apparently Fisheries Management does not consider the dollar generation from the sports fisherman sufficient and places a greater value on the commercial fishery. We tried to place a value on the Sport Fishery Statistics, Canada, Province of B.C. and Federal Fisheries could not give us any dollar values. We did find a private study that showed sporting good sales in Campbell River were between 2.3 and 2.5 million dollars (1977). If you add sales of boats, motors, gas, accommodation, meals, guides and other merchandise and services, it is a major economic base to Campbell River.



BREIF FROM THE TYEE CLUB OF BRITISH COLUMBIA TO THE SALMONID ENHANCEMENT  
PROGRAMME

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The Tyee Club takes this opportunity to examine some of the philosophical principles on which the Salmonid Enhancement Programme is based.

One of the aim's of this programme is to increase the present stocks to the levels at the beginning of this century. We are told that the monies spent on this will be returned tenfold in the increased harvest. The 200 mile limit is going to help. The restrictions on our commercial fishery will help and there will be new and increased licence fees on everyone partaking of this abundant harvest. New hatcheries, spawning channels and other modern technological measures will improve the output.

Now all of these add up to money, science and regulations. These are very symbolic of our present civilization.

We are well aware of the squabbling between nations about the harvesting. Federal and Provincial governments wrangle about methods and jurisdictions. Sportsmen and commercial men argue about restrictions. Other sectors of our civilization such as industry must look to profits before fish, civic governments will always put people ahead of fish and those promoting the tourist industry, in particular, find it hard to accept any limitation. One does not have to be a prophet to see that all of these problems are going to be with us for a long, long time.

The Tyee Club believes that this programme should be based on the philosophical principle of respect for the fish and their right to exist apart entirely from their value as food and money. We should also accept the fact that we are quite capable of wiping them off the face of the earth.

The two strongest measures we have at hand to do this are overfishing and destruction of habitat.

It would appear now that we are able to take some positive measures to control over-fishing. But what about the habitat? Many of us have the feeling that Federal Fisheries have given up on any attempt to restore lost habitat and are tending to rely more and more on hatcheries etc. This could be a dangerous game. There is no doubt that problems with the habitat are extremely complex involving differing jurisdictions, differing users and differing ownerships. However, in other complex problems, such as health care, Federal and Provincial governments have co-operated successfully. Two years ago at the first meeting of this programme here in Campbell River many people expressed concern about the fish habitat especially the smaller streams. There was one example of some volunteers improving a stream, although God knows, it will take more than volunteers to move some of the obstacles we have around here. What has happened in this area in the two years? Further encroachment has occurred mainly.

The Tyee Club makes the following recommendations -

1. Set up an inventory of all salmon habitat on this coast. Most of this material is already available. A concise account of their portion of such an inventory should be in the hands of every public body having any jurisdiction over salmon habitat. This to be done on a yearly basis. The same should apply to industrial users on a watershed, Fish and Wildlife Clubs, Tourist Bodies, Labour Groups, Indian Bands, etc.

2. Federal Fisheries Specialists such as biologists should be stationed permanently at strategic locations on the Coast to give on-the-spot immediate attention, to protect, promote and restore all aspects of fish and their habitat.

3. Federal and Provincial Governments should combine to enact legislation specifically designed to enhance salmon habitat.

*R. Murphy.*

*Pres.*

*Tyee Club of British  
Columbia*

*29<sup>th</sup> November 1978.*

Box 99  
Heriot Bay, B.C.  
VOP 1H0  
November 29, 1978

Chairman  
Salmonid Enhancement Hearing  
Campbell River, B.C.

Dear Sir:

Having been a resident of this area for the past 32 years and having participated in all phases of the commercial fishery for approximately 15 years, from buying fish to trolling, gillnetting and finally seining, I would like to offer some suggestions that I feel would benefit the fishery.

Firstly, a better liaison between the Federal and Provincial governments as regards stream protection. Up to date, in Areas 12 and 13, there seems to be a dismal situation according to the Department of Fisheries statistics - 12 and 13 particularly because I am more familiar with this area. Bad logging practices contribute the most damaging effect on our fishery. One case comes to mind when a stream supported a run of hump peaking at 20,000 to a low of 5,000. After logging this run was completely extinct. The last two cycle years no humpbacks have been observed. One wonders why the Fisheries keeps statistics unless they are acted upon. Surely it can't be to keep the field men busy. Would it not be better to upgrade our local streams by giving them better protection. After all, they have never cost us anything to build, and money spent on them would be returned a hundred-fold. All we have to do is give nature a hand, replant the depleted streams and hire men to patrol all streams not just for a few months but on a 12 month basis so the Fisheries Department will know immediately if anything goes wrong and takes steps to correct it. All the publicity and money go to the larger projects and the smaller local streams seem to be expendable. Therefore, Mr. Chairman, I would urge that a survey be carried out, or better still, just read the statistics that the Fisheries Department has now and see what these minor streams can support, bringing them up to the spawning capacity they once enjoyed.

I would suggest that sports fishermen be licenced, the money obtained to be used to compliment the Salmon Enhancement Program. At the last hearing I attended, sponsored by the Comox-Strathcona Regional Board, this suggestion was not well received - difficulty of enforcement being the excuse. The punch card system for steelhead has been enforced for a long time. Why not use this for other salmon species. The machinery is already in place. In talking to many visitors in all walks of life I find they cannot understand why a sports licence is not mandatory. Where else in the world can the common man enjoy salmon fishing such as we have on the B.C. coast?

As to the herring fishery, this directly affects many species of fish. Many people would like to see this fishery abolished. I personally think that herring are more than holding their own. Many of the people I represent receive some part of their living from this fishery. We have a bait operation on Quadra that employs at least 30 people and pound for pound generates more dollars per ton than any other section of the fishery. Who buys most of the bait? The sports fisherman, of course.

In conclusion, Mr. Chairman, I read recently of a plaque being dedicated to Mr. Haig Brown on the Adams River - very nice, very touching. I suggest, Mr. Chairman, a better way to honour a truly great man would have been to follow a few of the recommendations that he advocated all his life.

Thank you.

Respectfully,

Samuel J. Hooley  
Director Elect  
Area "J"  
Comox-Strathcona Regional District

1911 - 16th Avenue  
Campbell River, B.C.  
V9W 4L2  
December 8, 1978

Dear Sirs:

As a commercial fisherman who attended the Salmonid Enhancement Program meeting in Campbell River on November 29, 1978, I would like to present the following points.

One of the impressions arising out of this meeting indicated that commercial fishermen were guilty of overfishing. The 95% vs. 5% figures given are completely false and does not take into account the two very important factors of location and species. In vast areas on the fishing grounds, there is no competition whatsoever between commercial men and sports men. In the areas where there is competition (ie. Seymour Narrows south to Juan de Fuca) the sports men do quite well. In this context, only three species should be considered; coho, spring and steelhead. Undoubtedly, these species are taken by commercial men (particularly the net fleet) but more often than not, they are incidental in the harvesting of sockeye, chums and pinks.

In looking over the locations of your meetings, it is evident that they are based in reasonably large centres of population. From this it follows naturally that the sports fishermen will present a strong lobby. In areas on the coast, particularly the central area, where the population is small and commercial fishing is large, there is going to be no input. This is unfortunate as commercial fishing is the oldest industry on the coast and the people employed in it have been more concerned about overfishing and preservation of habitat for a lot longer than most sports men and environmentalists.

The commercial fishing industry is more regulated and closely watched and has been for the past several years than sports fishing. We bear very little animosity towards the sports fishermen as we recognize that we all eat out of the same pot and all of us have to accept limitations in order that the salmon survive.

An area licencing program will definitely cause a great deal of hardship and frustration for commercial boats. We are aware the fleet is too large and each year wonder where all the new boats have obtained licences. Perhaps another buy-back program would be a more logical procedure.

It is unfortunate that there is acrimony between factions relating to the harvesting of the salmon. The basic issue is fish vs. man in our civilization and it is this kind of philosophy that has led to the serious decline in the salmon stocks generally.

Yours truly,

Mike Murphy



Telephone 287-3066  
Area Code 604

# The Dolphins Resort

4125 DISCOVERY DRIVE  
CAMPBELL RIVER, B.C.  
CANADA



Recommended  
by A.A.A.

January 15, 1979.

Update '78,  
Salmonid Enhancement Program,  
Fisheries & Oceans Canada,  
1090 W. Pender St.,  
Vancouver, B. C. V6E 2P1

Dear Sirs:

re: Daily Salt Water Bag Limits  
Personal Salt Water Licences

Since the holding of the two meetings on fishing regulations here in Campbell River in late 1978, (the SEP meeting and the earlier one sponsored by the Chamber of Commerce), I have given further thought to the two items noted above and would like to restate my views on them. I remain most concerned that the short-stay tourist is fickle and the economies of Campbell River, British Columbia and Canada can ill afford to discourage tourism by making recreational fishing needlessly expensive and/or not worthwhile.

## Daily Salt Water Bag Limits

It is my understanding that some means of reducing salt water bag limits for Chinooks remains under active consideration. I still maintain that the need to reduce the sportsman's catch has yet to be demonstrated (especially when one considers that one seine-set can take more salmon than our resort takes in an entire season). However, if a Chinook limitation is politically necessary to placate the commercial fishing interests, then it should be on an annual basis BUT NOT ON A DAILY BASIS.

To reduce the daily bag limit from four salmon to three, for example, would deny the sportsman the opportunity to catch a fourth coho also - and there has certainly been no shortage of coho in recent years. A daily species limit (for example, 2 springs per day) is also a defacto reduced limit, if the springs are caught first, unless releasing is resorted to. I doubt that any of us would wish to encourage releasing with its high mortality rates.

I don't know what a reasonable annual limit would be except that it would be several times a day limit of four. Such a level would offer equity to residents

non-resident tourists alike - the former who fish all year and the latter who spend hundreds or thousands of tourist dollars to come to Campbell River for a few to several days of fishing.

#### Personal Salt Water Licences

I am now strongly opposed to personal licensing on an economics basis - such a program would only do one of two things:

- if sold by government bureaus, the public and our visiting tourists would be inconvenienced and the costs of collection, accounting and oversight would probably consume most, all, or even more than all the revenues collected. (In deference to the fisheries representative who stated to me that only the field and direct administrative cost of license sales were included in their cost/benefit analyses while the costs of the Vancouver and Ottawa bureaucracies which would be involved in accounting, oversight, etc. were not included, as a taxpayer of Canada, their cost to me would certainly not be zero!).
- if sold by private outlets such as ourselves, the costs of collection would be largely transferred to us, but the costs of oversight would probably be increased manifold. Certainly, the tourist would be less inconvenienced but the small businessman would be engulfed in yet one more paper burden of licence issuing, record keeping, money collection and reporting. In our case, we frequently have 18 or more people check-in as a group and we are extremely busy accomodating them, getting them settled and then out fishing - it may seem innocent enough to think that it would only take a few minutes per person to issue the licences but at 2-3 minutes per license, we're looking at ½ to 1 hour, for example, at a time when we don't have time. In addition, we would surely have more reports to prepare at month end, etc.

Either way, the net result of all of this effort is simply that, a lot of effort and little or nothing in the way of funds for salmonid enhancement to show for it - I say to heck with licences - we don't need them - we sports fishermen are already self-enforcing daily bag limits and good sportsmanship very effectively without them.

Yours very truly,



Colin Smith  
President

cc Mr. Wayne Harling, Representative, Sports Fishing Advisory Committee, Nanaimo  
Hon. Romeo LeBlanc, Minister of Fisheries and the Environment, Ottawa  
Hon. Hugh Anderson, MP Comox-Alberni, Ottawa  
Ombudsman, Office for the Reduction of Paper Burden, Ottawa  
Hon. Elwood Veitch, Minister of Tourism and Small Business Development, Victoria  
Hon. Sam Bawlf, Minister of Deregulation, Victoria  
Mr. Norm Lemon, Fisheries Officer, Fisheries Canada, Campbell River  
Mr. Norm McLaren, President, Chamber of Commerce, Campbell River

NANAIMO



Brief for  
THE SALMONID ENHANCEMENT PROGRAM  
Nanaimo Hearing - 30 November 78

My name is Edmond C. Packee, I have a Ph.D. from the University of Minnesota where I specialized in forest ecology, soils and fish and wildlife management. Two years ago I presented a brief to the Salmonid Enhancement Program hearing here in Nanaimo. In that brief I expressed some ecological concerns about the program:

1. What is the effect of salmon harvesting on the productivity of the stream?
2. What effects are the fishing techniques and certain enhancement measures going to have on fish populations (genetics and stocks)?
3. How much research is being done on real world salmonid management and ecology and how much of the laboratory work is applicable to the natural or management stream habitat?

Answers to these questions are essential if the Salmonid Enhancement Program is to be based on sound ecological principles. Now, two years later, I see hatcheries, stream cleaning programs, incubation boxes, and lake fertilization. But the answers or attempts to obtain answers to three basic questions, where are they? I am asking for an accounting of the Salmonid Enhancement Program - just

what degree of importance do the bureaucrats place on those questions? Lip service I've heard, but dollars to do the work, where? Also, has there been any real assessment of the financial return on enhancement techniques? Except for the lake fertilization program - and I am not sure what role if any the Salmonid Enhancement Program has had in funding the program - I wonder what we really know about enhancement?

To economize on time, I ask the Salmonid Enhancement Program directors to read for the first time, if they haven't, or reread my previous brief and comment on the points raised. In fact, I challenge them.

With reference to question 1 above, have there been any attempts to develop nutrient budgets for streams that take into account nutrient contents of carcasses and excess eggs? Is there any evidence to suggest that the current escapement approach to harvest is not affecting deleteriously the nutrient balance of many coastal streams? Do escapement figures take into account anything more than getting a limited number of fish to the spawning gravels? Is there any evidence to suggest that salmonid stocks are not being overfished in terms of the health and productivity of the freshwater ecosystem? What are the alternatives for maintaining the ecosystem (management) and maximum fish harvests?

With reference to question 2, I am distressed for lack of action. Carnation Creek in 1977 had a return of 180 adult coho and one male and one female steelhead; these returns "were the second lowest and lowest, respectively, yet recorded, a result probably of heavy commercial fishing."<sup>1/</sup> Ricker and Smith<sup>2/</sup> reviewed the history of the Skeena sockeye; they state, "It should not be assumed that the present relative abundance of these stocks reflects the primitive situation at all closely. Rather we should expect the least productive stocks to have decreased greatly..." Reisenbichler and McIntyre<sup>3/</sup> provide evidence for steelhead which shows wild steelhead have better survival than hatchery or hatchery x wild hybrids in natural streams, but that hatchery x wild hybrids have equal or better growth; in the hatchery pond, however, hatchery fish had the highest growth and survival! Is there any suggestion that overfishing of many stocks is not occurring in coastal B.C. waters? What evidence is there to suggest that escapement management or lack thereof (management) has resulted in or is resulting in the demise of stocks or whole runs in some streams? Can you prove that hatchery fish will not result in the decline of wild stock numbers of salmonids if cross breeding occurs? If you cannot demonstrate the safety of your program, then I suggest our heritage is threatened.

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<sup>1/</sup> Symons, P.E.K. (ed.) 1978 Carnation Creek Project Annual Report for 1977. Environment Canada Fisheries and Marine Service. 12 p.

<sup>2/</sup> Ricker, W.E. & H.D. Smith 1975 A revised interpretation of the Skeena River sockeye salmon (Oncorhynchus nerka). J. Fish. Res. Bd. Canada 32(8):1369-1381.

<sup>3/</sup> Reisenbichler, R.R. & J.D. McIntyre 1977 Genetic differences in growth and survival of juvenile hatchery and wild steelhead trout, Salmo gairdneri. J. Fish. Res. Bd. Canada 32(8): 1369-1381

And finally question 3 - since the inception of the Enhancement Program, funding for basic research into understanding stream and riparian ecosystems has decreased in real dollars. How do you manage real ecosystems if you don't have the basic data? Much management or protection is biased in favor of the aesthetic or personal likes and dislikes - "green slime" is a nuisance and besides it is slippery to step on; so one lady commented at a similar hearing up Island! Is this green slime not algae and does it not have an important ecological food role? We don't like things which appear "bad." Appearance is only skin deep, I was once told. What is being done by the Salmonid Enhancement Program to get below the cosmetics of the skin's surface.

In summary, I am demanding accountability - I want real answers to what are considered basic concerns. I believe 10% of the original Salmonid Enhancement Program budget should be spent on getting the necessary answers. Until then the Salmonid Enhancement Program borders on mis-management of funds and the resources for the future.

Submitted by:

E.C. Packee  
751 Bowman Avenue  
Nanaimo, B.C. V9R 4L4

## SALMONID ENHANCEMENT

My name is Edmond C. Packee, I have a Ph.D. in Forestry from the University of Minnesota where I specialized in forest ecology, soils and fish and wildlife management. I have spent 10 years on the Pacific Coast of North America from the redwoods of California to the spruce-hemlock forests of the Alaskan Panhandle. Nine years of that experience has been spent in coastal British Columbia working on ecological and silvicultural problems of the South Coast region and the Queen Charlotte Islands. This brief is a personal brief and does not necessarily represent the views of my employer nor the views of the forest industry.

The watersheds of the Pacific slope and coastal islands contain a large number of climax biogeocoenses which vary in productivity from some of the most productive (Fujimori 1971) to the least productive (Stockner and Shortreed 1976) in terms of net productivity at temperate latitudes. Due to natural and anthropic events or changes, these climax biogeocoenses vary in terms of successional status and productivity. Ecological change is an integral part of the ecosystem and is essential to the productivity of the ecosystem. Salmon enhancement will involve anthropic ecological change just as intensive forest management brings about ecological change.

Salmon enhancement will succeed or fail based on the cost/benefit ratio.

This cost/benefit ratio must be considered in two perspectives:

1. Social demands - how much is society willing to pay for each increment in benefits?
2. Ecological - how much change can be supported by the ecosystem in terms of ecosystem functions?

The first, I am unqualified to discuss in any detail. The second, I feel is most important and wish to present for consideration.

There has been much criticism of the forest industry concerning the effects of logging on the environment:

1. The effect of clearcutting (harvesting) on the future productivity of the land and associated waters.
2. The effect of planting programs resulting in the so-called monoculture.
3. The hazards of genetic manipulation and reduction of the gene pool.

I believe that the forestry profession and forest industry are actively seeking to manage the vegetation resource on an ecologically sound basis. It is still imperfect and we have renegades just like the fishing industry does.

I must question the Fishery Service and the fishing industry as to whether they are actively seeking to manage the fish stocks on a sound ecological basis... not on the basis of count them (to fill up the spawning area available) and catch the rest! This apparent policy creates a real dilemma when it comes to the salmonid enhancement program! Some pertinent questions regarding the ecological effects of salmonid enhancement:

1. What is the effect of salmon harvesting on the productivity of the stream? If the processes of the undisturbed ecosystem included a salmon run of 1,100,000 female fish, but only enough spawning area for 100,000 fish, what ecological role did those other 1,000,000 fish play since they are apparently surplus? Stockner and Shortreed (1976) suggest that at Carnation Creek, productivity is low because of "...extremely low nutrient concentrations (especially phosphate) and secondarily, poor light conditions under the canopy..."

I am told by fisheries people that alder along streams is important because, among other things, its litter which is rich in nitrogen is an important source of nutrients to the stream. I must now wonder about those 1,000,000 fish which are not needed and end up in the supermarket instead of the stream! Foerster (1968) summarized data concerning the role of carcasses as contribution to the nutrient salt supply of freshwater and concluded that decomposing carcasses were a significant and appreciable source. He presented evidence from Lake Karluk in Alaska where 1,000,000 sockeye carcasses provided an estimated 6,100 kg (6.7 tons) of phosphorous and 63,500 kg (70 tons) of nitrogen. Thus over a 50-year period, if on the average 1,000,000 fish were harvested, the stream or lake system has been deprived of 305,000 kg (336 tons) of phosphorous and 3,175,000 (3,472 tons) of nitrogen. The importance of carcass nutrients has been demonstrated through modelling at Lake Dalnee in Russia. This nutrient deprivation is ecological change and must be assumed as unacceptable to the ecosystem unless proven otherwise. Exploitation of the fish stock is being done at the expense of the ecosystem; in essence, the system is being overfished! How will the salmon enhancement program solve this apparent anthropic problem or will it be ignored?

2. What effects are the fishing techniques and certain enhancement measures going to have on fish populations? Seiners will select those fish which commonly school. Such selection is apparently happening and the result is a genetic change in the fish populations reaching the streams for spawning. In essence, the non-schoolers (the school drop outs) are being selected for breeding stock.

Hatcheries will result in large numbers of more genetically similar stock being released into the waters and will make up a large percentage of the overall fish population. When the harvest time arrives, more hatchery fish will be caught but also will escape for breeding while the natural populations will be more heavily caught. How do you separate one population from another at fishing time? It is only logical, if 1,000,000 hatchery fish are mixed with 100,000 natural fish and the populations are managed and fished equally, that the increased catch will come proportionately from the two major groups; in this case, if the catch is increased by 1,000,000, only about 10,000 native fish will escape while 90,000 hatchery fish will escape for breeding. If 1,000,000 hatchery fish are placed into a river or stream, what about interspecific competition?

In essence there is the distinct possibility of developing a monoculture at both the species level and a reduced genetic population within a species. This ecological change is certainly undesirable in the long term and cannot be ignored. One solution, an obvious one, is to develop a more efficient method of fish harvesting. The Indian method using fish traps appears most suitable.



3. How much research is being done on real world salmonid management and ecology and how much of the laboratory work is applicable to the natural or managed stream habitat? There are some notable examples regarding freshwater-terrestrial habitat interface work and salmon studies--Alsea, Carnation Creek, Babine Lake, and Hollis, Alaska. These are exceptions. Little is known about interspecific competition, diel temperature fluctuations, terrestrial inputs into the stream, and what is ecologically acceptable and unacceptable! The need for leave strips is commonly expressed by fishery managers - but there is seldom a statement as to what effect the strip should have or what it should consist of--it is there to "protect" the stream. Protect the stream from what? One answer was from direct solar radiation-- using this approach, there is no real need for a strip on the northside of a stream. Using Stockner and Shortreed (1976), it may actually be desirable to increase the light reaching the stream. Increasing stream temperatures may increase stream productivity and fish size. How do we interpret lethal temperatures for fish when the lethal temperatures are based on laboratory studies which largely ignored diel temperature fluctuations? A protection strip or more appropriately a non-management zone is a cop-out for, "We don't know, but nature did okay!" What is essential is positive management for those ecosystems within the riparian zone.

The salmonid enhancement program is a potentially great approach to meeting social demands. To be successful, it requires hard facts, a sound ecological base and cooperation between land managers and fishery management -- the last must not be an adversary game! There are land management practices which result in environmental changes which are ecologically unacceptable -- these changes can be considered damage. We must minimize or eliminate such practices -- examples include logging debris jams, skidding down or up stream channels or beds, "necessary" equipment doing essential work in streams at the improper time, excessive disturbance to streambanks. Care must be taken to ensure that what is aesthetically unpleasing or appears to be damaging in the eyes of a human, is not construed as ecologically unacceptable.

Certain forest management practices can enhance stream productivity and salmonid production at little or no cost to fisheries. Roads are essential to logging; during construction borrow areas are essential and if properly designed can provide more than gravel for the road -- when allowed to fill with water they can serve as a nursery area for juvenile salmonids. Yet, the only extra cost is to construct an inlet and outlet to the fish stream. Species management of the riparian vegetation can improve the stream habitat. Where stream temperatures are sub-optimal, clearing and managing for light-filtering species will improve productivity. Certain species of plants can provide badly needed nutrients to the stream as litter drop whereas others are not as efficient -- alder, rich in nitrogen, is a good source of nitrogen but a poor source of calcium and cottonwoods are a good source of calcium, but only a mediocre source of calcium. Some species are poor in terms of stabilizing the banks, others are excellent. There are some plant species

which harbor a large number of species and quantity of insects which can drop into the stream; this terrestrial insect biomass may be critical for the maintenance of the salmonid population of the stream. Good management of the terrestrial ecosystems along the shores and banks of lakes and streams can only enhance salmonid production.

Road construction should be minimized and opening size restriction should take into account road requirements and maintenance. This will minimize sediment sources. With the salmonid enhancement program in mind, road layout in each watershed can be developed to provide fishery benefits at low cost. Access to specific stream locations can then be had to remove natural debris dams, for fish ladder construction, nutrient leach tanks to maintain or enhance the nutrient status of the system, and incubation boxes. These roads also provide access for maintenance of fishery structures.

Elaborate and expensive enhancement structures should only be considered after the potential of the ecosystem has been developed through strong, positive management. The ecological ramifications of hatcheries must be reviewed and publicly approved because of the high ecological and monetary costs involved.

Research is needed now! Work on the role of different plant species as related to the stream must be investigated -- such activity was begun several years ago at the Pacific Biological Station and is now relatively inactive. Fertilization of streams and lakes must be further investigated -- can artificial fertilization replace those 1,000,000 fish carcasses? The ecological and physiological requirements of the fish in their natural

environment or in carefully controlled laboratory conditions that simulate stream conditions must be determined. Practicable limits or tolerances must be established for management of riparian ecosystems. The research results must be interpreted and used in the management program.

A positive approach to the management of stream ecosystems and their riparian margins is economically sound and ecologically most desirable. A similar statement concerning lakes is equally valid. Protection strips with the objective of maintaining the status quo, fish numbers management and hatcheries at this stage of fisheries management is simply not facing up to the reality that the systems for production are there but are not being properly utilized or are not being managed to reach the potential production.

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Edmond C. Packee  
751 Bowman Avenue  
Nanaimo, B.C. V9R 4L4



PORT ALBERNI

R.R. #1  
Somers Road  
Port Alberni, B.C.  
V9Y 7L5  
January 8, 1979

"Update 78"  
Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sirs:

I have read with interest your "Guideline for Land Development and Protection of the Aquatic Environment" but could find no actual reference to farming.

The proposed buffer zone along creeks and catch basins for run-off water may be fine for a land developer who will just pass the additional cost onto the buying public but what do you intend doing with creeks in farmland. This is land bought and paid for by people who will receive no direct benefit from the buffer zone, in fact will have fishermen's dogs chasing his cattle, beer bottles in his fields and paper everywhere. All this to a fellow who is already told what price he will get for his product.

I wonder if the Fisheries department is prepared to buy this land, which no farmer would want to sell, at fair market value and keep up all fences, etc.

I don't think any farmer wants to see the fish destroyed, in fact, most farmers do more for the fish than any of you - in keeping creeks open so fish can get up to spawning grounds. How many creeks do your men clean up and check? I would say most checking is done along roadways and creek banks that the farmer has made accessible.

The conclusion I would like to suggest, a team of two men - one from the Fisheries and one from Agriculture, each with equal say in the decision when work along creeks in farmland is being done.

Thank You,

A handwritten signature in cursive script that reads "Marjorie Emblem".

Marjorie Emblem

BRITISH COLUMBIA FARMERS' INSTITUTES

INCORPORATED UNDER THE  
FARMERS' AND WOMEN'S INSTITUTES ACT

FARMERS' INSTITUTE Alberni,  
c/o Mrs. Betty Cameron, Secty,  
R.R.#1, Comox Trail,  
Port Alberni, B.C.

January 8, 1979

"Update 78"  
Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

To Whom It May Concern:

SALMONID ENHANCEMENT PROGRAM

This Institute has followed with interest the formation and progress of the Salmonid Enhancement Program. We wholeheartedly support its aims and objectives.

In 1976, when SEP held a hearing in Port Alberni, this Institute attended and spoke on that occasion. We appreciate that the public at large has an opportunity for input.

This farming community recognizes the importance of stabilizing an industry as important as the fishing industry, at the same time, we feel strongly that we have an industry that is second to none in importance to the people of Canada. Of prime importance to the farmer is his land, as is the waterways to the fishing industry.

We feel that the small streams are the life blood of the fishing industry. These same streams are the life blood of the farming community. These streams provide the farmland with drainage for wet land and irrigation for dry lands, both of which are of major importance to the farmer. This Institute feels that when a stream or drainage ditch becomes plugged to the point where it creates a problem for the farmer, it at the same time, becomes less productive for the fishing industry. Therefore, we feel that we share a common problem. We urge SEP to adopt a policy of co-operation with the farming community. A policy which would give equal rights and responsibilities to the Fisheries department and the farmer on any land or stream that is jointly effected.

Yours sincerely,

cc. Ministry of Agriculture  
BCFA  
Hugh Anderson, M.P.  
Bob Skelly, M.L.A.



Ernest Burnett, President  
ALBERNI FARMERS INSTITUTE





DUNCAN

Cowichan Fish & Game Association

Salmonid Enhancement Brief

We basically support the programs aims and objectives to double salmonid production.

This association has initiated and carried through several small stream projects in this area and would like to continue our involvement and expand our endeavours.

In this portion of Vancouver Island, we have some basic priorities which we feel should be identified as a major part of salmonid enhancement.

1st Priority

Protection and enhancement of the Cowichan estuary with a phasing out of industry and domestic pollution. Establishment of permanent safeguards against any future developments. This should apply to all productive estuaries.

2nd Priority

More permanent trained staff for enforcement and other Fisheries duties. This is necessary for restricting illegal net fisheries, illegal sale of fish, and habitat protection problems.

3rd Priority

An education program should be initiated as soon as possible in the elementary and high schools. Benefits from education are widespread and long lasting.

4th Priority

More small stream projects and semi-natural fish production schemes could enhance natural production without large initial investments of capital. Pen-rearing and feeding of fry to produce larger healthy smolts could help offset natural disasters such as floods and droughts. Local professional assistance should be provided for in addition to funding for club projects.

5th Priority

Sport fishery should not be controlled before the Native food fishery, as Natives are sports fishermen also. Indian food fishery should be set at their required levels and subsequently cut off after quotas are met. Spearing or any other wasteful inefficient methods which wound a large number of fish should be discontinued.

This association feels that sport fishing is of higher economic value than commercial fishing. Therefore, more emphasis should be placed on production of sports species including chinook, coho, steelhead and cutthroat. The aims for sports fish management should be to optimize fishing opportunities were possible. Closures such as the extended Cowichan Bay closure are not in the best interest of sport fishing. If a serious problem arises, regulations should be included in the annual synopsis rather than be posted ad hoc by directors orders. If a licence is initiated for sports fishermen as part of the cost recovery scheme, then this organization will endorse such a licence only if sports fishermen are given a larger voice and a larger piece of the pie.

SUBMISSION TO THE SALMONID ENHANCEMENT MEETING

December 5, 1978

Duncan, B.C.

By The

COWICHAN ESTUARY PRESERVATION SOCIETY

## CEPS 1978 SALMONID BRIEF

On behalf of the 110 members of the Cowichan Estuary Preservation Society, we are pleased to appear and present our comments on your program that we welcomed in 1976.

At that time we cited 3 major concerns:

- competing uses;
- regulation;
- education - this has been demonstrated amply, especially through programs in some schools.

Unfortunately, our concerns are still much the same as then; now we are able to speak about them with greater authority.

Our main concern is environmental degradation. Since your program is in part dependent on the natural environments in existing streams, which often become much more productive after projects that you have stimulated, the future of those enhanced fish populations may depend greatly on factors beyond your control. We refer to urban sprawl, with the reduction in precipitation absorption and runoff attenuation, plus the amount and quality of pollution and sewage, and to industrial expansion in particular. We are only too familiar with the pressure on valuable and fragile estuarine environments as sites for industrial expansion. This pressure is mounting in the Fraser, where development may proceed while the capacity of the delta to absorb it, is still being "studied," in the estuaries of the Nanaimo and Campbell Rivers, plus our own Cowichan. We will concentrate on the latter, but our experience likely mirrors that in the others.

The Cowichan has been world famous for its salmon fishing, but it is no longer. This change has occurred in the lifetimes of local residents. Yet it was still considered of national and international significance

for some natural resources when the 1974 Estuary Task Force completed its brief review. The lack of specific details regarding the biological basis to those resources and the potential impacts by industry prompted the "status quo" recommendation, to be followed by studies of the complexity and stability of the ecosystems affected. To our knowledge, these studies were never done, and we are unable to compare our suspicions of environmental degradation to scientifically-defensible data and conclusions. Yet, we feel obvious considerable environmental degradation has occurred in the Cowichan Estuary from the following sources:

1. Log booming and storage in the intertidal area. Despite stated limits, one operator apparently and frequently exceeds the booming area allowed. This has been pointed out to Federal authorities more than once, and was commented on by a Federal fish scientist at the Shoreline Management Symposium in October as an example of habitat damage. But apparently nothing has been done to stop this, despite the powers of the Federal Fisheries Act.
2. Log movement and towing. Towing at all but extreme high tide has led to disturbance of the estuary bottom through dragging and propeller wash. Hang-ups often are removed during falling tide by tug or shore-based winches, causing more habitat damage.  
  
Log scraping and boom wash are wearing away portions of Marriner's (sic) Island - fish habitat again covered by Canada's Fisheries Act. Protests have produced the apparent inaction noted above.
3. Bark build up. Due to both physical smothering and leachate of sugars and volatile compounds, conifer bark accumulation is detrimental to most marine organisms. Log movement means bark deposition. Build-up may not be obvious due to covering by silt action, but anaerobic decomposition under silt is slower and produces more toxic chemicals than aerobic decomposition.
4. Leachate. Wood chips and hog fuel accumulations are further sources of toxic leachates. So are wood preservatives used in mill and dock construction and so are the volatile oils in asphalt. All such sources of leachate generally increase as industry concentrates near the shore.

5. Direct loss of habitat. Filling for buildings, storage areas and roads, dredging for ponds and docks all consume area that formerly were productive habitats for marine or terrestrial resources. As we have no "known mitigative techniques"\* to repair damage in estuaries, obviously we must wait while natural estuarine build-up occurs - another unknown.
6. Degradation of river water quality. The population build-up and concentration associated with "economic expansion", particularly industrial development, requires more water and generates more sewage. Local sewage treatment generally has lagged behind volume increase, causing pollution of our rivers and bays - N.B. the shellfish closures on the lower B.C. coast. Domestic and industrial water consumption may withdraw so much from tributary streams or aquifers that the potential for salmonid enhancement is lowered, or its security is threatened. (NB. Experience in Holland is that hi-volume pumping from aquifers is risky: sea water may creep in and ruin such wells for domestic and most industrial use. As groundwater is not regulated in B.C., Duncan and North Cowichan water supplies may be ruined by sinking of one major well in their aquifer, or by a preventable error during drilling or construction, perhaps far removed from the wells, or by the new bridge over the Cowichan River, which is very close to the wells.)
7. Degradation of shoreline quality. Increased boat activity for both industry and recreation means additional petroleum oils and gases in the water. Both are known to harm many marine organisms and surface life forms, particularly those at the bottom of the food chain.
8. Alteration of the flooding pattern and productive network by urbanisation. Buildings erected in floodplain areas need protection from floods, necessitating dykes. They alter stream flow and estuarine flushing and build-up patterns, and unless the dykes are very carefully designed and maintained well, they may increase the likelihood of flooding in the apparently - secure upstream areas. This seems particularly the case of Duncan city core, which is built on a floodplain, requiring regular and expensive channel dredging to protect it against river overflow or backup flooding from Cowichan Bay. The dredging reduces fish habitat, as a sluiceway through the city area is desired, and dyking action by high river flows backs up water in some tributary streams. eg. Somenos Creek, causing flooding of low-lying urban areas. That flooding might be aggravated by any more dyking along the Cowichan to protect the industrial or agricultural areas further downstream. Such dyking we believe is being considered by the current Estuary Task Force.

\* Lill, Marshall, Hooton, Conservation of Fish and Wildlife of the Cowichan - Koksilah Flood Plain. B.C. Fish and Wildlife Branch.



Your Enhancement Program includes both improvement of existing streams and fish production from hatcheries. Hatcheries offer the chance of immediate gains in fish crops and the potential benefits of genetic improvement of fish strains. Since genetic manipulation has yielded remarkable gains in all domestic food crops, the potential in fish stocks may be as high. But the gains in agriculture have been achieved in part through homogenizing the cultural environment and by major injections of energy through chemicals and machinery operation. Since you cannot control the environment in natural ecosystems as well, the temptation to plump for hatchery programs must be powerful. But fish-rearing programs involving enhanced feeding of young contain the seeds of disaster through mechanical failure (eg 1MM fish in Capilano) or biological calamity (eg 7MM fish lost in Cascade Locks, Oregon to bacterial disease). The best approach is to tailor the enhancement program to the local situation: stream and shoreline enhancement where potential productivity is high; hatcheries where shoreline productivity potential is good but stream dimensions, water flow regimes or water quality is limiting; and a mix, perhaps to increase fish crops rapidly while the benefits of stream and shoreline improvement in seriously degraded areas are mounting.

All the foregoing point to the need for effective law enforcement and monitoring of environmental quality. Our experience with "enforcement" of the Fisheries Act concerning log booming were mentioned. We feel similar frustration over environmental quality monitoring in the Cowichan. For example, a study of water quality in Doman's log pond was commissioned by the company and described by them and their contractor as indicating no environmental degradation. As probably it is a tax deductible business

expense, public money is involved. Yet it does not appear to meet acceptable standards of sampling, and particularly of data analysis and interpretation. Where a public resource is involved and public money is spent, the public must receive studies conducted to the proper standards: these must be set by the relevant government agencies, not by industry, or its consultants, who may be constrained by time and money.

## RECOMMENDATIONS

1. The Salmonid Enhancement program be continued because of the potential economic benefits relative to the cost and because of the suitability of the program to engaging and educating the public. The public has done considerable work for the Enhancement program and is willing to do more.
2. A policy of caution be followed in determining the proportion of the program devoted to hatcheries vs. stream and shoreline enhancement:
  - a. hatcheries where shoreline water quality is good but stream channel or water quality is poor and beyond improvement in the medium term.
  - b. shoreline and streamside improvement where water regime or quality is not limiting.
  - c. a combination of small (?) hatcheries and stream/shoreline improvement for quick fish-stock development where habitat improvement will require a long time.
3. Enforcement of laws and regulations be strengthened, otherwise, they are meaningless, as our experience in Cowichan Bay indicates. A growing program will require more money for development and enforcement as a growing community needs more policemen.
4. Enhancement of fish resources in one location be not accepted as justification for sacrificing another productive location. Salmonids need estuarine environments, whereas industry covets estuarine locations for their "economy", but not necessarily regional economies.
5. Information obtained by public agencies or paid for by public money must be available to the public immediately and without hindrance. Otherwise, developments may appear as "faits accomplis" and be difficult to change.  
(eg. A Provincial/Federal group investigating alternative industrial sites to Cowichan Bay identified many locations less sensitive environmentally than Cowichan Bay, but the public cannot obtain the report. Duke Point, Nanimo was identified through this process in 1974, yet our governments are again "studying" Cowichan Bay's potential for industrial development in our opinion, because of the political prominence of industry. Thus, the public paid for the survey, is paying for Duke Point development, yet is paying again to re-study a decision that no one has said is wrong.)
6. The principles and experience to date in salmonid enhancement stimulate enhancement of other marine resources - egs. shellfish and groundfish and the derivation of economic benefits they can bring.

Respectfully submitted,

*Robert Skinner*  
*President*

Cowichan Estuary Preservation Society.

REPORT BY MR. A.D. De LEEUW  
TO  
UPDATE '78 SALMONID ENHANCEMENT PROGRAM

Soole Road, R.R. #1  
Cowichan Station, B.C.  
V0R 1P0  
December 5, 1978

I would like to thank Mr. James Boland of the Salmonid Enhancement Program for giving me the opportunity to air my views regarding the development of the program.

It is my belief that the objective of the Salmonid Enhancement Program is to restore our present rapidly dwindling salmon stocks to their former levels of abundance. It is also my belief that such a goal can be attained in one of three ways, First, one can manage all our stocks in such a way that a maximum sustained yield is attained. This has been tried in the past and is in part responsible for the continued decline of the resource being managed. Or, secondly, one can just miraculously produce more fish to compensate for the decline. This is what the Salmonid Enhancement Program is attempting to do. The third option, as I see it, is some advantageous combination of both management and enhanced salmon production.

In general, the majority of present enhancement techniques center around the production of juvenile stages of salmonids in fresh water. It is taken for granted that the marine environment will transfrom these juveniles into adults. Our knowledge of the salmon's ocean feeding habits and general marine productivity related to the salmon is seriously lacking. The decline of salmon stocks might, in fact, be actually due to and as yet, unknown marine factor. If this is the case, no amount of fresh water related enhancement techniques is going to arrest such a downwards trend.

Paradoxically, this decline in fish numbers has had little effect on the income of fishermen. With fish numbers steadily dropping but their per pound value steadily rising, fishing is still, for some that work hard, an extremely lucrative business. The tremendous increase in catch efficiency of the commercial fleet, combined with this price increase of the catch, has resulted in an increase in returns to the fishermen, but not for the fish. This increased catch efficiency on a rapidly diminishing stock has not been adequately compensated for by the closures and openings which presently make up a large part of the management of our salmon resource. Furthermore, the enhancement of some individual species, in specific areas, and not of other species in the same rivers or inlets, has caused an alarming decline of those species not enhanced. We must remember that the commercial salmon fishery is largely indiscriminate in its catch. What I mean is that the fisherman can not, in his fishing, discriminate between species of salmon, or where they come from. This is less true of the troller who can change his lure style, but is especially true of the gillnet and seine fishery. In the latter case, literally all fish which are in the area are caught regardless of stock origin or species.

Another loss to the salmon and also, ultimately, to the fisherman, is the high number of fish lost in the ocean due to non-boated mortality. These fish are killed as a result of the method used to catch them, but are either too small and therefore released, or are never boated. This is especially true with trolling and gillnetting. Furthermore, the high seas fishery results in the taking of generally immature fish. If the fishermen waited five months or so, and caught fish near the river mouths, these same fish could have gained up to one hundred percent or more of their weight. Thus less fish would have to be caught, (a great benefit to the salmon species) and yet the same weight quota could be attained ... a benefit to both the fisherman and the fish.

In terms of management, the prevailing system leaves much to be desired.

More research funds should be allocated towards perfecting commercial harvesting techniques, thus enabling the fisherman to more accurately segregate his catching efficiency from species to species in various locations depending on which species are enhanced and where. It is, I feel, impossible to manage the salmonid resource along the entire British Columbia coast. It is more practical to intensively manage all stocks and species in a few areas which are easily havestable and close to river mouths. All streams and species within one locality should, in my opinion, receive equal enhancement. A great harvest of one enhanced species, to the detriment of all other species in the same area should not be permitted. Gillnetting and seining should be reserved only to such areas where all species are enhanced. In localities where all species of salmon congregate simultaneously from a great variety of different river systems, gillnet and seine fishing should be drastically cut back, or shut down altogether unless these types of fishing techniques can be altered to be more discriminating in their catch.

It is my view that the perfect method of harvesting and managing the salmonid resource is to use weirs in river systems. I realize, however, that to change from the present methods of harvesting to a complete weir fishery is politically not possible. On the other hand, in those areas where it is practical to enhance only one or a few select species to great numbers a weir fishery is, in my estimation, the only acceptable harvesting method. In a carefully controlled weir fishery, individual salmon species can both be harvested and managed more accurately than is possible with an open seas fishery. Such weirs could be placed in many rivers on our coast and be of benefit to the various communities there, Native as well as white. In a weir fishery, fish are mature, they have reached their maximum size, there is no non-boated mortality I spoke of earlier, and due to less handling, they become a more marketable product. Species which are low in numbers in any given year can be released unharmed. In combination with a reduced present fishery method, weirs could be installed on problem rivers like the Skeena or Stamp Rivers with more beneficial management results.

In terms of the actual enhancement technique, I am not in favour of hatcheries or any other large scale artificial method of fish propagation. If such a method is deemed necessary, the maintenance of genetically pure wild stocks will become increasingly important. The transplanting of salmonids from one system to a hatchery located on a different stream and back again must be done with the utmost care to avoid any possible disease problems. The world's largest salmon hatchery in southwest Washington State has fallen as much as ninety percent below its goals, largely due to disease related problems.

In rivers, a more rigorous control of our freshwater supply should be brought about through a more salmon oriented watershed management. Greater protection of streams from extensive water removal by dams, agricultural irrigation and industry, etc., is needed where such uses conflict with salmon production. Large parcels of land surrounding valuable salmon production areas such as spawning areas, freshwater nursery areas, estuaries, and salt marshes should be set aside as parks or purchased specifically to maintain salmon stocks for future generations. This would ensure their preservation from urban and industrial development. On the oceans also increased protection is needed to maintain or increase the quality of the marine environment. I hate to think what effect a two molecule thick layer of crude oil covering the entire surface of the north Pacific would have on primary production and ultimately on our salmon rearing in that area. Also essential to

the proper management and enhancement of our salmon stocks is an improved working relationship between industry (notably the forest industries) and Fisheries.

In summary, the following are the main points I wish to draw to your attention.

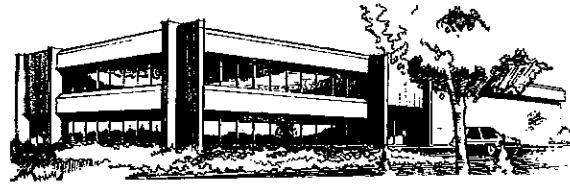
1. More research funds should be allocated towards a better understanding of salmon, especially in the oceans.
2. More research funds should be allocated towards obtaining more refined harvesting techniques.
3. Weir fishing in rivers should be implemented to augment or partially replace present and future ocean harvesting methods and make a more precise and effective management policy possible.
4. Indiscriminate harvesting techniques such as gill-netting and seining should be restricted to those areas where outside stocks are minimally effected. In such areas, intensive management should be implemented and all salmonid species effected by such a harvest enhanced.
5. In artificial propagation methods, all possible care should be taken to minimize the potential effects of disease, and all efforts should be made to maintain genetically wild stocks.
6. The environmental aspects associated with every phase of the salmonid life cycle should receive greater protection from other conflicting resource users.

Thank you very much for letting me speak this evening. I am hopeful that the views which I have just expressed will be of some benefit to you in making the goals of Salmonid Enhancement a reality.

SECHELT



# Cameo Industries Ltd.



P.O. Box 741

Sechelt, B.C. V0N 3A0

Tel: 885-5711

688-3314 Toll Free

December 6, 1978

ORAL BRIEF TO THE SALMONID ENHANCEMENT PROGRAM  
PUBLIC MEETING OF DECEMBER 6, 1978 SUBMITTED BY  
HENRY A. HALL OF CAMEO INDUSTRIES LTD IN THE  
INTEREST OF CREATING ADDITIONAL TOURISM DOLLARS  
THROUGH ENHANCEMENT OF THE SUNSHINE COAST AREAS  
SALMON SPORTS FISHERY.

## RESIDENT OR WILD ONE

The heading of this submission Mr. Chairman, may seem strange to your Committee, but it could well be one of the success ~~gauges~~ of your Salmon Enhancement Program.

Sportsmen in the sport of hunting use this term often, for it is quite common to have resident or wild game or fowl, yet this term is actually unknown to sports fishermen and especially unknown when we relate it to the Salmon Sports Fishery.

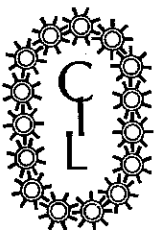
Residency in the Salmon Fishery is becoming a new word and it is hoped that further research and concentrated effort by leaders of the Salmon Enhancement Program will create a strong resident salmon population for the B.C. Salmon Sports Fishery.

In this regard Mr. Chairman, I take this opportunity of introducing your Committee to an area which is highly suited for resident salmon and is perhaps the best area in B.C. to explore, achieve and monitor a Resident Salmon Sport Fishery Program.

The Inland Sea is 50 miles of protected inland waterway with 150 miles of shoreline that has deep cold calm channels open to the main Jervis Inlet through the Skookumchuck, a natural control gate provided by Mother Nature.

The many creeks and rivers of the area could provide excellent resident salmon spawning channels inaccessible to vehicle traffic and accessible by the sea only, giving added security to spawning salmon and hatchery fingerlings.

. . . . . continued 2



Following are some of the benefits offered by our proposed Inland Sea Resident Salmon Fishery.

FOR THE SALMON ENHANCEMENT PROGRAM

1. A low cost experimental proving ground for a pilot resident salmon program within 30 miles of Vancouver, yet protected from the sometime destructive habits of man.
2. Controlled access to assist in monitoring the effectiveness of the program.
3. The single entrance at the skookumchuk creates excellent sport and commercial fishing control.

FOR THE PUBLIC AT LARGE

1. A tidal sport salmon fishery within a 2 gallon gas distance from Vancouver, or a pleasant 2 hour scenic bus and ferry ride, or 15 minutes by air.
2. An all season, all weather, wind protected sport salmon fishing area at a cost that the one plus million lower income residents of the Lower Mainland can afford.
3. A tidal sport fishery where 98% of the waterfront is owned and controlled by the Government, thereby providing a low cost fishing and nature experience for the boating public.

FOR THE SUNSHINE COAST ECONOMIC BASE

1. A \$1 Million Dollar addition to the areas annual tourist industry.
2. Increased provincial parks branch promotion and expansion of their Sechelt Inlet overnite marine park campsites.

Orient Committee with Park Program. (copy of Program enclosed).

In conclusion we strongly recommend that the Salmon Enhancement Program engage in a Pilot Salmon Residency Program for Sechelt Inlets.

Tourism in the Inland Sea is dependent on a good salmon sport fishery, we therefore ask that a Resident Salmon Program be started in 1979 to complement the Parks Branch Marine Campsite System.

When sports fishermen use the term Resident Salmon on a daily basis Mr. Chairman, we then will know that your Salmon Enhancement Program has indeed been successful.

*Henry A. Wall.*  
Thank you.

Henry A. Wall

SISHALS GARDENS  
RHODODENDRONS AND AZALEAS

4405 West 5th Avenue  
Vancouver, B.C.  
V6R 1S4  
December 8, 1978

Update "78"  
Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sir:

Unfortunately, I was away from Sechelt on December 6th and missed your meeting which I would have liked very much to attend.

Is there a printed copy of the Progress Report available? If so, I would very much like to receive one.

For the past 47 years I have enjoyed spending part of the year on my property on Sargeant Bay. At the head of the bay, above the gravel berm, there used to be a delightful little lake, fed by Colvin Creek on the east and a smaller stream on the western side of the bay. There were always lots of fish in this lake.

Unfortunately, the berm has been cut through and the lake level has dropped until now it's a very small pond. I am wondering what action can be taken so that this lake can be restored and again become a breeding ground for fish.

Thank you.

Yours truly,

A handwritten signature in black ink, reading "Kenneth P. Caple". The signature is fluid and cursive, with the first name "Kenneth" and last name "Caple" being more prominent than the middle initial "P.".

Kenneth P. Caple

Mr. D. Robilliard  
P.O. Box 367  
Sechelt, B.C.  
V0N 3A0  
December 7, 1978.

Salmonid Update '78  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1  
Attention: Les Edgeworth

Dear Sir:

I attended your meeting at Sechelt on Wednesday, December 6, 1978 and found it most interesting.

Being a concerned citizen and a 33 year resident of this community, I am very pleased with the basic overall salmon enhancement program.

I am, however, somewhat disturbed with a few of the items discussed and have a few comments I wish to make on both these and other items not discussed. The reason I did not speak at the meeting was, I did not want to prolong it with unnecessary argument. Perhaps some of what I have to say is non-applicable to the Salmonid Program and if not, I only hope you could steer me in the right direction.

I speak as a sportsman, concerned British Columbian and one who has a great deal of commercial fishing experience, although at present, I do not make my living fishing.

#### Roe Herring Fishery

Fact - The roe herring fishery is big business.

Falacy - Herring fishing is the rape of the sea.

Suggestion - The public be more informed as to how the actual fishing is done, what amounts of fish are caught, how much fish are left to reproduce, what this fishery does for our economy, etc., etc.

I hope that successful herring seasons continue but I feel perhaps a different approach to the fishing could be looked at.

My idea is that the total fleet be divided into two parts. One fishing one season, the other the next with half of existing quota being used each season. I feel that this may be a way to allow for a larger return of herring each year and everyone involved having a better chance for a good season.

#### Sport Fishing Licencing

I fully agree that a sports fishing licence for saltwater fishing is a good idea and might add long overdue. However, I only can agree if the sportsman's dollar is put back into salmon enhancement on the west coast, not into Ottawa's general fund.

### Spawning Grounds

I am of the opinion that much more effort from your program should be put into the cleaning out of small creeks and rivers throughout B.C., especially in remote areas where man is more uncommon. This will not only prove to be extremely beneficial to providing more fish but also it will improve the existing spawning areas we have at a low cost and should only help the overall program with the assistance of the million dollar rearing facilities.

### Sechelt Inlet Resident Fish Proposal

As to Mr. Hall's proposal, I have mixed feelings. His statement about safe, sheltered waters for the small boaters is only partially true - it can blow very hard and come up very quickly - just ask old time local residents.

I do agree that if his idea was carried out, that it would be financially rewarding to this area - especially in the tourist related industries and businesses. However, we do happen to have several commercial fishermen who do make a portion of their living fishing these inlets and I firmly believe that they should not be closed out of this area should Hall's proposal succeed.

One fact that was not brought out in Hall's proposal was that sportsmen (or so called) on a whole are greedy and it would not be long before the beaches will be stripped of shellfish.

### Reducing Sport Fishing Limit

I feel that to reduce the present limit from 4 fish is really not the way to go. You ask why? Several reasons - as pointed out by some of the commercial men - the fish that the sportsmen take is really only a drop in the bucket in the overall picture. Perhaps some sort of tag system such as that used in hunting should be considered, although I can foresee a policing problem.

Fishing derby's really are utter nonsense as so far as protecting fish is concerned. Many small salmon are caught and kept, then thrown overboard when a larger one is caught. I've seen this time and time again (not only during a derby). Last but not least, how many times does one actually limit out?

### Gear and Season Limiting

This is a sore subject with me. As far as I am concerned, the whole concept is wrong. A sportsman is supposed to be just that - a sportsman - so why in hell is he allowed a commercial type gurdie on his boat? Although it is not legally allowed to be powered, it is still a basic method of commercial fishing. All I can see is the dollar sign comes first, not sportsmanship or fish. So then why must a commercial man have to disconnect his power supply to gurdies or drum if he wants to go out and catch a fish to eat during the winter? After all, if the sports fisherman can use a hand gurdie during the peak of the fishing season as well as the winter, what's the difference if a power device is used provided, that you are limited to one lure per line and a sportsman limit of fish? Because when one uses a gurdie whether it be a hand or power type, the sport is gone anyway and the object to catch a fish to eat is gone.

Why not consider the closing of the blueback season in the gulf for the sports fishermen until the 15th of June. Don't you agree that this would provide a better quality fish and more to go around?

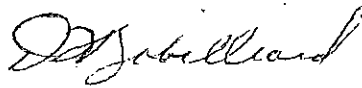
Also, I feel there should be some sort of a closed season for sports fishermen. Some of these new lures such as the "buzz bomb" are deadly on fish waiting at the mouths of rivers before their ascent to spawn.

I realize that perhaps my ideas are not all good for several reasons but I felt I must express them.

You people are doing a super job and I only hope that big business and government does not interfere and ruin a gallant effort.

May your endeavour in a "first" be long lived and super successful for all of us to benefit.

Yours truly,

A handwritten signature in cursive script, appearing to read 'D. Robilliard', written in dark ink.

D. Robilliard

Mr. A.W. Garland  
Sandy Hook  
R.R. #1  
Sechelt, B.C.  
December 15, 1978

Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sirs:

My attendance at the SEP meeting held in Sechelt on December 6th, 1978 prompts me to write of my concern. Firstly, your efforts so far with this program are to be commended and wish you every success for the future.

It was revealed at this meeting that the Sechelt Inlet, Jarvis Inlet areas are destined for Phase II of your program. If you do not update this area to Phase I, it will be a sad mistake. Persons familiar with this area are adamant that stream exploration, testing and enhancement should be carried out in 1979.

This deep water seaway encompasses over 120 miles of shoreline, with numerous streams, many of which already have chum salmon runs. These chums however, are of no value to the sports fishermen but indicate that local streams are suitable for spawning.

The Sechelt Inlet being protected from severe winds, is an ideal location for small boats and family outings. Deep sea shipping is non-existent and only the odd tug enters these waters, so pollution from these sources is minimal.

As the populace has become aware of these advantages, they appear in droves, particularly in the region south of Salmon Inlet to Porpoise Bay. This influx of people has increased the fishing pressure tremendously, with the result that today, instead of being able to catch salmon spinning from shore as was possible 10 years ago, a person is lucky to hook two salmon in a whole season by trolling or mooching.

1. Inlets not especially productive re: marine environment (Stockner Paper).
2. Over-utilized.
3. Enhancement is not necessarily going to improve all season abundance. Probably just during migration period and early spring.

Now what does a person do when there are no more salmon? They reach for the next sought after fish, namely the ling cod. During the last three years, these have become practically extinct, next in line are the rock cod and these are definitely on the decline also, so on it goes.

All we hear about are salmon, which is natural enough, but what about the ling cod? Which for some people is preferred over salmon as a food. Is there a place in your program for restoration of this valuable fish? Has consideration been given to reducing the seal herds to keep things in balance with the reduced fish stocks?

We hear about all the facilities on Vancouver Island such as the Big Qualicum, Puntledge, Nanaimo, etc., and the over abundance of salmon from the Capilano Hatchery, all of which increase fish stocks in Georgia Strait but what about Jervis Inlet and our forgotten area in Sechelt Inlet, which has grown by leaps and bounds, not only by permanent and summer residents, but by a major provincial campsite with six marine parks contemplated next year.

A hatchery on the Inlet would probably be too costly, but work done on stream improvement which would include rearing boxes supervised by Fisheries officials with local volunteer labour would be a great step in improving the present situation. Another approach would be to annually stock the Inlet with well developed surplus fingerlings from the Capilano to develop a resident population of both coho and spring salmon. This would in turn benefit both the sport enthusiast as well as the commercial fisherman.

Yours truly,

A handwritten signature in cursive script, reading "A.W. Garland". The signature is written in dark ink and is positioned centrally below the typed name.

A.W. Garland



L. E. Chapman  
Sandy Hook  
R.R. 1, Sechelt, B.C.

December 19, 1978

Update 78,  
Salmonid Enhancement Program,  
Fisheries & Oceans Canada,  
1090 West Pender Street,  
Vancouver, B.C. V6E2P1

ATTENTION: Mr. Les Edgeworth

Gentlemen:

Regarding the meeting of the Salmon Enhancement Committee, held in Sechelt, B.C. on December 6th, 1978, I would like to make the following points.

1. That the Sechelt, Salmon and Narrows Inlets be put into phase one (1) and as soon as possible.

2. I do not understand why this area should be relegated to phase 2 (2) when we have such a potential for real salmon enhancement in Sechelt Inlet.

When we came to live in this area fifteen years ago, we had no trouble getting salmon in Sechelt Inlet. Over the years fishing has deteriorated so that now you are lucky to get one or two. I believe there are several reasons for this:

1. Overfishing of the herring population.
2. Clearcut logging down to the stream bank.
3. Allowing commercial fishing in the above inlets.
4. A large dogfish population.

5. If we had a large salmon population in the inlets, the seals would be a small problem. As it is now, they compete for the few salmon we now have.

In summing up, I believe, with the present technology, the salmon population could be greatly enhanced in the inlets whether it be by planting eggs, fry, incubation boxes or spawning channels as there are many good creeks.

I believe the time is now, not seven years from now.

Sincerely,



L.E. Chapman  
Sandy Hook,  
R.R. 1 Sechelt, B.C.  
VON 3A0

LEC/nc

December 28, 1978

Update '78  
Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Attention: Mr. Les Edgeworth

Dear Sir:

I am writing to you in regard to your recent Salmonid Enhancement - Update '78 meeting held in Sechelt, B.C. and the subsequent briefs and correspondence submitted to your office.

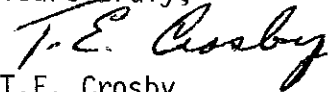
While not attending the December 6, 1978 meeting myself I have been advised of the presentation, question period and request for individual input.

My particular area of concern is the Sechelt Inlet. Its enclosed sheltered waters, spawning streams and recreational qualities would seem an ideal location for a Resident Salmon program.

This letter is intended to point out an area which, if environmentally acceptable, should be beneficial to both the Fisheries and Public alike.

Should the matter of assistance or manpower be a problem, I'm sure the local Fish and Game Clubs, Ratepayer Associations and other groups would be glad to give time and assistance in completing and administering the project.

Yours truly,



T.E. Crosby

Compartment 34  
Sandy Hook, R.R.1  
Sechelt, B.C.  
VON 3A0

10131 Freshwater Drive  
Richmond, B.C.  
V7E 4H6  
January 2, 1979

Dear Mr. Edgeworth:

As property owners and part-time residents at Sandy Hook in Sechelt Inlet, my husband and I are very disappointed to hear that our area and Jervis Inlet are not, at present, included in the first phase of the salmon enhancement program.

It seems short-sighted to leave out this protected inland waterway. We and many like us, do not have large boats suitable for the more open water of Georgia Strait and are thus confined to protected water on the rather rare calm days on the "outside".

With the present trend to smaller cars incapable of towing large boats, it would seem to be the proper time to increase the fishery in areas where these boats may safely operate. Small boats require less moorage space and use less fuel to operate as well.

We hope you will consider these points and provide for better fishing in Sechelt and Jervis Inlets.

Yours truly,

*Ethel Nielsen.*  
*(Mrs. Herbert Nielsen)*

Ethel Nielsen  
Mrs. Herbert Nielsen

SARGEANT BAY PROPERTY OWNERS' ASSOCIATION

P.O. Box 1562  
Sechelt, B.C.  
V0N 3A0  
January 1, 1979

Update '78  
Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C. V6E 2P1

BRIEF: SALMONID ENHANCEMENT UPDATE '78

The Sargeant Bay Property Owners' Association wishes to express its disappointment with the response it got from S.E.P. with respect to its application regarding Colvin Creek as addressed to Mr. Les Edgeworth and dated May 28, 1978. In a subsequent phone call on June 6 Mr. Edgeworth advised that he was "just getting organized", "thought that the letter had been acknowledged" and that he "would get in touch with us". On July 6 he said that he "was still working from a temporary office", but that a "survey would be made". The Hon. Romeo Le Blanc, in a letter dated July 14, advised that he was asking Mr. Edgeworth to reply directly to us with respect to this application.

I had the pleasure of meeting both Mr. Edgeworth and Mr. Dennis Deans at the public meeting in Burnaby on December 11, 1978. Mr. Edgeworth said he thought they "just had not gotten around replying to the request". The only letter we did receive was from Mr. W.J. Schouwenburg, dated July 17, which addressed itself primarily to the Department of Fisheries' having no objection to a marina in the mouth of Colvin Creek, rather than to Salmonid Enhancement.

I wish to make it clear that the foregoing is not meant as a personal criticism to Mr. Edgeworth or anyone on his staff. I appreciate that S.E.P. is a large and ambitious program and that everyone must be very busy. However, after reading the newspaper reports on the public meeting in Sechelt on December 6 and attending the one in Burnaby on December 11, I am inclined to think that it is symptomatic of S.E.P.'s assumed priorities: to produce myriads of salmon fry in whatever way possible and not to waste too much effort on these difficult small streams (twenty times as much money for Major Capital Works as for Small Stream Enhancement, according to the figures supplied at the meeting).

I noted from the submissions at your public meetings that the concern about too much emphasis on hatchery-produced fry and relative neglect of small stream enhancement is shared by many, both at the Sunshine Coast and in the

Vancouver area. At the Burnaby meeting at least five speakers expressed this view. I realize that hatcheries are necessary to off-set the effects of human settlement and industrialization. However, I consider the figure of 90% survival rate compared to 10% in nature deceiving. What matters is not the survival rate in the hatchery, but the survival rate during the whole life cycle. Surely the law of survival of the fittest will take its toll at some later stage. Natural stream enhancement, on the other hand, does not interfere with the process of natural selection.

It is important that the public-at-large (ie. not the selected few who attended your public meetings) is made aware of the need not to interfere with natural habitats. Therefore, your Small Stream Enhancement Program should direct itself to all streams in general, not just to those that appear to be of some economic importance. This applies in particular to areas where the population already expressed an interest in such enhancement. Such an area is for instance the Halfmoon Bay - West Sechelt area, as may follow from the first draft of their Official Settlement Plan, a copy of which is attached.

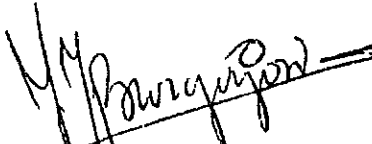
Our Association has proposed a regional park for the head of Sargeant Bay, which would include a Fish and Wildlife Reserve with beach access, parking, washrooms, trails, etc. as illustrated on attached sketch and sketch plan. The Provincial Biologist of Ducks Unlimited, Mr. R.T. Sterling, has looked over the area and D.U. are prepared to undertake waterfowl enhancement once the area becomes available for the proposed purpose. A copy of his letter of December 20 is attached. There is also an internal report by Regional Biologist Dr. Donald R. Guthrie, in which the area is recommended as a "green belt" or "natural" area for continued public use. I am not at liberty to send you a copy of this report, but you may be able to obtain a copy from Regional Land Manager, Don R. Hehn. A partial copy of a report by foreshore expert Wolf Bouer was already submitted with our original request to Mr. Edgeworth, dated May 28.

In summary, we submit the following recommendations:

- (1) That S.E.P. devote more effort to small stream enhancement.
- (2) That more of such effort be directed towards the Sunshine Coast, where the population is generally receptive to measures of habitat preservation.
- (3) That S.E.P. respond to our request of May 28 in a positive way. By this we mean constructive advice and action towards enhancement of salmon and trout habitat in Colvin Creek and marsh. While there is no salmon at present, there is no reason why there should not be after some initial seeding with fry. There is trout right now and my understanding is that salmonid enhancement includes trout. We submit that your effort in this area will not yield an economic return, but as pointed out above, should be judged in terms of public relations and education of the public-at-large.

- (4) That we may receive a reply to our request of May 28, 1978 and in due course, a copy of your report on "Salmonid Enhancement Update '78.

Sincerely,

  
Joop Burgerjon  
President

Enclosure:  
As referred to.

40 Years of Co-operation in Conservation

DUCKS UNLIMITED (CANADA)

#2 - 345 Victoria Street  
Kamloops, B.C. V2C 2A3  
December 20, 1978

Mr. J.J. Bergerjon  
Sargeant Bay Property Owner's Association  
P.O. Box 1562  
Sechelt, B.C.  
VON 3A0

Dear Joop:

Following my cursory inspection of Sargeant Bay Marsh with you on December 8, 1978 and after receiving the information you sent me with your letter of December 10, 1978, I can confirm that Ducks Unlimited would be interested in assisting with habitat preservation and enhancement developments for fish and wildlife as proposed by your association and as stated in the objectives of the Official Settlement Plan for Halfmoon Bay - West Sechelt.

The improvements that I visualize and that D.U. could be involved with are: the installation of a water control structure on the ditched outlet, clearing of invading shrub growth from former marsh bottom and the development of shallow ponds and low silhouette islands in the presently overgrown marsh area. The design, installation, operation and maintenance of these facilities would be with Fisheries and wildlife needs the prime concern, as agreed by the co-operating parties. The foregoing of course is on the assumption that the land is made available for such enhancement work, as Ducks Unlimited does not purchase or lease lands.

If we can be assured that there is high probability that the area will be preserved for fish and wildlife, and that the proposed enhancement would be desirable, D.U. would plan to obtain the necessary biological and engineering survey information for detailed planning and costing early in the coming year. Depending on the expected benefits to waterfowl, D.U. could then make a commitment for development and maintenance costs providing all legal aspects are cleared and a management plan is agreed upon by the co-operating parties.

I hope that our support will help you succeed in a very worthy project. Please contact me if we can be further help in your negotiations.

Sincerely,

R.T. Sterling  
Provincial Biologist

cc. Mr. T. Burgess  
Mr. J.A. Stephen, Jr.  
RTS/cd

FIRST DRAFT

HALFMOON BAY - WEST SECHelt OFFICIAL SETTLEMENT PLAN

1. Preamble

This plan covers all of the southern portion of Area "B" from Wood Bay to the western boundary of the Village of Sechelt. It is a rural residential area with a number of local centers but no major core area. It lies between the Regional Centers of Sechelt and Pender Harbour. This plan provides for the maintenance of the rural character of the area and the protection of the unique environmental qualities. It suggests a new community center facility and an expanded local commercial area for the residents. The plan recognizes that essentially strip development along the waterfront cannot be avoided as that is the most desirable properties in which to live. However, strip development of commercial services or a dense residential area along the highway is firmly rejected.

The population of the area is growing at a steady rate. The large lot size residential area is particularly attractive to families and that is reflected in the general age distribution of the population in the area. The plan encourages recreation use while protecting environmental values.

2. Purpose

The purpose of the Settlement Plan is to record a statement of broad social, economic and environmental objectives for the community of Halfmoon Bay - West Sechelt. The plan will contain a statement of the policies of the Regional Board respecting the general form and character of future land uses in the area. This will reduce the element of uncertainty by establishing strategies for the location and staging of growth as a guide for private and public investment as to where and under what conditions development activity will be encouraged within the plan area.

The plan must be approved by the community of Halfmoon Bay - West Sechelt, the Regional Board of the Sunshine Coast, and by Provincial agencies. The plan will represent a co-ordinated statement of Regional District and Provincial policies for the development, conservation and management of the area's physical resource. Although the plan is not binding upon Provincial agencies it will reflect provincial objectives and concerns for the planning and management of the area. The plan will indicate the Regional District's objectives for the comprehensive development of the area.



An Official Settlement Plan does not commit the Regional Board to undertake any of the projects suggested or outlined in it but does require that the Regional Board or trustees of an improvement district shall not adopt a by-law or undertake works contrary to or at variance to the Official Settlement Plan. The Plan will establish objectives and policies as the basis for the preparation and adoption of land use regulating by-laws such as subdivision and zoning by-laws. It will provide direction for the expenditure of public funds for capital works and individual projects such as water and sewage servicing or public buildings. It will establish development policies as a basis upon which elected officials and others having statutory approval authority may evaluate proposals which require amendments to implementing by-laws.

The Plan has no fixed duration but will be reviewed within 3 years of its implementation in order to assess its overall practicality and appropriateness in terms of the community. Thereafter, the Plan will be systematically reviewed every five years. Changes to the Plan shall only be made after extensive public discussion and only when it is clear the change enjoys widespread community support.

### 3. Planning Area

The area of this plan is illustrated on the plan map appendix A. It includes Area "B" from Wood Bay to the boundary of the Village of Sechelt, north to Piper Point and west to Wood Bay. Thormanby Island although in Area "B" falls under the jurisdiction of the Islands Trust and is not part of this Settlement Plan.

### 4. Goals

- 4.1 Areas with particular environmental, scenic or recreational values are to be retained in public ownership to provide for the use and the enjoyment of residents in the local area and the region generally.
- 4.2 Community services are to be provided in a way that supports the pattern of development defined in the plan.
- 4.3 The quiet rural community atmosphere is to be retained and all future development will be designed to reinforce the rural characteristics of the area.

### 5. Objectives and Policies

The following objectives and policies are designed to assist in achieving the goals of this plan. Further development under the plan shall comply with these stated objectives and policies.

Number designations are the same for the goals. For example, the objectives and policies for goal number 4.1 are presented under 5.1 and for 4.2 are presented under 5.2.

## 5.1 Preservation of environmental, scenic or recreational values.

### 5.1.1 To preserve the aquatic environment for fish and wildlife.

There are a number of areas that currently provide habitat for fish and waterfowl. These valuable resources are to be protected and wherever possible enhanced to maximize benefits for the wildlife population.

#### Policies

5.1.1.1 Valuable marshes found in areas such as Trout Lake, Sargeant Bay, Cooper's Lagoon are to be preserved and developed as nesting sites and waterfowl sanctuaries. Creation of new habitat is to be encouraged.

5.1.1.2 All creeks in the area are to be protected from pollution. For this reason setbacks for buildings, structures or drainage fields are to be rigidly enforced.

5.1.1.3 Spawning areas in creeks such as Wakefield Creek and Halfmoon Creek are to be enhanced. Where roads must cross creeks, culverts must be designed in consultation with Fish and Wildlife Branch. The culverts should assist rather than hinder fish movement through the creek.

5.1.1.4 All embayed areas are to be protected by ensuring that outfalls are located only in areas where there is sufficient flushing action.

5.1.2 To improve the opportunity for recreational boaters by expanding boating and docking facilities and by developing suitable areas for boat launch facilities and for attendant upland servicing.

#### Policies

5.1.1.2 An active program of developing new boat launch facilities is to be undertaken immediately in the area. These facilities are to be located where there is adequate space for parking and areas for storage of boat trailers. Boat launch sites are to have garbage containers to minimize any detrimental effect on surrounding land.

5.1.2.1 The expansion of the Halfmoon Bay Wharf is requested. To improve docking facilities in the area, privately owned marinas are to be encouraged. Marinas are to be located in places where minimal environmental disruption will result.

- 5.1.3 To provide opportunities for the quiet enjoyment of streams and lakes by obtaining wherever possible, easements in favour of the local government, or dedications to the Crown, upon which walking trails can be established.

The lakes and streams are an important part of the attractiveness of the area, public access to them is to be assured.

Policies

- 5.1.3.1 Walking trails are to be established in conjunction with wilderness parks or areas of particular interest, especially in built-up residential areas public pathways are to be encouraged.
- 5.1.4 To preserve buildings or structures for historic or cultural interest including archaeological resources and historic buildings.

Policies

- 5.1.4.1 The archaeological sites indicated on the plan map are to be protected. The use of these areas is to be compatible with the preservation of the particular archaeological value.
- 5.1.4.2 Historic buildings identified by the community are to be preserved if threatened by demolition or removal. Provincial funds are available for such a purpose and are to be utilized wherever justified.
- 5.1.5 To preserve all possible present or future watersheds.

Water supply is a limiting resource without which development cannot proceed.

Policies

- 5.1.5.1 All lakes and rivers in the area are to be considered possible water sources. Accordingly any recreational use, even such minor uses as walking trails, are to be designed in a way that will preserve the water quality.
- 5.1.6 To develop existing public land to increase the opportunities for use in a way compatible with the surrounding area.

Policies

- 5.1.6.1 Areas of particular interest such as the Limestone Caves should be developed to increase public access to them. Camp sites in the Forest Reserve are desirable and trails to those camp sites should be clearly marked.

5.1.6.2 Smugglers Cove Marine Park provides an opportunity for upland development that is not being utilized. The Cove and water resource is being used by the boating public and is an important recreation resource. A large amount of land is within the Park, however, and that land could provide additional recreation uses without infringing upon the water based use. Camp sites should be provided the motoring public with walking trails leading to ocean views and beaches.

5.2 The supply of community services.

5.2.1 To supply safe potable water from a central system to all residents within the area.

Policies

5.2.1.1 The Regional District water system is to be expanded to include all areas within Halfmoon Bay - West Sechelt. This expansion is to be done only when it is economically feasible. It seems evident that in areas such as Secret Cove it would be too costly and unjustified to require community Regional water at this time.

5.2.1.2 The expansion of the Regional District water system in no case is to be used as a reason to increase development. The density of development is to be decided firstly on a social basis as defined in this plan and only secondly is to be guided by the availability of water and septic capability.

5.2.2 To improve existing systems of sewerage disposal and to ensure future systems have a minimal environmental impact.

Policies

5.2.2.1 Sewer systems will be necessary in some parts of the area. Such systems are not to pollute existing water courses. All outfalls must be in areas with good flushing and all systems should be of the highest quality.

5.2.2.2 The accumulative effect of sewerage outfalls is to be carefully examined upon application for any new outfalls. Likewise the cumulative effect of septic systems on waterfront property is to be considered prior to approval for development.

5.2.3 To provide a community center for the entire area which will be a focus for the local community and offer services for the larger regional community.

### Policies

5.2.3.1 An auditorium is a desirable development for the area to facilitate meetings and other uses. Such a development requires the support of a larger community and it is anticipated that only one auditorium can be built for the Halfmoon Bay - West Sechelt area.

5.2.3.2 Tennis courts and small area parks are to be dispersed generally throughout the local area. The facilities for lawn bowling and tennis are to be established in a number of these area parks.

### 5.3 Retention of rural character.

5.3.1 To design residential roads that will maintain a quiet, pleasant neighbourhood atmosphere while assuring basic safety standards.

### Policies

5.3.1.1 Narrow winding roads are acceptable in individual subdivision developments with use of cul-de-sacs and loop roads wherever possible.

5.3.1.2 Connector roads between discrete subdivision areas should be 12 meters wide and provide a paved surface to accommodate two-way traffic.

5.3.1.3 The controlled access highway through the community should include passing lanes as an alternative to widening the entire highway. The passing lanes will provide for fast passage of through traffic.

5.3.1.4 Highway 101 is to be designated a scenic route once the new major through way is constructed.

5.3.1.5 The Ministry of Highways is requested to provide a bicycle lane along Highway 101.

5.3.1.6 View points are to be provided at appropriate areas along Highway 101. Such view points should be large enough to remove the cars from the travelled portion of the highway such as is found in other provincial highways in the province.

5.3.1.7 In residential areas the highways department is requested to leave a band of vegetation on either side of the shoulder. The vegetation at the edge of the shoulder decreases the noise generated on the route and creates a more rural visual aspect to the area.

- 5.3.1.8 To allow roads to follow natural contours, Regional by-laws are not to insist upon conventional rectangular shaped lots only. Some flexibility of design is to be provided.
- 5.3.2 To provide a variety of residential lot sizes throughout the area at a scale designed to maintain the rural character.

Policies

- 5.3.2.1 Lot sizes in the area should be generally an average of .2 ha.
- 5.3.2.2 Variation in lot size from an average .1 ha. to .2 ha. is to be consistent with existing development. The West Sechelt area particularly is identified as being suitable for .1 ha. average lot size in the established residential area. Properties north of Northwest Bay Road are to be in larger lot sizes to maintain the rural aspect to that area. No urban 5 meter wide lots are to be allowed to develop in strips along the residential roads.
- 5.3.3 To provide for uses of land consistent with a rural character.

Policies

- 5.3.3.1 All lots in the area should allow single family dwellings only. Apartments are not allowed anywhere in this area.
- 5.3.3.2 Home occupation and home industry is to be allowed on all lots where they satisfy the requirements for siting as defined in the land use by-laws.
- 5.3.3.3 A local commercial center is to be designated for Halfmoon Bay - West Sechelt area. Such a center should provide a local store and tourist oriented commercial enterprises. The center is to be oriented off Highway 101 but may back onto Highway 101 as shown on the plan map.
- 5.3.3.4 Only one dwelling is to be allowed on a lot to maintain an average density of 1 dwelling per .2 ha. In larger lot size areas secondary cottages may be allowed.



#1006 - 4758 Grange Street  
Burnaby, B.C.  
V5H 1R2  
January 2, 1979

Mr. Les Edgeworth  
1090 West Pender Street  
Vancouver, B.C.

Re: Salmon Enhancement Program

Dear Sir:

Further to information provided at the meeting held on December 6th, 1978, I would suggest that a slight adjustment in your Department's schedule might be considered.

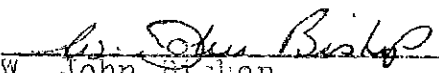
As I understand it, the "enhancement" of Sechelt Inlet is not scheduled to take place until "Phase 2" is implemented.

I and many others who own property on the Inlet, have been seriously concerned for many years that salmon fishing on that body of water has deteriorated seriously.

Therefore, I feel that early "enhancement" of the Inlet should be given priority and included in "Phase 1".

Anything that might be done to make this change will be greatly appreciated.

Yours truly,

  
W. John Bishop

P.S. I also wonder if your Department has any reaction to our observations that the increase of dogfish and seal populations has had a serious negative effect on "bottom fish", especially cod?



#203 - 8740 #1 Road  
Richmond, B.C.  
January 2, 1979

Mr. L. Edgeworth  
Salmonid Enhancement Program  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sir:

In reference to your comments at the meeting held in Sechelt on December 6th, it came as rather a shock to me to hear that your office had received very little input from the people in the Sechelt area to encourage any development under the Salmonid Enhancement Program.

We have been summer residents of the Sechelt Inlet area for twelve years and are very aware of the recreational possibilities of the inlet.

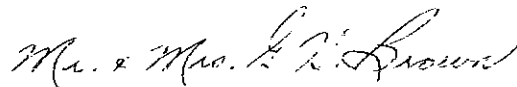
We enjoyed the fishing in the first few years but have sadly watched it diminish to the point now where very few people bother wasting their time.

We have heard the residents complain (possibly incorrectly) that the fishing has gone steadily downhill since the power dam was built at Choldm Lake, so felt that some complaints must have reached your office.

The Salmonid Enhancement Program is one of the most promising of the last decade and we would like to see it extended, especially if it could include some help for the beautiful Sechelt Inlet.

We would also like to recommend aims at depleting the dog fish and increasing the herring stock.

Sincerely,

A handwritten signature in cursive script, appearing to read "Mr. & Mrs. G.H. Brown".

Mr. & Mrs. G.H. Brown

P.O. Box 1066  
Sechelt, B.C.  
VON 3A0  
January 2, 1979

Update "78"  
Salmonid Enhancement Program  
Fisheries & Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sir:

In response to your newspaper advertisement inviting written submissions in regards to the Federal/Provincial Salmonid Enhancement Program, I would like to formally register the following concerns and recommendations. I would like to limit my remarks to the Sechelt Inlet and its immediate waters, although the comments may have more universal applications.

Starting with the promise that the Sechelt Inlet should be maintained and protected as a sport fishermen's paradise, I am concerned that there appears to be a decreasing number of salmon inhabiting these waters. Indeed, the population of ling cod, red snapper and even rock cod seem to be decreasing.

I believe this decrease is due to four major contributing factors.

1. Industrial and commercial developments resulting in some unnecessary pollution.
2. Overfishing by commercial and recreational fishermen.
3. Too many seals.
4. Too many dog fish.

I would like to offer specific recommendations with respect to each of these problems.

1. Industrial & Commercial Development should be restricted in the Sechelt area, with further development subject to stringent pollution control standards developed jointly by government and corporate representatives. Existing developments should be, over a reasonable period, subjected to the same standards. Heavy fines should be introduced for failure to meet standards. Appropriately trained government-employed pollution/environment inspectors should be employed to monitor private and public corporations/businesses. The continued monitoring of Aggregate Co. & Black Top Co. in Porpoise Bay will be explanatory.

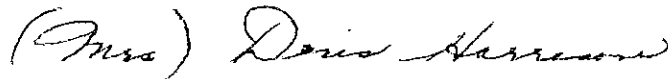
2. Overfishing should be attacked on a multi-dimensional program of reduced time limits to reduce the catch limits and increase size limits. The introduction of such limits would, I believe, conserve the present populations, permit the development of enhanced fish populations and reduce the obvious abuses presently occurring.
3. Seals should be controlled by either introducing a bounty or attempting a capture scheme transporting seals to other waters. If the former (bounty) idea is reintroduced, bounties should be under the direct supervision of a suitable trained local resident. Failing this, interested local residents should be consulted. With respect to the capture scheme, I believe this would be a costly program, although its merits should be considered.
4. Dog fish could be controlled if the Canadian and Provincial governments were to encourage the development of an export market for dog fish, trading with nations such as Japan and England, where dog fish is a staple food. Subsidies could be provided by commercial fishermen in the early stages. The resulting employment would compensate for reduced employment, resulting from the reduction of time limits (suggested under item 2).

Finally, increased emphasis on the transplantation of salmon roe into many creeks and rivers flowing into Sechelt Inlet waters would also enhance the salmon population.

I firmly believe that these steps must be taken immediately. I understand that programs aimed at Sechelt waters are not presently included in Phase 1 of the Salmonid Enhancement Program. I believe it is essential that a realization of priorities, giving immediate attention to these waters. Failure to do so may have a long term affect of greater costs when the area is finally addressed, a greater depletion of fish populations and more complex ecological problems of regaining a healthy and productive sports fishing area.

Thanks for the opportunity to comment.

Yours truly,

A handwritten signature in cursive script, reading "(Mrs) Doris Harrison".

Mrs. Doris Harrison

39 Coracle Drive  
Sandy Hook  
R.R. #1  
Sechelt, B.C.  
VON 3A0  
January 3, 1979

Mr. Les Edgeworth  
Salmonid Enhancement Program  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sir:

Approximately five miles from Sechelt there is an area known as Sandy Hook. The residents of this area were at one time known as the, "Happy Hookers" but since the fishing in the Sechelt Inlet fell off about ten years ago, they are no longer very happy.

I understand that you have a program underway which may increase the number of salmon in the Sechelt Inlet in future years.

May I point out that most of the residents in this area are retired people and may not have too many years left in which to enjoy one of the sports which in some cases, decided these people to settle in this area.

During the S.E.P. meeting held in Sechelt on December 6, 1978, we were informed that this area would not be looked at for possibly seven years which would mean approximately twelve to fourteen years before we can expect a return from your efforts. Twelve years or more on the age of these residents pretty well means that they will be too old to enjoy their favourite sport through the present schedule that you are operating on.

If this area was worked on now these people may still see the results of your program and be able to enjoy their favourite sport once again.

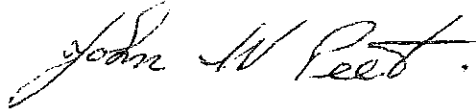
Unfortunately, I am not able to suggest what can be done to improve the Inlet other than ask that a strict check be kept on the industries which have opened up on the shores of the Inlet in the past few years (e.g. gravel pit and waterfront loading, black top plant and increased logging areas) to ensure pollution is kept to a minimum.

If it is impossible for you to bring this area into your stage one to get salmon back into the area in quantities, would it be possible as a stop gap measure, to look into the possibility of increasing the ling cod, rock cod, red snapper and oyster populations which have been decreasing drastically in the past three or four years?

Anything which your program does to increase the fish population in the Sechelt Inlet will be greatly appreciated by the people of this area and who knows, it may even make the unhappy hookers, Happy Hookers once again enjoying their favourite sport.

Please try to do something for us? Thank you.

Yours sincerely,

A handwritten signature in cursive script that reads "John W. Peet". The signature is written in dark ink and is positioned below the typed name "John W. Peet".

John W. Peet

#1002 - 1919 Bellevue Avenue  
West Vancouver, B.C.  
January 4, 1979

Mr. L. Edgeworth  
Salmonid Enhancement Program  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sir:

As a resident of Sandy Hook, located on the Sechelt Inlet, approximately four miles west of the town of Sechelt, I am extremely interested in the Salmonid Enhancement Program.

Unfortunately, I was unable to attend the recent meeting held in Sechelt but through discussions with other residents of the area, understand that the Sechelt Inlet is included in Phase 2 of your Update "78" program.

The purpose of this letter is to seek your consideration of including the Sechelt Inlet in Phase 1 of your program, rather than Phase 2, and in support of this proposal, respectfully offer the following reasons.

The Sechelt Inlet is a rather unique body of water in that it is one of the few unpolluted waters located so close to large urban areas. The waters are deep with partially controlled tides because of the limited passage at the head of the Inlet and yet the tidal currents are strong enough to create the desirable cleansing agents necessary to the healthy survival of sea life.

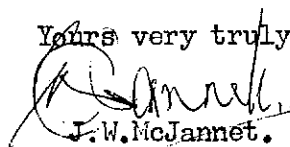
With a coast line of approximately 140 miles, the Inlet contains a large number of fresh water streams which could provide the necessary spawning grounds for the propagation of salmon. There is also an abundant natural supply of food for salmon in the heavy herring runs.

The Sechelt Inlet is readily accessible to sport fishermen and is an ideal small boat area due to its relatively sheltered position.

If, for reasons beyond my knowledge, it is not possible to include the Sechelt Inlet in Phase 1 of the Salmonid Enhancement Program, perhaps it would be possible and practical to "seed" the area with fingerlings or "wild" salmon during the time interval between Phase 1 and Phase 2 from surplus Capilano Hatchery stocks. In my somewhat limited knowledge of the subject, may I suggest the Porpoise Bay area of the Inlet as an ideal spot for such a planting.

The above is offered for your valued consideration.

Yours very truly,

  
J.W. McJannet.

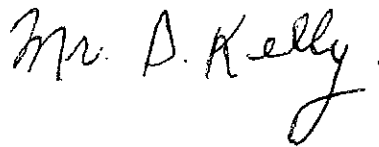
Mr. D. Kelly  
R.R. #1  
Sechelt, B.C.

Dear Sirs:

I live on the Sechelt Inlet and am very concerned with what is happening to our fish. Due to the increase in population and tourists, we seem to have an extreme lack of coho, spring salmon, herring, ling cod, and cutthroat trout. Compared to a few years ago, when we could go to the few spawning streams we have and see hundreds of fish, now we see nothing. We also have a very bad problem with seals, if we could get some kind of control on them and improve and protect our spawning streams, we could be sure of a lasting supply of fish.

Our area is now scheduled for Phase II on the planned update, but I feel we should move onto Phase I. I think if we wait any longer, we might find ourselves with no fish at all.

Thank You.

A handwritten signature in cursive script that reads "Mr. D. Kelly." The signature is written in dark ink and is positioned above the printed name.

Mr. D. Kelly

Mr. & Mrs. C.N. Stephens  
Sandy Hook  
P.O. Box 102  
Sechelt, B.C.  
VON 3A0

Dear Sir:

We wish to protest against the phasing of Sechelt, Salmon and Narrows Inlets and would like to see this area in Phase 1 and as soon as possible.

This area is made up mostly of retirement couples, who like to go out fishing and it is just impossible to get a salmon in our bay and now the cod are fast disappearing as well.

We are just overrun out here with dog fish and seals, and a strong effort should be made to bring these under control.

At one time we could go out fishing 20 years ago and never come home without a salmon, now there just aren't any and if your committee delays for seven years more, it will be worse than ever. Please give this matter great consideration and thank you.

Please send us your monthly bulletins, etc? We would also like to be put on your mailing list.

*Mr. & Mrs. C.N. Stephens*

Mr. & Mrs. C.N. Stephens



Mr. Len Herder  
P.O. Box 13  
R.R. #1  
Sechelt, B.C.  
VON 3A0

Mr. Les Edgeworth  
Salmonid Enhancement Program  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sir:

In regards to the meeting on December 6th, 1978, at Sechelt, B.C., I feel the following points are pertinent.

I would like to see Sechelt Inlet placed in Phase I, rather than in Phase II for the following reasons: If the first Phase is to be 7 years and this area to be later in Phase II, plus the 4 years it takes to bring a run of salmon from egg to maturity, it could well take 11 to 15 years to enhance this area. Lets do it now.

We should either place smolts in now barren streams, (this should not result in a disease problem as there is no natural run there). Or place rearing pens at the mouth of above streams to create a resident fish population in the Inlet.

Clowhom Lakes and River - build a fish ladder to overcome the B.C. Hydro dam and again give salmon access to these waters as they had in the past. This was at one time a great spawning situation. It again could be done with smolts for a starter.

Grey Creek - the placing of smolts above the falls or a fish ladder would be of great value to salmon enhancement in this area.

Earle Creek - the same applies as with Grey Creek.

Seals - I agree to the control of seals as a salmon enhancement program. Fish for humans not seals.

Herring - keep a close watch on the amount harvested as this is an important food for the salmon.

Dogfish - again salmon enhancement could be greatly improved by extensive fishing of the dogfish. I believe there are many markets such as Japan, Germany, etc., for this type of fish. I have personally seen the fry or smolts from Black Creek on Vancouver Island eaten by the dogfish as they come into the ocean.

Send all Fish and Game Clubs in the province the knowledge on how to adopt a stream and enhance it. Also, give them pamphlets on this to pass out to others and get them interested too.

If you wish further information on my thoughts, I am available at 936-1927 or 885-2878 in Sechelt.

I am yours sincerely,

A handwritten signature in cursive script that reads "Len Herder". The signature is written in dark ink and is positioned centrally below the typed name.

Len Herder

Mr. P. Juntunen  
R.R. #1  
Sechelt, B.C.

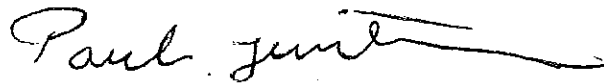
Dear Sirs:

I am writing to you in regards to the Sechelt Inlet. We live in a beautiful and protected waterway and now because of the growing population and increased fishing pressure due to the ever growing campsites in the area. We have found ourselves with an extreme lack of coho and spring salmon, herring, ling cod and cutthroat trout. We have a few spawning streams but we need more and the ones we do have should be made safer and more protected for spawning. A few years ago we could watch salmon spawn up the streams. They were so abundant you could walk across the streams on a mat of salmon without getting your feet wet. Now you can't see any salmon at all. If we could get some kind of control on the seals, we could insure a lasting supply of fish, especially around the mouths of the spawning streams.

At the moment, our area is scheduled for Phase II of the planned update, but I feel we should be moved into Phase I to help bring back the fish now. I feel, if we wait the number of years it would take for the completion of Phase II, we could find ourselves without any fish at all.

Thank you for your attention.

Yours truly,

A handwritten signature in cursive script, reading "Paul Juntunen". The signature is fluid and extends to the right with a long, sweeping underline.

Paul Juntunen

CHILLIWACK



# THE STEELHEAD SOCIETY OF BRITISH COLUMBIA

P.O. BOX 33947 - STATION D - VANCOUVER, B.C. V6J 3J3  
UPPER FRASER VALLEY COMMITTEE

December 4, 1978

Update '78  
Salmonoid Enhancement Program  
Fisheries & Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

During the past two years, we have seen S.E.P. publications containing the statement "Steelhead as Constraints to Salmon Enhancement!". We cannot over-emphasize the importance of this trophy fish. We in the Steelhead Society would most certainly rephrase this statement using: logging, pollution, hydro dams, highway construction, habitat destruction, etc. as constraints to steelhead and salmon enhancement.

Included in the S.E.P. Public Inquiry II pamphlet of Oct. 78 was the statement "The future direction of S.E.P. must take into account the concerns of British Columbian communities". With this in mind, please consider the following recommendations:

## COQUIHALLA RIVER

The sport angling closure for approximately the last fifteen years has done little for the comeback of the historical abundance of Summer Steelhead in this system. We recommend that sufficient Coquihalla egg stock be taken, and with the resulting smolt plants, achieve a return of 5,000 adult Summer Steelhead by 1983 and 2,500 Winter Steelhead by 1983-84.

The new and unnecessary Hope to Merritt highway poses the threat of an environmental disaster. We recommend strict supervision during highway construction by Federal and Provincial Fisheries Departments to ensure no fish habitat is destroyed.

Guaranteed public access during and after the highway construction is one of our concerns. We recommend that S.E.P. initiate a program to set aside public access corridors to the river. In the case of private lands, the buy-back of an access corridor may be required.

### SILVER-HOPE CREEK

The summer run steelhead in this system virtually disappeared. A float count in 1978 counted six Adult Fish. We recommend sufficient Coquihalla egg stock be taken to obtain a return of 3,500 Adult Summer Steelhead and 1,500 Winter Steelhead by 1983-1984.

We recommend there be in-stream improvements (removing falls, jams, etc.) to facilitate upstream migration. Guaranteed public access corridors are required in the lower section of the creek.

### CHEHALIS

One of the reasons for the decline of salmonoid stocks in this system is the yearly flooding caused by the logging in the upper watershed. In order to stabilize the flow, we recommend the immediate reforestation of the upper watershed including streambank deciduous plantings. Future logging operations require stricter supervision. The protection and enhancement of fish habitat must be the priority. A Salmonoid stock inventory, including summer steelhead, on this system is required immediately. All means of natural or semi-natural propagation must be explored before any artificial methods are considered. It is unlikely artificial propagation is required, but if any is undertaken, Chehalis egg stock must be used. The genetic integrity of Chehalis stock must be maintained. Guaranteed public access is required. The interception and eventual sale of steelhead from the Indian Food Fishery located at the mouth of this river is one of our concerns. Stricter enforcement of existing laws regarding the sale of food fish is required.

### WEAVER CREEK

The present Federal salmon spawning channel seems to overshadow the very existence of steelhead in this system. Immediate steps must be taken to protect and enhance this small population of steelhead.

The encroachment of the Hemlock Valley Recreation Development further necessitates the protection of fish habitat on this creek. Consideration should be given to taking egg stock and with resulting smolt plants, increase the present population of steelhead.

### RUBY CREEK

To facilitate future enhancement, we recommend a stock inventory be done on this creek. Consideration should be given to expand on the existing cutthroat fishery in the lower sections of the creek. We recommend negotiating guaranteed access to sportsmen

through Indian lands which border this creek.

#### SUICIDE CREEK

We recommend a stock evaluation on this stream. Enhancement should be concentrated on improving habitat by upgrading of existing spawning areas and reforestation throughout the watershed. Deciduous streambank planting is a must. Natural propagation should be concentrated on. Consideration should be given to the Historical presence of Summer Steelhead in this creek. With regards to cutthroat, we recommend a stock evaluation, upgrade of spawning areas, and if required, a Cutthroat Trout smolt plant.

#### VEDDER

The protection and enhancement of fish habitat in this system cannot be over-emphasized. Logging operations in the upper watershed require stricter supervision. Restrictions must be implemented with regards to future residential or commercial development on the river. The same width of green belt area as the one that is within the set back dykes should be considered for the entire system. This would definitely provide the guaranteed public access that is required, as well as protecting fish habitat.

The proposed pool and riffle type hatchery at Slesse Creek must use Vedder egg stock only. Methods of propagation must retain the November to May fishery with an even balance of returns each month. Also the quality of fish with regards to size must follow the natural degree of variation of existing wild stock. The release of smolts from the wet bridging area results in returns that cannot be utilized by the sports fishery. This practice must stop.

The introduction of Vancouver Island stock into Marshall Creek, a tributary of the Vedder, presents the potential for a genetic disaster. The use of Vancouver Island stock simply because of a surplus of that stock on hand is appalling. This practice must stop immediately. Vedder stock is available to provide adult fish for the public viewing ponds at the Abbotsford Hatchery.

The incidental catch of November steelhead by an enhanced commercial chum fishery in the Fraser concerns us. These large early run steelhead must be protected. Closures of the Fraser commercial fishery should be considered. We recommend a complete closure to sports fishing on the Vedder from May 1 to July 1 to protect smolts and spawning adults. We would like to see the target date for the enhancement of spring salmon brought forward.

### CUTTHROAT TROUT

The Cutthroat enhancement program should be enlarged. The protection and enhancement of habitat should be concentrated on. SEP should encourage more cutthroat oriented enhancement projects to public groups who wish to undertake a small scale, but valuable project.

### INDIAN FOOD FISHERY

The illegal sale of Indian caught steelhead and salmon and their roe, in the area from Abbotsford to Yale, is a serious threat to any proposed salmonoid enhancement project. We recommend an immediate increase in Federal enforcement staff and an increase of the minimum fine for the buyer to \$1,000.

### REGULATIONS

We, the Steelhead Society, reluctantly condone catch and release regulations concerning our trophy steelhead. We fully realize the importance of protecting the last of the spawning adults, and thus we support the Fish and Wildlife Branch on this last resort measure. However, it is truly sad that we, the sportsman, who have the least impact on the fishery, have to further restrict ourselves to compensate for the destruction of our fishery by industry. What compensation will SEP receive from logging and gravel mining operations who have virtually destroyed entire river systems in the past? What will stop this destruction in the future?

Why is the commercial fishery allowed additional fishing time during a period when they are most likely to intercept our catch and release steelhead?

Catch and Release regulations will not return the abundance of steelhead, it is but a small bandaid on a large wound. Any river system that has a catch and release fishery should be considered a disaster area by SEP and thus be an emergency priority.

It seems that we, the sports fishermen, have now become an endangered species and as conservationists, we feel this is unacceptable.





BURNABY

Mr. Grant Steele  
3976 West 27th Avenue  
Vancouver, B.C.  
V6S 1M2

The Salmonid Enhancement Program  
of Fisheries and Oceans Canada

Dear Sirs:

The return and multiplication of fish to the waters of B.C. must be looked at from the obstacles encountered by the fish. Unconcerned scalping of the forest in some areas has left barren areas, easily eroded and washed away by rains and run-offs, to clog and pollute the lakes and streams. Man-made dams placed in strategic areas has locked in and prevented either the escape or additional fish from entering the trapped areas. The influence of towns and uncontrolled waste disposal further hampers migrating fish. Nature with some degree of success has attempted to flush free the polluted areas, but it appears to be a losing battle. Success to any extent appears to only be in the virgin areas.

Man, in the name of progress, adventure, etc., has through hard rock mining, adverted and in some situations, eliminated spawning streams.

I propose, if the laws of logging and mining were changed, we may stand a chance to repropagate the fish areas.

Logging areas should be required to leave sufficient vegetation to bind and protect the soil and also to sustain wildlife. Further, all resulting debris capable of clogging waterways to be cleared.

Placer miners do more good than harm in the water systems. Small dredge operations wash and sift the stream gravels, returning the dredged material to the stream, finer sands filtering through the water as nature intended. The result, cleaner gravel beds, free of clays and sharp sands. As evidenced in the State of California, U.S.A., placer laws permit miners to operate in a stream or river, with no harmful effects to the fish population. In fact, a marked increase has been noted in the fish population in those dredged areas.

Government interest in people's employment, ie. increased employment and less social assistance in remote areas, and in the best interest of nature both fish and wildlife, could be the resultant factor of a change in the placer laws.



Grant Steele

cc. Dept. of Fisheries  
Victoria/Ottawa

Second Submission to

THE SALMONID ENHANCEMENT PROGRAM

BY

Eric Carlisle  
1095 Inglewood Avenue  
West Vancouver, B. C.

November, 1978  
Vancouver, B.C.

Gentlemen:

In my present brief, I wish to comment on the achievements of the Salmonid Enhancement Program since the public hearings were held two years ago. At that time, I presented a brief which focussed on SEP from the viewpoint of a river angler. Today, with two exceptions, I will cover different ground.

#### 1. Natural and Semi-Natural Enhancement

At the public inquiries held in November, 1976, seventy-three presentations favoured natural enhancement, semi-natural enhancement and small-stream work. A mere handful of commercial fishermen and one group of foresters spoke in favour of hatchery construction. In the last two years, a considerable amount of valuable stream improvement work and natural and semi-natural enhancement has taken place. Examples include the British Columbia Institute of Technology upwelling gravel incubators and semi-natural rearing channel on the upper Seymour River, the Squairetailers Rod and Reel Club's fish ladder on Hastings Creek, watershed improvement on a tributary of the Serpentine River, experimental habitat improvement work on Colquitz and Craigflower Creeks, the Cowichan Fish and Game Association's fish passage work and fry feeding on various tributaries of the Cowichan River, the "Community Fisheries Enhancement" stream improvement projects taking place on nearly forty Nanaimo area creeks, the opening and restoration of a Chemainus River side-channel by the Hallalt Indian Band, the excavation of a bypass channel around a massive log jam on the Nahatlatch River, the installation of a new culvert on Tenderfoot

Creek, remedial work on Ain River waterfalls, the temporary dismantling of ten beaver dams on Toboggan and Cathlyn Creeks, the planting of steelhead smolts in two underseeded Chilliwack River tributaries, the radio tagging and subsequent tracking of Thompson River steelhead, and the recolonizing of the Yiddle Shuswap River with chinook salmon. However, much, much more natural and semi-natural enhancement is required.

Unfortunately, it is apparent that the bulk of the enhancement money is to be spent on the construction of hatcheries. While I realize that hatcheries have a definite place in the Salmonid Enhancement Program, like many other sportfishermen I feel that the Fisheries and Marine Service is not heeding the wishes of the public and is placing far too much emphasis on hatchery construction. For example, at the 1976 hearing in Chilliwack, not one person spoke in favor of hatcheries and eight out of eleven briefs favored natural enhancement or stream protection. Yet, plans are proceeding for the construction of a hatchery on the Chilliwack River. This apparent refusal to follow the wishes of the public as expressed two years ago raises a question -- are these public inquiries just window dressing and a waste of everyone's time, or does the Fisheries and Marine Service intend to pay attention to the desires of the public regarding increased emphasis on natural and semi-natural enhancement?

## 2. The Native Indian Fishery

Sport fishermen, commercial fishermen and the general public will not benefit from the Salmonid Enhancement Program if most of the migrating salmon are harvested in rivers by Indians. Above all, the Fisheries and Marine Service MUST retain control of the fishery. Clause #6 of the Squam-

ish Indian Band Bylaw #10 (signed last Fall by Indian and Northern Affairs Minister Hugh Faulkner) says, "Members of the Squamish Indian Band shall be permitted to engage in fishing upon Squamish Indian Band waters at any time and by any means except by the use of rockets, explosive materials, projectiles or shells." Immediate court action must be undertaken to repeal this controversial bylaw, otherwise Squamish Indian Band members potentially could eliminate salmon and steelhead runs in the Squamish system and in the Capilano and Seymour Rivers. If this bylaw is not repealed, other Indian bands throughout British Columbia will pass similar bylaws, resulting in a chaotic situation which will seriously harm, if not destroy, the salmonid resource.

Assuming conservation is not an issue, I fully support the right of native Indians to fish in rivers for food, using traditional native methods and equipment. However, British Columbia Indians, together with sport fishermen and commercial fishermen, must recognize that THE FISH HAVE A RIGHT TO SURVIVE. It does not matter whether a run is down because of poor management of the commercial fishery or simply because of low natural survival, Indian bands must accept the fact that their fisheries may be curtailed at certain times in order to allow adequate spawning. I believe that each Indian band should be allowed a reasonable yearly quota of fish, based on the needs and population of the band. If a band cannot achieve its quota through its food fishery, the Federal Government should supply that band with surplus salmon from one of the hatcheries or spawning channels. If at all possible, each band should be allowed to catch its salmon; a shut-down of a food fishery should be undertaken as a last resort in order to protect a river's salmonid populations.

Only licenced commercial fishermen are allowed to sell their catches; neither sport fishermen nor Indian food fishermen can sell their fish legally. While I realize that Fishery Officers cannot be everywhere at once, I believe that strict enforcement of Indian food-fishing permits and further drives against all poaching are necessary. Stiff penalties should be imposed upon buyers as well as sellers of bootlegged salmon. In the future, enhancement of specific chum, pink and sockeye runs (especially enhancement projects undertaken by Indian bands) may make feasible some Indian commercial river fisheries for surplus chums, pinks and sockeyes. Fishing methods should allow the release of cohoes, chinooks, steelhead and trout and, in addition, adequate spawning escapements of the food salmon.

Indian spokesmen frequently talk about their respect for the fish. Yet, when observing the gaffing fishery on the lower Capilano, I see little, if any, respect shown for the fish. The youthful gaffers lunge at their quarry with laughter, impaling and retaining some, jeering at those which escape--totally unconcerned about the substantial number of cohoes which thrash their way off the gaffs only to die of their wounds in the river. Whenever the river is fairly low, any fish which tries to swim upstream must run a gauntlet of gaffs. Tragically, fish other than the surplus cohoes are gaffed. Summer steelhead, cutthroat trout, pinks, chums and chinooks--all of which are needed for spawning, either in the river or at the hatchery--have been gaffed and, on occasion, left to rot because the gaffer wanted cohoes only.

### 3. The Commercial Fishery

Unquestionably, summer steelhead runs in various Alberni Inlet rivers have suffered due to clearcut logging of the rivers' watersheds.



However, the misfortunes of these summer steelhead have been augmented; their runs have been decimated by the increased commercial netting which has taken place in Barkley Sound. Successful lake fertilization experiments have enhanced sockeye production, resulting in an increase in commercial fishing pressure. I hope such irresponsible action by the Fisheries and Marine Service will not be repeated, but I fear that future enhancement of sockeyes, pinks and chums will bring about the extinction of many summer steelhead runs in British Columbia. It is possible that constant hatchery production, with all the costs and difficulties involved, may be the only means by which summer steelhead in this province can be preserved.

I hope the Fisheries and Marine Service will take every possible step to separate enhanced runs from all other runs so that commercial exploitation of these other runs can be kept to a minimum. In particular, I should like to see the Fisheries and Marine Service explore the concept of terminal fishing. Fish traps could be constructed at the mouths of certain enhanced rivers, and commercial fishermen could brail the fish from the fish traps rather than net them and assorted wild fish at sea. Trout, sport salmon and a good escapement of food salmon could be released from the traps.

#### 4. River Salmon Angling Regulations

On the basis of conversations held with Fisheries personnel, I understand that the latter can open and close rivers on the basis of time or fish species. However, a change in the Fisheries Act is required in order to allow an opening on the basis of fish size. I believe that such a change in the Fisheries Act should receive top priority. This year, the Capilano River was open to coho fishing from July

15th and closed to all chinook fishing through November 30th. Thanks to unseasonal rainfall, the river was in reasonable to good fishing condition throughout much of September. A few adult cohoes and jack cohoes were caught, but most successful anglers caught and released three to six pound jacksprings. In fact, I rated the fishery as a no-kill jackspring fishery, not a coho fishery. Adult chinook returns were not large and required protection, but anglers should have been allowed to retain any jacksprings they desired. These jacksprings were not required for brood stock and will be sold to a fish packer. A regulation requiring the release of chinook salmon over 28 inches in length would have protected the adult chinooks and would have allowed a harvest of the jacksprings by river anglers. Such a regulation is in effect on many Washington State rivers; we need a similar regulation in British Columbia.

#### 5. Catchability of SEP Produced Salmon

In the past five years, I have found that hatchery raised cohoes in the Capilano River are far less responsive to an angler's offerings than were the wild cohoes present in the river years ago. Eventually, I should like to see the Fisheries and Marine Service conduct experiments designed to find and develop strains of coho and chinook salmon which will respond to the river anglers' and rivermouth anglers' lures --at least as readily as do wild salmon. If nothing else, I feel that hatchery personnel who are choosing their coho and chinook brood stock should exercise one form of selection. Any coho or chinook which has a lure or a bait hook in its jaw should be used to supply eggs or milt. This fish with a hook in its jaw was a "taking" fish. It possessed a certain element in its makeup which made it hit a lure or bait; perhaps

its progeny will be more willing to strike than will the progeny of non-taking salmon.

#### 6. SEP Catch Predictions

In spite of all the talk of SEP doubling present salmon stocks (and, by implication, doubling catches) the sportfishermen's share of the harvest is minute. In the September, 1977 Proposed Cost Recovery Scheme for the Salmonid Enhancement Program, predicted annual incremental production at full production (1986-87) is: sockeye - 3,110,000; chum - 2,170,000; pink - 620,000; coho - 190,000, and chinook - 120,000. The recreational share is just one-half of the cohoes and chinooks-- 95,000 and 60,000, respectively, or 2.5% of the total production of 6,210,000 salmon. I believe that excessive emphasis is placed on the production of pinks, chums and sockeyes and that the anglers' share is far too small. At present, there are more than 200,000 sea anglers in British Columbia, plus an unknown number of river salmon fishermen. Other catch estimates which I have seen do not even mention already established river salmon fisheries and forecast small catches by salt water sport fishermen.

Several months ago, the Fisheries and Marine Service announced that sport catches for Georgia Strait were underestimated by 50 - 65%. Now the Fisheries people are stating that anglers in Georgia and Juan de Fuca Straits are catching 800,000 - 1,000,000 plus cohoes and chinooks per year instead of the 1972-1976 combined species average of an estimated 380,000. Assuming that this statement is true, an additional 155,000 salmon does not represent a doubling of sportfishermen's catches and, obviously, the amount is insufficient. Due to SEP publicity, a greatly augmented number of anglers will absorb this increased catch all too quickly.

I still believe the Salmonid Enhancement Program can and will produce vastly increased runs of salmonids which will benefit anglers, commercial fishermen and the Native People. However, I also believe that the topics which I have discussed should be applied to SEP. Without changes in production methods, enforcement, commercial fishing methods and angling regulations, SEP may fall short of its desired benefits for all Canadians.



PRINCE RUPERT

January 9, 1979

Salmonid Enhancement Brief

Alot of what I have to say has been mentioned before but I will present this anyway to further instate the necessity of these items in future projects.

I am a representative from the Senior Secondary Marine Science class. In the past year the government supplied our project with a grant of \$5,000 to install rearing boxes over the summer. After this was installed, fry were caught and left in the rearing boxes to mature and be tagged for Fisheries records. In all, a total of 1700 fry were returned into the creek.

We feel that if there had been or could be in the future, (1) more professional help to work with the students; (2) on-the-job training for the students; (3) more available information concerning this topic, that this could have or could be even more successful.

Having professional help ensures that the fish population in these areas is not destroyed completely by errors that could have been avoided by a more learned person. The grants could have been used to employ only professional help but we feel it would be better to work with students as far as branching out the knowledge now and in later years is concerned. For example, the High School students could in turn go to the Elementary schools and show them at their most influential age the importance of taking care of the creeks.

While doing other projects related to enhancement programs, we found that there was limited access to information on this topic. That is why there should be more information in the form of pamphlets, posters, instruction booklets, or books, available to the general public.

We experienced a certain amount of vandalism to our project over the duration, this brings about the need for greater emphasis to the general public on why we should keep the creeks clean and open. This also brings a need for better enforcement of some type in this area.

If there was also more recognition placed on the success or attempts of these projects, we feel that there could be a great deal more people involved in this type of idea.

Although, it is a difficult subject to solve at this time, as everyone knows, we need to control the amount of pollutants going into the creek. One suggestion might be the filtering of the creeks. Also, we feel that there is a need for a sewage bypass for the Civic Centre Ice Arean. The Civic Centre cleans the ice and puts a small amount of the waste in the creek. Since the waste contains amonia, although it may be a small amount, it is enough to damage the fish population. This is also a factor because their cleaning time seems to be right around the time that the fish are spawning.

One last point, is the need for a combining of the governing body for sport and commercial fishing. We feel that in relation to sport fishing, there should definitely be mandatory fishing licences.

By

Charles Menzies

A BRIEF TO  
SALMONID ENHANCEMENT UPDATE '78

PRESENTED BY:  
PRINCE RUPERT FISH DEALERS



## OPENING REMARKS

The views of the Prince Rupert Fish Exchange remain unchanged from those expressed in a brief presented to Salmonid Enhancement two years ago. While we support the concept of Salmonid Enhancement we must say that in the Northern Areas 1 to 6, enhancement is somewhat conspicuous by its absence and we are still asking when is it going to happen. We also express our disappointment that none of the recommendations presented in our last brief have been considered or even acknowledged and that the only consultation with this group has been with regards to the method of payback for Salmonid Enhancement.

### Area 6

We realize that the chinook population in the Kitimat system has declined to a dangerously low level and that the Enhancement of other systems in the area cannot begin until these stocks are increased to a level where they can stand the increased incidental catch that could be expected while fishing other enhanced stocks. Considering that the inside waters of Area 6 have been closed to commercial fishing for twelve years and that the sports fishing has increased dramatically during this period, it is difficult to conclude that the proposed hatchery on the Kitimat River is not of the highest priority.

It is our understanding that because of water problems the construction of this hatchery has been postponed at least one year by Salmonid Enhancement and completion prolonged two years by Treasury Board cutbacks. We would like to suggest as an interim solution that the present temporary facility located at the Eurocan site be enlarged to the point where it is capable of releasing one-half to three quarters of a million smolts each year until a suitable location is found and a permanent hatchery is completed. If this is done perhaps some enhancement on other species and other systems can begin earlier than it appears possible at the present time.

### Area 5

Area 5 is recognized as a difficult area to enhance. With the exception of lake fertilization, probably the only method available is stream clearance and restoration. It is our hope that the lake fertilization programme in Area 5 will proceed without delay. While results of this programme will have some impact on the commercial fishery, it will probably make a greater contribution to the Indian food fishery and may help reduce the food fishing pressures on the Skeena system.

### Area 4

Area 4 had received some attention this past year by way of stream clearance, etc. While we agree these projects are worthwhile and necessary we would prefer to see a programme of preventative maintenance whereby some party or department was charged with the responsibility of maintaining our fish habitat as opposed to restoration after the natural reproductive cycle has been impaired. The only major project that we are aware of on the Skeena system in the near future is a hatchery for chinook and coho on the Kalum River which we understand is meeting with some opposition from those representing the recreational and sports fishery. As the situation on the Skeena is much the same as that on the Kitimat system with regards to chinook and the enhancement of other species, it is our hope that the construction of this facility will proceed as soon as possible.

### Area 3

The problem created by the foreign interception of fish from the Nass and other systems of Area 3 do not appear much closer to being solved now than they were two years ago. While the policy of no enhancement of stocks that may be intercepted by foreign fishermen is generally accepted, we wonder if the situation in Area 3 does not deserve a second look. It appears that enhancement in South East Alaska will probably come on line before an interception agreement between the two countries can be ratified. If this happens there will be an increased fishing effort in South East Alaska that will presumably result in increased incidental catches of unenhanced stocks returning to Area 3 streams. This situation would lead to further fishing restrictions, lowering the fishermen's earnings and possibly a decrease in employment for workers in other areas of the industry.

### Areas 1 & 2

Areas 1 and 2 encompass the entire Queen Charlotte Islands and represent enormous potential. So far there is only one meaningful enhancement project underway and that is on Pallant Creek in Cumshewa Inlet. This installation will undoubtedly make a considerable contribution to the fishing industry. However, it is a long ways away and full production cannot be expected until facilities are completed on Mather's Creek and accurate information on local stock interception is made available.

An immediate problem on the Queen Charlotte Islands involves the forest industry. Because of a combination of physical factors such as steep slopes, heavy rainfall, etc., certain areas are naturally unstable and methods such as "clear-cut" logging are simply not acceptable. In Rennell Sound for example, the three major producing streams may be saved if logging were to cease right now. It is however, doubtful that it will happen so we suggest that 1978 may have seen the last significant returns to this area.

Several major spawning streams on Lyell Island and Louise Island face the same fate if the method of "clear-cut" logging is not reconsidered. Detailed surveys and analysis of some areas on the Queen Charlotte Islands have been made and are available from the Ministry of the Environment and areas of concern are well known by the Department of Forestry and the forest industry. Permission to log these areas have been given to the forest industry and logging has proceeded with full knowledge of the consequences.

It is difficult to understand how the provincial government can commit itself to a programme of Salmonid Enhancement on the one hand and sacrifice fish habitat on the other.

In summary, our major concerns are the lack of enhancement projects in the north. While we are aware there have been many small streams clearance and restoration projects and some small incubation boxes installed by various interested groups and individuals and we acknowledge the necessity and importance of these projects. We also know that if the goal of doubling the returns of salmonids is to be attained through the Salmonid Enhancement Programme we are going to have to see alot more activity in the area of larger facilities and we cannot understand why these types of projects are going ahead in the south and not in the north.

We are very concerned with the consequences of certain logging operations on the Queen Charlotte Islands and as the Department of Fisheries seems powerless, we appeal to the provincial government to investigate the situation. With proper management, fishing and forestry can be compatible, however, we cannot accept the concept of sacrificing one industry for the other.

When your group passed through Prince Rupert two years ago the Prince Rupert Fish Dealers made a presentation to you. A short time later a follow-up paper was sent to Mr. Sinclair. As we did not receive any acknowledgement of receipt of that paper we have attached a copy to this brief.

In closing, we would like to express our enthusiasm for salmonid enhancement and hopes that some of our suggestions will be considered.

The major salmon stocks on the Queen Charlotte Islands are pink and chum, although several rivers produce fairly substantial runs of coho and at least two systems produce steelhead and chinook. While pink returns have been erratic, with the occasional large return, the general trend has been a drastic decline. Chum stocks on the other hand have shown a steady and drastic decline in recent years despite intense conservation.

We feel that the rehabilitation of these stocks should be at the top of the priority list of the Salmonid Enhancement Programme and list below some of the reasons why:

1. Generally speaking, the salmon stocks on the entire Queen Charlotte Islands, are in a drastically depleted state.
2. The timing of these runs would serve to lengthen the fishing season.
3. The area is capable of accommodating a large fishing fleet, thereby lessening the intense pressure on other fishing areas.
4. The accessibility of enhancement sites.
5. The availability of Indian labour.
6. The relative ease of managing the enhanced stock.
7. The probability that the interception of these stocks is minimal.

On the following pages, we will briefly examine the Queen Charlotte Islands by area pointing out the locations and priorities of Salmonid Enhancement as seen by the Prince Rupert Fish Dealers.

#### Area 1

The salmon production in Area 1 comes mainly from two locations, the streams of Naden Harbour and Masset Inlet. Both areas are productive in pinks and chums.

The biggest single producer of pinks on the Queen Charlottes is the Yakoun River located in Masset Inlet. In 1966, this system produced a commercial catch of one million pinks and received an escapement of 800,000. In 1968, it produced a catch of 600,000 and an escapement of 750,000. By 1970, this run had diminished to an escapement of 250,000 and only a token catch. Although there has not been a fishery on these stocks since 1970, the largest escapement since that year was recorded in 1976 when only 150,000 spawners returned.

It is our opinion that a spawning channel should be built or incubation boxes installed for the enhancement and stabilization of pink salmon in this system. This river also produces significant numbers of coho, steelhead and chinook and we propose that a hatchery for the production of coho be built in conjunction with the spawning channel/incubation boxes project and that early consideration be given to the production of steelhead and chinook.

As the Yakoun River has an early sockeye run which makes a major contribution to the Indian food fishery, we feel that a study should proceed immediately to ascertain the possibility of a lake feeding program to enhance this stock.

Most areas of this river are accessible by existing roads and can be reached easily from the village of Masset, where there is an Indian reservation from which a labour force can be drawn to provide 100% Indian participation. As these stocks migrate into Masset Inlet, the enhanced stocks can be managed with considerable ease.

For the remainder of Area 1, we feel that a program of general stream upgrading and restoration should be implemented as soon as possible for the enhancement of both pinks and chums.

Because the timing of chum returns to the Athlow-Oirt Louis systems are the same as Naden Harbour and Masset Inlet, we propose that any enhancement of chum stocks in one area be carried out in the two remaining areas as well.

### Area 2 West

Area 2 West has suffered a tremendous decline in chum salmon stocks in recent years and stringent conservation measures have not been effective in halting this decline. The same trend is apparent in the returns of pink salmon to West Coast streams.

The West Coast may be broken down into three major producing areas; Tasu - chums, Engelfield Bay - pinks and chums and Rennel Sound - pinks and chums.

### Tasu

The stocks returning to Tasu Sound are produced mainly from three streams; Tasu, Botany and Fairfax, which have enjoyed escapements of up to 17,000, 50,000 and 15,000 respectively. Two of these streams have suffered very little environmental damage, however, Tasu Creek has been decimated by logging.

Because Tasu Sound is capable of accommodating a large fleet of boats and the ease of which the enhanced stocks could be managed, we propose that production hatcheries of the type so successful in Japan, be proceeded with immediately for the enhancement of chum salmon.

We feel that the enhancement of stocks in Tasu would have a very favourable impact on the fishing in Area 3 and also in Area 2 East by drawing a considerable part of the fishing effort away from areas of depressed stocks and areas where parallel enhancement is difficult because the stocks are returning to several smaller streams.

The major concerns with the remaining areas of 2 West is logging - Security Inlet in Engelfield Bay which has been a good producer of pinks and chums presently has its watershed under the eyes of the forest industry. If logging of this area is permitted, the streams of Boomchain Bay and Kaisun will also be effected. Other areas of Engelfield Bay have been logged causing heavy damage to spawning beds and if restoration is not carried out immediately, the stocks returning to these streams will be completely annihilated.

The streams of Rennel Sound are not quite as fortunate as logging operations are presently being carried out in the watersheds of the three major pink producers of this area and undesirable logging practices are already evident.

The two big chum producers of the area, Seal Inlet and Sartu Inlet are environmentally very fragile. It is known that the forest industry wants to log these areas but we believe it should not be allowed without a meaningful guarantee that the productivity of these streams will not be impaired as result of logging practices or activities - ie. the posting of a substantial bond or an advance contribution to Salmonid Enhancement.

It may be pointed out that the timber being cut in Rennel Sound is exported in its raw state, therefore, produces no secondary industry to Canadians.

#### Area 2 East

Most streams on the East Coast of the Queen Charlotte Islands seem to be capable of accommodating large returns, however, they are extremely fragile and often produce failures from what are considered successful seedings. In most cases, these streams are situated in logged-off areas.

The Deena River in Skidegate Inlet is one of the largest on the East Coast and has recorded escapements of 200,000 pinks, 100,000 chums and 15,000 coho. It is our opinion that a spawning channel and hatchery for the enhancement of all three species should be proceeded with immediately.

This system is easily accessible from Queen Charlotte City and the Indian village of Skidegate and, as in many other areas, is an ideal situation whereby the entire project from construction to protection can be 100% Indian.

Other areas on the East Coast require immediate attention with regard to stream maintenance and upgrading and an orderly planned project should be scheduled at the earliest possible date and work should commence without any further delay.

We support the project presently planned in Cumshewa but must confess we are dismayed with the speed at which it is proceeding. It is our opinion that this project could have and should have been completed and stocked for the enhancement of chum salmon in 1976.

When considering enhancement of salmon stocks in the Nass River system probably the first thing that comes to mind is the problem of "Interception of Canadian Fish by American Fishermen".

It would appear we are a long way from reaching an agreement to abstain from intercepting each others fish. This situation is not improved by the fact that we cannot even agree where our International boundaries lie.

In Alaska, we know there is an enhancement program considerably larger than the one proposed for British Columbia. The effect of this project will be an increased effort by American fishermen to catch American fish, which will undoubtedly result in an increased catch of Canadian fish in American waters. The results will be even less fish reaching areas where they can be exploited by Canadian fishermen and ultimately the streams of their origin. With the decline of our natural stocks, we feel very strongly that we cannot allow the fact that some of our fish is being intercepted by American fishermen to stand in the way of the rehabilitation of salmon stocks in the Nass River system.

There is no shortage of streams and rivers with excellent potential for salmon enhancement in this area. Areas, that in the past have been looked at, studied and recognized as having good potential and finally shelved for lack of whatever. There are sites where the entire project could be constructed, stocked, maintained and protected, by local Native Indians who are presently unemployed and actually living on the site.

The late run pink and chum runs have been selected for enhancement because coupled with the fact they are in a severely depleted state, the exploitation of these stocks would not conflict with the much troubled Meziadin and Bowser sockeye runs which occur early in the season.

The sites selected because they have the necessary qualities required to support incubation boxes and/or spawning channels - ie. good gravel, good water accessibility, etc.

Construction and maintenance costs would be minimal at all but the Quinamass and Khutzemateen site, however, because Quinamass and Khutzemateen are presently the largest producers (although they do not provide a fishery) and have the greatest potential, it is felt that spawning channels are the only method of achieving full production of these systems.

In order to properly exploit pink and chum salmon stocks in this area it is necessary that the proposed sites be developed jointly.

Attached is a list of the sites, species and method of enhancement along with the reason that each site would be a viable project.

As far as priorities are concerned, it is our opinion that the enhancement of Quinamass and Khutzemateen would have the largest impact and produce the greatest returns and should jointly, be number one on the list. However, because of the availability of Indian labour in the village of Kincolith, we see no reason that the money presently being spent on U.I.C. and Welfare could not be used to install incubation boxes and perhaps build a small hatchery on the Kincolith River.

#### Quinamass - Khutzemateen

Species - pinks, chums, coho and chinook.  
Method - spawning channel and incubation boxes.

The Quinamass River along with the Khutzemateen system produce the bulk of the pink salmon in Area 3. It has significant returns of chums, coho and chinook as well. The river is some 13 miles in length and has many miles of good spawning gravel. Despite the capabilities of this river, it never seems to reach more than 50% of its potential. It has good gravel available as well as adequate water to support a spawning channel/incubation boxes.

#### Khutzemateen

Species - pinks, chums, coho and chinook.  
Method - spawning channel and incubation boxes.

This river compares favourably with the Quinamass River in size and potential although there may be some difficulty in obtaining water supply to safely support a

water channel. Traditionally this river does not produce as many pinks as the Quinamass but it does produce considerably more chums.

Both systems produce a significant number of "off-year" pinks and could conceivably be enhanced to the point of producing a full-scale fishery every year.

The timing and production of these two systems is approximately the same, therefore it is imperative that they be enhanced jointly.

### Kincolith River

Species - pinks, chums and coho.

Method - incubation boxes and possibly small hatchery.

This river is situated on the Kincolith Indian Reserve. It has returns of pinks, chums and coho. It is a fairly large river of considerable length and is accessible to some species for almost its entire length. It has a good stable water source and has lots of good gravel for spawning. This river has never been an extremely large producer but has recorded pink returns up to 25,000 and they seem to return equally as well in "off-years" as in "even years". The timing of this river is approximately the same as the Quinamass and the Khutzemateen.

Because this river is situated on an Indian Reserve and actually flows through the village of Kincolith, there is the opportunity for 100% Indian participation from construction to protection, and could be a valuable tool in the local education system.

### Chambers Creek - Iceberg Bay

Species - pinks and chinook.

Method - incubation boxes.

This is an excellent stream which simply does not produce fish the way it should, although it has accommodated up to 15,000 pinks, it is felt that with incubation boxes it could become a much better and consistent producer. It also is frequented by some chinook with as many as 800 being recorded (1967).

Because this stream is close to the village of Kincolith, it also offers 100% Indian participation.

### Bear River

Species - chums.

Method - incubation boxes.

This is a large river with capabilities of producing large numbers of chum salmon. It has good water supply and good gravel. The town of Stewart is situated at the mouth of this river so personnel accommodation, etc. is not a problem. This is an excellent opportunity for salmon enhancement to enjoy the benefits possible by making the project accessible to the education system of Stewart.



Because of enhancement planned by Alaskans on other streams in the Portland Canal area, it is most important that this river be placed high on our priority list in order to prevent the decimation of these stocks by Alaskan fishermen.

Because of the success of "feeding" in the Great Central Lake system of Vancouver Island, we would like an immediate study of Meziadin and Bowser Lakes to determine whether or not any benefits could be derived from implementing the same type of program in these systems and if so, should be introduced immediately.

#### Area 4

We would like to suggest that the Skeena River Advisory Board become more active in dealing with future enhancement in Area 4 and also in dealing with the problems that presently plague salmon enhancement on the Skeena River.

#### Area 5

This area has no large salmon producing systems but rather is made up of 53 smaller salmon producing streams. We realize the implications of attempting to enhance an area such as this. However, we feel that a great deal can be done to increase productivity by stream maintenance and the restoration of spawning beds.

Of the 53 streams in Area 5, 14 produce early run sockeye. We feel that a study should be done to determine whether or not a lake feeding program would enhance these stocks. If it could be done, we think it would serve to lessen the pressure of the Indian food fishery on the Skeena River.

#### Area 6

Area 6 has, in recent years, suffered natural disasters which have reduced the salmon stocks to such a level as to prohibit a commercial fishery.

The late run chums are depleted to the point where if something is not done now there will be nothing left to enhance. The later run pinks returning to the outside streams of Area 6 appear to have recovered with the help of extensive conservation measures. However, the early pinks returning to interior rivers such as Kitimat and Camano systems, do not appear capable of restoring themselves and obviously require enhancing.

The importance of pink and chum stocks to this area cannot be overemphasized. Not only has the depletion of these stocks seriously curtailed the earnings of local fishermen, especially local Indian fishermen, but because extensive closures were necessary to save what stocks remained, a large fleet of boats were forced to move to other areas which could ill afford the pressure of an increased fishing fleet.

For these reasons, the Prince Rupert Fish Dealers feel that spawning channels should be built on the Kitimat and Camano Rivers for the enhancement of pink and chum salmon and that these projects be started immediately and that their early completion be regarded as urgent.

We support the proposed hatchery for chinook on the Kitimat River and urge its early completion.

### Summary

With regards to priorities, we feel that the following sites are all classed as urgent:

TASU  
SKIDEGATE  
YAKOUN  
QUINAMASS  
KHUTZEMATEEN  
KITIMAT  
CAMANO

Each and every one of these represent an area where no commercial fishery can be allowed because of depleted stocks or, the fishery on a viable stock has to be curtailed because of incidental catch of depleted stocks.

The methods proposed to enhance these stocks have all been proven successful in other areas of British Columbia and we do not feel that lengthy plans and studies are required in order to begin these projects now.

The subject of stream clearance and restoration is important. In many cases, it is probably the only method of enhancement available. We feel that a program whereby regular inspection and maintenance of all streams is necessary and that this should take place each summer before fish enter the streams to spawn. We suggest a summer works program for University students could work very well.

The question of "Interception of Canadian Fish by Foreign Fishermen" has to be viewed in a different light since the announcement of an enhancement program by Alaska. At this point, there is no indication of an International Agreement to protect these fish and it is obvious that we must enhance just to stay alive.

The only criticism we have of Salmonid Enhancement is that industry has not been involved to the degree that we want to be. We have many people, rich in knowledge acquired through experience who we think should be making their contribution to Salmonid Enhancement.

That is why in the case of Area 4, we suggest that the Skeena River Advisory Board become involved immediately with any enhancement of that area. Further, we would suggest that any further Advisory Boards of Management Committees should have an active part in Salmonid Enhancement.

In closing, we would say that we view the Salmonid Enhancement Program with much enthusiasm and we thank you for allowing us to present our views.

PRINCE RUPERT FISH DEALERS

# SKEENA-QUEEN CHARLOTTE REGIONAL DISTRICT

## INDUSTRIAL DEVELOPMENT COMMISSION

SUITE 2      342-3rd AVENUE WEST      PRINCE RUPERT, B.C.      V8J 1L5  
TELEPHONE 624-3106

January 9, 1979

Jamie Boland  
Salmonid Enhancement Program  
Department of Fisheries  
and Environment  
Pacific Region  
1090 West Pender Street  
Vancouver, B. C. V6E 2P1

Dear Mr. Boland:

The Skeena-Queen Charlotte Industrial Development Commission has reviewed the Prince Rupert Fish Exchange's brief to the January 9, 1979 Prince Rupert Salmonid Enhancement public meeting and is in full agreement with statements contained in the brief.

Sharing the Exchange's concern over lack of Enhancement projects in this region, the Commission supports the call for a greater emphasis on North Coast and Queen Charlotte Islands enhancement.

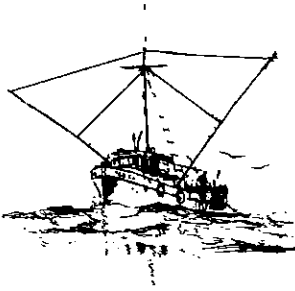
The Skeena-Queen Charlotte Industrial Development Commission, therefore, urges the Salmonid Enhancement Program management to consider the suggestions put forward by the Prince Rupert Fish Exchange to undertake expansion of Enhancement work on the North Coast.

Sincerely,



Reed Clarke  
Co-ordinator  
Industrial Development  
Commission

RC/mf



# Northern Trollers Association

Box 713  
Prince Rupert, B.C. V8J 3S1

January 9th, 1979

Update '78  
Salmonid Enhancement Program  
Fisheries & Oceans Canada  
1090 West Pender Street  
Vancouver, B. C. V6E 2P1

We are a newly formed organization formed a year ago. Therefore we missed the first salmonid enhancement public meeting.

If you look at the last ten years for coho and chinook salmon production, you could see that both species are decreasing badly in especially Northern Areas. But nothing has been done for the Northern Areas.

1. First of all we are strongly against distributions of your salmon enhancement for Sockeye 50%, Chum 35%, Pink 10%, Coho 3%, Chinook 2%.

Because when fishing is concentrated on the three major species. I am afraid that the two weaker species will not survive. Because only a 5% enhancement will not maintain the balance let alone increase the species. Therefore we would like to see a higher percentage distribution for coho and chinook salmon.

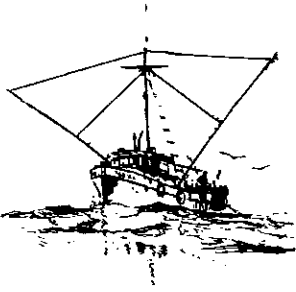
2. More financial and manpower input into stream clearance and hatchery program in small rivers have a positive effect on the salmon runs.

A. Salmonid enhancement co-ordinator for Skeena Area and Queen Charlotte Islands. Full time resident technician, full time biologist in both Areas.

B. The most important one is stream and river protection from improper logging.

## example

Mr. Jim Hart, Habitat Protection Officer (Liason Logging - Forestry) from Queen Charlotte City was at the advisory meeting in Masset, Nov. 21, '78, with a slide show of washout, slides, etc. caused by logging.



# Northern Trollers Association

Box 713  
Prince Rupert, B.C. V8J 3S1

He also stated there was a Provincial Government study done for the Charlotte, and in many areas logging and fishery could not co-exist.

- C. We are looking forward to the proposed portable chinook salmon hatchery or spawning boxes on the Yakoun River.

We would like to see a permanent counting fence for Yakoun River constructed so it would not wash out.

- D. We like to get involved in Local Creek Project especially for Kloiya Creek which used to be a good chinook salmon producer.

We think it would be an ideal location for it. As it has easy access and would be in the public eye.

An increase in enhancement effort for Central Coast and Northern Areas including Queen Charlotte Islands has greater chance for stocks to increase due to less human and industrial pressure.

### 3. For sport fishing

We would like to see a punch card system for tidal and non tidal water. So that a realistic catch figure can be obtained.

Frank Kanno  
President

KITIMAT



TERRACE



BRIEF TO SALMONID UPDATE '78

BY

THE NORTHWEST CHAPTER OF THE  
STEELHEAD SOCIETY OF B.C.

4120 Highway 16 East  
Terrace, B.C.  
V8G 3N3

## SUMMARY ASSESSMENT OF SEP TO DATE

Our overview of SEP is admittedly a harsh one because the 15 year proposed enhancement program fails badly in meeting the needs of the sports fishing public. The great threat of this program is to bio-engineer every watershed in the province to produce maximum financial returns to the commercial fishing industry. There is precious little evidence to show that Federal Fisheries is concerned enough about declines in our steelhead to veer away from Babine and Great Central Lake type sockeye projects -- projects which have devastated the once great summer steelhead runs of the Skeena and Stamp/Ash systems. Even now, plans are underway to compound this past folly by enhancing the Morrison River sockeye stocks of the Babine system; a development which could very well seal the fate of our magnificent Copper, Kispiox, Babine and Morice summer steelhead. It is no secret that this sockeye gillnet interception is exploiting our steelhead at a rate which has already placed these runs in serious jeopardy. Yet Federal Fisheries continues to forge ahead as if we have learned nothing from this ten year "enhancement" fiasco.

In spite of this sad past experience, SEP wishes the general public to support a new 15 year program which threatens to spread this mismanagement to nearly every river system in the province. A close look at the Babine project as a prototype of future enhancement projects provides some insight into the problems we can expect:

1. Millions of tax dollars continue to subsidize the Babine project during a time of sorely needed fiscal restraint.
2. Returns of adult sockeye approaching two million fish fall far short of the returns needed to justify increased investment in the project.
3. Biological constraints have placed a ceiling on returns in spite of an enormous investment in increasing sockeye fry production to the 200 million level.
4. The artificially high ocean harvest of enhanced sockeye has eroded wild sockeye and other salmonid stocks causing an increasing dependancy on man-made spawning channels.
5. The Babine project has decimated our invaluable summer steelhead stocks through increased commercial fishing pressure at the Skeena River mouth.

The Auditor General of Canada has outlined several criteria for identifying wasteful expenditures such as those SEP proposes for large hatcheries. Firstly, SEP has failed to convince us that many of these projects are necessary to achieve long-term enhancement goals. Secondly, SEP has failed to evaluate these alternative methods of enhancement of natural fish stocks in its hurry to get "fish back quick to the canneries" philosophy. Thirdly, as we have already pointed out, the entire SEP program as proposed will fail to meet the ostensible purpose of enhancing commercial stocks without destroying sport fishing and our tourist industry in the bargain.

The actions of SEP to date have ignored public opinions as expressed in the first round of public meetings. A massive preponderance of speakers asked for environmental protection and conservation to protect and enhance wild fish, yet the government has defied this opinion and continues to give marginal funding to Habitat Protection while SEP grows on fat budgets, giving away \$42,000 for such irrelevant purposes as "Development and Operation of a Viable Native Indian Social and Economic Development Program" and "Co-ordination and Preparation of Photostories on the Salmonid Enhancement Program", (\$23,400).

SEP has proceeded with an obviously pre-conceived plan for massive hatchery projects for the primary benefit of the commercial fishing lobby. Thus, the actions of SEP speak more loudly than its propaganda espousing public input into the planning process. These public hearings, therefore, must be seen as an elaborate public relations gimmick to garner support for the give-away of hundreds of millions of tax dollars primarily for the benefit of the commercial fishing industry. We are especially concerned about increasing foreign control of this industry with at least 40% of West Coast processing Japanese owned.

Our position is that the entire 15 year program must undergo major surgery. The top-heavy first phase emphasis on commercial species must stop; expensive hatchery projects must be cancelled; stocks of all species must be rebuilt using strong conservation measures in conjunction with environmental protection and upgrading. SEP itself as a separate arm of the Department of Fisheries and Oceans should be abolished and merged with Federal Fisheries per se. This action would halt the expensive and unnecessary duplication of bureaucracy that is now producing a jungle of paperwork and propaganda unmatched by the production of fish. Moreover, the scope of the entire program should be reduced while a lengthened and more flexible timetable is introduced. Introduction of these crucial changes will reverse loss of public confidence in SEP and lay the groundwork for a true reconnaissance of our salmonid resource.

## Opposition to Large Hatchery Projects

SEP should emphasize environmental protection and strict conservation measures rather than wasteful megabuck hatchery projects such as those planned for the Kitimat and Kalum Rivers. In a time of universally recognized financial restraint SEP continues to forge ahead with plans for million dollar monuments to the cement industry. Not one cent of taxpayers money should go for hatcheries conceived and designed under pressure from the commercial fishing lobby for quick financial returns. We fail to see why the general taxpaying public should increase its subsidy to the commercial fishing industry. Therefore, hatcheries designed to produce commercial fish should be paid out of profits accruing to that industry.

There is good evidence to show that patience and strict conservation will yield high salmon returns in the foreseeable future. In Alaska, Susitna River chinook runs hit a low of 17,000 fish in the early 1960's, followed by a rebound to 130,000 fish in 1977 after near total closure of the Cook Inlet sports and commercial fishing. In our own area, there has been a phenomenal increase in feeder springs in the Douglas Channel in response to closure of the inside commercial fishery in Area 6 along with restriction on freshwater catches in the Kitimat River. Therefore, there is every reason to believe that we will soon see a progressive increase in escapement of spawning fish to the Kalum and Kitimat Rivers.

In our view, a lower Kalum River master hatchery at the Deep Creek site would do incalculable damage to the local tourist industry. It is no secret that most tourists come to this area for the purpose of taking giant spring salmon indigenous to the Kalum system. Currently the run of wild chinook to the Kalum system lasts from May through August -- creating a tourist fishery attraction spread out over a four month period. On the other hand, human genetic selection at a hatchery will tend to create a very brief fishery with a large number of fish returning over a short period of time. The net effect would be to greatly diminish the time period for tourists to fish and spend "golden dollars" in our area. In addition, hatchery-bred fish which are less desirable as a tourist attraction.

## Condemn Industrial Encroachment on Lakelse System

We condemn the concept of the proposed Terrace/Kitimat Industrial Corridor as a direct threat to the Lakelse system which produces a major share of the valuable Skeena pink salmon runs and supports a vigorous sports fishery for coho, steelhead and cutthroat trout. This so called "plan" produced by the Kitimat Stikine Regional District would see an industrial slum with a copper smelter and sulfuric acid plant located in the proposed Dubois industrial site—a site adjacent to the immensely productive Angeles Creek and Clearwater Creek drainages which feed Lakelse Lake. Habitat Protection Officers should veto this proposal as incompatible with the current and potential value of the Lakelse and Kitimat River systems as salmonid producing resources.

A review of the brochure promoted by the Kitimat Stikine Regional District reveals a proposed oil pipeline along the route of the defunct and discredited K.P.L. scheme. This oil port and pipeline proposal has been denounced on environmental and economic grounds by two high level government commissions. This project is clearly not in the public interest nor is it consistent with maintenance and enhancement of our salmonid resource.

Our view is that establishment of a sockeye spawning channel in Williams Creek could have a triple positive effect. Firstly, such a project would be a highly visible attraction to tourists and could be dovetailed with the provincial parks naturalist program. Secondly, a run of 30,000 to 50,000 sockeye were maintained by earlier hatcheries with a similar potential expected from a new program. Furthermore, this run would be an early run of valuable commercial fish not in conflict with steelhead runs. Our third reason for supporting a Williams Creek spawning channel is a belief that sockeye fry form a key element in the Lakelse Lake food chain for juvenile steelhead and cutthroat. These species, therefore, could be greatly enhanced as a positive sportfishing side effect of a commercially oriented project.

### Jesse Lake Sockeye Project

We would like to see a strong sockeye run established in the Jesse Lake system. Landlocked sockeye or kokanee are already abundant there, indicating that a sockeye run has existed during past higher sea levels. Perfect spawning gravel and rearing conditions exists in the upper lake.

In order to establish a run there, it will be necessary to establish fishways up the right hand side of the falls. It appears that this could easily be done by blasting, creating natural appearing pools without hurting the beauty or flow over the main falls. Transplantation of brood stock and lake fertilization will be necessary, at least initially. Other species of salmon and steelhead could be easily introduced later, once the initial fishway problems have been overcome.

We believe that the Jesse Lake project should be part of an overall SEP strategy to de-emphasize the late Babine sockeye fishery in order to relieve pressure on steelhead and coho stocks. This would be an alternate sockeye fishery for the commercial fishing industry. Excess sockeye escapement to Babine Lake could be harvested upriver by a freshwater commercial fishery.

### Continue and Expand Research

We would like to see an accurate stream catalogue for all of the lower Skeena tributaries, including non-commercial species. Without an accurate data base proper management of the resource will be impossible. The extent of steelhead runs into the tributaries below the Lakelse drainage is a virtual unknown. It is difficult to see how we can protect and enhance this resource without some baseline knowledge for planning purposes.

Specific priorities for data gathering should include the pristine Gitnadoix Valley which is very high on the provincial list for a class "A" park status. Knowledge of exactly what fish populations we are trying to protect with a "wild river" designation could help strengthen the case for a park reserve.

Radio tag experiments must continue on a widespread basis until we know precisely when and where steelhead pass through the lower Skeena sockeye net fishery. Such knowledge may enable Fisheries to impose an accurate netting closure to spare the steelhead runs without significantly hurting the valuable sockeye fishery. But until such accurate data is available we are asking for a two week netting closure around the end of July to allow steelhead stocks to recover.

Radio tagging of steelhead, springs, coho at the mouth of the Copper River is mandatory to determine the timing of upstream runs past the falls in the canyon. We strongly suspect that fish passage is greatly delayed or blocked entirely at certain high water flows. If it is proven that such an obstruction exists, blasting or fishways may be necessary to remove it.

Copper River system radio tagging is also necessary to determine where steelhead spawn in the system and where they overwinter. The extent of steelhead migration up the Clore River remains a complete mystery as cataracts in the Nimbus Mountain area may or may not prevent access to Burnie Lake. Again fishways and/or blasting obstruction may significantly help these runs. We will know the answers to these questions only if we get a systematic radio tag program started below the first canyon.

### Clean Up Kitimat Estuary

We wish to encourage Habitat Protection officials to press for application of critical environmental protection measures authorized by Fisheries Bill C-38. It will be wasteful to spend millions on expensive hatchery and other SEP projects planned for the Kitimat River while extensive log booming operations continue to degrade the Minette Bay salmonid rearing area -- an area identified in several studies as a key rearing area for juvenile salmon from the Kitimat River. Removal of these booming operations to a deep water site should be a prerequisite for any upstream salmonid enhancement investments. Therefore, we urge Federal Fisheries to immediately begin an accelerated program to end log booming in Minette Bay by 1980 as an integral part of the Salmonid Enhancement Program. In addition, booming operations should be restricted to log storage only with log bundling and sorting done exclusively on land.

It is time for SEP to come to grips with environmental degradation which has contributed to the decline of our salmonid species in the past and which threatens future upstream enhancements. Adequate legislation is now in place to deal with this problem; but it remains to be seen whether Fisheries officials will vigorously apply Bill C-38 in the public interest.



## Regulations and Conservation

In our view, Fisheries Service has been glacially slow in mandating needed restrictions on abusive sport fishing activity in certain watersheds. There has been a strong tendency to allow wholesale meat fishing to continue until stocks are endangered on specific rivers; the panic response to this has been complete closure of many of our finest fishing rivers such as the Kalum and Ischkenich chinook fishery.

We would rather see graded restrictions aimed at weeding out unsportsmanlike activity which threatens salmon runs. First and foremost is the abuse of treble hooks and roe on certain heavily over-fished rivers such as the Pseax and Cranberry Rivers. We need a roe ban and treble hook ban on these rivers before the need arises for a complete closure to all fishing. But in view of the parlous state of the chinook and coho runs in the entire Skeena and Nass systems a system-wide treble hook and roe ban should be strongly considered.

Season catch limits are far too generous with no limit for salmon and an unrealistic ten fish per river steelhead kill. We desperately need a fresh and saltwater salmon punch card system similar to those in Oregon and Washington States. A season limit is obviously unenforcable without such a punch card. Such a salmon punch card would also cut down on the illegal catch and can meat fishing practiced by many tourists. In summary, sports men must accept some reasonable regulations to help manage the resource before we lose our fishing privileges altogether.

## Native People and S.E.P.

We wish to emphasize our belief in scientific rather than political management of the fishery resource. An adequate data base is an obvious prerequisite of sound management which requires detailed enumeration of numbers and species of salmon and steelhead caught in the Native net fishery. Such data collection must be undertaken by qualified Fisheries personnel; tabulation by the Indian Bands involved in the freshwater fishery is unacceptable because of conflict of interest between the subsistence fishery and probable strict conservation measures that may arise from accurate tabulation of the catch. We believe that Native people must share the burden of tough catch limitations in order to rebuild depleted stocks.

No tax dollars should be spent on hatcheries that will primarily benefit one tribal group. An example of this is the recent Federal give-away of fishing rights to the Squamish Indian Band with the result that hatchery fish migrating through tribal land are intercepted exclusively by this one Band. A similar outcome here would see the Kalum Band as the primary benefactor from a Kalum River chinook hatchery; this scenario would be devastating to the local sport fishery and tourist industries which are key facets in our local economy.

Overall control of all salmon fisheries must remain with the Federal government; any delegation of controls or "self regulation" by individual Bands can only signal disaster for our remaining stocks of wild fish. Only Federal Fisheries personnel are in a position to ascertain the percentage interception of Skeena and Nass salmon by the U.S. fishery. Only Federal Fisheries can rationally balance and allocate the catch to ensure adequate escapements. Native peoples, sport fishermen, and the commercial fishing industry must all share in the restrictive regulations mandatory for recovery of the salmonid resource.

### Reforestation Should Be Accelerated

Included within the Steelhead Society's belief in environmental protection as a way of enhancing fish is the concept of accelerated reforestation. It is no secret that bad logging practices and large clear-cuts have contributed greatly to siltation and decreased salmonid populations in our area. This folly is compounded by the huge backlog of reforestation which sees our denuded hillsides continue to ooze tons of mud and silt because of inadequate forest restocking. In view of the fact that we spend about 1/100 as much on reforestation as similar U.S. jurisdictions, it would seem reasonable that SEP should act as a catalyst to speed replanting of key river drainages such as the Zymacord River, Cedar River and Coldwater Creek clear-cuts.

Roadbanks along U.S. Forest Service logging roads are reseeded with grass immediately upon construction. The same standards should be applied here, especially with the recognition that siltation is increased 25 times by clear-cut logging but that 80% of this salmon-smothering mud comes from logging roads alone. Again, funding and manpower provided by SEP should provide impetus for this crucial job - creating activity.

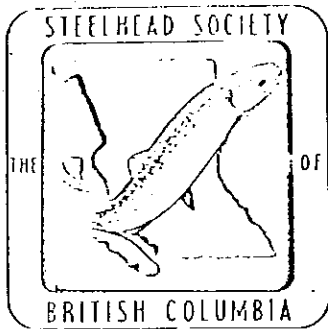
### Clumsy SEP Public Relations

We are dismayed at the lack of publicity for local SEP projects. For instance, the local SEP stream clearing crew has been a complete mystery.

SEP does not even contact resident local Fisheries Officers when arriving to undertake specific projects. We have had the experience of calling a local Fisheries official for information about a SEP project only to be informed that Fisheries was not informed that any such project was in the works.

Members of the Regional District of Kitimat - Stikine have asked us, "What is SEP doing with our tax money?" The implication being that SEP was doing nothing. Of course, this is not true, but from the local public's standpoint, SEP has been a big fat zero.

We believe that a public enlightened as to the research and natural enhancement activities of local SEP projects will be less likely to accept panic solutions such as high impact hatchery projects. There should be write-ups in the local paper, presentation on outdoor T.V. programs to outline this natural enhancement activity.



# THE STEELHEAD SOCIETY OF BRITISH COLUMBIA

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NORTHWEST CHAPTER

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(November 16, 1976)

## BRIEF TO THE SALMONID ENHANCEMENT PROGRAM - PUBLIC MEETING, TERRACE, B.C.

The Northwest Chapter of the Steelhead Society appreciates the opportunity to present its views on how the Fisheries Service and the Fish & Wildlife Branch, with help from the general public, should attempt to bring back the populations of salmon and trout in our region to the historic numbers of the past.

The manner in which we would like to make our presentation is by starting at the estuary of the Skeena River - working our way upriver, discussing the type of program we would like to see take place on the various tributary streams. As well, we would like to devote some of our time to a discussion on a number of points relative to Fisheries Management and enforcement. And we would like to make some mention of rivers running into Douglas Channel and the Nass River.

For a variety of reasons, the Skeena River estuary has escaped serious environmental damage and we can only hope that it will stay that way. Pulp mill effluent from the Kraft Mill on Watson Island is the only industrial pollutors within close proximity of the estuary. While the effluent from the mill is dispersed into Chatham Sound and is apparently not causing serious harm because of the strong tidal movement diluting the effluent with the saltwater - we are however, concerned with the construction of a second Kraft mill adjacent to the existing mill. We urge district surveillance and monitoring of the effluent from both mills in the future.

We can see no practical enhancement work in the estuary. The only recommendation that we can make is that industrial or urban development must not be allowed to gain a foothold adjacent to or within this valuable fishery area.

We are not too familiar with the lower tributaries of the Skeena, ie. the Prudhome Lake system, the Ecstall River, Johnson Creek, Khyex and Kloiya Rivers, other than to say that there appears to be no obvious environmental damage other than normal logging problems which can be overcome. We hesitate to recommend any particular program for these rivers, although we strongly suspect that artificial enhancement is not needed. Possibly the use of incubation boxes along with rearing ponds to supplement natural production could be a visible tool to be used to improve the chinook stocks. The clean-up of small streams, spawning areas and other improvements to the natural system is the direction which we urge to be taken in any enhancement work for the lower Skeena system.

Moving upriver, we want to next talk about the Kasika, Gitnaidoix, Exchamsiks and Zymacord - the latter three rivers have logging damage, particularly the Zymacord, where it is fairly severe. The Kasika and Gitnaidoix, on the other hand, are untouched by mans activities.

We know that these rivers are lightly exploited and for that reason see no need to enhance steelhead. What we do recommend is the collection of population data and pertinent information that is essential for good management of steelhead in these rivers.

As with all the Skeena tributaries, the chinook runs to these rivers are suffering. Once again, logging probably has had some effect on their decline. For this reason, the clean-up of known spawning and rearing areas and tributary creeks and improved logging practices would assist these fish.

We can make a specific recommendation for enhancement of these rivers, other than the possibility of using incubation boxes to supplement natural spawning, over four or five cycles, in those rivers which appear to have the greatest potential for improvement.

Coho appear to be holding their own, except in the Zymacord, which requires the cleaning up of a number of important small tributaries. We see no need for the artificial propagation for coho.

While not a sports fish, sockeye are the bread and butter fish for the commercial fisherman. If the sockeye populations to the Gitnaidox migrate earlier than the summer run (July) steelhead to the Skeena, we see some potential for the natural enhancement of these fish; we hesitate to recommend artificial propagation on this system because it may be classed as a wild river, the two things are not compatible.

Continuing along to the Kalum River, we see the need for some improvements, particularly help for the chinooks. A number of years ago, when logs were floated down the Kalum, many very important and productive side channels were blocked off to make the river a sluiceway or canal. We believe that by opening up these side channels and possibly using incubation boxes for a number of cycles to rebuild these runs; along with continued cutback in commercial openings, as well as the closure of the entire Kalum to chinook angling after July 31; and a reduction in daily and possession limits, would see the re-establishment of the large populations of the past.

As well, we would like to see the clean-up of spring and/or other small tributary creeks which would benefit all species in the system. We see no need for a major enhancement program for steelhead; they appear to be in good shape and require only sound Fisheries management and protection to maintain their present numbers. Since the largest runs of steelhead to the Kalum are late fall and winter fish, there is very little conflict with the commercial fishery.

Coho numbers are down, probably reflecting the destruction of many of the small tributaries by logging and urban development on the lower river, as well as the blockage of the side channels. Improvements in these areas could make a big difference in rehabilitating the coho runs.

The early run sockeye to the Kalum appear to have declined considerably from historic numbers. The reasons for this decline are probably related to damage to the spawning habitat by logging and overfishing by the commercial fishery. The

answer to rebuilding these runs may involve a combination of things; such as less commercial fishing time, habitat improvement and the possibility of a spawning channel or a hatchery, if habitat improvement does not fulfill the objective. We will not support enhancement of the sockeye on a large scale if their timing through the commercial fishery interferes with steelhead migrations.

We know of two specific environment problems on this river which should be rectified, the Terrace Municipal Dump on the lower Kalum, while not causing serious pollution to the river, is contributing to the decline of the water quality, while at the same time, it is a blight on the landscape. We urge it be phased out over the next year. Further upriver we recommend erosion control and the control of silt on Goat Creek be attempted at the earliest possible time. During heavy rains this stream pours out tremendous quantities of silt into the Kalum system.

Copper River - We do not want to see anything done with any of this species in this river, other than habitat improvement work, ie. cleaning up small tributary streams, as well, we would like to see an in-depth study take place on the feasibility of constructing a fishway at the 9 mile falls on the Copper.

We want to see very careful management of this river and enforcement of the Fisheries Act. The populations of steelhead and salmon are still relatively healthy in the Copper. By practicing good management and strong enforcement, we believe these runs can be increased in size to their historic numbers.

Lakelse River - The Lakelse River is still in reasonably good shape as a natural producer of salmon and trout. One major tributary, Coldwater Creek, has suffered extensive logging damage and for this reason, is in need of habitat improvement. The chinooks are in tough shape in the Lakelse, in part, the reason could be the deterioration of Coldwater Creek which in the past was a major spawning area for these fish. We support habitat improvement work in Coldwater Creek, as well as a supplemental incubation box; semi-natural rearing pond proposal to build up these runs. For steelhead we see no need for major enhancement work. Our choice is good management of the fishery with strong enforcement of the Fisheries Act.

For coho, we believe their environment is intact and in relatively good shape. We believe greater restrictions on the sports fishery is necessary to build up these runs, ie. reduction in the daily limit (to one fish); implementation of an annual limit (for salmon of 20 fish); banning the use of treble hooks on lures; and close the upper river spawning area to coho fishing (from the C.N. trestle to Lakelse Lake).

At this time, we do not support a hatchery system for coho on the Lakelse, because of its high quality capability of producing large numbers of these fish. We would want to see very comprehensive and accurate information compiled saying that the natural environment of the Lakelse can no longer sustain historic numbers of coho before we would support a hatchery proposal.

The sockeye runs now in the Lakelse are a far cry from the population of the 1930's, which probably were the result of the hatchery production for sockeye during that time. If it can be substantiated that Lakelse Lake cannot produce more sockeye naturally, and further can be shown that the peak of the 1930 sockeye runs arrived

at Lakelse Lake in the middle of June, which in turn does not interfere with the summer migration of steelhead in the Skeena River, then we would support the building of a sockeye hatchery or spawning channel.

We would like to see the hiring of a Guardian to patrol this river because of the extreme angling pressure it receives.

Kitimat River - Has had its lower section severely altered by floods which probably has something to do with large clear-cut logging areas in the watershed. It appears as though the chinook populations have declined in numbers because of these environmental changes and the overly generous daily and annual limits over a number of years. The possibility of incubation boxes and semi-natural rearing ponds used to build up the chinook runs, would have our support.

For steelhead, cutthroat and coho, we believe that they are holding their own - but would benefit if habitat improvement work took place on the smaller tributary streams. We see no need for artificial enhancement of these species.

On the larger upper Skeena tributaries, the Babine, Morice, Kispiox, Suskwa, Kitsequecla, Bear, Sustat and Bulkley - The Northwest Chapter of the Steelhead Society see the decline of the salmon and steelhead runs as the result of poor logging practices, much too liberal commercial openings, particularly for chinooks and sockeye; some abuse by the Indian net fishery; the primitive Indian gaff fishery on the Bulkley (should be replaced with dip nets); and at times, too liberal catch limits for sports fishermen.

We are of the opinion, as with the middle and lower tributaries, that the fishery habitat in the upper tributaries is in relatively good shape. We believe that habitat improvement is the best enhancement tool on these upper rivers where logging has caused problems, ie. highway culverts have blocked fish runs. Along with improved management of the fishery and enforcement of the Fisheries Act, most of the runs can be brought back.

With one exception, we believe that another attempt should be made to rehabilitate the Nanika River sockeye runs. This is an early run of sockeye which does not conflict with steelhead migrations and could be of great benefit to the commercial fishery if it were substantially increased in size.

The Steelhead Society is of the opinion that artificial propagation should not be used to replace the natural productive capacity of our northwest rivers, rather it should only be used when natural propagation has been destroyed by mans activities. We support the replacement of fish stocks to historic numbers, but we do not want to see the size of the runs increased to levels which creates unethical behaviour amongst sports fishermen; over-crowding of anglers on the rivers and conflict between the commercial and the sports stocks of fish. The fact is that it is impossible to satisfy the demand of either the sports fishery or the commercial fishery (this fact has been proven in Washington and Oregon with their hatchery program), therefore, we suggest that all concerned strive for a production level which meets the constraints we have mentioned.



We in the Steelhead Society conceive sports fishing as being a "quality recreation" and not as a means of filling the freezer or smokehouse.

To assist in promoting an understanding amongst fishermen and the general public for a better appreciation of quality angling, conservation of the resource, identification of species and Fisheries management, we urge the following:

- a) A knowledge test on Fisheries and conservation of those people obtaining their first fishing licence.
- b) Education included in the school curriculum on - conservation, Fisheries management and fish identification.
- c) Colour plates to be distributed to sports shops, etc., identifying the various species of salmon and trout in pre-spawning and spawning colours.
- d) Showing films, slides, displays, etc., at shopping centres, exhibitions, etc., explaining Fisheries management, conservation principles, etc.

In respect to habitat protection, we are concerned about river bank protection from activities created by logging, road building, pipeline construction, urban and industrial development.

We understand the Fisheries Act is to be ammended to make it a stronger and a more powerful Act in controlling environmental damage and pollution. (could the chairman elaborate on this)

In wrapping up our brief, we would like to discuss a number of points - the first is condemning the past methods of Salmonid Enhancement in the Skeena watershed, which were designed to increase the numbers of a single species with no regard for the effects on other species of salmon or trout. Specifically, we are referring to the Babine sockeye populations. Unfortunately, these fish migrate at the same time as the Morice, Babine, Copper and Kispiox steelhead, (which amounted to approximately 10,000 steelhead this year). In order to reduce the catch of steelhead, we urge a reduction in production of sockeye in Babine Lake or use different methods of harvesting them, ie. fish traps at the mouth of the Skeena and at Babine Lake itself).

We recognize the number of commercial fishermen who would be affected by this drastic action, but we believe it is necessary if the Skeean River steelhead are going to be saved.

The Steelhead Society believes that the recreation fishery of the Skeena system is equally as valuable as the commercial fishery when both the economic and social values are considered. We strongly urge that whatever enhancement programs are enacted, that no future conflict between species occurs.

In regards to present and past Fisheries management, we condemn the manner in which entire river systems are being closed to sports angling. We recognize the

necessity for this drastic kind of action, when a population of fish are nearly depleted. However, we ask the question, "Why didn't the Fisheries managers recognize the decline in these populations and take less stringent action at an earlier date"? Reductions in daily limits, possession limits, canning of salmon and gear restrictions are some of the things that could have been and should be done in the future.

- We recommend the establishment of a punch card for salmon at the earliest possible date. The establishment of a punch card would make it much easier for protection officers to enforce regulations and it would provide much more accurate data on the freshwater salmon fishery.
- Canning of salmon should be made illegal, except at one's permanent residence. We believe that it should be made legal to catch sockeye and pink salmon by sports fishermen in the mainstem Skeena River.
- The Northwest Chapter would like to see the establishment of auxiliary protection or conservation officers similar to that which our police forces have.
- Moving away from Fisheries management, we would like to ask the question of how the Task Forces were selected and why conservationists who have a long shown interest in Fisheries were not included in the committee for our area?
- On the subject of the Nass River watershed, it is our belief that sound Fisheries management and protection is what is required, along with habitat improvement where logging damage has taken place.

In concluding our presentation, the Northwest Chapter of the Steelhead Society wishes to go on record as being opposed to 50% of the Fisheries resource of any river or lake system in British Columbia being controlled by the Native Brotherhood or any Indian tribe in our province.

It is impossible to manage an anadromous fishery or to allocate any portion of it on a fair or equal basis, to separate users, when there is no known way to determine the number of returning adult salmon or steelhead.

In Washington State, the Boldt decision, which granted Indian tribes in that State, 50% of the fishing resource, has created an explosive situation. In most instances, the Indian tribes have caught far in excess of the allotted 50%, in fact, the figure is probably closer to 80%. As an example, on the Green River, up to December 1975, Indian netters took 4,839 steelhead, while 12,000 non-Indian sports fishermen during that time, caught 50 steelhead.

We know of instances where shots have been fired at Fishery Officers and that the sports fishermen have almost reached a point where they will be taking action on their own. The last thing that the Boldt decision has created is racial harmony in Washington State.

The Northwest Chapter recognizes that the Indian people of B.C. have certain prior rights to fishery resources in the province. However, the fact remains that non-Indians in B.C. also have rights. We believe the fishery resource must be shared equally by all and if it is going to be managed in an efficient and practical way, the fishery managers cannot be answering to two separate groups of people.

We believe that there are ways in which the Indian people can receive justice. As an example, a percentage of fishing licence fees could be allocated as a royalty to them.

The Northwest Chapter of the Steelhead Society is prepared to work towards a just settlement of the fishing rights question and would welcome the opportunity to work with the Indian people to reach this end.

Our position is that all people in British Columbia, regardless of race, colour or origin, must abide on an equal basis to laws of their land. In this particular instance, we are referring to the Fisheries Act.

We agree that the Indian people resident on a particular river or lake have a priority right to the fish in that system for their own sustenance. We do not believe that they should be able to sell the fish, nor do we support the position that tribes many miles away from a river or lake should have special fishing rights.

It could be argued that our views on Indian fishing rights is totally out of context with the theme of this public meeting. We say to that, if the Native people in British Columbia are granted 50% fishing rights, then the Salmonid Enhancement Program, as it presently is being planned, is an academic exercise. We ask the question, "Is it really possible to plan such a grand plan for Fisheries management in British Columbia when it is not known at this time how and who is going to be harvesting the fish"?

If the Salmonid Enhancement Program is going to be a success, co-operation and understanding amongst all concerned is imperative. The biggest challenge is not enhancing salmon and trout, but rather it is finding a way for all of us to agree on how they should be enhanced.

RESPECTFULLY SUBMITTED BY THE NORTHWEST CHAPTER OF THE STEELHEAD SOCIETY OF  
BRITISH COLUMBIA

FURTHER NOTES RE: SALMONID ENHANCEMENT PRESENTATION

1. The lowly position of the Fish & Wildlife Branch is always subserviant to other government department's, both Federal and Provincial. We would like to see the Branch elevated to a position whereby they could have more control over their areas of responsibility; particularly when environmental decisions are being made. We believe that they should have the final say when fish and wildlife under their jurisdiction could be effected. We do not agree with the Pearse Royal Commission, that the Fish & Wildlife Branch should come under the same Ministry as the Forest Service.

We beleive they would become even less effective with the Forestry interests dominating that Ministry. Fish & Wildlife needs a single Minister as it now has. It is much too big a responsibility to come under the Minister of Forests, Lands and Water Resources, who is now overloaded with responsibilities.

2. Ban treble hooks in all tributary streams of the Skeena; as well as the Cranberry-Kiteen system and the Tseax River tributary to the Nass River.
3. Present Fisheries staff are unable to effectively control snagging and the taking of overlimits on rivers in the Nass system. We would like to see Guardians posted on the Cranberry-Kiteen and the Tseax Rivers.
4. We would like to see the re-evaluation of regualtions for salmon in fresh-water. Such as the establishment of annual limits, reduction in possession limits and some daily limits; introduction of a trophy fee and tag system for chinooks, etc.

We would like to be more specific and would hope that we could get an opportunity to present our views at another meeting, specifically with the Fisheries Service.

5. The Northwest Chapter would like to see more accurate live counts of salmon, ie. swimming rivers with scuba gear, before rivers are closed to angling because of declinding stocks, as opposed to protection officers live counts by other methods. (Ishkeenishk River as reference)
6. We are concerned about money being wasted on research and projects which are questionable or not really on top of the priority list. Our preference is strong imaginative management which leans towards conservation, rather than gambling on over-harvesting but we ask that it be fair and reasonable.
7. In reference to the proposed Kitimat to Edmonton Oil Pipeline - We question the practicability of the oil pipeline going ahead in conjunction with the Salmonid Enhancement Program. Not only is the coastline from Kitimat to Dixon Entrance, along with its fish & wildlife, threatened by the potential of a massive oil spill from 300,000 ton oil tankers but as well existing jobs in the commercial fishing industry and the tourist industry are also being jeopardized.

Along with the social values, ie. sports fishing by local residents and yours and my tax dollars are possibly being wasted on enhancing the fishery resource, when all could be lost should an oil spill ever take place.

And something no one wants or cares to discuss, is the extensive environmental implications of the construction period of the pipeline. When you install a 30 to 35 inch pipeline, you are digging a very large ditch and moving a lot of material. Silting and destruction or disruption of spawning and rearing areas in river environment, such as the Copper River, is a very serious and certainly puts a question mark on preservation and protection of fish stocks.

RICHMOND

SUBMISSION TO SALMONID UPDATE '78

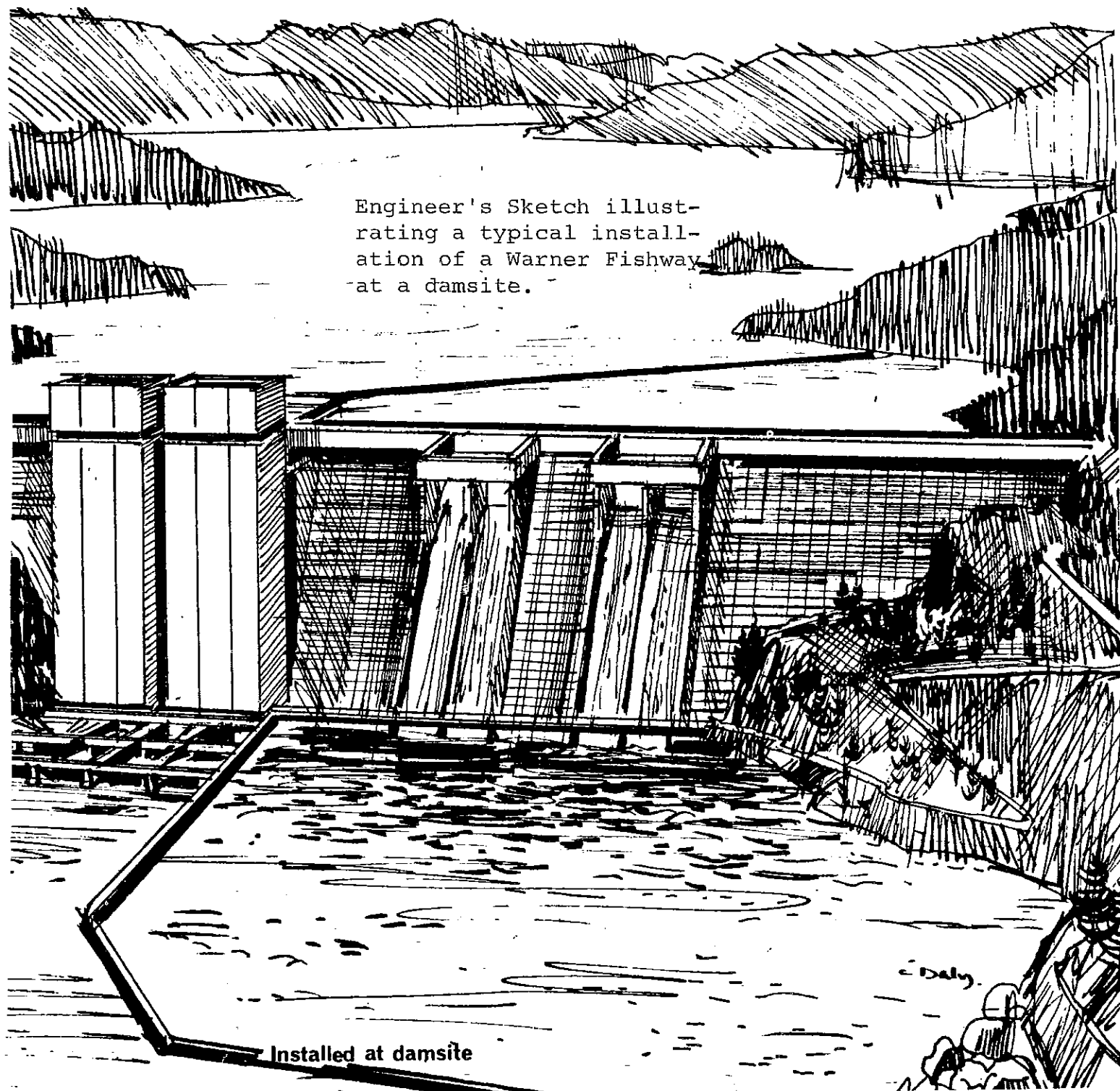
BY

J.W. Warner  
Warner Fishways Ltd.

Richmond  
January 18, 1979

DEVICE TO ASSIST FISH TO TRAVEL  
PAST OBSTACLES IN WATERWAYS

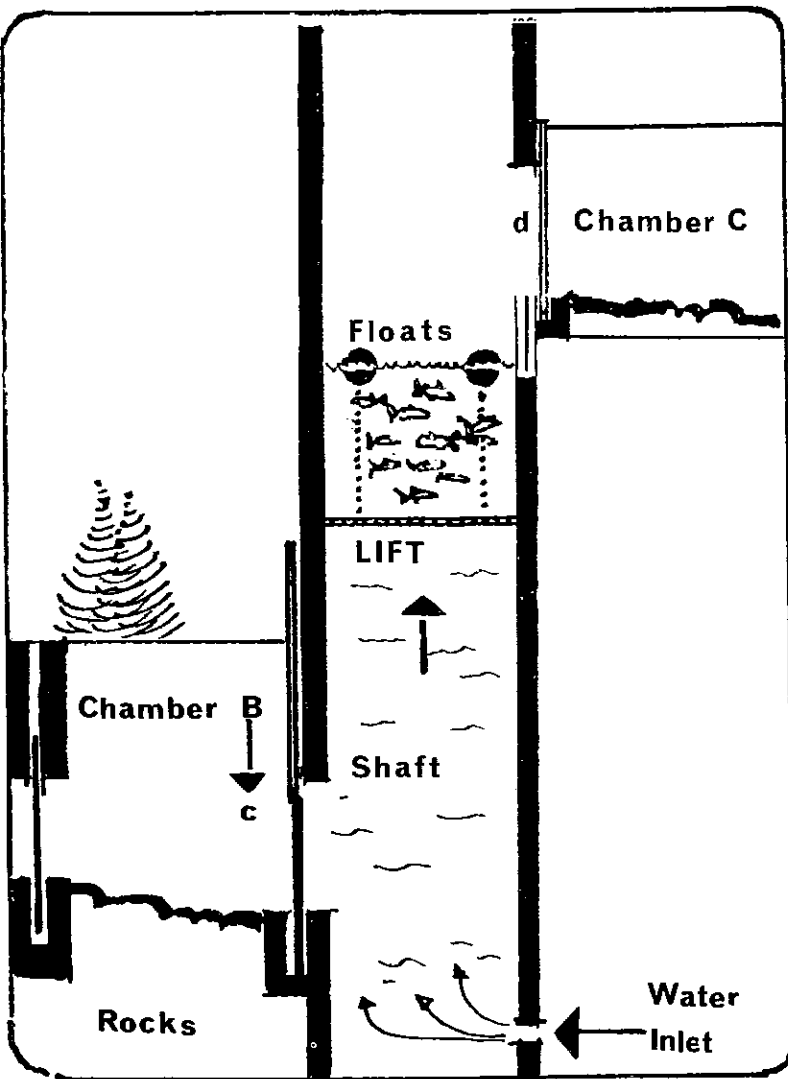
Engineer's Sketch illustrating a typical installation of a Warner Fishway at a damsite.



A system for assisting fish to circumnavigate obstructions, such as dams in rivers or streams, which impede or block their migration route to their spawning areas. The fish are diverted from the main stream of water into a series of locks terminating in a passageway containing a semi-buoyant cage which acts to raise or lower the fish by altering the level of the water in the passageway. The fish, attaining the new level, are released through a series of locks to continue their migration.



# NEW SYSTEM MOVES FISH EASILY OVER DAM



This invention relates to a novel method and device for safely assisting fish to overcome and circumnavigate obstructions in rivers and stream. More particularly this invention is concerned with safely assisting fish over and around obstacles which impede or block their migration route from the ocean to the spawning areas and from spawning grounds or areas to the ocean.

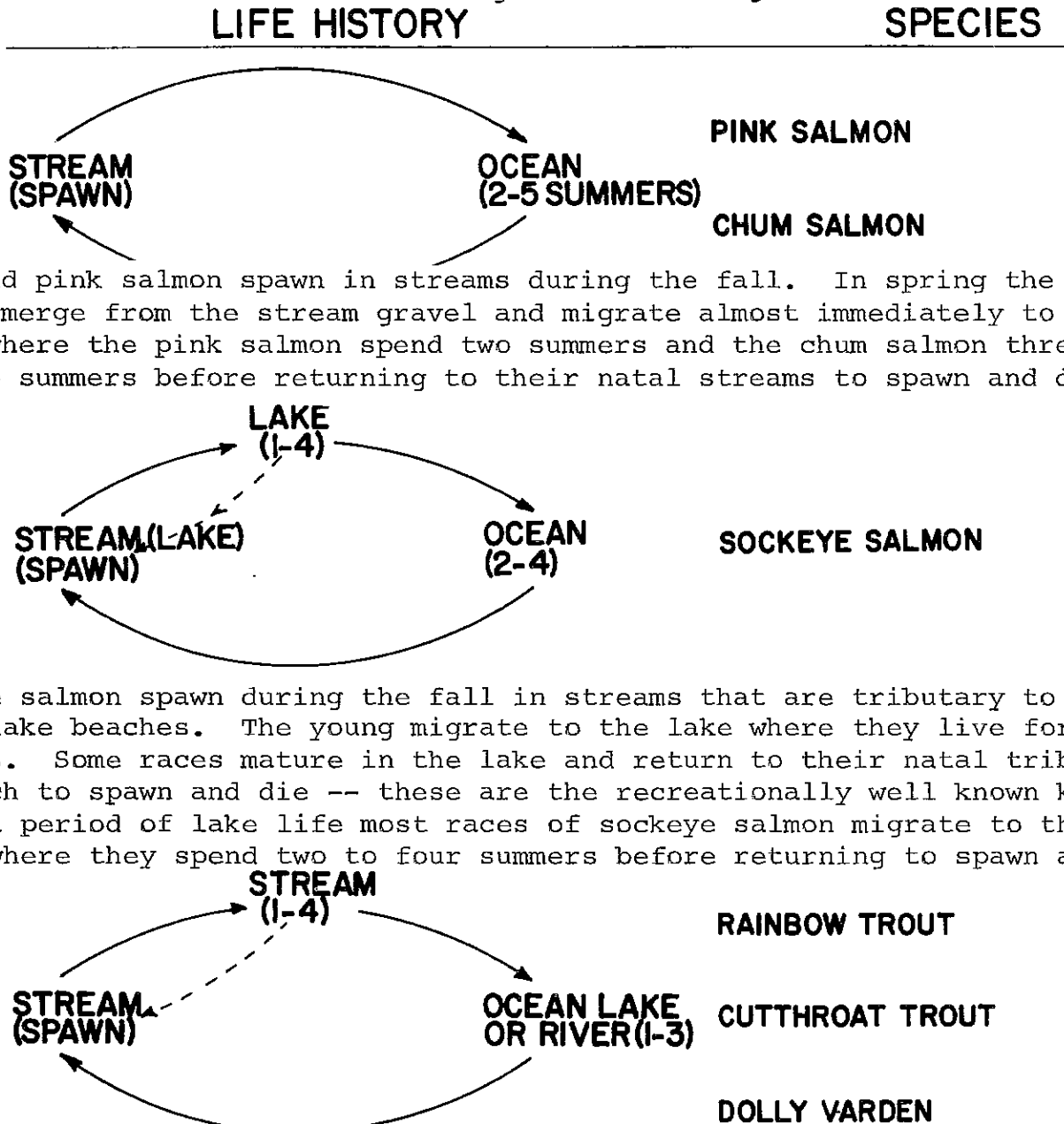
This type of fish with which this invention is concerned is primarily the anadromous fish, such as the salmon, which spend the greater part of their lives at sea and return to fresh water to spawn. During this migratory spawning period, literally millions of these fish enter the fresh water coastal rivers and start their journey upriver towards the quieter streams and tributaries. In attempting to go up the river, the fish encounter many natural hazards impeding their progress such as rock slides and natural waterfalls. With the demand for more and more electrical power and the need for water conservation, man-made obstacles in the form of hydroelectric power developments and dams

are becoming more plentiful on the rivers used by these fish. Because of the increased efficiency of the fishing industry coupled with the many obstacles blocking the fish from the spawning areas, these fish are slowly becoming more depleted. In order to attempt to maintain good yields of this type of fish it is necessary to allow the mature fish access to the spawning areas and also to permit the young spawn to escape to the seas and oceans.

Prior to this invention, by-passes in the form of fish ladders or fish-passes have been employed to overcome such obstacles as dams. These ladders and passes, usually comprising a series of pools of water constructed in the form of steps, have limitations. They are costly to install and must be laid out on a gradual rise, each step limited to the height that the fish are able to naturally overcome. These types of by-passes also require diverting part of the waterflow thereby losing a percentage of the total possible hydro development. In cases of very rugged terrain it is, at times, not practical to install these known types of systems and the river is made impassable to the fish. Under certain circumstances, trucks are at times used to transport fish around obstacles wherein they have been trapped but this again is a very costly and time consuming manoeuvre.

THE FISHER RESOURCE BASE

We are concerned with eight species of salmon and trout. These comprise the most important recreational and commercial fisheries resource on the west coast of North America from Oregon to Alaska. All of these species depend on streams for spawning and embryonic development and six of them require either lakes or streams for juvenile rearing.



Chum and pink salmon spawn in streams during the fall. In spring the young emerge from the stream gravel and migrate almost immediately to the ocean where the pink salmon spend two summers and the chum salmon three to five summers before returning to their natal streams to spawn and die.

Sockeye salmon spawn during the fall in streams that are tributary to lakes or on lake beaches. The young migrate to the lake where they live for 1 to 4 years. Some races mature in the lake and return to their natal tributary or beach to spawn and die -- these are the recreationally well known kokanee. After a period of lake life most races of sockeye salmon migrate to the ocean where they spend two to four summers before returning to spawn and die.

Rainbow trout (steelhead), cutthroat trout and Dolly Varden (a species of char) spawn in streams that are often tributaries of lakes or larger streams (trout spawn in the spring and char in the fall). After emergence the young usually reside in the natal stream (from 1 to 4 years) before migrating to the ocean, a lake or a large stream. After 1 to 3 years they return to their natal stream to spawn (but these do not necessarily die). Obviously natural streams are still of great importance in the production of salmon and steelhead in Oregon, Washington and British Columbia.

Some of the problems with hatcheries are the high capital and operating costs; harvesting the excess supply of hatchery salmon or steelhead without overharvesting wild stocks with which they mingle in the ocean; the potential for disease problems both within the hatchery and by contamination of wild stocks; and the high probability of creating major genetic changes both within the hatchery stock and through their interbreeding with wild stocks. Potentially, the genetic problem is much more serious with hatchery than with spawning channel operation. Within a few generations a hatchery operation can change substantially the genetic composition of a salmon or steelhead population, making the production entirely dependent on the artificial environment of the hatchery. These fish become well adapted to freshwater rearing in hatcheries but may do poorly in the natural stream. Another problem is "genetic pollution" of native salmon and steelhead populations by straying hatchery fish. Thus once a program is started, a permanent commitment is made.

Assuming that the biological problems of disease and genetic changes can be controlled, a dangerous assumption since our ignorance in genetic changes is great, let us examine the costs of both a centralized and decentralized hatchery operation.

Most hatchery programs depend on large hatcheries each contributing fish to a large number of streams. Taking the extreme case, if such a program were required to replace all the coho production in British Columbia the annual expense would be immense. (Of course this would be an impossible task). We saw earlier the annual sport and commercial catch of coho is about 4.3 million with a total run of at least 5.2 million. If an average survival of coho from seaward migrant to mid-summer adult is 5%, then 104 million smolts are required annually. If operational costs plus the amortization of capital investment results in smolts costing about 30¢ each, it would cost \$31.2 million per year to replace the natural production of coho salmon in British Columbia. IT IS OBVIOUS THAT SUBSTITUTION OF ARTIFICIAL FOR NATURAL SPAWNING GROUNDS VERY QUICKLY BECOME PROHIBITIVE. These fishes all have a great dependence on a high quality freshwater environment -- healthy watersheds in general -- so it is not surprising that biologists, commercial fisherman and sport fisherman are among those extremely concerned when faced with the impairment of that environment by any other activity.

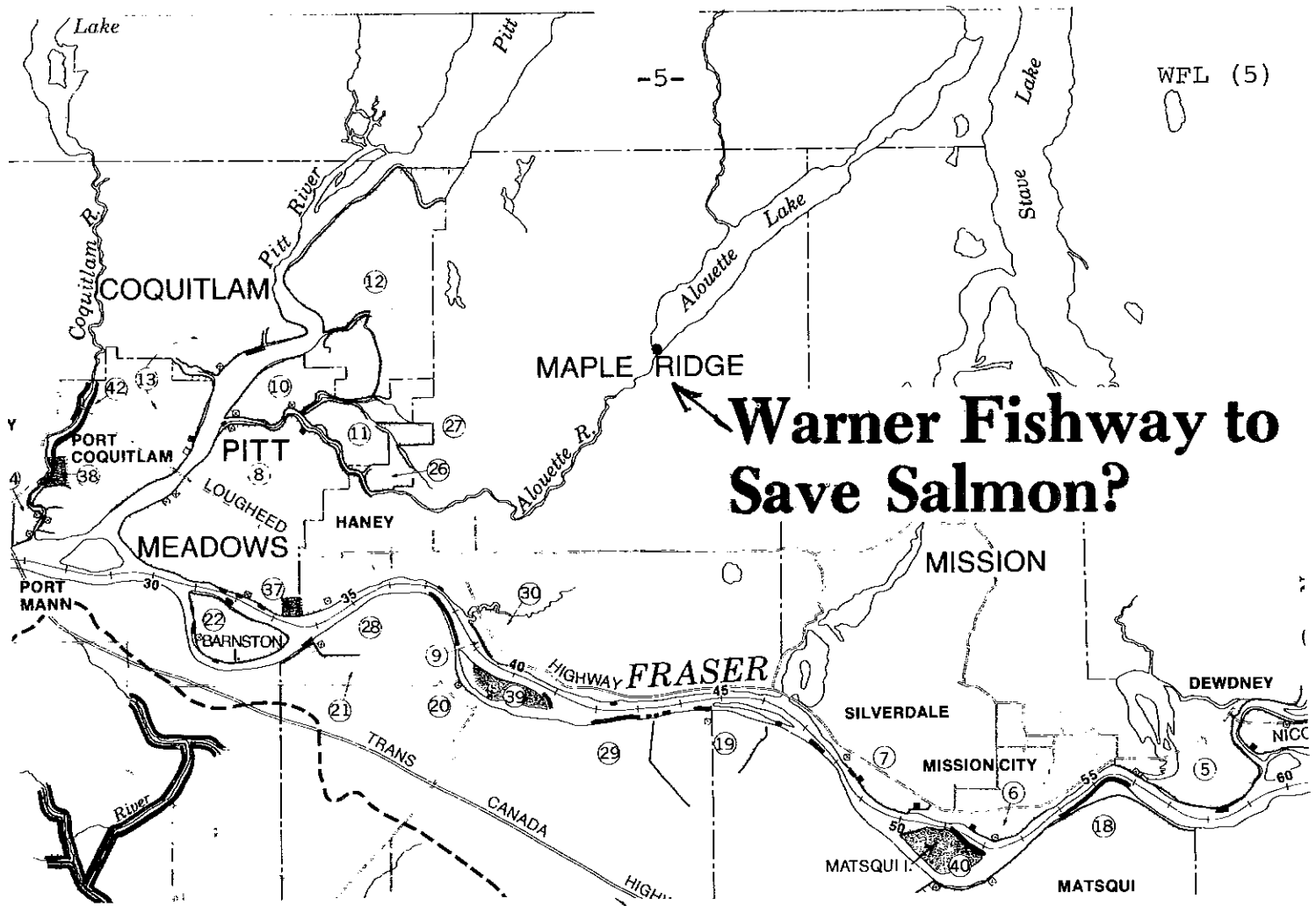
#### CONCLUSION

I emphasize again that hatcheries and spawning channels can play an important role in maintaining stocks of salmon and trout in the face of environmental change such as dams, extremely intensive fisheries, and in those limited situations where Mother Nature can be improved.

I have argued that the total fishery values associated with forested streams are large and increasing, and that direct and indirect costs to society are probably much less if stream protection measures are employed than if the stream is not protected and hatcheries or spawning channels are constructed to replace lost salmon and trout production. There are no alternatives to strict and rational stream protection measures.

Quote: David W. Narver, Circular 93, Fisheries Research Board of Canada, March 1973.

"Are hatcheries and spawning channels alternatives to stream protection?"



QUOTE FROM HANSARD, PROVINCE OF BRITISH COLUMBIA, PARLIAMENT BUILDINGS, VICTORIA, B. C., DATED 78.04.28

MR. MUSSALLEM: Mr. Chairman, as a change of pace and something of interest to this house, I'd like to draw the attention of the minister - and I hope he knows about it already - the Warner fishway proposal. Mr. Warner, a very enterprising citizen who lives in our constituency of Maple Ridge, had developed a system of bringing fish over dams without damage and taking them out the same way

If this can be resolved - and engineers tell me it's absolutely feasible and may be done and can be done - I would enlist the minister's assistance and help in this proposal. I would like to see him authorize that a virtual experiment be made at the Alouette River Dam, which was closed up in 1920 and which landlocked salmon in the alouette River and almost killed the Alouette River for fishing since that period, although there has been a slight return in the meantime, I do not feel, Mr. Chairman, if the minister would take this under special consideration, a great breakthrough could be made in the matter of fish and dams.

It would not only be a breakthrough for this country, but it would be a worldwide breakthrough. The system is simple, but novel. It requires no power to operate, and is operated by the pressure of the water itself. Engineers of note and ability tell me it is most workable, and why hasn't someone thought of it before? If it's that good, I think the minister should take special time in giving it an assessment. I know his biologists have gone into the situation, and they've found out it would be most workable. I hope he would give it special consideration, and bring up in this forum because I think it needs the impetus because dams and fish have always been a problem. If this will work, and it seems to be possible, it would solve this very difficult problem.

WARNER FISHWAYS LTD.

Maple Ridge, B. C.

Dated: 78.01.12

A PROPOSAL  
FOR THE INSTALLATION  
OF A WARNER FISHWAY  
AT ALOUETTE DAM

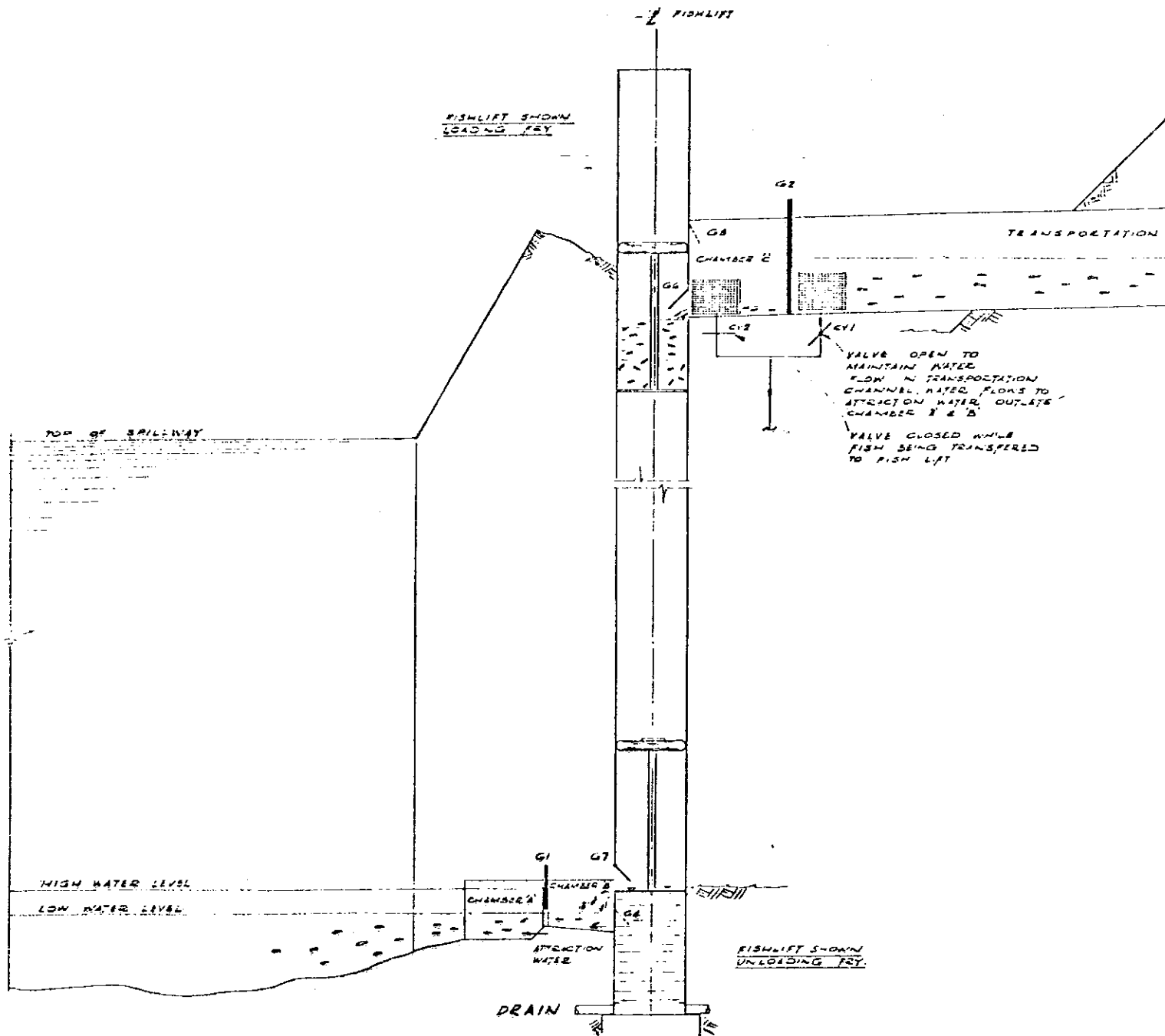
Dear Mr. Minister:

My M.L.A., Mr. George Mussallem, has suggested that I contact you regarding the following:

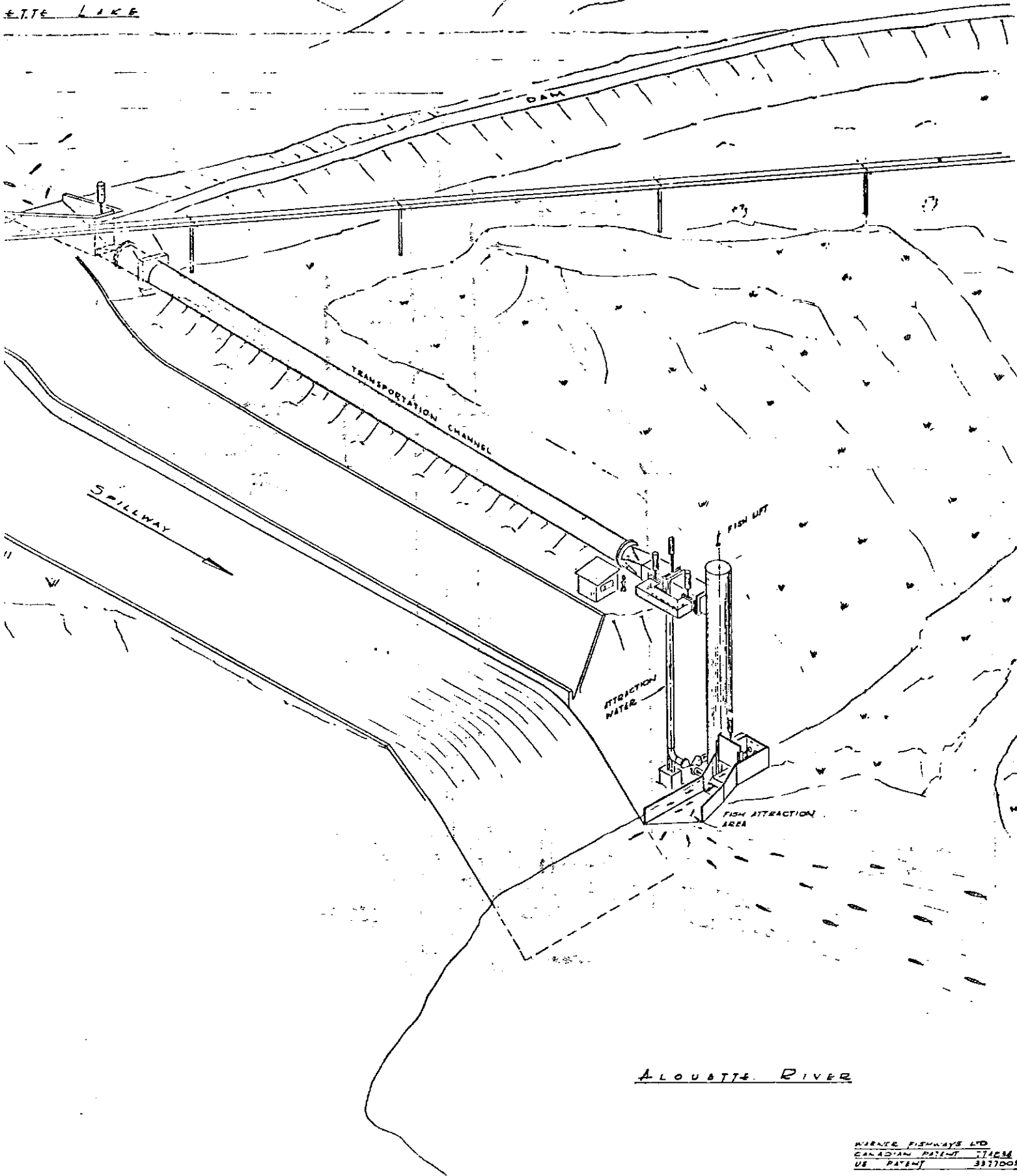
Warner Fishways Limited asks your favourable consideration of this proposal, that your Ministry provide the funds necessary to carry out a Feasibility Study of the Installation of a Warner Fishway at the Alouette Dam on the South Alouette River near Maple Ridge, B.C.

The Alouette Dam, constructed in 1923 by the B.C. Electric Co. Ltd., effectively blocked the spawning run in the South Alouette River of a number of species of salmon and destroyed the salmon production of the river above that point. The diversion of the inflow to the newly formed Alouette Lake via a tunnel to Stave Lake for power production purposes effectively destroyed any salmon spawning below the damsite.

The recent practice, established in 1973, of providing a continuous flow of water past the Alouette Dam has led to the reappearance of salmon all the way up the river to a pool just below the dam.



ETTE LAKE



ALOUBETTE RIVER

The Warner Fishway is a device which would permit passage of salmon upstream past the dam to Alouette Lake and would allow the re-establishment of an anadromous fishery in the lake and its tributaries. The Warner Fishway will also provide for the passage of the fingerlings downstream at the appropriate time. WFL (7)

The principles of operation of the Warner Fishway are indicated in the drawings accompanying this proposal. A model is also available for demonstration of the fishway's operation.

The operation of the Warner Fishway needs to be proven in practice and further development can only be carried out by construction of an experimental prototype at a site such as Alouette Dam. Prior to building a prototype, a feasibility study is required, which would consist of:

1. The gathering of the pertinent plans and drawings of Alouette Dam;
2. Obtaining the operational rules and conditions for Alouette Lake and River;
3. Limited site surveys and investigations to supplement existing data;
4. Preparation of a preliminary design of the installation in sufficient detail for scheduling and estimating purposes;
5. A schedule for construction and installation;
6. A cost estimate for construction, installation and operation.

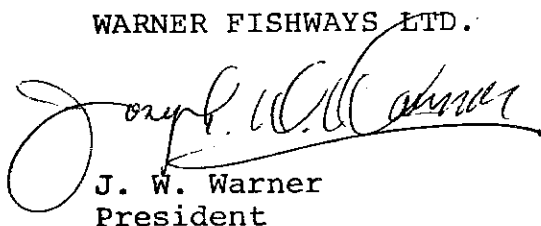
CBA Engineering Limited of Vancouver, B.C. has prepared the accompanying Terms of Reference for the feasibility study and a cost estimate and schedule for carrying out the study which accompanies the proposal.

I ask your favourable consideration of this proposal and request that your Ministry provide the necessary funds to carry out the Feasibility Study of the Installation of a Warner Fishway at Alouette Dam.

Representatives of Warner Fishways, supported by CBA Engineering, will be pleased to meet with you or members of your Ministry at your convenience to discuss this matter and provide any further information you may require.

Yours very truly,

WARNER FISHWAYS LTD.



J. W. Warner  
President



## ANGLER ACCESS

I'm fourteen years old, I live on the nineteenth floor of an apartment in Burnaby. I would like to catch a salmon but all I have got is a bike and a rod. Where can I go? Or perhaps I am sixty seven years old and I don't have a bike . . . but I have the desire and patience to do a bit of fishing now and then. Where can I go!!!

85% of the population of British Columbia squat here in the metropolitan Vancouver area and short of owning thousands of dollars in transportation and boating equipment, there is just nowhere to fish.... Mind you there used to be. At one time there were over a hundred names for bars from Mission downstream where one could placidly cast bait and line for migrating salmonids.

Today, where these bars once were, are log booms, endless miles o' log booms, or a dock for unloading cars from Japan, or maybe a seaplane base, or a graveyard for junk cars. You'll find everything there . . . except fishermen. Well they don't catch many fish anymore you'll hear and quite right . . . what the hell can you catch if you snag sunken log boom debris on every cast. Or if the police interrupt your fishing to charge you with trespassing.

**Access to angling water is as valuable as the fish in that water.** It doesn't matter how many Steelhead or Coho you propagate in the river . . . you can't catch them easily over a barbed wire fence. . . or with a doberman hanging from your butt. Log booming leases, private property, population and industrial encroachment have taken a heavy toll.

Efforts have been made in the past to organize fishing reserves on the lower Fraser . . . these have all met with failure due to lack of focus. The Public Involvement Section of S.E.P. is an ideal vehicle to bring together a program that would assist in preserving what is left and restoring a few lost sites to this time honored pastime. What is needed is the clout to be dealt with fairly.

**Access** to those areas already in use should be preserved or established.

**Boom Free areas** to fish in. . . not large areas 100 meters is sufficient in most cases, too much in others.

**Facilities**, latrines, garbage collection and perhaps the luxury of a place to safely park a car . . or a bike. Anything that is needed in the way of building could be built at minimal cost at one of the Penal Institutions woodwork shops such as Agazziz mountain or Matsqui. These could be put in place and maintained by local groups.

Contact your District offices, ask about places with names like Brownsville, gypsum, Block, Savage, Tilbury, Annacis, and Glendale, see if these places really existed . . . validate my issue. Then act quickly before there isn't a place left to cast a line on the lower Fraser River.

If this should happen you have lost many thousands of people that would otherwise not be touched by your program. There aren't many streams that can be rehabilitated and fished in this area. Fish from Enhancement sites on the Upper Fraser are destined to be caught by ocean commercial or sport fishermen. Because there are few good sites left to fish from, let's create angler opportunity with angler access as well as with the enhanced fish of your program.



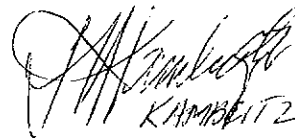
KAMBEITZ 18/1/79

J. Kambeitz

## QUALICUM

Anyone who has seen, taken part in, or understands the recent Steelhead Fishery on the Big Qualicum Rivers "meat Hole" will say it is an unbelievable spectacle in the truest sense. The Salmonid Enhancement Program has proven its point in producing steelhead in such commendable numbers, however the travesty of sportsfishing that accompanies the river harvest is nothing but damaging to those that try to practice angling as the "gentle art". The entire river is open to angling, however sometimes up to 50 fishermen will cram themselves onto this 75 foot long man made hole to fish, jig, cuss and kill nearly 1000 steelhead. It is the understatement of the year to say that fish here are easy. I propose that regulations be immediately implemented to allow access to this pool only to anglers under 16 years of age and over 60 years of age.

These two groups are "left outs" of steelhead fishing. . . . sure there are exceptions but by inlarge most have a hard time competing with your average dyed in the wool steelheader who would have the rest of the river to angle in. It is time we provided a steelhead angling opportunity to those who have gone before us . . . . it is done in several areas now for trout. If we give this little, - it could mean so much to a youngster and a few old folks. Why, just this morning I noticed how grey my hair was getting. . . .



KAMBEITZ 18/1/79

J. Kambeitz

## The Salmonid Enhancement Program: A Costly Capitulation

### The Steelhead Society of B.C.

It seems only yesterday Roderick Haig-Brown warned that reliance upon fish hatcheries would signal British Columbia's failure to manage its Fisheries resources. Today, failure seems imminent. Tomorrow, except for some fortuitous intervention of conscience and courage, this province's rivers and streams will be deeply scarred by hatcheries and spawning channels - grim and concrete reminders that we cared too little, too late.

It's sad that our leaders -- both political and professional -- have chosen the easy way, the quick way, instead of what they know in their hearts is the right way. It's even sadder that those who steer the Salmonid Enhancement Program have set a dangerous course. Saddest of all is the fact that if SEP planners do have their way, the public will pay hundreds of dollars for a scheme it deems both unwise and unnecessary.

That SEP planners have not truly listened to the public needs to belabouring. But it might be helpful to know why SEP is not listening. One distinct possibility is that SEP is simply too busy re-enforcing its own wishes. Certainly the Salmonid Enhancement Program Report on the Public Inquiries -- published in March Of 1977 -- shows that the professionals and the public view Fisheries management responsibilities and opportunities from opposing sides of a wide gulf. The report shows that the public is not anxious to have the province's landscape dotted with hatcheries. A page 14 paragraph -- entitled "Summary of Major Inquiry Input" -- lists 27 of the public's major concerns. Support for hatcheries was last on the list.

Tied for first on the list were "more enforcement" and "more education". Fourteen times more people wanted enforcement and education increased than wanted hatcheries constructed. And, not only has there been no increase in enforcement and little increase in education, there's no sign that enforcement will be increased, and no sign that the education package will be directed at the appropriate target -- the governments and industries of the province.

Similarly, natural enhancement, small stream enhancement, and inter-agency co-operation were wanted nine times more than were hatcheries. A few pallid gestures have been made, but overall, SEP has paid scant attention to the public's concerns. Meanwhile, SEP's hatchery vision, and its primary aim -- to produce more fish -- have remained unaltered.

That SEP planners intend to have their way -- no matter what -- is not easily dispelled. Certainly a three page handout, distributed at the public meeting held in Vancouver in December of 1978, helps keep the public's suspicion alive. In that handout, SEP planners brazenly placed hatcheries at the top of the list of "Techniques (SEP) Can Use".

It's important to note that SEP prides itself in possessing the technological know how to produce more fish than nature can produce. It seems a little silly to get puffed up by such stuff, but the puffery signifies a deep-seated mischief; that

SEP sees itself as Creator. The public should be forgiven if it believes that the more SEP reverses its own power over life, the less it will respect nature.

SEP's aim to produce more fish -- primarily for the commercial industry, it should be noted -- and its intention to rely on hatchery production -- signify a costly capitulation (its promise to produce a few additional fish for sport fishermen is a transparent attempt to bribe public support, and does not deserve deep reflection). And if SEP is permitted to proceed, it will doubtless find expedient ways to rationalize both its methods and the consequence of its methods.

That politicians agree with SEP's sermon on the supremacy of technology is not surprising; the sermon can be used to support what too many politicians still want to believe; that man can trifle with nature. Thus SEP's hatchery campaign helps condition the public to accept tampering with the environment, to honour technology as mankind's Saviour, and to ignore (even detest) nature. In its myopia, SEP is paving the way for continued -- if not accelerated -- habitat abuse. In view of all this, it should surprise no one if future plans to dam even the Fraser River rouse little public protest.

While there's still time, the public should protest. And it should protest loudly. It should protest against SEP's main aim and SEP's methods, against the continued disregard for the environment, against the lack of determination to restore habitat health, against the worship of technology, against the abysmal shortage of conscience, courage and vision, and against Federal Cabinet constraints.

In and of themselves, Federal Cabinet constraints are enough to make both SEP and the habitat mere playthings to commerce. In effect, the Federal Cabinet has said, "Go ahead only if you can make a profit". It's a repugnant limitation, though an understandable one. After all, Federal fiscal successes have not been numerous of late (witness the predictable mega-buck failure of the bilingual program). The profit motive is not automatically evil, but when it over-rides all other considerations, the price is too high. Three of the Federal Cabinet's main constraints are that SEP must generate wealth, create jobs and recover costs. Clearly SEP can prove its profit-making capacity only by serving the commercial industry. But making SEP a commercial industry hireling is hardly in the best interests of either the public or the environment.

By definition, profits are the commercial industry's primary objective. The higher the profits, the happier the industry. At best, the industry's concerns for the habitat, the fish and the public are secondary. In a rather characteristic display of remote wisdom, the Federal Cabinet is obliging the public to assume a multi-million dollar burden in order that SEP can conveniently subsidize the industry. In effect, the industry is given a grip on the public purse and a strangle-hold on Fisheries management. To think the industry will learn to restrain itself is to be blind to commercial realities. The more industry gets, the more it will want. And the more it wants, the more it will twist the tail of its latest servant, SEP. The more SEP is pressured to produce profits, the more it will rely on hatchery production. It is inevitable. And the more SEP concerns itself with hatcheries, the less it will concern itself with habitat. That, too, is inevitable.

That SEP will demonstrate only an incidental concern for the environment is predictable enough. Not only has the Federal Cabinet decreed that SEP shall serve the commercial industry, but the Cabinet, like the industry, has only a passing interest in the habitat. And, for all its mishandling of public funds, the Cabinet -- like politicians of every stripe and at every level -- tends to be economically, not ethically, centred. Politicians also tend to have short-term interests -- their primary one being to get elected or re-elected. They too often seem to subscribe to three environmentally dangerous premises; that man can always bail himself out through technology, that growth is good and that whatever provides profits now is good.

Their short-term, profit motive point of view illustrates why politicians are far more likely to sell the environment than they are to save it. That's largely why the habitat has been up for grabs for so many years. That's also why the environment will continue to be sold out to the commercial interests -- hydro, logging, mining, housing -- which promise immediate profit. Politicians are clearly prepared to leave a legacy of habitat destruction trailing after them. SEP is destined to become part of the legacy.

It's bad enough that politicians believe that growth is good, but when a Fisheries spokesman utters the cliché that "without growth we die", the public can only shudder. That mindless nonsense was uttered at an early SEP meeting in Vancouver, and it demonstrates why those versed in the biological sciences cannot always be trusted. The cliché can, of course, be countered by pointing out that cancer, too, is growth, and that cancer kills.

It's bad enough that SEP is counter productive in terms of environmental enlightenment. What's worse is that SEP is an enormous give away program, and that the public is being obliged to bail out the very people who directly or indirectly allowed or caused the now decreed fish stock depletions. Had caution, conscience and courage been exercised in the past, B.C.'s salmon and trout runs would not now be imperilled. We now need what we've always needed. But, by ordering SEP to generate wealth, create jobs and recover costs, the Federal Cabinet all but guarantees that caution, conscience and courage will still not be exercised and that the public's concern for the habitat will still be ignored. Quite clearly, the intention is merely to modify the game plan. Under such circumstances, it is senseless to proceed with SEP.

The public knows -- all too well -- that politicians, bureaucrats, industrialists, and even Fisheries managers, have for years remained indifferent to the damage they did or condoned. Wasting of the environment coupled with over-harvesting at sea has brought about the present fish-poor circumstances. So who is to be blamed? And who will pay? The public should be excused for believing that its elected representatives should be held accountable. The public elects them to manage public resources. But count on it, from the top down, politicians will plead innocence. And even as they do so, they will take no serious measures to prevent continued rape and pillage of the habitat. Nor will expenditures of huge sums of public money save the day. What's required is a total overhaul of attitudes -- both political and professional.

To set matters right, we must take determined measures to safeguard the environment, and to restore much of what we have so willfully destroyed. If setting matters right costs us dearly, then we must pay dearly. But let us not hear some sanctimonious politician or bureaucrat claim he is doing the public a favour, or that he'd like to get government money back. And let us not permit public money to be wasted on schemes hatched in empty spaces. Let us reject the easy, quick, profit-centred way. And let us make certain that what's done is done in the public's long-range interests.

Though a change of attitude at the Federal government level is imperative, it is not enough. Provincial and Municipal values must also be totally re-evaluated. All levels of government have sacrificed habitat to profits. All have been anxious to attract industry and development, and in so doing have permitted -- if not encouraged -- habitat damage. Few politicians or bureaucrats have either the capacity or character to see -- or to say -- that the habitat should not be placed on the sales block, that this development or that industry should not be permitted adjacent to this river or that stream. Still fewer realize that water-courses should be treated as the land's natural parks and as such, should be set aside for the public's use -- in perpetuity.

Rivers and streams are the land's veins and arteries. Fish are the pulse which tells whether the freshwater habitat is healthy. Today's shortage of salmonids says, quietly but firmly, that a rich but fragile resource has been seriously mismanaged. Accidents and ignorance have taken their toll, but the damage has more often been willful. It has long been known that it is wrong to permit roads, industries and homes to encroach on streams, that it is wrong to pollute streams, or rob them of gravel, that it is wrong to abuse river environments simply because quick and easy profits can be made. Politicians have known these things for a long time too, yet they continue to sanction them.

Knowingly or otherwise, politicians have long been profiteer's ally. And the profiteer -- the land speculator or developer -- has no allegiance to the people, the land or the future. Once he profits here, he moves on -- to profit there. It's the nature of the beast. And, wherever riverbank development occurs, damage -- often extensive damage -- results. The public is left to pick up the tab for much of the flood control and property damage. In effect, selling riverbank land obliges the public to subsidize real estate interests (which sell the land) and home owners and businessmen (who use the land). And the public is often denied access -- to what should be public lands -- by the very people who have been subsidized. That the public has accepted such treatment for so long is one of life's truly astonishing ironies.

We doubt that we could ever discover how much the public has paid for damages which never could have occurred had river bank development been prevented in the first place. Certainly millions have been spent on the North Shore alone. The costs of road work, bridge construction, dredging and dyking could be calculated. The costs to the commercial industry could be calculated. The loss of sport fish could also be calculated. But how can the loss of joy be calculated?

Where is the politician with conscience and courage enough to truly fight to safeguard the environment? Where is the fishery manager with conscience and courage enough to admit that until the environment has been safeguarded, the salmon fishing

industry will demand more and more of the public's support. To shell out millions of dollars for salmon hatcheries with one hand while still maltreating fish habitat with the other is down right criminal. And SEP's political and professional pushers know it.

The public is understandably suspicious about SEP. Consider, for example, the fact that SEP is trying to sell the public on subsidizing the salmon fishing industry when the public can ill afford to eat salmon -- at \$5.00 plus per pound for troll caught fish. Or that SEP is so determined to give huge transfusions to a terminally outmoded industry. It does seem a bit silly to spend too much on boats, gear, fuel, and support services -- not to mention time -- to chase salmon around on the high seas when the salmon return to the very doorstep anyway. It's also silly to subsidize an industry which has the fish catching capacity to wreck even enhanced salmon runs. Or consider that SEP is prepared to forge ahead even when the management of many salmon streams might suddenly be handed over to the Indian people. And consider, finally, SEP's intention to provide jobs for Native Indians. It might be right to provide Indians with fish producing jobs. After all, fish centred activities are an important part of the Indian's heritage. But SEP's motives might not be pristine. It is possible that SEP is attempting to bribe Indian co-operation -- a contemporary equivalent to beads and muskets; keep the Indian's happy, keep their minds off land claim issues. It might go deeper, deeper even than trying to atone for past injustices. It might bespeak a subconscious contempt for Indians, a contempt which says, in effect, "We'll do this specifically for you since you cannot do much for yourselves". This much is certain, to encourage Indians to remain hewers of wood and drawers of water does not help them realize their potential in the contemporary world, and to give job preference to any special group does not help the cause of national unity. Job opportunities should be equally available to all citizens.

Before the public can be expected to support any mega-buck Fisheries schemes, the public must have evidence of many things. Among them: that the scheme planners and promoters truly listen to the public, that all levels of government are genuinely committed to safeguarding the habitat (and even to restoring much of the habitat which has been so wantonly destroyed), that full disclosures be made of the extent and type of financial assistance the salmon fishery has received, and is expected to receive, that the extent of foreign ownership of fishing companies be revealed, that the salmon fishing industry gradually be replaced by a river mouth, fish collecting system (which in the long run, would make better economic and biological sense), that Native land claim, food fishery, and illegal fish sales issues be resolved, that salmon and trout runs will not be made dependant upon man's technology (which would, incidentally, saddle the public with needless installation and operational costs), and that wealth generating, job creating and cost recovering constraints will not be demanded. Were SEP not so set on serving the commercial industry and were politicians not so set on abetting habitat abuse, Fisheries's bills would be far, far smaller.

And if our leaders -- both political and professional -- truly intend to do what's right for the habitat and for the people, they must spend time, energy and money to restore habitat health where the habitat is sickest and where people are most heavily concentrated -- here on the Lower Mainland. To do what's right, our leaders will have to aim at the highest common factor, not the lowest common denominator. They will have to heed those who have no ulterior motives. They could do worse than to heed the angler.

The angler has no desire to pillage that which provides him with so much. Nor does the angler want the habitat -- or the fish -- to be rendered dependant upon man's technology. The angler derives much of his joy from seeking nature's fish in nature's habitat. Man-produced fish are simply not adequate. The angler's first concern, then, is for the habitat; his second concern is for the fish themselves; his third is for angling opportunities; his fourth for the catch; his fifth for the taking of fish. Fish killing is the business of the commercial fisherman, not the angler. An angler might kill, but he is not driven by the need to kill. The angler takes or spares fish according to a deep inner sensitivity which defies explanation. Angling is more quest than conquest. And that's why the angler is not even driven by the need to catch fish. He enjoys catching fish, but he is not compelled to do so. But the angler does need to know that there is something beyond man, or man's works. And he needs to visit the places where he can find it, in a habitat which is free and wild.

The angler does not oblige the habitat to provide his wants. He respects the habitat for what it is, and accepts the bounty it offers. The angler does not expect the wanton destruction of the environment to cease, but he hopes it will. The public, too, must keep its hopes alive. The honourable management of this province's Fisheries resources might be too formidable a task for our present leaders, but we must keep hoping. The task would require much caution, conscience and courage. But we know it could be done. And though we have behaved stupidly and cowardly in the past, we are not a foolish or fragile people. Our house will not tumble merely because we reject the easy way and the quick profit. And, by resolving to do what's right, we will earn something far greater than a few extra fish -- we will earn some self respect.

Delta - Airport Inn, Richmond, B.C.  
January 18, 1979



SEP Update '78

Chairman:

I would suggest that upon observation of tonights anti-Indian feeling that you embark upon an information campaign to dispel ill feeling based upon misinformation. By doing so, I feel you may be able to promote co-operation between all groups to aid in conservation.

RICHARD SMITH  
#102 2291 W. 1ST  
VAN. B.C.

Richard Smith  
#102 - 2291 West 1st Avenue  
Vancouver, B.C.

10725 - 127th Street  
Surrey, B.C. V3V 5K4

November 24, 1978

Update '78  
Salmonid Enhancement Program  
Fisheries & Oceans Canada  
1090 West Pender Street  
Vancouver, B.C. V6E 2P1

Dear Sirs:

Re your recent advertisement requesting more input into salmon enhancement.

For a small project that will increase spawning many times over for all species of salmon, with the exception of sockeye you need look no further than Sweltzer Creek which drains Cultus Lake near Chilliwack.

Aside from the tremendous increase in spawning of salmon, with little additional cost it could be made into one of the better attractions for the public, being utilized virtually year round.

Up until the early 1950's Sweltzer Creek could be travelled from Cultus Lake outlet to the Vedder River by rubber boat.

However, it has now become nothing more than a collecting area for all the floating trash out of Cultus Lake plus the natural accumulation from nature. Today I suspect the water flowing from Cultus Lake has a difficult time reaching the Vedder River, also the useable spawning gravel has been reduced by at least 50 percent.

Prior to the mid 50's Sweltzer Creek had substantial runs of chum and pink salmon, these being drastically reduced partially due to the deterioration of the spawning areas.

I would suggest that by removing the junk and debris, log and trash jams etc. and replacing and cleaning of the gravel, Sweltzer Creek could be brought back to life.

Also, with a little thought a walking trail, such as at the Adams River could be added to each side. This in turn could create a situation whereby thousands of people would utilize the area during the time of spawning salmon and also during the summer months when tourists flock to the campsites on Cultus Lake.

...../2

During the time of the spawning salmon it could if desirable be made into a major tourist attraction rivalling all other fish viewing areas in the lower mainland.

Cost for such a project would approach \$100,000 however there is an Indian reservation on the lower portion of Sweltzer Creek, therefore by utilizing manpower from those people, funding from that department would also be a possibility.

A second project, a little further from home would be on Anderson Creek near Boston Bar.

This small river draining into the Fraser River prior to the mid 1960's had access from the Fraser River. However during the early part of 1960 a cement apron was constructed beneath the railway trestle. This proved to be most successful in blocking off all migration of salmon and trout into the lower spawning areas of Anderson Creek.

Unless it has already been done a small fishway could be constructed and once again allow spawning fish to migrate from the Fraser River into Anderson Creek.

A third small project that would open up many miles of spawning areas to salmon and trout lies just upstream of Ashcroft; the Bonapart River draining into the Thompson.

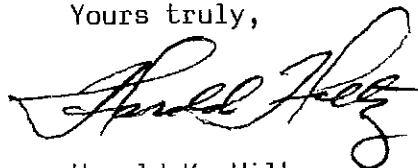
On the Bonapart, approximately 2 miles upstream from its mouth a small cement dam was constructed prior to the 1950's. This no longer serves any useful purpose, however it also creates a 100 percent block to fish moving upstream from the Thompson River. This could be readily removed and thus provide access into many miles of spawning gravel.

I have personally walked the lower portions of every stream entering into the Fraser River as far upstream as Quesnel and also all streams entering the Thompson River as far upstream as Kamloops Lake.

Therefore I would suggest there is no lack of good and meaningful projects to be done on the Fraser River drainage.

I have enclosed a slide of the cement apron on Anderson Creek. Please return this slide as soon as possible.

Yours truly,

A handwritten signature in dark ink, appearing to read "Harold K. Hiltz", written in a cursive style.

Harold K. Hiltz

16203 - 8th Avenue  
Surrey, B.C.  
Tel: 531-5220  
January 18, 1979

Government of Canada  
Fisheries and Oceans (Pacific Region)  
Salmonid Enhancement Program

Re: Placing of Culvert 8th Avenue  
Surrey Contract No. MS - 1-21-78, Project #4926

Dear Sirs:

First I would like to lodge a complaint that everyone on our street with the exception of one other neighbour, received a letter from Willis, Cunliffe and Tait, advising them of this work being done two months prior to the start of the project. It seems to me we are the property owners most concerned since our property is directly involved.

When we first learned of the culvert extension, my husband phoned Federal Fisheries in Vancouver. They insisted there was nothing "west of the freeway". When my husband said he had seen 34 inch carcasses many times, they decided they were interested. We also contacted Lorna Giles of Fish and Wildlife who assured me they were concerned. Ed Kendall of Semiahmoo Fish and Game came to see us in response to a protest there. He suggested contacting Wayne Lowden. Mr. Lowden came when the contractors for Columbia Bitulithic were actually dumping the fill and had them stop. He was impressed with the creek and remarked that, "if any amount of development goes on in the area, that the present size of the culvert would be inadequate".

My question is, why have they extended a culvert which is inadequate in size, when they know the area is going to develop into city lots in the near future? Why have they placed the culvert at the worst possible time for disturbing the creek? You are the experts, yet I understand Mr. MacFarlane gave the okay on my husband's statement that he had seen the fish in January.

Three days after the placement of the culvert, my husband saw the fish go up and took pictures. Mr. Don McDermott of B.C. Wildlife Federation, was also present two days after that when more pictures were taken. He estimated about 30 fish went up while he was present. Is this not a great enough amount to have our creek designated an area of concern? I am concerned about this "blanket areas of concern" in Surrey manager's report #8837 - November 20, 1978. My worry is that in eliminating "red tape" and referrals we may also eliminate some very valuable spawning grounds. Ed Kendall has taken fish counts sporadically here for years. He says the creek area on our property is in good condition for spawning but could be improved by putting in pools and rock. I would like to suggest this as a possibility under your program. Also, I would like to suggest that Fisheries Officers check the velocity of the creek further upstream on adjacent property. There has been a recent diversion and narrowing of the creek there, from four feet and more to an 18 inch ditch. In the past the fish have spawned right up to tenth Avenue.

One final point, had we been notified along with the rest of the street, it is possible that we would have had an injunction to stop work, as to this date, we have not had satisfactory proof of purchase from Surrey. This under review with our solicitor. My point is this, under our ownership, we would be entitled to apply for help under your program for help with a retaining wall which could be done with rip rap. At least some consideration would have been given to our wishes. I would appreciate your checking into these questions for me and request a written reply for our information as to what constitutes a sufficient number of fish to be of concern.

Perhaps, luckily no harm has been done to the fish. Since they are our mutual concern, it would be nice in future if we could work together on this and not be at loggerheads.

Yours truly,

*Lillian Long*  
*G. Roy Long*  
*16203 - 8th Ave*  
*Surrey B.C.*

Lillian Long  
G. Roy Long  
16203 - 8th Avenue  
Surrey, B.C.  
V4A 1A3

cc. Benno Freison  
Willis, Cunliffe & Tait  
Cheboyar Roads Eng., Surrey

Additional comments after listening to your meeting.

I congratulate you all on your efforts and your patience with all the confrontations. We are the owners of property with a large ravine and lively spawning creek. Our family has preserved this area for over 70 years. I feel that small streams are important. Our creek has been taught about in Earl Mariott school and some of the children have come down to see the fish spawn. I cannot understand how your Fisheries office two years ago would say it is a dried up creek. There is not much point in our preserving our property if adjacent properties are not also preserved. We are enclosing some of the pictures mentioned in my brief. We would be glad to do some of the work ourselves and not necessarily be after funding. I think you would find many willing to volunteer labour for these programs. We would welcome discussing this with any of your representatives as you mentioned. I feel Surrey's main objection to a retaining wall in our ravine was economics. There are other acceptable alternatives, such as a log wall or rip rap which are inexpensive. I hope it is not just a political decision as I feel we have a valuable natural spawning area.

Thank you.

January 10, 1979

Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Dear Sirs:

Although I intend to try and be present at your meeting on January 18, 1979, I am forwarding my submission in the event that I may not be present.

The purpose of my submission is to bring a real awareness to all concerned, that we may be seriously over-exploiting our herring resource. Having spent over sixty years on the B.C. coast, I can say that a great deal of my working years exposed me to both sport and commercial fishermen.

I have talked to a highly respected cannery manager in Tofino. A man my own age with many years experience. I asked him if we were foolishly overfishing the herring. His comment was that in his opinion, there was no doubt about it. According to this man, we are grossly overfishing the herring and as important, we are wasting a valuable resource. The facilities to properly recover the flesh of the herring simply doesn't exist. The day I talked to this man, the dock we stood on was full of dozens of containers full of rotting herring that would be taken out to sea for dumping. The plant did not have sufficient capacity to process this catch.

Surely no civilized nation should carry on such a wasteful practice in the interest of putting herring roe, a delicacy, on the table of the wealthy Japanese. We gag when we read about the slaughter of the buffalo for their tongues and hides but are we doing any better in handling the resources.

Once upon a time we had a large Pilchard catch in British Columbia. How that's gone? Overfishing. No controls. Simply mans greed.

Some years ago the Minister of Fisheries, Jack Davis allowed a large catch of herring in Jervis Inlet. Despite the hue and cry from sports fishermen and charter boat operators, the catch was allowed. We were told to relax, everything would be okay. The experts patted the little boys on the head and told them that the scientists knew best. Anybody familiar with that area knows that what occurred was a disaster. Salmon stocks declined dramatically and a few years passed before the situation stabilized.

During the year of 1978, we have seen a situation of bedlam, concerning the herring situation in the Pender Harbour area. It reminds one of the Keystone cops. At one stage there was to be a closure on herring fishing and then the pressure groups went to work and suddenly the closure was no more. Did the herring suddenly return to cause this turn around. I think not.

It is my firm belief, based on our past track record, that we are groping in the dark. We simply don't know what the true facts are. For anybody to argue otherwise, they must first provide answers to the above facts that I have stated.

1. What happened to the large herring catch that was predicted for the West Coast of Vancouver Island in 1978? What do the professed experts have to say?
2. What occurred to bring about the disaster in Jervis Inlet a few years back? What have we learned to prevent that from happening again?
3. What has caused the depletion of herring in the Pender Harbour area? Will we see wall to wall boats in Nanoose Bay fishing herring while Pender Harbour goes without?

It is my belief that fishing herring for the roe is lunacy and it is unlikely that any other country would allow it to happen as it occurs here. I would like to see a substantial roll-back in tonnage until such a time that we can be positively assured that over-exploitation is not occurring. If we are to curtail the commercial fishing of herring then I feel that the sports fishermen (I'm one) should also participate by outlawing the use of herring while trolling. Herring could then only be used mooching, strip casting and drift fishing.

Thank you for the opportunity to express my views.

Yours very truly,

K. G. Clarke

1756 Ottawa

West Vancouver, B.C.

V7V 2T7

K.G. Clarke  
1756 Ottawa  
West Vancouver, B.C.  
V7V 2T7

R.R. #4  
19743 - 64th Avenue  
Langley, B.C.  
V3A 4P7

Update "78"  
Salmonid Enhancement Program  
Fisheries and Oceans Canada  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Update "78":

You have asked for people's opinions, recommendations and concerns, so here are mine. It concerns the Fraser River system. I share the same concerns as the ones voiced in the Salmonid Enhancement Newsletter, Vol. III No. 5, 1978.

It concerns the Fraser River system. There are a number of projects that if they go ahead, could severely hurt the present runs of salmon (all species), steelhead, oolichans and any other type of fish in the river. The projects I am most concerned about are the proposed McGregor Dam project (which is not dead as we have been lead to believe) and the Harbours Board planned training wall program.

The opinion shown in it, the newsletter, was that enhancement of upper river springs and enhancement of steelhead should go ahead right away before the proposed projects can go ahead or the enhancement might not be worth it.

This would at least give us something else to fight the proposed projects with until enhancement of sockeye and pinks could take place when an agreement with the Americans can be worked out.

Yours truly,

A handwritten signature in cursive script that reads "Lorrie John Muench".

Lorrie John Muench  
Commercial Fisherman  
Fraser River



8311 Fairfax Place  
Richmond, B.C.  
V7C 1Y2

Salmonid Enhancement Program  
1090 West Pender Street  
Vancouver, B.C.  
V6E 2P1

Attention: Mr. Les Edgeworth

Subject: Meeting held at the Delta Airport Inn in Richmond, B.C.  
on January 18, 1979.

Dear Mr. Edgeworth:

Despite the very garbled and inept presentation of slides (which bordered on an insult to the audience) I enjoyed the meeting very much and am looking forward to the next one.

I would like to congratulate you upon the variety of enhancement methods displayed and discussed, and hope that you will continue along this road.

Personally, I hope that many more creek cleaning operations such as "Jenkins Creek" will be undertaken as they will use local labour.

Please disregard the ramblings of the well-heeded fat cats who refer to the fishing boat fleet as archaic. In actual fact our building overseas sales depends on a supply of fresh "silver-brite" fish and these salmon are only taken before they reach even partially freshwater. Therefore, any fish caught by traps or staked nets are, in effect, second class fish.

During the meeting, the fact that this program has to pay its way was mentioned and later a saltwater fishing licence fee was discussed. Inasmuch as everything is an expenditure of energy and that energy must show a return, I am in favour of all sectors benefiting from this program paying something towards it. This includes all commercial fishermen, shoreworkers, tendermen, processing plants, fishing companies, etc. as well as recreational fishermen.

I congratulate you upon your stand about the expansion of the Roberts Bank complex.

However, do you realize that although the Port of New Westminster is now virtually non-existent, yet the Trifurcation plan to flush out the Fraser River is still going forward and will result in the river speed being raised by probably one to two knots and the entrances to the sloughs blocked by training walls.

This is opposite to what is needed - move slough areas (non-drying) in which to rear young fish.

To summarize:

1. We need more jobs in B.C.
2. There is a big potential market for B.C. fish and an expansion of the B.C. fishing industry would produce more jobs.
3. Only "silver-brite" fish command a good price and a high priority position.
4. If we expand the numbers of fish, we must also expand the food stocks for these fish.

Yours truly,

A handwritten signature in cursive script, reading "J. A. McDowall". The signature is written in dark ink and is positioned centrally below the typed name.

J.A.M. McDowall



### **PUBLIC PARTICIPATION**

Salmonid Enhancement's Public Involvement Program exists in order to:

1. promote public awareness of concerns to, and commitment to the protection of stream systems and estuaries.
2. provide the concerned public with information on the goals, strategies, methods, plans, costs and benefits of SEP.
3. develop a two-way communication system to ensure that SEP reflects the reasonable views and desires of those who will be affected.
4. provide opportunities for the public to actively participate in Salmonid Enhancement projects and activities.